

**Unravelling the Role of the Dopamine D1 Receptor
in Anhedonia, Asociality, and Avolition**

By

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Abstract

Despite major advances in neuroscience, there has been limited progress in improving pharmacological treatment for neuropsychiatric disorders. Neuropsychiatric disorders are heterogenous with variance in symptoms *within* disorders and partial overlap in symptoms *between* disorders, leading to symptoms that remain untreated. To improve treatment outcomes, the Research Domain Criteria (RDoC) was created to provide researchers a framework for examining the neurobiological mechanisms underlying individual components, or dysfunctions, across disorders. Anhedonia, asociality and avolition are three excellent examples of prominent symptoms associated with poor functional outcome across disorders.

This thesis aimed to use an RDoC approach to investigate the neurobiological mechanisms underlying anhedonia, asociality and avolition. To this aim, rats with a selective genetic reduction in the dopamine D1 receptor (DAD1^{-/-} mutants) underwent a series of five studies across their lifespan that examined the role of the D1 receptor in anticipatory and consummatory pleasure, social play behaviour, socio-emotional communication, and motivation. Additionally, previous research with DAD1^{-/-} mutants used only males; this thesis expanded on that work by including both sexes and explored the possibility for sex differences in each study. The overall hypothesis was that the D1 receptor is implicated in anticipatory anhedonia, asociality and avolition, therefore DAD1^{-/-} mutant rats were predicted to exhibit reduced socio-emotional communication, anticipation of rewards, engagement in social play and exertion of effort across their lifespan.

The results showed that genetic reduction in the dopamine D1 receptor led to blunted motivation and diminished anticipatory pleasure, while consummatory pleasure and social play were not impacted. There were marked sex differences in some measures such as stability of social play across sessions and exertion of effort. Ultimately, this thesis furthered pharmacological research, implicating a clear role for the dopamine D1 receptor in motivation and anticipatory pleasure which could be targeted for treatment. In addition, this thesis supported the vital importance of including both sexes in animal research to address the strong focus of males in neuroscience research and in the role of D1 receptors in general.

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List of Abbreviations

ACC	Anterior cingulate cortex
ADHD	Attention-Deficit/Hyperactivity Disorder (ADHD)
ANOVA	Analysis of variance
APA	American Psychiatric Association
ASD	Autism Spectrum Disorder
cAMP	Cyclic AMP
CNS	Central Nervous System
CS	Conditioned Stimulus
D1R	Dopamine D1 Receptor
DA	Dopamine
DAD1 ^{-/-}	Homozygous genotype of the dopamine D1 mutant rats
DAD1 ^{+/-}	Heterozygous genotype of the dopamine D1 mutant rats
DAD1 ^{+/+}	Wildtype genotype of the dopamine D1 mutant rats
DSM	Diagnostic and Statistical Manual of Mental Disorders
EEfRT	Effort Expenditure for Rewards Task
ENU	N-ethyl-N-nitrosourea
FAA	Food anticipatory activity
fMRI	Functional Magnetic Resonance Imaging
fMRI	Functional Magnetic Resonance Imaging
HPS	Hippocampus
ICD	International Classification of Diseases
kHz	kilohertz
KO	Knockout
MDD	Major Depressive Disorder
mPFC	Medial prefrontal cortex
MRI	Magnetic Resonance Imaging
Nacc	Nucleus accumbens
NIMH	National Institute of Mental Health
OFC	Orbitofrontal cortex
PFC	Prefrontal cortex

PND	Postnatal day
RDoC	Research Domain Criteria
SD	Standard Deviation
SEM	Standard error of the mean
SPSS	Statistical Package for the Social Sciences
SSRI	Selective serotonin reuptake inhibitor
STS/STG	Superior temporal sulcus/gyrus
TPJ	Temporoparietal junction
USV	Ultrasonic vocalisations
vmPFC	Ventromedial prefrontal cortex
VTA	Ventral tegmental area
WHO	World Health Organization

Chapter 1

General Introduction

Chapter 1

General Introduction

1.1 Thesis Overview

Despite major advances in neuroscience in the last fifty years, there has been minimal progress in improving pharmacological treatment for neuropsychiatric disorders (Cuthbert, 2022; Hyman, 2013; Miller & Raison, 2023). Neuropsychiatric disorders, such as schizophrenia, major depressive disorder, and bipolar disorder, are diagnosed based on symptoms defined in the Diagnostic and Statistical Manual of Mental Disorders or International Classification of Diseases, and are currently treated with antipsychotic, antidepressant, and mood stabilising medications respectively. These medications have improved in safety and tolerability, but their molecular targets remain largely unchanged since the 1950s, and subsequently their efficacy has not substantially improved (Cuthbert, 2022; Hyman, 2013). Importantly, current pharmacological treatment does not address all symptoms present in disorders. For example, second-generation antipsychotics possess fewer side effects than their original counterparts, but they still only treat the positive symptoms of schizophrenia (e.g., hallucinations, delusions) while the negative symptoms associated with poor functional outcomes remain untreated (e.g., amotivation, anhedonia) (Barnes et al., 2014; Kirkpatrick, 2014; Strauss et al., 2019).

Attempting to address the standstill in pharmacological treatment, neuroscience research is shifting from a disorders-based approach to a domain-based approach that acknowledges the clustering of symptoms across disorders (Cuthbert, 2022; Kozak & Cuthbert, 2016). The Research Domain Criteria (RDoC) were created to improve diagnosis, prevention, and treatment of neuropsychiatric disorders (National Institute of Mental Health, 2023). Instead of conducting research based on diagnostic criteria for disorders as defined in the Diagnostic and Statistical

Manual of Mental Disorders, RDoC looks at biological and behavioural components that range from normal to abnormal and present across a multitude of disorders (Cuthbert, 2022; Kozak & Cuthbert, 2016; National Institute of Mental Health, 2023). In using a domain-based approach, we can examine the neurobiological mechanisms underlying disruptions in normal function (i.e., symptoms) that occur across disorders.

There are multiple prominent symptoms that overlap across neuropsychiatric disorders such as anhedonia, asociality, avolition, abulia and alogia (Gorwood, 2008; Hyman, 2013; Miller & Raison, 2023; Nestler & Hyman, 2010; Whitton & Pizzagalli, 2022). These symptoms are of interest because they have a significant impact on daily functioning leading to a decreased quality of life (Barkus, 2021; Kirkpatrick, 2014; Strauss et al., 2016). Importantly, these symptoms are currently not improved by pharmacological treatment; therefore, if we can determine the neurobiological mechanisms of these symptoms, then we are one step closer to improving pharmacological treatment and functional outcomes across disorders.

This thesis utilises a domain-based approach to examine the role of the dopamine D1 receptor (D1R) in anhedonia, asociality and avolition using rats with a genetic reduction in D1R. It is important to note that neuropsychiatric disorders are complex with polygenic and environmental risk factors; therefore, alterations in one neurotransmitter receptor, such as D1R, does not account for the entire picture. However, examining one receptor can give us insight into a potential mechanism underlying specific symptoms and may open opportunities for future research into environmental and polygenic interactions with the D1 receptor.

1.2 Neuropsychiatric Disorder Diagnosis and Treatment

1.2.1 A Brief History of Neuropsychiatric Disorder Diagnosis

Neuropsychiatric disorders are generally diagnosed using the Diagnostic and Statistical Manual of Mental Disorders (DSM) or the International Classification of Diseases (ICD) which provide psychologists and psychiatrists criteria for classifying and diagnosing disorders. Development of the ICD began as early as 1893 as a global guide for classifying diseases; however, psychiatric disorders were not included in the ICD until its sixth version in 1948 (Hirsch et al., 2016). The DSM was introduced in 1952 as a variant of the ICD-6 after the United States military wanted a classification system solely for psychiatric disorders for statistical purposes. The ICD is under the World Health Organization's (WHO) responsibility while the DSM is under responsibility of the American Psychiatric Association (APA). Both systems are reviewed frequently to incorporate new research and improve diagnostic criteria with current versions ICD-11 and DSM-5-TR (Tyrer, 2018; World Health Organization, 2023).

One of the key approaches of the DSM is its emphasis on diagnosing neuropsychiatric disorders based on specific symptoms. Each disorder has a set number, duration, and severity that the symptoms must reach for individuals to meet diagnostic criteria. For example, major depressive disorder (MDD) has nine symptoms (e.g., depressed mood, loss of pleasure/interest, weight change, fatigue, etc.) and an individual must have five (or more) of these symptoms present for a minimum of two consecutive weeks to meet diagnostic criteria (American Psychiatric Association, 2013). This approach provides clinicians with a standardised tool to diagnose disorders and establishes a common language which promotes consistency and reliability across clinical practices globally (American Psychiatric Association, 2013; Buckley, 2014; van Heugten – van

der Kloet & van Heugten, 2015). Although the DSM and ICD aim to provide a standardised tool for diagnosis, there are critical limitations that are discussed in the next section.

1.2.2 Limitations of Neuropsychiatric Disorder Diagnosis and Treatment

As mentioned, diagnosis criteria for neuropsychiatric disorders are based on a set number of symptoms, which means individuals do not have to display all symptoms for diagnosis. For example, as previously noted, diagnosis of MDD requires five out of nine symptoms, meaning symptoms can vary substantially across diagnosed individuals. One person diagnosed with MDD might experience hypersomnia, weight gain and psychomotor retardation while another person might experience psychomotor agitation, insomnia and weight loss. Although these symptoms are opposite in nature, both persons are diagnosed with the same disorder which is associated with heterogeneity among symptoms within disorders (American Psychiatric Association, 2013; Goldberg, 2011).

In addition to variability in symptomology *within* disorders, comorbidities occur frequently *between* disorders (AL-Asadi et al., 2015; Allsopp et al., 2019). For each disorder, the DSM outlines which other disorders frequently co-occur (American Psychiatric Association, 2013). For example, under the major depressive disorder section, the DSM-5 states that MDD frequently co-occurs with substance use disorders, panic disorder, obsessive-compulsive disorder, anorexia nervosa, bulimia nervosa, and borderline personality disorder. The common occurrence of comorbidities can make it difficult for clinicians to distinguish between disorders and suggests that disorders might not be as discrete as the DSM and ICD categorise them to be (AL-Asadi et al., 2015; Allsopp et al., 2019; Klein & Riso, 1993).

Although neuropsychiatric disorders are diagnosed as separate syndromes by clinicians, it is now commonly accepted within the neuroscience field that neuropsychiatric disorders are not

fully independent syndromes and represent clusters of symptoms that partially overlap. There are several lines of research to support this notion (Newson et al., 2021). Bromet and colleagues (2011) found that diagnoses were not stable long-term; diagnoses shifted between schizophrenia, MDD and bipolar disorder over a ten-year period, supporting the notion that there is substantial overlap between disorder symptomologies. In addition, genome-wide association studies (GWAS) show polygenic overlap across neuropsychiatric disorders including schizophrenia, major depressive disorder, bipolar disorder and autism spectrum disorder (ASD). This suggests that pathophysiology is not confined to individual disorders, instead mechanisms underlying symptoms cluster across disorders (Gandal et al., 2018; Schulze et al., 2011).

Pharmacological treatment for disorders also supports the idea that disorders are not separate syndromes. Current medication for neuropsychiatric disorders falls into several classes (e.g., antipsychotics, antidepressants, mood stabilisers, stimulants, etc.) that were created for specific disorders but are not actually specific in their usage. Antipsychotics, typically D2 antagonists, were developed to treat schizophrenia, antidepressants, generally serotonin and/or norepinephrine reuptake inhibitors, were created for major depressive disorder, and mood stabilisers were developed for bipolar disorder (Hyman, 2013; Miller & Raison, 2023). Even though these medications were developed for specific disorders, they are nonspecific in their prescription by clinicians. For example, antipsychotics were developed for schizophrenia but are used to also treat major depressive disorder, anxiety disorders, PTSD, OCD and personality disorders which is seen across other drug classes (Miller & Raison, 2023). Additionally, current pharmacology does not treat all symptoms present in disorders. The second-generation antipsychotics prescribed to treat schizophrenia only alleviate the positive symptoms (e.g., hallucinations, delusions) while the symptoms associated with poor functional outcomes remain

untreated (e.g., amotivation, anhedonia) (Barnes et al., 2014; Kirkpatrick, 2014; Strauss et al., 2019). The non-specificity of drug prescription across disorders and lack of efficacy for all symptoms in a disorder are likely caused by the heterogeneity of neuropsychiatric disorders; disorders are not fully independent syndromes and represent clusters of symptoms that partially overlap (Feczko et al., 2019).

1.3 A Shift in Neuroscience Research: Research Domain Criteria (RDoC)

1.3.1 Introduction to RDoC

To resolve the limitations of disorder-based research discussed above, the National Institute of Mental Health (NIMH) created the Research Domain Criteria (RDoC) in 2009 to provide a multidimensional conceptualisation of neuropsychiatric disorders (Hakak-Zargar et al., 2022; National Institute of Mental Health, 2023). RDoC is not meant to serve as a diagnostic guide for clinicians or replace the DSM and ICD systems; it provides researchers an organisational structure that examines psychopathology in the context of behaviour and functioning “domains” rather than individual disorders with concrete symptoms. Using a domain-based approach to understand the neurobiological mechanisms of different components of disorders can lead to more targeted treatments (Insel et al., 2010; Vilar et al., 2019).

The multidimensional approach of RDoC incorporates lifespan development, environmental factors and fundamental biobehavioural processes to assess six domains of human functioning: (1) arousal/regulatory systems, (2) positive valence systems, (3) negative valence systems, (4) systems for social processes, (5) sensorimotor systems and (6) cognitive systems (see figure 1.1). These six behavioural domains are examined on a continuum ranging from normal to abnormal rather than the DSM-based approach of a set of present or absent symptoms (Clark et al., 2017; Hakak-Zargar et al., 2022; Morris et al., 2022; National Institute of Mental Health, 2023).

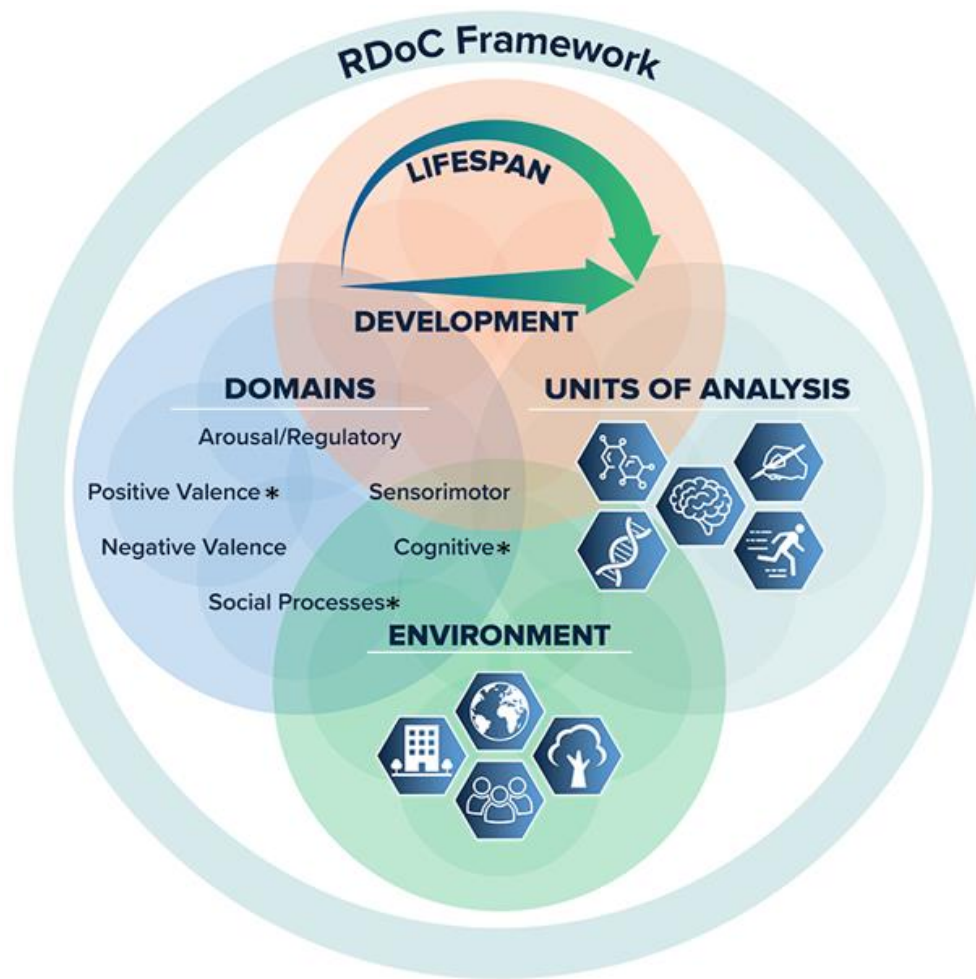


Figure 1.1. The Research Domain Criteria (RDoC) Framework retrieved from <https://www.nimh.nih.gov/research/research-funded-by-nimh/rdoc>. *indicates the domains that anhedonia, asociality and avolition fit into.

The six domains are broken down into constructs and their subconstructs which consists of different behavioural and cognitive elements, processes, mechanisms, and responses (Clark et al., 2017; Hakak-Zargar et al., 2022; Morris et al., 2022; National Institute of Mental Health, 2023). The six domains consist of the following components (see figure 1.2 for a visual depiction).

The *positive valence systems* domain focusses on approach behaviour, reward processing and the experience of pleasure through three constructs and their subconstructs. The reward valuation construct examines how individuals assign value, make decisions about rewards, and evaluate effort requirement. The reward responsiveness construct examines an individual's

response to anticipated rewards, to the presentation of a reward for the first time, and to repeated deliveries of rewards. Finally, the reward learning construct explores how individuals acquire information about stimuli and reward outcomes, and how habitual responses influence responses to reward-stimuli (Clark et al., 2017; National Institute of Mental Health, 2023).

The *negative valence systems* domain focusses on responses to aversive situations or stimuli such as fear, anxiety, and loss through five constructs. The acute threat (fear) construct examines responses to immediate threat and perceived danger while the potential threat (anxiety) construct includes responses to potentially harmful situations or stimuli. The sustained threat construct examines prolonged exposure to aversive or threatening situations over time. The loss construct focusses on deprivation of a significant object, person or situation which can be acute or chronic. Lastly, the frustrative non-reward construct involves the withdrawal or prevention of rewards (Clark et al., 2017; National Institute of Mental Health, 2023).

The *systems for social processes* domain focusses on social interactions, communication, and perceptions of others and self through four constructs. The affiliation and attachment construct examines engagement in positive social interactions and subsequent social bonds developed through interactions. The social communication construct explores receptive and productive components used in exchanging social information. The perception and understanding of self-construct focusses on self-awareness, self-monitoring, and self-knowledge while the perception and understanding of others construct focusses on perceiving and understanding others' actions and mental states (Clark et al., 2017; National Institute of Mental Health, 2023).

The *arousal and regulatory systems* domain focusses on processes involved in homeostatic regulation such as metabolism and sleep. The arousal construct examines how individuals respond to stimuli with varying levels of salience such as the startle reflex. The circadian rhythms construct

explores the biological processes that interact with the environment to regulate the sleep-wake cycle while the sleep and wakefulness construct examines the actual states of sleep and wakefulness themselves (Clark et al., 2017; National Institute of Mental Health, 2023).

The *cognitive systems* domain focusses on processing information through attention, perception, memory, language, and cognitive control. The attention construct examines how individuals focus on relevant information while filtering out irrelevant information through awareness, higher perceptual processes, and motor actions. The perception construct involves the reception and interpretation of sensory information via visual, auditory, olfactory, and somatosensory responses. The declarative memory construct focusses on the encoding and storage of facts and events while working memory involves the active maintenance and updating of task-relevant information. Lastly, the cognitive control construct involves the executive functions involved in goal-directed behaviour, response selection and performance monitoring (Clark et al., 2017; National Institute of Mental Health, 2023).

Lastly, the *sensorimotor systems* domain focusses on the control and execution of motor behaviours through four constructs. The motor actions construct examines how individuals plan, initiate, execute, and inhibit responses related to movement. The agency and ownership construct involves how individuals sense and are aware of their motor actions. The habit construct examines learned responses specifically relating to motor systems while innate motor patterns construct examines unlearned responses (National Institute of Mental Health, 2023).

Arousal and Regulatory Systems Arousal Circadian rhythms Sleep-wakefulness	Negative Valence Systems Acute threat (“fear”) Potential threat (“anxiety”) Sustained threat Loss Frustrative nonreward
Cognitive Systems Attention Perception • Visual perception • Auditory perception • Olfactory/ somatosensory/multimodal perception Declarative memory Language Cognitive control • Goal selection; updating; representation; maintenance • Response selection; inhibition/suppression • Performance monitoring Working memory • Active maintenance • Flexible updating • Limited capacity • Interference control	Positive Valence Systems Reward responsiveness • Reward anticipation • Initial response to reward • Reward satiation Reward learning • Probabilistic and reinforcement learning • Reward prediction error • Habit Reward valuation • Reward (probability) • Delay • Effort
Sensorimotor Systems Motor actions • Action planning and selection • Sensorimotor dynamics • Initiation • Execution • Inhibition and termination Agency and ownership Habit – sensorimotor Innate motor patterns	Systems for Social Processes Affiliation and attachment Social communication • Reception of facial communication • Perception of facial communication • Reception of non-facial communication • Perception of non-facial communication Perception and understanding of self • Agency • Self-knowledge Perception and understanding of others • Animacy perception • Action perception • Understanding mental states

Figure 1.2. Overview of the Research Domain Criteria (RDoC) framework highlighting the six main domains (coloured boxes), comprising 25 main constructs (bold text) with 10 constructs broken down further into 33 subconstructs (bullet points). This overview was created using the matrices outlined on the National Institute of Mental Health website.

RDoC research examines these six domains and their constructs through eight units of analysis including genes, molecules, cells, circuits, physiology, behaviour, self-reports, and paradigms (Cuthbert, 2022; National Institute of Mental Health, 2023). RDoC incorporates

lifespan development into the framework since biobehavioural elements of human functioning change throughout our lifetime; therefore it is essential that research into the neurobiological mechanisms of these functions includes developmental aspects (National Institute of Mental Health, 2023).

One advantage of the RDoC framework is that it allows researchers to move beyond disorder-defined mechanisms to examine commonalities and overlaps in symptoms (Cuthbert, 2020; Morris et al., 2022). Instead of investigating the neurobiological mechanism of major depressive disorder or schizophrenia, researchers can examine the mechanisms underlying reward processing or social communication that are implicated in these disorders. This allows for identification of shared neurobiological mechanisms that could be targeted for pharmacological treatment across disorders (Cuthbert, 2020; Morris et al., 2022). Moreover, because the RDoC framework examines domains across a continuum of functioning, we can capture the variability seen in symptoms within and across disorders (Clark et al., 2017).

1.3.2 RDoC Framework in this Thesis

As outlined previously, there are multiple prominent symptoms that overlap across neuropsychiatric disorders (e.g., schizophrenia, major depressive disorder, and bipolar disorder) including, but not limited to, anhedonia, asociality and avolition (Gorwood, 2008; Hyman, 2013; Miller & Raison, 2023; Nestler & Hyman, 2010; Whitton & Pizzagalli, 2022). Anhedonia is a decreased ability to experience pleasure, asociality is reduced interest and engagement in social interactions, and avolition is the inability to initiate and engage in goal-directed behaviours (American Psychiatric Association, 2013). These symptoms are associated with, and can even predict, poor functional outcomes in occupational and social domains. These poor functional outcomes include social isolation, unemployment, and poor quality of life (Barkus, 2021;

Kirkpatrick, 2014; Strauss et al., 2019). Most importantly, these three symptoms are not improved with current pharmacological interventions for their respective disorders (e.g., antipsychotics, antidepressants and mood stabilisers) which warrants more research into the underlying mechanisms driving the symptoms themselves, not the individual disorders (Cerveri et al., 2019; Hyman, 2013; Kozak & Cuthbert, 2016; Miller & Raison, 2023; Remington et al., 2016).

This thesis used an RDoC framework to investigate the neurobiological mechanisms of anhedonia, asociality and avolition. Since the RDoC framework does not fit neuropsychiatric disorders into discrete categories, there are not dedicated domains within the framework for the three symptoms of interest. However, we can examine symptoms within the context of the domains they correspond with the most (Morris et al., 2022; National Institute of Mental Health, 2023). Anhedonia involves the loss of pleasure during anticipation and/or consumption of reward (i.e., hedonic processing), asociality involves the loss of social behaviour, and avolition involves the loss of motivation and goal-directed behaviour. Therefore, from an RDoC perspective, this thesis examines the following domains and their associated components. Hedonic processing corresponds to the *positive valence systems* domain and its subconstructs reward responsiveness, reward learning and reward valuation. Social functioning relates to the *social processes* domain and its subconstructs affiliation and attachment, and social communication. Motivation and goal-directed behaviour correspond to both the *positive valence systems* and *cognitive systems* domains under the reward responsiveness and goal selection constructs respectively.

In terms of neurobiological elements contributing to these domains and constructs, this thesis focussed on the role of the dopamine (DA), specifically its D1 receptor. The dopamine D1 receptor (which is discussed in detail in the subsequent sections) fits into the RDoC framework's molecule unit of analysis. Additionally, in accordance with RDoC's emphasis on lifespan

development, this thesis examined the domains and their relevant constructs from the neo-natal period through adulthood. The upcoming sections explore elements and potential mechanisms underlying hedonic processing, motivation behaviour, and social processes.

1.4 Anhedonia: Dysfunction in Hedonic Processing

1.4.1 An Introduction to Hedonic Processing

Anhedonia is a decreased capacity to experience pleasure from positive stimuli or rewards, in other words, a dysfunction in reward (hedonic) processing (American Psychiatric Association, 2013). Dysfunctions in hedonic processing are seen across a multitude of neuropsychiatric disorders including schizophrenia, major depressive disorder, bipolar disorder, substance use disorder, and personality disorders (American Psychiatric Association, 2013; Lambert et al., 2018; Pizzagalli, 2022; Whitton & Pizzagalli, 2022). Additionally, deficits in hedonic processing have been reported in adolescents and children as young as three years old which highlights the need for studies across lifespan development (National Institute of Mental Health, 2023; Prabhakar et al., 2022). To understand dysfunctions in hedonic processing and their potential underlying mechanisms, we need to examine the components involved in normal hedonic processing.

Hedonic processing involves assigning value to rewards, predicting potential rewards, initiating behaviour to receive the reward, exerting effort for the reward, and experiencing pleasure (Kieslich et al., 2022). These processes can be broken down into two distinct phases based on time: an anticipatory phase *prior* to reward delivery and a consummatory phase *during* reward delivery (Berridge & Robinson, 2003; 2016; Nusslock & Alloy, 2017). The experience of hedonia, otherwise known as pleasure, differs across the phases: the anticipatory phase involves anticipatory pleasure which refers to the pleasure associated with anticipation or “wanting” of a predicted reward, while the consummatory phase involves the in-the-moment experience of pleasure in the

presence of a reward or “liking” a reward while consuming it (Berridge & Robinson, 2003). It is important to note that dysfunction in either phase can reduce reward-seeking and goal-oriented behaviour (Barch & Dowd, 2010; Sherdell et al., 2012). If an individual does not experience consummatory pleasure, then they are less likely to seek out that reward in the future and/or be motivated to work for the reward (see section 1.5 for a detailed description of motivation). Similarly, if someone does not experience anticipatory pleasure, then they are also less likely to seek out or exert effort for that reward (Sherdell et al., 2012; see figure 1.3 for a visual depiction).

As previously discussed, schizophrenia, major depressive disorder and bipolar disorder are associated with dysfunction in hedonic processing or anhedonia. The DSM and ICD do not distinguish between anticipatory anhedonia and consummatory anhedonia, but evidence suggests there might be differences in hedonic phases between disorders. Evidence from a multitude of studies found that schizophrenia is associated with intact consummatory pleasure and deficits in anticipatory pleasure. Individuals with schizophrenia experienced in-the-moment pleasure but had diminished pleasure in anticipation of future events (Gard et al., 2007; Mote et al., 2014; Smuncy et al., 2021). The distinction between these phases has not been as extensively studied in individuals with major depressive disorder and bipolar disorder, and existing studies have conflicting results. Several studies found that patients with major depressive disorder (MDD) and/or bipolar disorder reported decreased levels of both anticipatory and consummatory pleasure (Li et al., 2019; Long et al., 2022; Smuncy et al., 2021; Wu et al., 2017) while other studies found impairments only in anticipatory pleasure (Sherdell et al., 2012). Since the DSM and ICD do not distinguish between anticipatory and consummatory anhedonia in general, most studies report on anhedonia levels rather than distinctions between the two phases (American Psychiatric Association, 2013). The potential differences in hedonic processing phases between disorders

highlights the need for examining the neurobiological mechanisms of individual components of hedonic processing, rather than disorders themselves. The following sections discuss the two phases of hedonic processing and their potential neurobiological mechanisms.

1.4.2 A Deeper Dive into Anticipatory and Consummatory Phases

The anticipatory phase occurs *prior* to reward delivery and involves three major components: predicting potential rewards (“reward prediction”), an increase in behaviour prior to the predicted rewards (“anticipatory behaviour”) and engaging in reward-seeking and/or goal-oriented behaviour (“motivated behaviour”) (see figure 1.3 for a visual depiction). Reward prediction involves the acquisition of reward-related predictive cues and the ability of these predictive cues to cause expectation of a reward. Reward prediction is strongly associated with reinforcement learning as predictive cues are updated as individuals learn new reward-related information and/or are presented with mismatched expectations and outcomes (Kieslich et al., 2022; Phillips & Ahn, 2022). When a predicted reward differs from the delivered reward, this is coded as a “reward prediction error” (Hollerman & Schultz, 1998). If a large reward is predicted but the outcome is a small reward, a negative prediction error is coded; whereas, if the outcome exceeds the expected reward, a positive prediction error is coded (Hollerman & Schultz, 1998).

Anticipatory behaviour is an increase in behaviour prior to predicted events which is strongly associated with pleasurable feelings known as anticipatory pleasure (Berridge & Robinson, 2003; Kieslich et al., 2022; Phillips & Ahn, 2022). Anticipatory behaviour and subsequent anticipatory pleasure are strongly linked with reward prediction as they can both change as reward predictions update (Kieslich et al., 2022; Phillips & Ahn, 2022). For example, if an individual learns that a reward is more likely to occur, then they might engage in more anticipatory behaviour and experience more pleasure in anticipation of that reward. Dysfunction

in reward prediction and anticipatory pleasure, are seen across neuropsychiatric disorders. For example, a growing number of studies show that individuals with MDD, schizophrenia and bipolar disorder have impaired reward prediction error signals (Bakic et al., 2016; Juckel et al., 2006) and blunted anticipatory pleasure (Hägele et al., 2015; Schwarz et al., 2019).

The third component of the anticipatory phase is motivated behaviour which involves engaging in reward-seeking or goal-oriented behaviour aimed at receiving a reward or achieving a goal (Ikemoto & Panksepp, 1999; Reddy et al., 2016). Motivated behaviour and its subcomponents will be discussed in further detail in section 1.5. Of note, dysfunction in reward prediction or anticipatory pleasure can reduce motivated behaviour (Linnet, 2014). If a subject cannot predict that a cue is associated with reward delivery, they are less likely to seek out the reward, and if they do not experience pleasure in anticipation of the reward, they are also less likely to seek out the reward.

The consummatory phase occurs *after* reward delivery and involves two major components: consummatory behaviours and consummatory pleasure. The initial response to the reward in which individuals engage in consummatory behaviours targeted towards consumption of the reward such as eating, drinking, taking drugs or engaging in sexual activity (Berridge & Robinson, 2003; 2016; Nusslock & Alloy, 2017). Consummatory pleasure refers to the experience of pleasure or “liking” that occur when consuming the reward (Berridge & Robinson, 2003; 2016). Accumulating evidence from human and animal studies shows that although anticipatory and consummatory phases can influence each other, these phases are controlled by distinct neural mechanisms and neurotransmitters which are discussed in the following section (Bressan & Crippa, 2005; Berridge & Robinson, 2003).

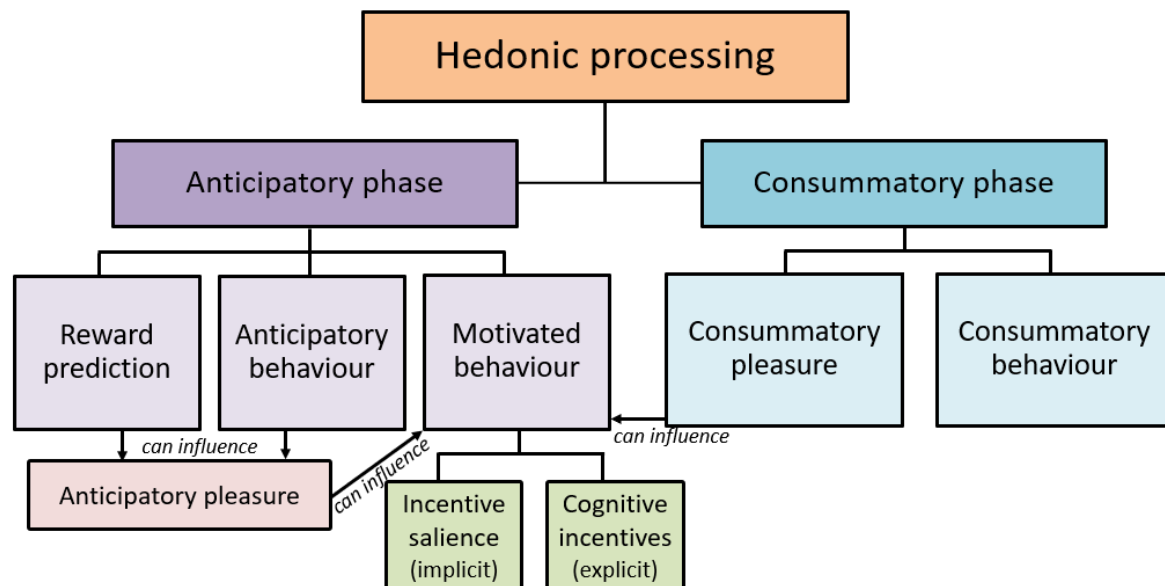


Figure 1.3. Overview of hedonic processing.

1.4.3 Neurobiological Mechanisms of Hedonic Processing

Animal studies allow researchers to manipulate neurotransmitter systems to directly examine their role in hedonic processing. In animal studies, subjects cannot explicitly rate their subjective experiences of pleasure like in human studies, therefore, researchers commonly examine anticipatory behaviour as an indirect measure of anticipatory pleasure. For rodents in particular, anticipatory behaviour includes increased locomotor activity and rearing which are *suggestive* of the subjective experience of anticipatory pleasure. However, more researchers are adopting a direct measure of pleasure in rodents known as 50kHz ultrasonic vocalisations which reflect a positive affective state (Anderson et al., 2020; Simola & Granon, 2019). It is important to note that anticipatory behaviour and pleasure are often used interchangeably in reference to the subjective experience of pleasure. For clarification, in this thesis, “anticipatory behaviour” refers to indirect measures such as locomotor movement or rearing while “anticipatory pleasure” only refers to the direct measure of ultrasonic vocalisations (Anderson et al., 2020).

Prior to the distinction between anticipatory and consummatory pleasure, it was thought that the dopaminergic system mediated all components of hedonic processing because dopaminergic neurons are activated in the presence of rewards and dopamine antagonism (or depletion) leads to diminished experiences of pleasure (Berridge & Robinson, 2003; Moran et al., 2022). However, when anticipation and consumption are investigated as distinct processes, it is clear that midbrain dopaminergic pathways mediate anticipatory processes but not consummatory processes (Berridge & Robinson, 2003; Höflich et al., 2019). Dopamine depletion and administration of dopamine antagonists (i.e., haloperidol) in rodents led to reduced anticipatory locomotor activity, while reward consumption remained intact (Salamone et al., 1991; Blackburn et al., 1987).

In terms of anticipatory processes, reward prediction, anticipatory behaviour, and subsequent anticipatory pleasure are all mediated by midbrain dopaminergic pathways, particularly projections from the ventral tegmental area (VTA) to the nucleus accumbens (Nacc) in the ventral striatum, amygdala, hippocampus (HPS) and prefrontal cortex (PFC) (Robinson et al., 2016; Reddy et al., 2016). Regarding reward prediction, imaging research in humans shows that the ventral striatum responds to predictive-cues and its activity changes depending on the type of reward prediction error experienced. A positive prediction error (e.g., when the outcome exceeds the prediction) is associated with increased striatal activity while negative prediction error is associated with decreased striatal activity (Höflich et al., 2019; Linnet, 2014). Regarding anticipatory behaviour and pleasure, neurochemical and electrophysiological studies show that extracellular dopamine signals increase during anticipation of rewards, drugs (e.g., cocaine, heroin) and sexual encounters which suggests a correlation between dopamine and anticipatory behaviour (Berridge & Robinson, 1998). Moving beyond correlations, manipulation of

dopaminergic systems with lesions, drugs, and electrical stimulation in animals alters anticipatory behaviour which shows a causal role between dopaminergic activity and reward anticipation (Berridge & Robinson, 1998; 2003; Glimcher, 2010; Höflich et al., 2019; Linnet, 2014). Nacc, anterior cingulate cortex (ACC), and ventromedial prefrontal cortex (vmPFC) activation prior to reward delivery was reduced in individuals diagnosed with schizophrenia, major depressive disorder, and those experiencing the depressive phase of bipolar disorder, suggesting that phasic striatal dopamine signalling is crucial for anticipatory processes (Whitton et al., 2015).

On the other hand, studies have shown that consummatory processes are not affected by methods that reduce or diminish anticipatory behaviour; activation or suppression of midbrain dopaminergic systems via dopamine antagonists, Nacc lesions, and amphetamine injections do not alter consummatory behaviour or pleasure (Berridge & Robinson, 1998; 2003; Höflich et al., 2019). Instead, accumulating evidence suggests that consummatory behaviour and pleasure is mediated by serotonergic, opiate and endocannabinoid systems in the Nacc, orbitofrontal cortex (OFC), and ventral pallidum. Opioid agonism in animals increases “liking” of rewards such as sweets, social interaction, and sexual encounters, and serotonin blockage results in reduced consummatory pleasure (Berridge & Robinson, 2003; Höflich et al., 2019; Li et al., 2015). Serotonin’s role in consummatory pleasure could explain why some selective serotonin reuptake inhibitors (SSRIs) slightly improve anhedonia symptoms; however, because anhedonia involves two components controlled by distinct mechanisms, SSRIs do not significantly improve hedonic processing dysfunction (Hägele et al., 2015; Li et al., 2015).

In summary, it was previously thought that dopaminergic systems mediated hedonic processing as a whole; however, it is now commonly accepted that midbrain dopaminergic pathways originating in the VTA moderate anticipatory processes, while serotonergic,

endocannabinoid and opioid systems regulate consummatory processes. The following section takes a closer look at dopamine's involvement in the anticipatory phase.

1.4.4 The Role of the Dopamine D1 Receptor in the Anticipatory Phase

While it is now well accepted that dopamine is involved in anticipatory processes, it is less clear which dopamine receptor mediates this effect. Dopamine exerts its action by binding to one of its five G-protein coupled dopamine receptors: D1-like (D1 and D5) and D2-like (D2, D3, D4) which initiates secondary signalling cascades that activate corresponding functions. Dopamine binding can cause both excitatory and inhibitory synaptic transmission depending on the receptor it binds to (Jones-Tabah et al., 2022; Nicola & Malenka, 1997; Zhang et al., 2009). D1-like receptors activate adenylyl cyclase and enhance cyclic AMP (cAMP) production, ultimately resulting in excitation of postsynaptic cells. D2-like receptors, on the other hand, inhibit adenylyl cyclase and cAMP production, resulting in inhibition of postsynaptic cells (Bhatia et al., 2022). Dopaminergic neurons fire via two modes: tonic (background, sustained) firing or phasic (bursting, transient) firing. Phasic dopamine activity occurs in response to external stimuli, such as reward-cues, and leads to greater dopamine release. Tonic activity, on the other hand, appears to be involved in basal functions and leads to a smaller dopamine release than phasic signalling (Beeler et al., 2010; Budygin et al., 2020; Dreyer et al., 2010). Phasic and tonic dopamine signalling might differentially activate D1-like and D2-like receptors; tonic activity appears to activate D2-like receptors while phasic activity mainly activates D1-like receptors, but more research is needed to elucidate these patterns (Budygin et al., 2020; Dreyer et al., 2010; Grieder et al., 2012).

Dopamine receptors are present throughout the body, including the central nervous system, kidneys, heart and gastrointestinal tract. The D1 receptor is the most abundant dopamine receptor in the central nervous system with high density in the Nacc, caudate-putamen, substantia nigra and

olfactory tubercle, and lesser density in the amygdala, PFC, thalamus and hypothalamus (Bhatia et al., 2022; Cadet et al., 2010; Martel & McArthur, 2020). Given the differences between receptor signalling cascades, modes of dopaminergic firing and location, it is not surprising that that receptors are implicated in different cognitive and behavioural functions. Regarding the anticipatory phase, emerging evidence suggests that the dopamine D1 receptor is involved in mediating reward prediction, anticipatory behaviours, and anticipatory pleasure.

In terms of pharmacological manipulations of the D1 receptor, pre-treatment with varying doses of the D1 antagonist SCH23390 significantly reduced anticipatory behaviour in expectation of a food reward (i.e., food anticipatory activity (FAA)) in mice (Liu et al., 2012). Administration of SCH23390 directly into the Nacc of rats reduced locomotor activity and rearing, while food intake and latency to begin food intake was unaffected (Baldo et al., 2002). Additionally, anticipatory pleasure for palatable food, measured via 50 kHz ultrasonic vocalisations, were dose-dependently reduced by SCH23390 in rats (Buck et al., 2014).

In terms of genetic manipulations of the D1 receptor, dopamine D1R knockout mice (D1R KO^{Vgat-Cre⁺}) displayed attenuated anticipatory behaviour with limited jumping, rearing, and walking compared to control animals (Assali et al., 2021; Gallardo et al., 2014). Gallardo et al. (2014) also showed that pharmacological activation of the D1 receptor produced anticipatory behaviour in control animals but not in knockout animals. Additionally, D1R knockout mice showed no reward prediction behaviour and no Nacc activity during the anticipatory phase, but there was activity similar to the control animals during reward consumption (Tran et al., 2005). Together, these studies suggest a role of the D1 receptor reward prediction, anticipatory behaviour, and anticipatory pleasure during the anticipatory phase which is investigated further in this thesis.

1.5 Avolition: Dysfunctions in Motivation

1.5.1 An Introduction to Motivational Processes

Avolition is a reduced motivation to initiate and engage in goal-directed behaviour, in other words a dysfunction in motivational processes (American Psychiatric Association, 2013). Dysfunction in motivational processes is seen across neuropsychiatric disorders including schizophrenia, major depressive disorder, bipolar disorder and attention-deficit/hyperactivity disorder (ADHD) (American Psychiatric Association, 2013; Oram et al., 2020; Yang et al., 2021). Deficits in motivation have been reported in children and adolescents which highlights the need for studies across lifespan development (National Institute of Mental Health, 2023; Robichaud et al., 2020). To understand dysfunctions in motivational processes and their potential underlying mechanisms, we need to examine normal motivational processes.

As introduced in the previous section, motivated behaviour involves engaging in reward-seeking or goal-oriented behaviour aimed at receiving a reward or achieving a goal (Ikemoto & Panksepp, 1999; Reddy et al., 2016). This behaviour includes behavioural activation, exertion of effort, and task engagement (e.g., pressing a lever for a sugar pellet). Reward-seeking behaviour in this phase is driven by incentive salience, the implicit motivational component of reward, and cognitive incentives which are the explicit desires for reward. Both incentive salience and cognitive incentives are strongly associated with reinforcement learning as desires for rewards are updated following reward outcomes (Berridge & Robinson, 2003; Berridge et al., 2009). It is important to note that incentive salience differs from anticipatory pleasure; anticipatory pleasure is the subjective hedonic feeling experienced in anticipation of a reward while incentive salience indicates how much an individual wants a reward (Berridge & Robinson, 2003; Berridge et al., 2009). Although different components of the anticipatory phase, these two components can

influence each other; anticipatory pleasure can alter motivation to obtain rewards and motivation for a reward can impact the anticipatory pleasure experienced (Berridge & Robinson, 2003; Sherdell et al., 2012).

Incentive salience and cognitive incentives provide individuals with the capacity to choose between outcomes. For instance, if presented with choosing between monetary or food rewards, the incentives attributed to each reward impact the outcome someone chooses (Berridge & Robinson, 2003; Berridge et al., 2009). Moreover, motivation extends to decisions between outcomes of varying values. Generally, when making decisions, individuals prefer choosing larger rewards over smaller rewards. However, the decision becomes more complex when effort is required to achieve the outcome (effort-based decision-making) (Kurniawan et al., 2011). Effort refers to the cognitive (e.g., paying attention to a stimulus, computing a math problem) and/or physical (e.g., pressing a button on a keyboard) demand an individual must exert to obtain an outcome (Chong et al., 2016).

Effort-based decision-making involves cost-benefit calculations in which an individual weighs the amount of effort exertion required (cost) against the reward value (benefit) (Chong et al., 2016; Reddy et al., 2016). As previously discussed, exertion of effort occurs during reward-seeking and goal-oriented behaviours, and is driven by explicit and implicit incentives, meaning that motivation influences how much effort individuals are willing to exert for a desired goal or reward (Kurniawan et al., 2011; Reddy et al., 2016). An individual who has less motivation for a reward is less likely to exert effort and engage in tasks targeted towards that outcome which can lead to reduced goal-driven behaviour. For example, if someone is not highly motivated to get a good mark on their assignment, then they likely would not spend a lot of time on the assignment;

whereas someone who really wants to do well is more likely to put a lot of effort into completing their assignment.

As noted earlier, schizophrenia, major depressive disorder and bipolar disorder are associated with dysfunction in motivation, specifically seen in decreased willingness to exert effort (Lee et al., 2015). Schizophrenia, MDD and bipolar disorder are associated with decreased exertion of both physical and cognitive effort leading to reduced engagement in goal-directed behaviour (Reddy et al., 2016, Barnes et al., 2014, Calabrese et al., 2014). When examining disorders separately, patients with schizophrenia (Barch et al., 2014), major depressive disorder (Treadway et al., 2012) and bipolar disorder (during the depressive phase) (Hershenberg et al., 2016; Zou et al., 2020) displayed decreased exertion of effort compared to healthy controls in Effort Expenditure for Rewards Tasks (EEfRT). When comparing disorders, participants with either schizophrenia, MDD, or bipolar disorder (during the depressive phase), all reported lower levels of explicit motivation than healthy controls which was correlated with a decreased exertion of effort in highly demanding tasks. On tasks with low- or no effort requirement, participants with MDD and bipolar exerted similar amounts of effort as controls; whereas patients with schizophrenia exerted less effort in both low- and high-demand tasks (Barch et al., 2014; Yang et al., 2021). The dysfunction in motivation resulting in reduced goal-directed behaviour, highlights the need for using an RDoC framework to examine the neurobiological mechanisms of motivational elements. The following section discusses the potential mechanisms of motivation and effort-based decision-making.

1.5.2 Neurobiological Mechanisms of Motivation

Like reward prediction and anticipatory behaviour, animal studies allow researchers to manipulate neurotransmitter systems to directly examine their role in motivation. In animal

studies, subjects cannot explicitly state their desire to obtain a reward or achieve a goal like in human studies, therefore, researchers commonly examine initiation of reward-seeking behaviour or exertion of effort for a primary reinforcer such as food, sexual or social encounters, and drugs. For rodents in particular, researchers examine how much effort subjects will exert via lever pressing, barrier climbing or wheel running (Berridge & Robinson, 1998; Salamone et al., 2022). Additionally, subjects are often given choices between tasks with varying effort levels such as pressing a lever for a sucrose treat or eating standard chow out of a bowl on the floor. These measures of effort exertion give researchers insight into the animals' motivational state (Salamone et al., 2022; Treadway et al., 2012; Young & Markou, 2015).

The midbrain dopaminergic system is implicated in reward-seeking and goal-oriented behaviour. Dopamine antagonism and depletion, primarily in the Nacc, impairs reward-seeking behaviours (e.g., behavioural activation, effort exertion), while consummatory behaviours (e.g., eating, drinking) remain intact (Berridge & Robinson, 2003; Salamone et al., 2015). Administration of dopamine antagonists reduced entering a niche with food (Baldo et al., 2002; Blackburn et al., 1987), pressing the lever for food and drug rewards (Salamone et al., 2015), and preparing for sexual encounters (Pfaus & Phillips, 1991) but did not alter amount of food or drugs consumed nor the engagement in sexual behaviour.

In terms of specific reward-seeking behaviours, behavioural activation, exertion of food and task engagement are all mediated by midbrain dopaminergic systems originating in the VTA and projecting to the Nacc and PFC, and pathways from the substantia nigra to the ACC (Berridge & Robinson, 2003; Ikemoto & Panksepp, 1999; Lee et al., 2015). Dopamine depletion and antagonism in the Nacc reduces activation of reward-seeking behaviour (Salamone et al., 2007, 2015), exertion of effort (Salamone & Correa, 2012; Salamone et al., 2015) and transfer of

incentive salience to instrumental responding (Lex & Hauber, 2008; Salamone et al., 2015). Additionally, decreasing dopamine activity in the Nacc impaired responding on high fixed ratio schedules of reinforcement (Aberman & Salamone, 1999), biased rodents toward low-effort decisions in a T-maze effort-choice task (Mai et al., 2012; Salamone et al., 2015), and reduced the willingness to work for the preferred food reward in an effort-choice paradigm (Salamone et al., 1991; Salamone & Correa, 2012).

Regarding neuropsychiatric disorders, individuals with schizophrenia showed abnormal ventral striatum activation; they showed more activation towards irrelevant stimuli than motivationally salient stimuli (Ziauddeen & Murray, 2010) while individuals with MDD showed overall reduced activation towards motivationally relevant stimuli (Gradin et al., 2011). Parkinson's disease is associated with a loss of dopaminergic neurons, specifically in the substantia nigra, which gives researchers the opportunity to examine how dopamine depletion in humans impacts cognition and behaviour (Cramb et al., 2023). In terms of motivated behaviour, unmedicated Parkinson's disease patients (depleted dopamine levels) exerted significantly less effort and over-estimated their exertion levels compared to medicated patients with supplemented dopamine levels which suggests that dopaminergic neurons are crucial for motivated behaviours, specifically exertion of effort and behavioural activation (Padmanabhan et al., 2023). These results provide strong support for midbrain dopaminergic pathways regulating motivated behaviour including behavioural activation, exertion of effort and task engagement. The following section takes a closer look at dopamine's involvement in motivated behaviour.

1.5.3 The Role of the Dopamine D1 Receptor in Motivation

As noted in section 1.4.4, dopamine exerts its action through five receptors and evidence suggests that the D1 receptor is implicated in motivated behaviour. Pharmacological manipulations

of the D1 receptor with D1 antagonists and injection of D1 antagonists directly into the Nacc reduced lever pressing for sucrose rewards while leaving the consumption of sucrose intact (Salamone et al., 1991; Salamone & Correa, 2012; Smith-Roe & Kelley, 2000). Injection of D1 antagonists into the core or shell areas of the Nacc decreased the transfer of incentive salience to instrumental responding (Lex & Hauber; Salamone et al., 2015). When choosing between low effort/low reward and high effort/high reward options in T-maze barrier-choice and lever pressing choice tasks, rats reliably choose the high effort/high reward option. However, D1 antagonism shifted subjects towards choosing the low effort/low reward choice (Cagniard et al., 2006; Salamone et al., 2015; Salamone & Correa, 2012).

In regard to genetic manipulations of the D1 receptor, D1 knockout mice showed diminished motivation to drink water when water deprived (Nakamura et al., 2014), reduced behavioural activation with food reward (Smith et al., 1998), and deficits in acquisition of lever pressing (El-Ghundi et al., 2003). D1 knockout mice took significantly longer to acquire lever pressing for sucrose reward and overall pressed the lever significantly less than heterozygous and wildtype subjects (El-Ghundi et al., 2003). Caine et al., (2007) found that D1 knockout mice failed to acquire self- administration of cocaine which suggests that the D1R regulates motivation to exert effort for rewards.

Taken together, these pharmacological and genetic manipulation studies suggest a role of the D1 receptor in motivated behaviours that are related to incentive salience, specifically behavioural activation, exertion of effort and task engagement which is investigated further in this thesis.

1.6 Asociality: Dysfunction in Social Processes

1.6.1 An Introduction to Social Processes

Asociality is a diminished engagement in social interactions and reduced desire for social contact, in other words a dysfunction in social processes (Brown et al., 2007; Strauss et al., 2019). As with anhedonia and avolition, asociality is seen across a wide variety of neuropsychiatric disorders including, but not limited to, schizophrenia, MDD, bipolar disorder, ASD and anxiety disorders (American Psychiatry Association, 2013; Kennedy & Adolphs, 2012). To understand dysfunctions in social processes, and their potential underlying mechanisms, we need to examine normal social processes. Social processes refer to how individuals perceive, recognise, interpret and respond to social cues and stimuli (Beaudoin & Beauchamp, 2020). Social processes can be broken down into three interdependent components: social perception, social cognition, and social behaviour (see figure 1.4 for a visual depiction).

Social perception involves detecting and identifying social stimuli and cues, such as facial expressions, body language, movement, and affective states, and is foundational for higher-level social processes. Social cognition refers to the mental processes involved in interpreting and understanding social cues and stimuli. The most studied elements of social cognition are theory of mind, empathy, social cue perception, judgment, and face and emotion processing (Adolphs, 2010; Andrić & Bojović, 2015; Kennedy & Adolphs, 2012). Social behaviour refers to the actions that individuals partake in when engaging with others or in other words the manifestation of cognitive and affective processes. It encompasses social communication, cooperation and prosocial behaviour (e.g., helping someone in need) (Adolphs, 2001; Beaudoin & Beauchamp, 2020). These three components, social perception, cognition and behaviour, are interdependent. Social perception provides information which undergoes cognitive processes and in turn impacts how an

individual acts socially. However, all components can impact each other; how an individual interacts with another being can influence their later perception of social cues and how they interpret the cues (Hanegraaf et al., 2021; Kennedy & Adolphs, 2012; Lobo et al., 2023).

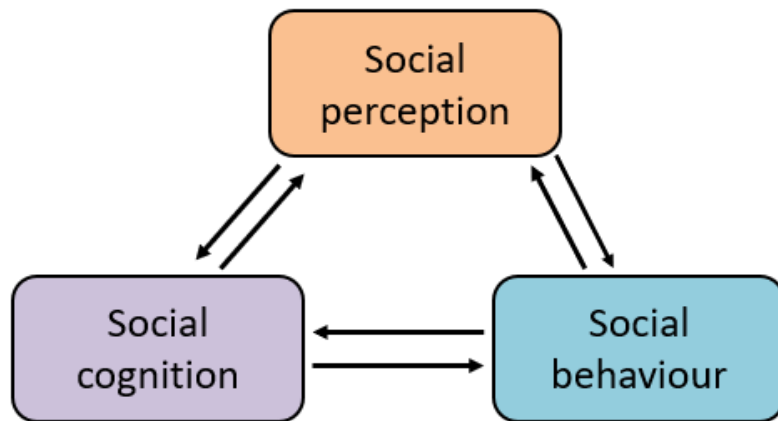


Figure 1.4. Overview of social processes.

Social perception, cognition and behaviour begin developing in infancy with a focus on infant-parental interactions, shifting to peer play in childhood and continuing throughout adulthood with a focus on peer and romantic relationships (Burnett et al., 2011; Kilford et al., 2016; Nelson et al., 2016). Social development is interdependent; there are correlations between childhood social cognition and behaviour and adult social functioning. Children who engage in cooperative play and prosocial interactions have more positive social interactions and socio-emotional cognitive skills (e.g., empathy, communication) in adulthood (Nelson et al., 2016; Soto-Icaza et al., 2015). Moreover, there are associations between children who display deficits in social play behaviour, such as avoidance or aggression, and dysfunction in adult social processes such as isolation and withdrawal (Nelson et al., 2016; Soto-Icaza et al., 2015). More recently, it has been proposed that there are distinct periods (“sensitive periods”) during development that contribute to long-term patterns of neural circuits underlying social processes which subsequently impact lifelong social perception, cognition and behaviour. One example involves face processing in

which brain regions associated with face processing learn to respond to social stimuli during infancy and early childhood, and these responses become more fine-tuned in early adulthood (Nelson et al., 2016). If animals are isolated from conspecifics during their infancy and early childhood period, then their face perception is delayed in adulthood, suggesting there are sensitive periods for development of social processes (Nelson et al., 2016).

Concerning neuropsychiatric disorders, schizophrenia, major depressive disorder and bipolar appear to have deficits in social cognition which subsequently alters their social behaviour (Andrić & Bojović, 2015; Kennedy & Adolphs, 2012). Patients diagnosed with schizophrenia showed deficits in monitoring others' intentions (theory of mind) and tended to misinterpret social cues as negative, which was associated with social isolation (Andrić & Bojović, 2015). Additionally, individuals with schizophrenia displayed lower levels of social cognitive skills on self-reports and clinician assessments in terms of attribution, nonverbal sensitivity, intentional bias, emotion recognition and awareness of social inferences (Silberstein et al., 2018). Individuals diagnosed with MDD, and bipolar disorder showed deficits in emotion recognition, theory of mind and social judgment. They were more likely to classify neutral social cues as negative reflections of themselves and misinterpret neutral facial expressions (e.g., interpreting a neutral facial expression as anger towards themselves) (Andrić & Bojović, 2015; Gillissie et al., 2021; Grove et al., 2016). Deficits in social processes in these disorders is associated with poor social outcomes, most notably decreased ability to maintain peer and romantic relationships (Grove et al., 2016).

The dysfunction in social cognition resulting in reduced interest and engagement in social interactions, highlights the need for using an RDoC framework to examine the neurobiological mechanisms of social processes. The following section discusses the potential mechanisms of social processes.

1.6.2 Neurobiological Mechanisms of Social Processes

The neurotransmitters and brain regions involved in social processes have been deemed the “social brain” by neuroscientists (Beaudoin & Beauchamp, 2020; Homberg et al., 2016). Key regions of the social brain that communicate with each other include the amygdala, medial prefrontal cortex (mPFC), temporoparietal junction (TPJ), superior temporal sulcus/gyrus (STS/STG), posterior cingulate cortex, temporal pole and insula (Beaudoin & Beauchamp, 2020; Tso et al., 2018). The “social brain” is composed of interconnected regions and lesion and fMRI studies show that regions are implicated in specific components of social processes. For example, the vmPFC is associated with processing the mental states of others, face processing and eye gaze are involved with activity in the STS/STG, and the fusiform gyrus and TPJ are involved with mental perspective of others (e.g., theory of mind) (Adolphs, 2009; Sherwin et al., 2019, Tso et al., 2018).

Regarding neuropsychiatric disorders, most studies involving social cognition and behaviour examine face processing due to the relative ease of neuroimaging during face processing tasks. Individuals with schizophrenia displayed decreased activity in the mPFC and amygdala during theory of mind, emotion recognition and self-perception tasks (Brunet-Gouet & Decety, 2006). Additionally, they showed dysfunction in communication between the PFC and amygdala during face processing, specifically for recognising fearful faces (Mier & Kirsch, 2016). Patients with MDD displayed increased activity in the ACC and amygdala when processing negative faces, which was associated with a negativity bias (Cusi et al., 2011). There was decreased connectivity between the OFC and ACC suggesting a dysfunction in cognitive control of processing emotional faces. On the other hand, patients with bipolar disorder showed increased amygdala activity in response to happy and sad faces (Cusi et al., 2011). Taken together, there is dysfunction in brain

regions associated with social cognition, specifically processing emotional faces in individuals with schizophrenia, MDD and bipolar disorder. More research is needed to elucidate potential alterations in “social brain” regions in individuals with social dysfunctions.

As the “social brain” is comprised of networks across brain regions, it is not surprising that multiple neurotransmitter systems are implicated in social cognition and behaviour including dopamine, serotonin, oxytocin (Duerler et al., 2022; Kennedy & Adolphs, 2012). Dopamine is involved in emotion recognition, affiliation and attachment, and the motivational value of social interactions (Kennedy & Adolphs, 2012; McCall & Singer, 2012; Rosenfeld et al., 2011). Amphetamine, which increases dopamine levels, is associated with increased talkativeness and improved detection of facial emotion processing across emotions (Wardle et al., 2012). A polymorphism in the dopamine transporter (DAT) gene that results in increased striatal dopamine levels was related to increased approach towards happy faces compared to participants without the polymorphism (Enter et al., 2012). Unmedicated Parkinson's patients with depleted dopamine levels, were worse at recognising facial emotions than individuals with supplemented dopamine levels (Prenger et al., 2020). Additionally, patients reported improvements in social cognitive dysfunction with levodopa treatment that supplements dopamine levels (Wootton, 2018).

A multitude of animal research shows that dopaminergic activity is crucial for social processes. In a prosocial paradigm for testing empathy in animals, rats showed increased dopamine release when their conspecific was spared from harmful foot shocks and decreased dopamine release during cues predicting harm towards the conspecific. This showed that prosocial empathetic behaviour was associated with dopamine release (Lichtenberg, et al., 2018). Gunaydin and colleagues (2014) found that VTA pathways to the Nacc encoded social interaction and optogenetic stimulation of VTA neurons increased social interaction. Individual differences in

mice social activity were mirrored by differences in midbrain dopaminergic signalling; mice who engaged in more social interactions had larger dopaminergic firing (Torquet et al., 2018). Recent evidence shows a distinction between phasic and tonic dopamine release in relation to social processes. In a social learning task for rats, there was increased phasic dopaminergic firing in the Nacc and mPFC when presented with novel social stimuli (De Leonibus et al., 2006). Sotoyama et al. (2022) also found that an acute increase in dopamine signalling was associated with increased social interactions while sustained increase in dopaminergic activity reduced social interactions. These results support the role of dopamine in social processes and highlight a potential distinction in modes of dopaminergic firing during social processes.

In summary, social processes are regulated by the “social brain” which is comprised of a complex connection between brain regions with involvement from multiple neurotransmitter systems. Dopaminergic signalling is associated with multiple social processes including empathy, face emotion processing and engagement in social interaction. There appears to be distinctions between phasic and tonic dopaminergic release which calls for a closer look at the potential receptors mediating dopamine’s role.

1.6.3 The Role of the Dopamine D1 Receptor in Social Processes

The dopamine D1 receptor has been implicated in social cognition and behaviour. In a positron emission tomography (PET) study in humans, D1 receptor binding in the amygdala and mPFC regions associated with social cognition, was positively correlated with social desirability and negatively correlated with physical aggression scores, suggesting a role of the D1 receptor in affiliative social processes (Plavén-Sigra et al., 2014). There are some contradictory results between animal studies examining the D1 receptor and social processes. D1 receptor activation in the Nacc prevented pair-bond formation in prairie voles, but once pair bonds were established D1R

signalling in the Nacc was necessary for maintenance of the bonds (Aragona et al., 2005). Pair bonding in monkeys was associated with increased D1R binding in the lateral septum but not Nacc, caudate, putamen or ventral pallidum which contradicts the finding in prairie voles (Hostetler et al., 2016). Optogenetic stimulation of D1 receptors in the Nacc increased social interaction in mice (Gunaydin et al., 2014). D1 agonism with SKF81297 and SKF82958 dose-dependently increased social recognition and the D1 antagonist SCH23390 reduced social recognition (Di Cara et al., 2007). Looking closer at types of social behaviour during interaction, D1 agonism increased aggression and submission while antagonism also increased submissive behaviours (Antunes et al., 2022).

There are few studies that have examined how genetic alterations of the D1 receptor impact social processes. Dopamine transporter (DAT) knockout mice have reduced D1-like receptors in the striatum and displayed dysfunctions in social interaction, specifically in increased reactivity and aggression with stereotyped behavioural patterns (Rodríguez et al., 2004). Homberg et al. (2016) utilised rats with a 30-50% genetic reduction in D1R and found that they displayed deficits in social cognition, specifically with decreased active social interaction, sociability, interest in novel social stimuli, scent marking and social communication. Taken together, these pharmacological, optogenetic, and genetic manipulation studies suggest a role of the D1 receptor in social processes linked to social cognition and behaviour such as social interaction, empathy, emotion processing, and bond formation.

1.7 Animal Models to Investigate the Dopamine D1 Receptor

1.7.1 RDoC Approach to Animal Models

Animal models, especially rodent models, provide researchers the opportunity to examine causal relationships between genetic and environmental manipulations and subsequent changes in

cognition and behaviour which cannot be studied in humans (Phillips & Roth, 2019). Animal models can elucidate the neurobiological mechanisms (e.g., epigenetic factors, neurochemical imbalances, neural circuitry disruptions) underlying various dysfunction seen in neuropsychiatric disorders (National Institute of Mental Health, 2023). In utilising animal models, it is important to focus on examining components of functioning rather than modelling distinct neuropsychiatric disorders. As neuropsychiatric disorders are heterogenous within human populations and symptoms often include subjective experiences (e.g., hallucinations, feelings of worthlessness), animal models cannot represent entire disorders (Nestler & Hyman, 2010). For example, rather than modelling schizophrenia with animals, researchers can look at the mechanisms underlying specific functions across disorders such as motivation, anticipation, or social communication.

In line with the RDoC framework's focus on dysfunction in behaviour and cognitive processes, rather than neuropsychiatric disorders, transgenic animal models should be used to examine how genetic alterations in specific neurotransmitters and neural circuits impact behavioural and cognitive processes (Anderzhanova et al., 2017). It is important to note that neuropsychiatric disorders are complex with polygenic and environmental influences, therefore, alterations in one neural circuit or receptor in an animal model, does not account for the entire pathophysiology of the disorder. However, examining one neural circuit may give us insight into a potential mechanism underlying specific dysfunctions and provides direction for future research into environmental influences and polygenic interactions with this circuit. In summary, animal models can provide insights into the neurobiological mechanisms underlying human functioning.

1.7.2 Limitations with D1 Pharmacological and Knockout Strains

As discussed in the preceding sections, studies using pharmacological manipulations and genetic knockout animals suggest a role of the dopamine D1 receptor in anticipatory pleasure,

social processes, and motivated behaviour, however, there are limitations. There are currently no drugs available that exhibit substantial selectivity for the D1 receptor over the D5 receptor, moreover some D1R - binding drugs also show affinity for non-dopamine receptors (Homberg et al., 2016; Jones-Tabah et al., 2022; Ramos et al., 2005). Due to the selectivity limitation of pharmacology, genetic manipulations of the D1 receptor provide the opportunity to isolate the role of the D1 receptor. Most of the research to date that genetically reduced D1 receptor levels, used D1 knockout rats or mice that have complete inactivation of the receptor. Knockout animals address the pharmacological selectivity issue, however there are still several limitations to consider. Complete inactivation of a gene can lead to unexpected compensatory mechanisms during development which makes it difficult to interpret whether functional consequences are due to the knocked down gene itself or compensatory mechanisms (Nelson, 1997). Additionally, complete inactivation does not fully translate into humans because complete inactivation is rare in humans (Homberg et al., 2016; National Institute of Mental Health, 2023). To date, five single nucleotide polymorphisms (SNPs) have been identified in human D1 receptors (hD1R; T37P, T37R, R50S, S199A, and A229) which are associated with alterations in receptor expression. Little is currently known about how these SNPs impact cognition and behaviour, but it is proposed that they are linked to dysfunctions associated with neuropsychiatric disorders (Al-Fulaij et al., 2008). These SNPs are not a complete inactivation of the D1R gene which is important when translating animal research to humans.

Given the limitations of pharmacological and knockout research on the D1 receptor, investigating the role of the D1 receptor in anticipatory pleasure, social processes and motivated behaviour using animals with a genetic reduction, not complete inactivation, in the D1 receptor is warranted and might be more translational to humans.

1.7.3 An Introduction to the DAD1^{-/-} Rat Model

To overcome limitations, Smits et al. (2006) used N-ethyl-N-nitrosourea (ENU) mutagenesis to develop a rat with a genetic reduction in the dopamine D1 receptor (DAD1^{-/-} mutants). The ENU mutagenesis resulted in a single point mutation which changed hydrophobic isoleucine into hydrophilic serine at position 116 (the second transmembrane domain) which leads to instability in the receptor. These DAD1^{-/-} mutant rats have 50% reduction in transmembrane insertion of D1 receptors in the PFC, substantia nigra, hippocampus, caudate putamen, and 20% in the olfactory tubercle (Homberg et al., 2016). Due to the reduced D1R function rather than complete inactivation, DAD1^{-/-} mutants provide a model of down-regulated D1R which is likely more translational to humans and leads to fewer compensatory mechanisms than knockout models.

DAD1^{-/-} mutants have normal general motor, sensory, and neurological characteristics. Likewise, they have normal exploratory and anxiety-like behaviours exhibited in the open-field and elevated plus maze paradigms respectively (Homberg et al., 2016; Squire, 2016; Vanden Broeke, 2016). DAD1^{-/-} mutants have lower bodyweight than wildtype rats but do not differ in home cage food consumption when accounting for body weight. In regard to hedonic processing and social processes, adult DAD1^{-/-} mutants displayed impaired anticipatory pleasure, normal consummatory pleasure, and reduced social interaction (Homberg et al., 2016; Squire, 2016; Vanden Broeke, 2016). Hedonic processing was examined with an anticipatory locomotor paradigm that measures anticipation of standard lab chow (Vanden Broeke, 2016). It is possible that DAD1^{-/-} mutants need a higher incentive value to induce anticipatory pleasure, so it is important to examine anticipation and consumption of high-value rewards such as sucrose solution, sweet treats, or social interaction. Regarding motivation, DAD1^{-/-} mutants did not acquire

lever pressing, therefore further investigation into motivational processes (e.g., incentive salience, exertion of effort) was not examined (Squire, 2016).

It is also important to acknowledge that all previous research with DAD1^{-/-} mutants used only male subjects. There are sex differences in prevalence, symptom severity and age of onset across neuropsychiatric disorders, and sex hormones, specifically estradiol, have a facilitating effect on dopaminergic signalling, therefore it is paramount that future research incorporate both sexes (Barth et al., 2015; Hidalgo-Lopez & Pletzer, 2017; Jacobs & D'Esposito, 2011). In line with the RDoC framework, research using animal models, in this case DAD1^{-/-} mutants, should examine a continuum of behaviour and cognition across lifespan development (National Institute of Mental Health, 2023).

1.8 Overall Aims and Predictions

This thesis investigates neurobiological mechanisms underlying anhedonia, asociality and avolition via a RDoC approach. In a series of five studies, this thesis examines the role of the dopamine D1 receptor in anticipatory and consummatory pleasure, social processes and motivation. The overall hypothesis is that the D1 receptor is implicated in these three functions; therefore it is predicted that rats with a genetic reduction in the D1 receptor, DAD1^{-/-} mutants, will exhibit reduced socio-emotional communication, anticipation of rewards, engagement in social play, and exertion of effort across their lifespan.

The hypothesis is examined in a series of experiments across the lifespan of DAD1^{-/-} mutants and DAD1^{+/+} wildtypes (see figure 1.5 for a visual timeline). Additionally, to address the strong focus on males in neuroscience research and in the role of D1 receptors in general, both sexes are included in all experiments in this thesis. Chapter 2 aims to examine offspring development across various factors and replicate previous findings that DAD1^{-/-} mutants display

decreased socio-emotional communication, specifically when isolated from their dam and litter. Chapter 3 examines the role of the D1 receptor in motivation via exertion of effort for sucrose rewards in an operant-chamber lever pressing paradigm. After subjects acquire lever pressing, effort requirement for rewards increases and subjects make choices between high effort-high reward and low effort-low reward options. In Chapter 4, potential explanations for the autoshaping impairments seen in Chapter 3 are explored in a Pavlovian Instrumental Transfer paradigm. Chapter 5 examines how DAD1^{-/-} mutants and wildtype controls anticipate and consume a sucrose solution. Building upon this, Chapter 6 explores how DAD1^{-/-} mutants and wildtype controls anticipate and engage in the highly rewarding activity of juvenile social play. The analysis includes the quantity and quality (e.g., sequence of play, vocalisations during play, etc.) of social play.

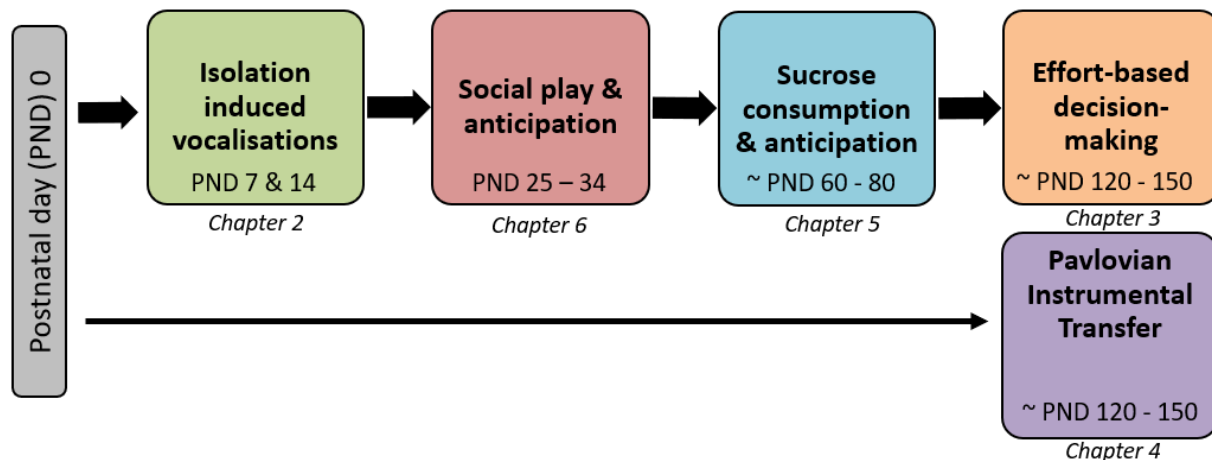


Figure 1.5. Visual timeline of experiments. To reduce the number of animals used per ethical guidelines, subjects completed multiple experiments throughout their lifespan with experimentally-naïve subjects used in the Pavlovian Instrumental Transfer. To minimise a potential extraneous order effect (Shoji & Miyakawa, 2021), subjects were randomly assigned to experiments across their lifespan. For example, subject A completed isolation induced vocalisations and effort-based decision-making whereas subject B completed social play and sucrose anticipation experiments, etc.

Chapter 2

Pup Development and Isolation- Induced Vocalisations

Chapter 2

Pup Development and Isolation-Induced Vocalisations

2.1 Background

This chapter explores the effects of a genetic reduction in the dopamine D1 receptor on pup development and socio-emotional communication during brief isolation.

Rats communicate their social, emotional and motivational states via ultrasonic vocalisations (USVs) which are high frequency vocalisations inaudible to humans (Simola & Granon, 2019; Vogel et al., 2019). Vocalisations can be categorised into three types defined by their average frequencies: 22 kilohertz (kHz), 50 kHz and 40 kHz. 22 and 50 kHz vocalisations are emitted by juvenile and adult rats, and reflect opposing affective states (Brudzynski, 2009). 22 kHz calls are an index of negative affective states associated with distressing, aversive, or potentially harmful situations such as foot shocks presentations, confrontation with an aggressive rat and startling noises. 50 kHz vocalisations, on the other hand, are an index of positive affective states associated with appetitive or neutral (e.g., nonaggressive, non-aversive, friendly) situations such as sexual contact with a mate, anticipation of rewards (e.g., drugs, food, non-aggressive social interaction) and during social play for juveniles.

The third category of vocalisations, 40 kHz, are emitted by pre-weaning pups when separated from their dam and littermates (Simola, 2015; Wöhr & Schwarting, 2008). The acoustic characteristics of isolation-induced vocalisations change with age. The number of vocalisations emitted by pups follows an inverted U-shaped curve with vocalisations increasing from PND 2, reaching a peak around PND 9, and then steadily declining through weaning (Bölükbas et al., 2020; Brudzynski et al., 1998; Elsner et al., 1990; Möhrle et al., 2023). Call duration, or length of the vocalisation, steadily increases as pups age. Principle frequency is the average frequency (kHz) of

a vocalisation while bandwidth is the minimum frequency subtracted from the maximum frequency. Both principle frequency and bandwidth also typically increase as pups age (Brudzynski et al., 1998; Wöhr et al., 2011).

The isolation-induced vocalisations reflect affective states and are reward-seeking in nature for both the pups and dams (Simola & Granon, 2019). Dams respond to their pups' calls (more than alien pups) by searching for the pups and retrieving if possible. Upon return of the *briefly* isolated pup, dams initiate maternal behaviours such as nursing, burrowing, and grooming which is rewarding for the pups (Boccia & Pedersen, 2000; Kosten & Kehoe, 2010; Rincón-Cortés & Grace, 2021), and subsequently pup vocalisations decrease (Shair et al., 2018). Upon return of the pups, dams also emit 50kHz vocalisations reflecting the rewarding properties of pup return (Stevenson et al., 2009).

Pup contact stimulates dopamine release in the dams' mesolimbic dopaminergic system and long-term removal of pups diminishes mesolimbic dopamine function seen in VTA activity (Rincón-Cortés & Grace, 2021). Return to the dam is rewarding for isolated pups and dopamine D1 blockade and D2 antagonism eliminates the typical reduction in vocalisations upon pup return, suggesting that a balance of D1 and D2 activation is necessary for the rewarding effects of pup return (Shair et al., 2018). In line with this, Homberg et al. (2016) found that when briefly isolated from their dam and littermates, pup with a genetic reduction in the dopamine D1 receptor emitted less vocalisations than wildtype pups. Although DAD1^{-/-} mutants emitted less calls, the *quality* of their calls was similar to wildtypes with no difference in average frequency (kHz) or duration of vocalisations. These pharmacological and genetic manipulations of dopamine and its D1 receptor, implicate the D1 receptor in the rewarding properties of isolation-induced vocalisations.

2.2 Overall Aims of Chapter Two

This chapter investigates the role of the dopamine D1 receptor in isolation-induced vocalisations and aims to replicate previous findings that DAD1^{-/-} mutants display altered *quantity* and intact *quality* of isolation-induced vocalisations (Homberg et al., 2016). Subjects, DAD1^{-/-} mutant and DAD1^{+/+} wildtype pre-weaning pups, are briefly separated from their dam and littermates to examine the quantity and quality of socio-emotional communication.

In line with Homberg et al. (2016), it is predicted that:

1. DAD1^{-/-} mutants will emit fewer vocalisations relative to wildtypes. All subjects will have fewer calls on PND 14 compared to PND 7.
2. DAD1^{-/-} mutants will not differ in quality of vocalisations with similar call durations and frequency characteristics (principle, bandwidth) as wildtypes.
3. Vocalisations did not differ between sexes (Homberg et al., (2016), but the sample size was small, so we might find sex differences in quality and quantity of vocalisations with the large sample size in the present study.

Pup development was analysed via pup weight, pup growth, and litter size. Homberg et al. (2016) did not include these factors; therefore, these variables were exploratory in this chapter.

4. DAD1^{-/-} mutant *adults* weigh significantly less than wildtype adults, so we predict that this genotype difference will already be seen in pre-weaning pups: DAD1^{-/-} mutant *pups* will weigh less than wildtype pups on PND 7 and 14. It is unclear whether this genotype difference in weight will extend to the growth rate (e.g., weight gain).
5. There is typically a negative relationship between litter size and pup weight with pups weighing less as litter size increases (Chahoud & Paumgarten, 2009),

therefore we predict that pups in large litters will weigh less than pups in small litters. It is not clear if genotype will interact with litter size.

2.3 Methods

2.3.1 Animals

There were 214 offspring from 16 DAD1^{+/-} heterozygous breeding pairs. Five DAD1^{-/-} mutant (not sexed) pups died prior to the experiment which resulted in 209 total offspring used in the present study: 25 DAD1^{-/-} mutant males, 20 DAD1^{-/-} mutant female, 24 wildtype males, 23 wildtype females, 60 heterozygous males and 57 heterozygous females.

Offspring were housed with the dam and littermates in standard polycarbonate cages in the breeding room until weaning on postnatal day (PND) 21. Food and water were available *ad libitum* at all times except during the experiment. Their individually ventilated polycarbonate home cages were lined with saw dust and contained a red enrichment tube. The housing room was on a reverse 12-h light/dark cycle (lights off between 7 am and 7 pm). All experimental procedures took place during the rats' dark cycle, between 8 am and 6 pm. The housing room was kept at controlled temperature (19–21 °C) and humidity (55%).

2.3.2 Apparatus

The experiment took place in a mesh cylindrical container (8.5 cm x 20 cm) lined with clean bedding (see figure 2.1B). To improve recording quality, the container was placed into a sound attenuated chamber made from Styrofoam (23.5 cm × 23.5 cm × 25.5 cm, lid: 2.5 cm, wall thickness: 3.5 cm) (created by Wilkin-Krug et al., 2022). The interior recording chamber measured 12 cm x 12 cm x 19 cm and a hole was cut in the chamber lid to fit the microphone (see figure 2.1A).

An UltraMic (250K 16 bi; Java Sound) microphone was suspended over the cylindrical container to record the vocalisations using Audacity recording software (Audacity Team, 2020). One Dell processor laptop was used to operate Audacity. All experimental testing occurred during the animals' dark cycle under red light.

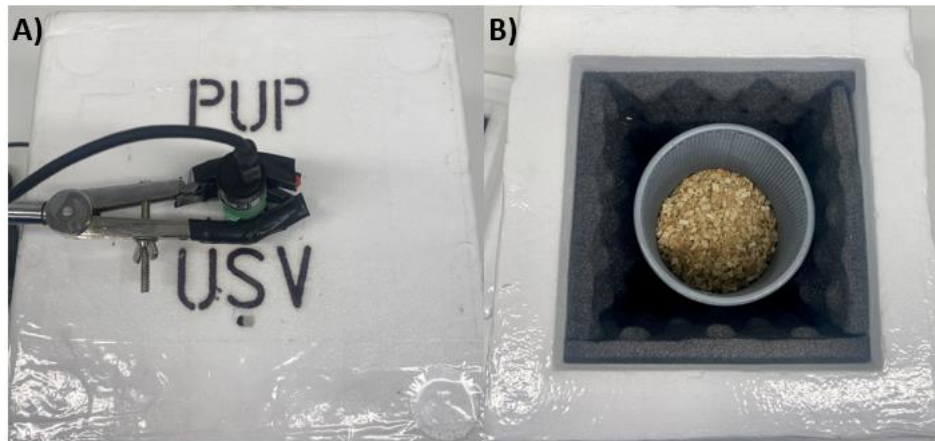


Figure 2.1 Experimental set up for isolation induced vocalisations. (A) Styrofoam recording chamber with the microphone inserted into the lid. (B) Chamber interior with the mesh cylindrical container each pup was placed in.

2.3.3 Procedure

2.3.3.1 Ultrasonic vocalisations

ON PND 7, the home cage, including dam and litter, was transferred to the experimental room 15 minutes prior to the session. Pups were individually removed from the home cage and placed in the cylindrical container in the recording chamber for 5 minutes. After the recording, pups were sexed, weighed, marked with a permanent marker on their back for later identification, and returned to their home cage. The next pup was selected, and the process repeated until all pups from the litter were recorded. The USV recording and weighing steps were repeated on PND14.

On PND 8, tail tips were collected from each pup. Each pup was individually held in the palm of the hand and an approximate 1-2 mm section of the tip of the tail was removed using scissors then immediately placed in a microcentrifuge tube for genotyping via qPCR (see section

2.3.3.2 below for a detailed genotyping procedure). After tip removal, pups were placed in a temporary cage while sampling littermates continued, to reduce maternal stress from any blood on the pups. Once all pups were sampled, the litter was returned to the home cage. Tail tip collection occurred the day following PND 7 recordings to prevent an association between the recording chamber and potential distress during tail tip collection.

Marker identifications were reapplied as needed every 2 – 3 days with the marker location transferred from their back to the tail at approximately PND 12-13 when the tails roughened out and held ink. Reapplication continued until the pups were weaned and ear tagged on PND 21.

2.3.3.2 Genotyping

2.3.3.2.1 Primer Design. Primers for genotyping by High Resolution Melt Analysis (HRMA) were previously designed in this laboratory to target the single nucleotide polymorphism on mutated allele of (Drd^{11116S}) (see Table 2.1).

Table 2.1

Primer sequences used for genotyping

Target	Primer Sequence	Melt Temp (°C)	Amplicon Size (base pairs)
DRD1 1	FW: CTA CGG CGT CCA TTC TGA AC	55.7	92
	REV: GGG GGT CAT CTT CCT CTC AT	56.0	

Note. Primer sets were designed and validated by Dr Bryony Thorne for DAD1 animal research.

2.3.3.2.2 Polymerase Chain Reaction. To prepare samples for polymerase chain reaction (PCR) amplification, tail tips taken on PND 8 were submerged in 25µL DNA Extraction Reagent (QuantaBio, Extracta DNA Prep Kit #95091-025). Submerged samples were incubated at 95°C for 30 minutes, vortexed, and placed on ice for subsequent addition to PCR reaction.

PCR reactions were created in 10 μ L volumes. Each reaction contained 1 μ L of extracted DNA, 5 μ L Accumelt HRM SuperMix (Quantbio, #95103-250), 0.2 μ M concentration of forward and reverse primers, and the remaining volume was nuclease-free water. All PCRs contained a negative, no template control with no DNA to ensure contaminated DNA was not a confounding result. Then, reactions were placed in a Bio-RadCFX96TM Real-Time PCR Detection System thermocycler (Bio-Rad, #185-5096) for amplification. PCR conditions were as follows: initialisation at 95°C for 2 min, preceding cycles of 15s at 95°C (denaturation), 15s at 52°C (annealing), and 15s at 72°C (elongation). Melt curves were generated after amplification by increasing the temperature across a 70°C-95°C range at 5s for each 0.2°C.

2.3.3.2.3 High Resolution Melt Analysis. Using the melt curves produced, subtle differences can be detected in PCR products between reactions by shifts in melt temperature (Lister, 2021; Tanaka et al., 2019). Melt temperature is determined by the peak in fluorescence resulting from denaturation of PCR product strands.

The T to G transversion present in the mutated allele of the DRD1 gene results in the replacement of a T-A bond with a G-C bond. The latter requires more energy to denature, resulting in a product with a subtly higher melt temperature. By using a high-resolution melt temperature range with minimal increments (0.2°C) and a short PCR product (<100bp), this subtle difference in alleles is detectable by melt analysis. DAD1^{+/+} wildtypes (WT) appear as the sharp peak with the lowest temperature, DAD1^{-/-} homozygous mutants (HOM) as a sharp peak with the highest temperature and DAD1^{+/-} heterozygous (HET) with a broad peak at a middle temperature. Analysis on Bio-Rad Precision Melt Analysis Software (Bio-Rad Laboratories, version 4.0.52.0602) was carried out to identify the differences between genotypes for pup identification. Cluster analysis of normalised melt curves identify all pups in this study as WT, HOM, and HET (see figure 2.2).

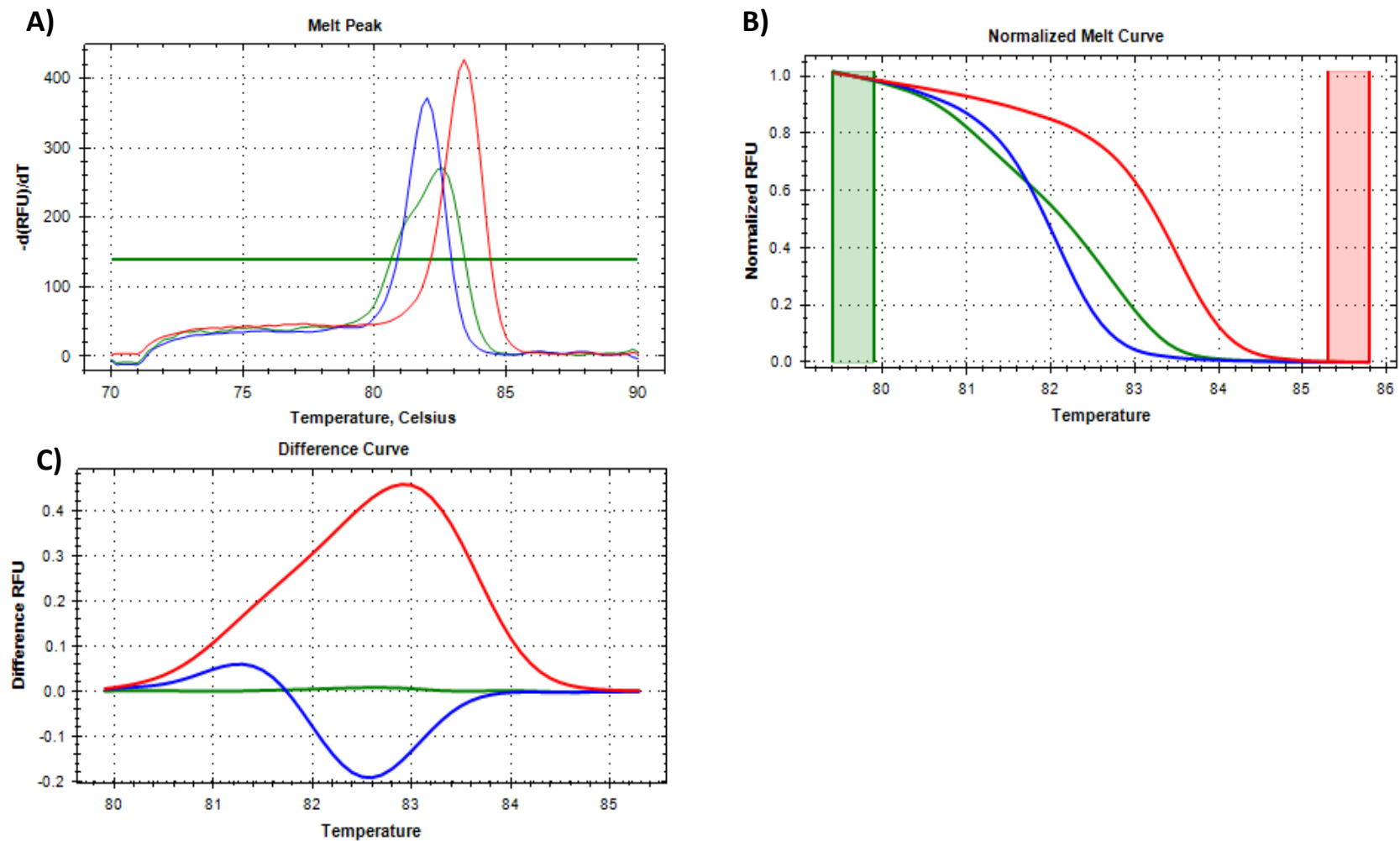


Figure 2.2. Genotyping by High Resolution Melt Analysis. All melt curves shown result from a random sample of pups, with results representative of all pups in this study. Colouring is consistent across the graphs: green as HET, red as HOM and blue as WT. (A) Example melt curves generated by qPCR show a sharp WT peak at approximately 82°C, a sharp HOM peak at approximately 84°C and a wider HET peak at approximately 83°C. (B) Normalised melt curves show 3 clusters across the litter (C) Difference curves derived from normalised melt curves show separated clustering of all pups tested in the litter, normalised with the green HET cluster at 0.

2.3.4 Data Analysis

Due to self-noise from the microphones, audio files were cleaned using Squeaky Clean, which uses a band-pass filter and spectral gating with a noise-threshold to remove noise (Grabham-Madden, 2023; see figure 2.3). Cleaned files were processed with DeepSqueak v3, which utilises a post-hoc de-noising network to automatically discriminate calls from noise (Coffey et al., 2019). Processed files were manually reviewed to account for missed calls (60– 80% accuracy with DeepSqueak). After manual review, the number of calls, average call duration, average peak frequency, and average bandwidth (range of frequencies) were exported for each file. Heterozygous pups were removed from analysis due to time constraints and some pups were removed due to issues with the audio files. A total of 66 pups were used in the analysis (21 DAD1^{-/-} male, 17 DAD1^{-/-} female, 12 wildtype male and 15 wildtype female) with two recording per pup (at PND 7 and at PND 14).

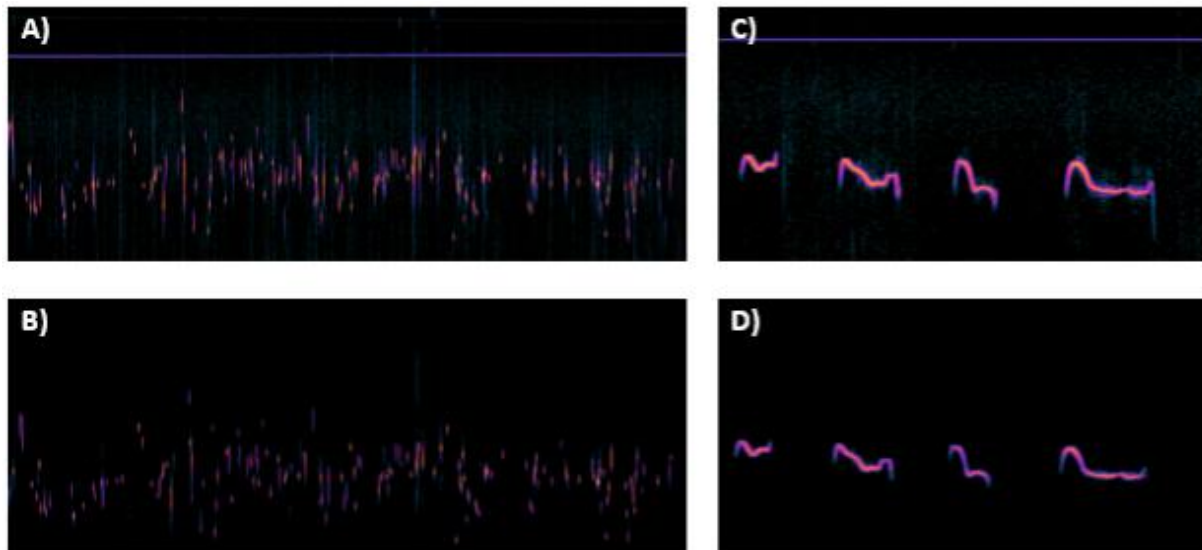


Figure 2.3. Before and after cleaning audio files with Squeaky Clean (Grabham-Madden, 2023). DeepSqueak was unable to accurately detect calls due to self-noise (straight line seen in A and C) and other background noise. Squeaky Clean uses a band-pass filter and spectral gating to remove most noise from files (segment from cleaned audio files in B and D).

USV variables (number of calls, average call duration, frequency measures) were examined with repeated measures ANOVAs with genotype (DAD1^{-/-} mutants, DAD1^{+/+} wildtypes) and sex (male, female) as the between-subject factor and PND (7, 14) as the within-subjects factor. Mauchly's test of sphericity indicates that the assumption of sphericity was not violated ($p > 0.05$), therefore no corrections were used. Pup developmental factors (pup weight and growth) were examined using a variety of statistical analyses which are noted in the relevant sections below.

All variables were analysed for statistical significance using Jamovi (version 2.3) and SPSS (version 27). The alpha level for statistical significance was $p < 0.05$ and partial eta squared (η^2) was used for the effect size. All significant interactions were followed up with post hoc Student's t-tests.

2.4 Results

2.4.1 Litter Size

Across the 16 litters, litter sizes ranged from 5 to 18 pups with an average size of 13.31 pups (see Appendix A, Table A1.1 for descriptive statistics). The average sex ratio (proportion of males per total litter) for all 16 litters was 0.53 ± 0.15 which did not significantly differ from the expected 0.50 sex ratio (Cameron, 2004; Romano et al., 2012; see Appendix A, Table A1.2).

There is typically a negative relationship between litter size and pup weight with pups weighing less as litter size increases (Chahoud & Paumgarten, 2009). We observed marked differences in weight between genotypes (see figure 2.4), therefore, the following analysis investigated whether DAD1^{-/-} mutants had a higher likelihood of runt classification.

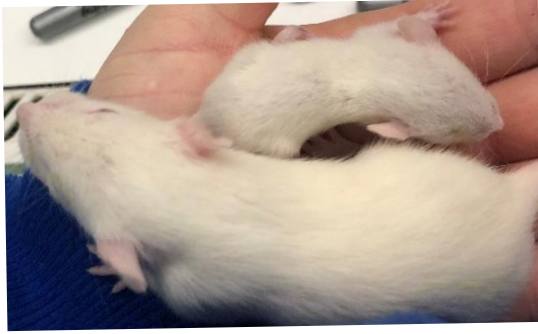


Figure 2.4. An example of a “runt” and a littermate. On PND 7, the $DAD1^{-/-}$ mutant (top) runt weighed 7.5 grams while its $DAD1^{+/+}$ wildtype littermate (bottom) weighed 17 grams.

A pup was classified as a runt if they weighed less than 90% of the mean weight for their litter on PND 7 (Woods & Weeks, 2004). Out of 214 total pups, 31 were classified as runs (17 $DAD1^{-/-}$ mutant, 4 wildtype, 10 heterozygous; see figure 2.5). $DAD1^{-/-}$ mutants had a significantly higher probability of runt classification than wildtype ($p < 0.001$) and heterozygous pups ($p < 0.001$) (genotype effect: $F_{2,162} = 10.45$, $p < 0.001$, $\eta_p^2 = 0.11$). Wildtype pups did not significantly differ from heterozygous pups in runt classification ($p = 0.91$). There was no sex effect on runt classification ($F_{1,162} = 0.06$, $p = 0.81$, $\eta_p^2 = 0.001$).

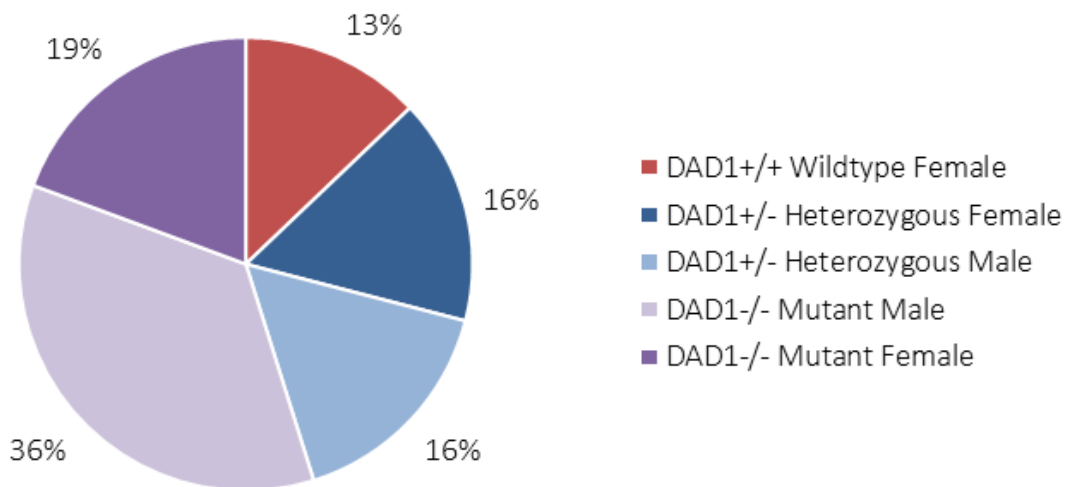


Figure 2.5. Runt composition across all litters. There were significantly more $DAD1^{-/-}$ mutant runs than wildtype and heterozygous runs. There were no male wildtype runs.

To examine if litter size impacted the runt classification, we analysed the number of pups per litter and litter size categorisation. Dams have 12 nipples to feed their offspring, so we divided litters into “ ≤ 12 ” and “ > 12 ” with 56.25% of the total litters categorised as “ > 12 ” (Azzam et al., 1984; Nazligül, 2000). Litter categorisation (≤ 12 , > 12) did not affect the probability of runt classification ($F_{1,195} = 0.86$, $p = 0.36$, $\eta_p^2 = 0.01$) (see Appendix A, Table A1.2). Additionally, the number of pups per litter did not affect the probability of runt classification ($F_{10,186} = 0.81$, $p = 0.62$, $\eta_p^2 = 0.04$).

2.4.2 Pup Weight

A linear mixed model examined the effect of sex, genotype, and postnatal day (PND) on pup weight with litter ID and litter size as covariates (Baayen et al., 2008). Regardless of sex or PND, wildtypes weighed significantly more than $DAD1^{-/-}$ mutants (genotype effect: $\beta = -3.33$, 95% CI [-4.53, -2.13], $t(390) = -5.44$, $p < 0.001$; see figure 2.6). Heterozygous pups did not significantly differ in weight from $DAD1^{-/-}$ mutants ($p = 0.12$) nor wildtypes ($p = 0.77$). Subjects, regardless of sex or genotype, weighed significantly more on PND 14 than PND 7 which is indicative of normal growth (PND effect: $\beta = 12.25$, 95% CI [11.23, 13.28], $t(390) = 23.48$, $p < 0.001$; see Appendix A, Table A3.1 for descriptive statistics). Sex did not have a significant effect on weight (sex effect: $\beta = 0.47$, 95% CI [-0.72, 1.67], $t(390) = 0.78$, $p = 0.44$), which supports findings that sex differences in weight emerge from puberty (Shi & Clegg, 2009; Slob & van der Werff ten Bosch, 1975).

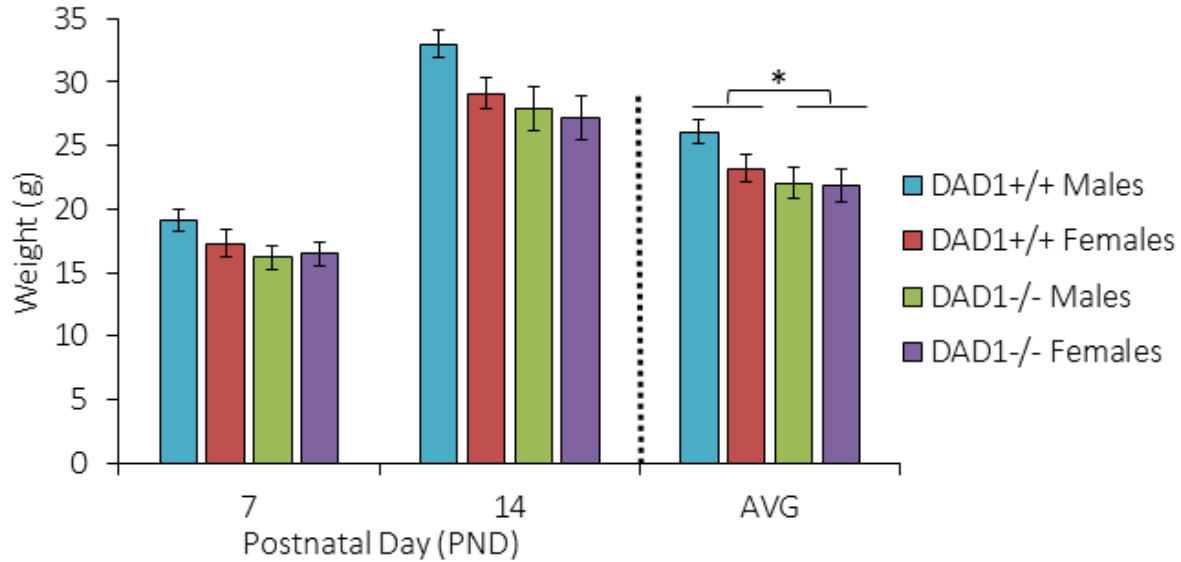


Figure 2.6. Pup weights across postnatal days. Across days, wildtypes weighed significantly more than DAD1^{-/-} mutants. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

There was a significant association between litter size and pup weight (beta = -1.32, 95% CI [-1.53, -1.10], $t(390) = -12.16$, $p < 0.001$) with pups from small litters (≤ 12 pups per litter) weighing significantly more than those from large litters (> 12 pups per litter). There was a significant negative correlation between litter size and pup weight (Pearson's correlation: $r(170) = -0.52$, $p < 0.001$); pups' weight decreased as litter size increased which is typical for multiple offspring mammals (Chahoud & Paumgarten, 2009) (see figure 2.7). There was not a significant association between litter ID and weight (beta = -0.01, 95% CI [-0.17, 0.14], $t(390) = -0.19$, $p = 0.85$), which indicates that the differences between litters were due to the size of the litter not the litter itself.

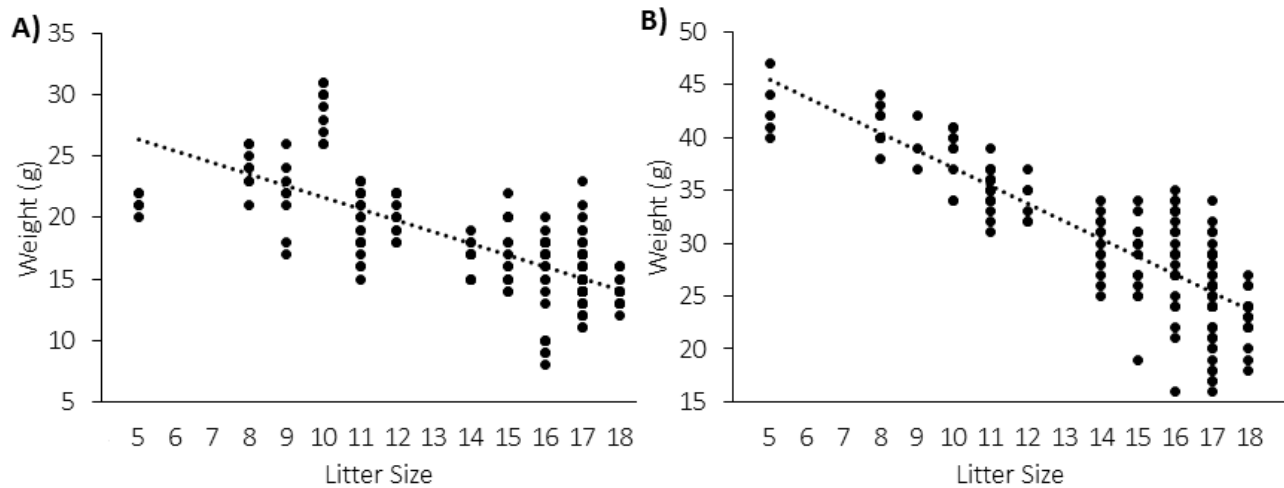


Figure 2.7. The negative relationship between litter size and pup weight on PND 7 (A) and PND 14 (B). Dots represent each pups' weight.

Regarding litter categorisation, pups in small (≤ 12) litters weighed significantly more than pups in large (>12) litters on PND 7 ($p < 0.001$) and PND 14 ($p < 0.001$) (litter categorisation * PND: $F_{1,91} = 20.85$, $p < 0.001$, $\eta_p^2 = 0.20$; see figure 2.8).

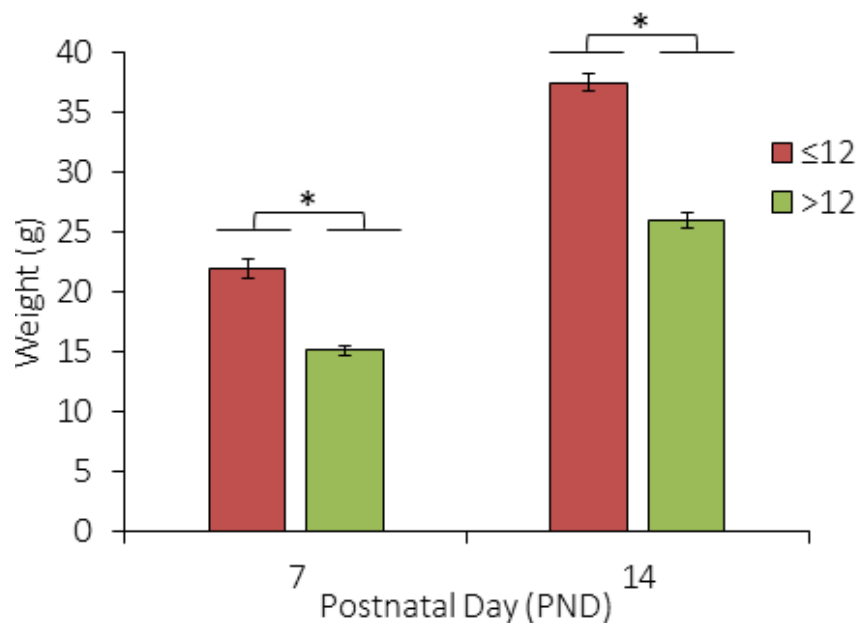


Figure 2.8. Pup weight based on litter categorisation. Pups in small litters weighed significantly more than pups in large litters on PND 7 and 14. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

2.4.3 Weight Change

There was no genotype effect for weight change from PND 7 to PND 14 ($F_{1,169} = 0.19$, $p = 0.83$, $\eta_p^2 = 0.001$), indicating that although DAD1^{-/-} mutants weighed significantly less than wildtypes, they grew at a similar rate between PND 7 and 14 (see Appendix A, Table A3.2 for descriptive statistics). There was also no sex difference in weight change ($F_{1,169} = 0.10$, $p = 0.79$, $\eta_p^2 = 0.001$) which aligns with sex differences in growth emerging later during puberty (Shi & Clegg, 2009; Slob & van der Werff ten Bosch, 1975).

There was not a significant effect of litter size on growth ($F_{1,169} = 0.65$, $p = 0.77$, $\eta_p^2 = 0.04$) nor a correlation between litter size and growth, $r(169) = 0.002$, $p = 0.98$ (see figure 2.9). In terms of litter categorisation, there was not a significant effect on weight change ($F_{1,169} = 0.06$, $p = 0.81$, $\eta_p^2 = 0.001$); ≤ 12 litters did not differ from >12 litters in growth.

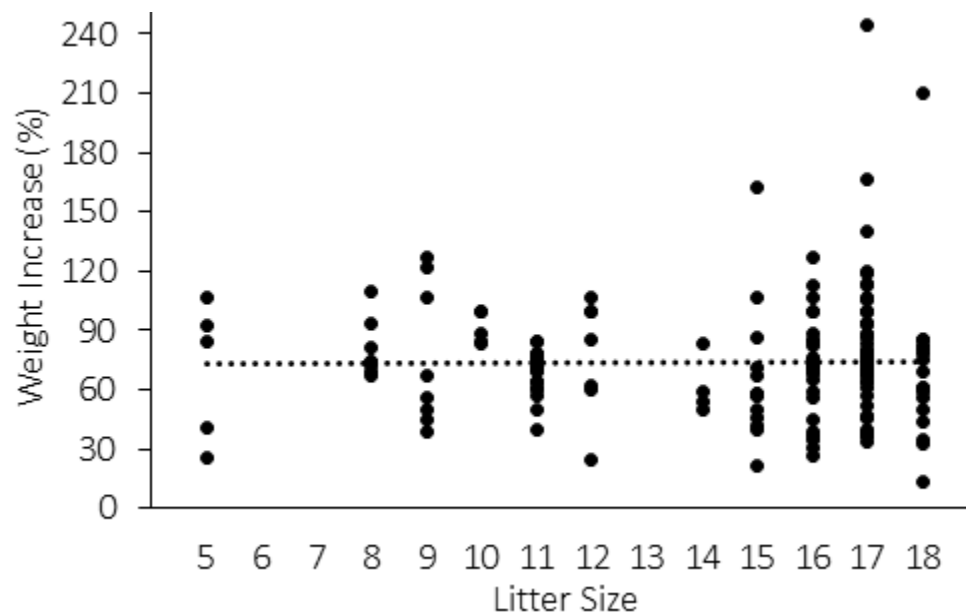


Figure 2.9. Weight increase for each pup from PND 7 to PND 14 based on litter size. There was not a relationship between litter size and weight gain.

2.4.4 Ultrasonic Vocalisations

2.4.4.1 Number of Ultrasonic Vocalisations

In line with Homberg et al., (2016), $DAD1^{-/-}$ mutants emitted significantly fewer vocalisations on PND 7 relative to wildtypes (genotype effect: $F_{1,65} = 5.65$, $p < 0.05$, $\eta_p^2 = 0.09$), but this genotype effect did not extend to PND 14 ($p = 0.82$ for the post hoc test). As predicted, $DAD1^{-/-}$ mutants ($p < 0.001$) and wildtypes ($p < 0.001$) emitted significantly fewer vocalisations on PND 14 relative to PND 7 (PND * genotype: $F_{1,65} = 4.11$, $p < 0.05$, $\eta_p^2 = 0.06$; see figure 2.10). There was no sex difference in number of vocalisations emitted ($F_{1,65} = 1.04$, $p = 0.31$, $\eta_p^2 = 0.02$).

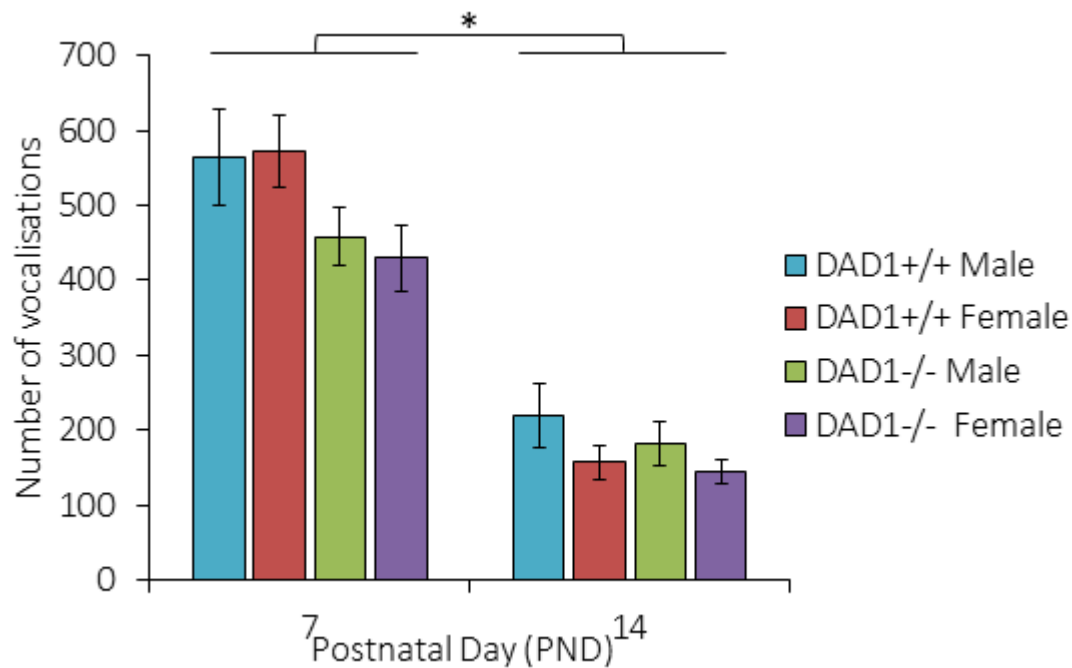


Figure 2.10. Average number of vocalisations emitted across days. $DAD1^{-/-}$ mutants emitted significantly fewer vocalisations on PND 7 relative to wildtypes. All subjects emitted significantly fewer vocalisations on PND 14. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

2.4.4.2 Duration of Ultrasonic Vocalisations

In line with Homberg et al., (2016), average duration of vocalisations did not differ between genotype or sex (genotype effect: $F_{1,65} = 0.67$, $p = 0.42$, $\eta_p^2 = 0.01$; sex effect: $F_{1,65} = 3.08$, $p = 0.09$, $\eta_p^2 = 0.05$). Subjects, regardless of genotype or sex, emitted longer calls on PND 14 relative to PND 7 (PND effect: $F_{1,65} = 9.02$, $p < 0.01$, $\eta_p^2 = 0.14$; see figure 2.11).

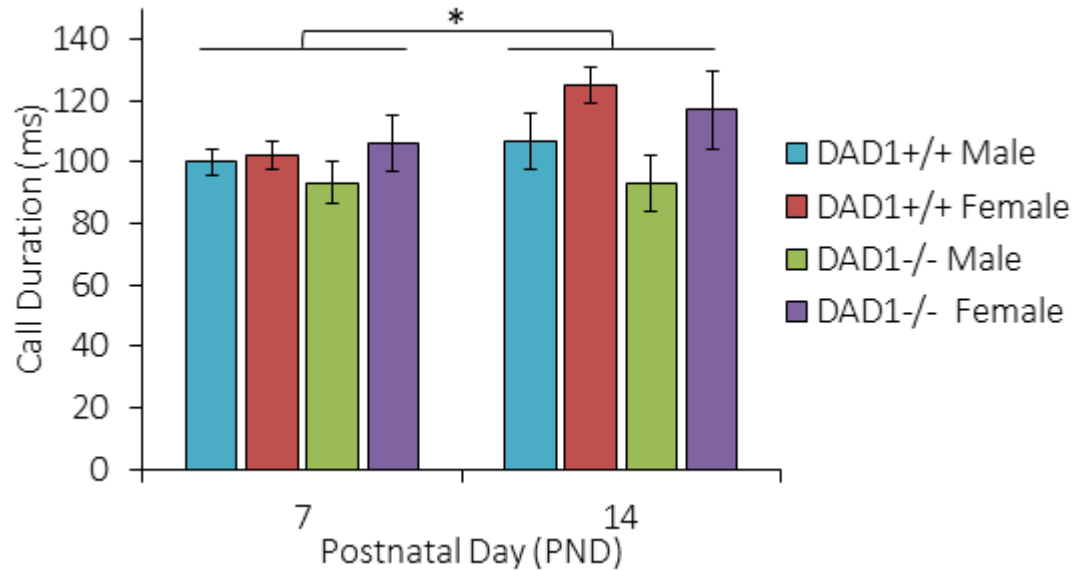


Figure 2.11. Average duration of vocalisations emitted across days. DAD1^{-/-} mutants emitted significantly fewer vocalisations on PND 7 relative to wildtypes. Subjects emitted longer calls on PND 14 relative to PND 7. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

2.4.4.3 Principal Frequency (kHz) of Ultrasonic Vocalisations

Contrary to Homberg et al. (2016), DAD1^{-/-} mutants emitted vocalisations at a higher average frequency (kHz) than wildtypes ($F_{1,65} = 4.99$, $p < 0.05$, $\eta_p^2 = 0.08$) and males emitted vocalisations at a higher frequency than females ($F_{1,65} = 4.44$, $p < 0.05$, $\eta_p^2 = 0.07$; see figure 2.12). There was no significant interaction between sex and genotype ($F_{1,65} = 2.16$, $p = 0.15$, $\eta_p^2 = 0.04$). Subjects, regardless of genotype or sex, emitted vocalisations at a higher average frequency on PND 14 relative to PND 7 ($F_{1,65} = 39.13$, $p < 0.001$, $\eta_p^2 = 0.41$; see figure 2.12).

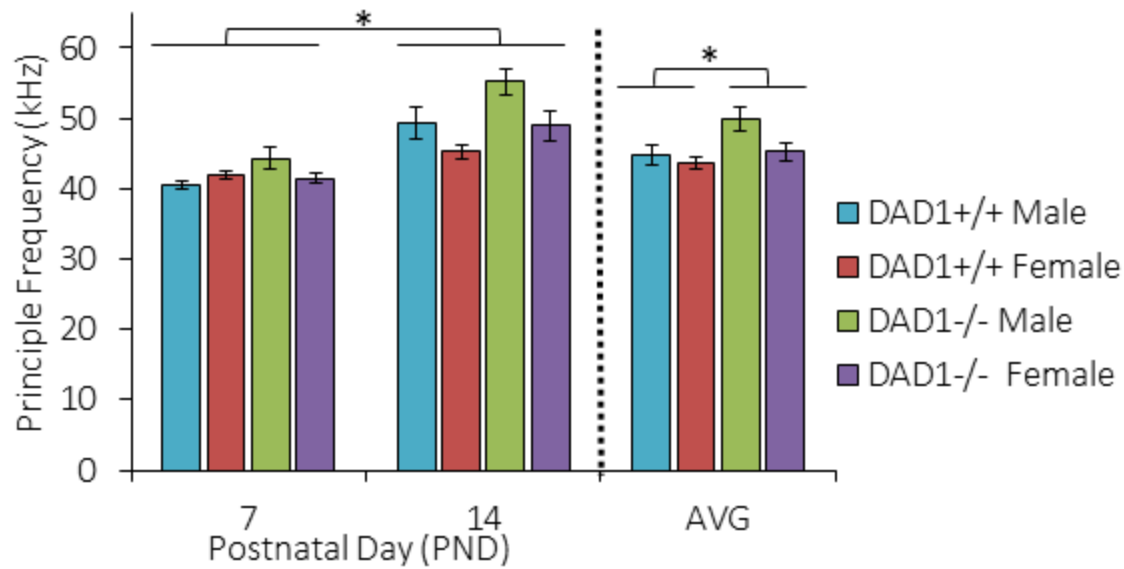


Figure 2.12. Average principle frequency (kHz) per day. DAD1^{-/-} mutants emitted vocalisations at a significantly higher frequency than wildtypes, and males emitted higher frequency vocalisations than females. Subjects emitted vocalisations with higher frequency on PND 14 relative to PND 7. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

2.4.4.4 Bandwidth of Ultrasonic Vocalisations

In line with Homberg et al., (2016), the average bandwidth of vocalisations did not differ between genotype ($F_{1,65} = 2.66$, $p = 0.11$, $\eta_p^2 = 0.05$) or sex ($F_{1,65} = 0.01$, $p = 0.99$, $\eta_p^2 = 0.001$). Subjects, regardless of genotype or sex, vocalisations on PND 14 had a greater change in frequency (kHz) relative to PND 7 (PND effect: $F_{1,65} = 17.05$, $p < 0.001$, $\eta_p^2 = 0.23$; see figure 2.13).

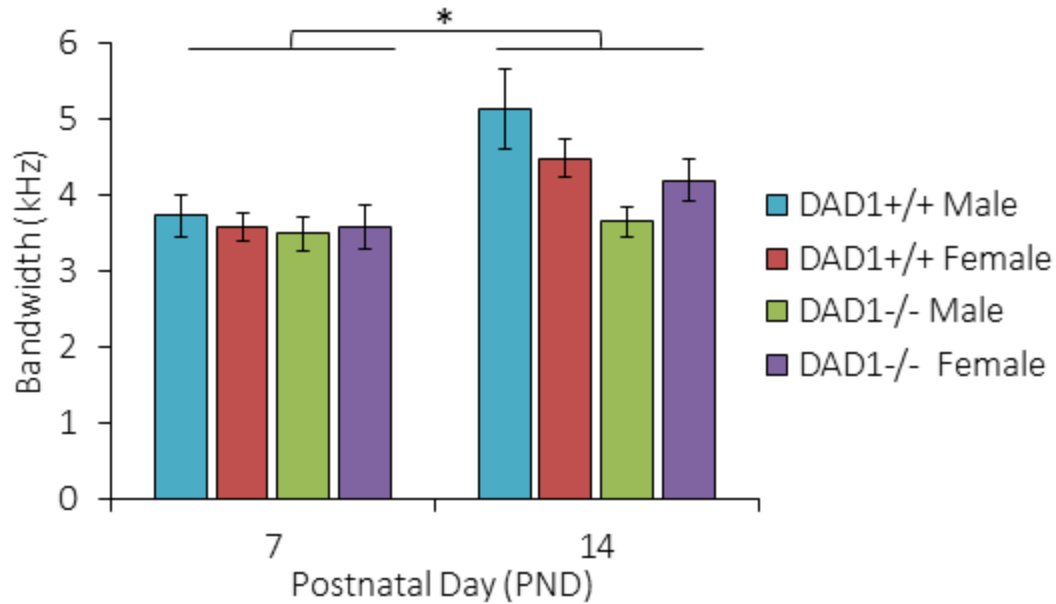


Figure 2.13. Bandwidth across days. Subjects had a larger change in frequency on PND 14 relative to PND 7. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

2.4.4.5 The Effect of Litter Size and Pup Weight on Vocalisations

Since litter size affected the pups' weight, we examined if litter size and pup weight impacted quality and quantity of vocalisations emitted. Litter size categorisation (≤ 12 , > 12) did not affect the number ($F_{1,65} = 2.75$, $p = 0.10$, $\eta_p^2 = 0.05$), average duration ($F_{1,65} = 0.67$, $p = 0.42$, $\eta_p^2 = 0.01$), average principle frequency ($F_{1,65} = 0.16$, $p = 0.70$, $\eta_p^2 = 0.003$) nor average bandwidth ($F_{1,65} = 0.11$, $p = 0.74$, $\eta_p^2 = 0.002$) of pup vocalisations. Additionally, there were no significant correlations between pup weight and vocalisation variables, nor between litter size and vocalisation variables (see Appendix A, Table A2.4). Although litter size affected pup weight, it did not alter pup vocalisations.

2.5 Discussion

The aim of this experiment was to investigate if a genetic reduction in D1R impacts pup development and to replicate the reduction in quantity of PND 7 isolation-induced vocalisation in DAD1^{-/-} mutants from Homberg et al. (2016). The acoustic parameters of isolation-induced calls

change with age; the number of vocalisations decreases while duration, principle frequency and bandwidth typically increase with aging (Bölükbas et al., 2020; Brudzynski et al., 1998; Elsner et al., 1990; Möhrle et al., 2023). Therefore, this study also expanded on Homberg et al. (2016) to examine if the differences found in quantity of vocalisations on PND 7 are stable as pups age.

It was hypothesised that $DAD1^{-/-}$ mutants would emit fewer vocalisations than wildtypes across days and that all subjects would have fewer vocalisations on PND 14 relative to PND 7. As predicted, $DAD1^{-/-}$ mutants emitted significantly fewer vocalisations than wildtypes on PND 7 which replicates Homberg et al. (2016). Additionally, as predicted, all subjects emitted fewer vocalisations on PND 14 compared to PND 7. However, the genotype difference found on PND 7 did not extend to PND 14 which suggests that the effect of a genetic reduction in D1R on number of vocalisations might change as pups develop. It was also hypothesised that $DAD1^{-/-}$ mutants would not differ in quality of vocalisations with similar call durations and frequency characteristics (principle, bandwidth) to wildtypes, which was mostly supported. In line with Homberg et al. (2016), the average duration and bandwidth of vocalisations did not differ between genotypes. Additionally, in line with vocalisation changes across development (Brudzynski et al., 1998; Elsner et al., 1990; Wöhr et al., 2011), all subjects emitted vocalisations with longer durations, and greater principle frequency and bandwidth on PND 14 relative to PND 7. Contradictory to predictions, $DAD1^{-/-}$ mutants emitted vocalisations at a higher principle frequency than wildtypes. Lastly, it was hypothesised that males and females would not differ in quality or quantity of calls with similar number of calls, call duration and frequency parameters. Results support this hypothesis with no sex difference in the number of vocalisations, duration, nor bandwidth of vocalisations. Contradictory to predictions, males emitted vocalisations at a higher principle frequency than females.

There are several possibilities for why DAD1^{-/-} mutant pups emitted fewer vocalisations on PND 7. Isolation-induced vocalisations reflect the pups' socio-emotional state and are reward-seeking in nature since dams respond to their pups calls by searching and retrieving the pups, if possible. On return, dams initiate maternal care behaviours such as nursing and grooming which is rewarding for the pups and dam, and consequently pups vocalisations decrease upon return (i.e., “contact quieting”) (Boccia & Pedersen, 2000; Kosten & Kehoe, 2010; Rincón-Cortés & Grace, 2021). The D1R is implicated in the reward-seeking phase of motivated behaviour specifically in assigning incentive salience to rewards (Salamone et al., 1991; Salamone & Correa, 2012; Shair et al., 2018; Smith-Roe & Kelley, 2000). It is possible that the reduction in D1R impairs incentive salience assigned to rewarding stimuli (e.g., maternal care upon return), therefore the DAD1^{-/-} mutants were less likely to emit vocalisations for maternal care. Since contact quieting upon return to the dam is reflective of the incentive salience assigned to the rewarding experience (Stevenson et al., 2009), future research could record vocalisations when pups return to the dam. If DAD1^{-/-} mutants have a smaller magnitude of contact quieting upon return, it would suggest an impairment in assigning incentive salience to the return, which could explain the reduction in reward-seeking behaviour (i.e., isolation-induced vocalisations).

Although isolation-induced vocalisations are reward-seeking in nature, other factors outside of reward-seeking processes can influence isolation-induced vocalisations. As mentioned, pup return to the dam elicits maternal care behaviour targeted towards the returned pup. Additionally, studies suggest that maternal behaviour *prior* to separation influences the number of vocalisations pups emit, with greater levels of maternal care leading to fewer isolation-induced vocalisations emitted (D’Amato et al., 2005; Wöhr & Schwarting, 2008). Subsequently, it is possible that DAD1^{-/-} mutants *might* receive greater care from the dams than their wildtype

littermates resulting in fewer vocalisations. To examine this potential cause, future research could examine if dams provide different levels of care between genotypes and whether this maternal care influences the number of vocalisations emitted.

It was hypothesised that DAD1^{-/-} mutants would weigh less than wildtypes in accordance with weight differences found in adult DAD1^{-/-} mutants (Homberg et al., (2016). As predicted, DAD1^{-/-} mutants weighed significantly less than wildtypes and interestingly, they had a higher likelihood of being runts. It was also predicted that litter size would be negatively correlated with pup weight which we found in the present study. However, it is important to note that litter size did not affect the likelihood of runt classification for any genotypes. The weight difference seen between genotypes could have influenced maternal care; smaller pups, especially runts, might have received more early care as the dam tried to ensure survival of her pups (Champagne et al., 2003; Enes-Marques & Guisti-Paiva, 2018; van Hasselt et al., 2012). The higher probability of DAD1^{-/-} mutants being runts could mean that, regardless of their litter size, they received greater maternal care in early life which subsequently led to fewer isolation-induced vocalisations.

There are several ways future research could examine whether maternal care and retrieval behaviour differs based on pup genotype. Firstly, we could measure maternal behaviours (e.g., nursing, grooming, retrieving, and nest-building) prior to isolation, analyse if maternal care differs based on the pup's genotype, and subsequently examine whether this influences their isolation-induced vocalisations (Beery & Francis, 2011; Meyza & Knapska, 2017). Secondly, playback of recorded pup vocalisations elicits search behaviour in dams. We could examine if searching behaviour varies based on the genotype of the recording (Winters et al., 2022, 2023). Lastly, we could use the Pup Retrieval Test to measure if dams retrieve pups differently based on the pups' genotype (Han & Lavooy, 2005; Winters et al., 2022, 2023). These three paradigms would give

information on whether maternal care and retrieval behaviour differs based on the pups' genotype and whether this accounts for the results in the present study.

In summary, DAD1^{-/-} mutants emitted fewer vocalisations than wildtypes on PND 7 and a majority of USV characteristics did not differ between genotypes which replicates Homberg et al. (2016). As seen in adult DAD1^{-/-} mutants, pre-weaning pups also weigh significantly less than wildtypes (Homberg et al., 2016). Building upon previous studies, this study found that DAD1^{-/-} mutants had a higher likelihood of being runts than wildtypes and although they weighed less, their growth from PND 7 to PND 14 did not differ from wildtypes. Litter size and pup weight did not alter quality or quantity of vocalisations, but more research is needed to examine if maternal care behaviour differs based on genotype and subsequently influences isolation-induced vocalisations. The higher likelihood of DAD1^{-/-} mutants being classified as runts warrants further investigation and could be due to impairments in motivation, which reduced the exertion of effort for milk and ultimately led to smaller weights (Mennella et al., 1990; Parra-Vargas et al., 2023). The following chapter addresses the role of the D1 receptor in motivation by examining the exertion of effort across multiple operant chamber experiments.

Chapter 3

Effort-Based Decision-Making

Chapter 3

Effort-Based Decision-Making

3.0 Prologue

This chapter, in its entirety, was published in *Progress in Neuropsychopharmacology & Biological Psychiatry* (Witt et al., 2022) with author rights giving permission to include the publication in a thesis (see Statement of Authorship for confirmation). Formatting of the headings, subheadings, and figures were changed for consistency with the rest of this thesis per the “Guidelines for including publications in a PhD thesis” from the Wellington Faculty of Graduate Research (WFGR 14/19A). Additionally, use of “Drd^{1H116S}” in the publication was changed to DAD1^{-/-}, a description of the subjects’ previous experimental experience was added, and descriptive statistics on the subjects’ weight was added for consistency with the rest of the thesis.

In Chapter 2, DAD1^{-/-} mutants had a significantly higher likelihood of “runt” classification relative to wildtypes, which could be due to impairments in motivation. DAD1^{-/-} mutants might display decreased willingness to exert effort for milk, which ultimately leads to smaller weight (Mennella et al., 1990; Parra-Vargas et al., 2023). This “runt” observation prompted the current chapter which explores the role of D1R in motivation and effort-based decision-making across a variety of operant-chamber experiments.

3.1 Background

When making decisions, individuals unequivocally prefer choosing larger rewards over smaller rewards. However, the decision becomes more complex when effort is required to achieve the reward (effort-based decision-making) (Kurniawan et al., 2011). Effort-based decision-making involves cost-benefit calculations in which an individual weighs the amount of effort exertion required (cost) against the reward value (benefit) (Chong et al., 2016; Reddy et al., 2016).

Motivated behaviour is defined as how much effort one will exert to obtain rewards. Individuals exert effort for a desired reward, however the preference for this reward decreases as effort required increases (Kurniawan et al., 2011; Reddy et al., 2016). The neurobiological mechanism underlying effort-based decision-making is gaining attraction largely because several neuropsychiatric disorders involve decreased willingness to exert effort (Walton & Bouret, 2019). For example, schizophrenia and major depressive disorder are both associated with decreased exertion of physical and cognitive effort which leads to diminished motivation to seek out and engage in potentially rewarding activities (Reddy et al., 2016, Barnes et al., 2014, Calabrese et al., 2014). Understanding the neurobiological mechanisms underlying effort-based decision-making is critical for developing treatment for amotivation in disorders such as schizophrenia and major depressive disorder.

Research has aimed to identify neurotransmitter systems involved in effort-based decision-making, and data strongly supports the dopaminergic system as one of the critical mediators. Dopamine antagonism and depletion, primarily in the Nacc, reduces the exertion of effort for food rewards while leaving appetite intact (Salamone et al., 2016). Dopamine depletion in the Nacc impaired responding on high fixed ratio schedules of reinforcement (Aberman & Salamone, 1999), biased rodents toward low-effort decisions in a T-maze effort-choice task (Mai et al., 2012; Salamone et al., 2015), and reduced the willingness to work for the preferred food reward in the FR-5/chow procedure in which they chose between pressing the lever for more-preferred sugar pellets (on FR-5 schedule) or free feeding on less-preferred lab chow (Salamone et al., 1991). Systemic administration of haloperidol (DA antagonist) biased rodents toward low-effort decisions in a T-maze effort-choice task (Salamone et al., 1994), and reduced the willingness to work for the preferred food reward in an operant-based choice procedure (Salamone et al., 1991). Thus, a

multitude of data strongly supports the role of dopamine in effort-based decision-making; however, it is not yet clear which dopamine receptor is mediating this.

Dopamine exerts its action through five G-protein coupled dopamine receptors. The D1 and D5 receptors represent one group of dopamine receptors (D1-like receptors) while the D2, D3, and D4 receptors represent the other group (D2-like receptors) (Zhang et al., 2009). Accumulating pharmacological evidence suggests that the dopamine D1-like receptors are involved in effort-based decision making. In two operant-based choice procedures (FR-5/chow and PROG/chow), rodents consistently pressed the lever for a high carbohydrate food pellet (preferred reward) over consuming freely available standard chow. After treatment with the D1-like antagonist, ecopipam, rodents consumed the freely available, less preferred chow over lever pressing for the preferred food reward (Yohn et al., 2015; Nunes et al., 2010; Randall et al., 2014). In a T-maze effort-choice task, rodents consistently climbed over the barrier for the larger food reward, but intra-ACC infusion of the D1-like antagonist, SCH-23390, biased rodents towards the low-effort, smaller food reward (Schweimer & Hauber, 2006). Together these findings suggest that the dopamine D1-like receptor is implicated in effort-based decision making.

A major limitation of the previous research is that available pharmacological manipulations do not selectively act on the D1 receptors—they show an affinity for D5 receptors and several, including SCH23390, also bind to serotonin receptors (Zhang et al., 2009; Homberg et al., 2016; Undieh, 2010; Squire et al., 2020). To specifically examine the role of the dopamine D1 receptor, the present study utilised rats with a genetic reduction in the D1 receptor, dopamine DAD1^{-/-} mutant rats (Homberg et al., 2016; Undieh, 2010; Squire et al., 2020). These rats have a point mutation in the third transmembrane of the D1 receptor, resulting in a reduction in D1 receptor function and expression, while not impacting the other four dopamine receptors (Homberg et al.,

2016; Squire et al., 2020). The DAD1^{-/-} mutant rats allow us to examine the specific role of the dopamine D1 receptor in effort-based decision-making.

A second limitation of previous research is that almost without exception, the pharmacological studies have all been performed in male rats. Effort-based decision making is impaired in several psychiatric disorders including major depressive disorder (Treadway et al., 2013) and schizophrenia (Culbreth et al., 2018), and both neuropsychiatric disorders exhibit sex differences in prevalence, symptoms, and age of onset (Brzezinski-Sinai & Brzezinski, 2020; Salk et al., 2017). Additionally, estradiol has an overall facilitating effect on dopaminergic transmission by increasing synthesis, release, and reuptake of dopamine via estradiol membrane receptors which could contribute to these sex differences (Barth et al., 2015; Hidalgo-Lopez & Pletzer, 2017; Jacobs & D'Esposito, 2011). It is therefore paramount that the role of the D1 receptor in effort-based learning is investigated in both sexes. To address this, we have investigated effort-based decision-making in male and female DAD1^{-/-} mutant and wildtype control rats (Schweimer & Hauber, 2006; Yohn et al., 2015).

3.2 Overall Aims of Chapter Three

The central hypothesis of the present study is that the D1 receptor mediates effort-based decision-making, which we investigated through a range of effort-based decision-making experiments. The purpose of the present study was to examine if DAD1^{-/-} male and female mutant rats can acquire lever pressing, and if so, examine how much effort they exert for food rewards. After acquisition, subjects were trained on FR-2 and FR-5 schedules of reinforcement, followed by a progressive ratio schedule in which the number of lever presses required for pellet delivery increased after each successful delivery. The final ratio a subject completes is the breakpoint which determines the amount of effort a subject is willing to exert for the reward (Sharma et al., 2012).

Next, subjects completed an operant-choice task in which they chose between pressing the lever for a preferred reward (sugar pellet) and consuming less-preferred, yet freely available standard lab chow. We predicted that the DAD1^{-/-} mutant rats would press the lever significantly less than controls across all experiments, have lower breakpoints, consume more freely available food, and have a larger decrease in breakpoints with free food available, suggesting that the dopamine D1 receptor is implicated in exertion of effort for rewards.

3.3 Methods

3.3.1 Experimental Animals

Forty-four adult Wistar (13 homozygous DAD1^{-/-} male, 11 homozygous DAD1^{-/-} female, 10 wildtype male, and 10 wildtype female rats) were used in this experiment. To minimise an extraneous order effect (Shoji & Miyakawa, 2021), some subjects were used in previous experiments while others were experimentally naïve (randomly assigned) at the start of the present study. The DAD1^{-/-} mutants were originally developed using N-ethyl-N-nitrosourea (ENU) mutagenesis (Harper, 2013; Jacobs & D'Esposito, 2011). The current experiments were performed with offspring that were backcrossed onto wild type rats for at least 10 generations.

The rats were housed in groups of two to three based on matched sex and genotype. Their weights were maintained at 80 – 90% of their free feeding weights (see Table 3.1 for descriptives) with food pellets delivered to their home cages once a day, after experimentation was completed for the day. Water was available ad libitum.

Their individually ventilated polycarbonate home cages were lined with saw dust and contained a red enrichment tube. The housing room was on a reverse 12-hour light/dark cycle (lights off between 7am and 7 pm). All experimental procedures took place during the rats' dark

cycle, between 8am and 6 pm. The housing room was kept at controlled temperature (19-21°C) and humidity (55%).

Table 3.1

Free feeding and food restricted weights per group

Subjects	Free feeding weight (g)*	Food restricted weight (g)*
DAD1 ^{+/+} males	470.44 (63.67)	394.16 (53.45)
DAD1 ^{+/+} females	254.56 (15.07)	228.54 (13.06)
DAD1 ^{-/-} males	329.33 (40.55)	297.29 (41.13)
DAD1 ^{-/-} females	204.08 (27.13)	184.88 (24.37)

*Mean weight per group with standard deviation presented in parenthesis

All experimental procedures were conducted in accordance with Victoria University of Wellington's animal care principles and were approved by the Victoria University of Wellington's Animal Ethics Committee (0000028350).

3.3.2 Apparatus

All experiments were conducted in standard operant chambers (Med Associates Inc., St. Albans, VT, USA) measuring approximately 31 cm x 31 cm x 25 cm. There were two non-retractable levers on the front interface panel, 5cm above the floor. A sugar pellet dispenser was located at floor level in the centre of the front panel between the two levers. Standard 100 mA white lights were situated above every lever and were turned on during specific parts of the experiments (see Procedure below). MED-PC® IV and V instrumentation and software were used to run experiments and for data collection. White odourless sugar pellets (45 mg Bio-serv® Flemington, NJ, USA) were used as reinforcement.

3.3.3 Procedure

Prior to training, all rats underwent a single habituation session. Five sugar pellets were placed into the mouth of the pellet dispenser of each chamber. Subjects were placed into the

operant chamber for 30 min in which they could explore the chamber and freely consume the pellets. Each rat was allocated to a specific operant chamber which was maintained across all sessions and chambers were cleaned between subjects.

Following habituation, subjects began a modified autoshaping procedure (Harper, 2013). During an autoshaping session, a light above one of the two levers (randomly determined) was illuminated for 15 s. A single reinforcer was delivered and the light turned off, if the rat made a response on the corresponding lever or the 15 s had elapsed. Thirty seconds later, a new trial began by one of the lever lights randomly turning on. If the rat made a single response during the presence of the light, then that light immediately turned off and a sugar pellet was delivered. Each rat was exposed to a minimum of nine autoshaping sessions with each session lasting for 30 min. Responding was unexpectedly low across all subjects during the first three sessions, therefore, from session four, the light illumination was increased from 15 s to 30 s to incentivize subjects to respond instead of waiting for a free reinforcer. Responding continued to be low during sessions 4 - 6 for DAD1^{-/-} mutant males, so the light illumination was increased from 30 s to 45 s during sessions 7 – 9, and to 60s for sessions 10 – 14, only for DAD1^{-/-} mutant males.

Subjects progressed to the fixed ratio phase once they had a minimum of 20 responses per autoshaping session for three consecutive days. During the fixed ratio phase, subjects completed one 30 min session on a FR-1 schedule of reinforcement in which one response delivered one reinforcer, followed by two 30 min sessions at FR-2 in which two responses were required for a reinforcer, and finished with two 30 min sessions at FR-5 in which five responses were required for a reinforcer. The number of lever presses performed during every FR session was recorded for each subject.

Next, subjects completed the progressive ratio phase, in which the number of lever presses required for pellet delivery increased after each reinforcer delivery. The number of lever presses required to deliver the reinforcer increased according to the following response ratio: 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 28, 32, 36, 40, 44, etc (Richardson & Roberts, 1996). The final ratio a subject completed is the breakpoint which determines the amount of effort a subject is willing to exert for the reward. Subjects underwent three progressive ratio sessions which lasted for 60 min each. The breakpoint was recorded for each subject.

Subjects then completed the operant effort-choice task in which the subjects chose between lever pressing for highly preferred sugar pellets or consuming less preferred but freely available standard lab chow (Randall et al., 2012, 2014). Subjects underwent six total effort-choice sessions that lasted for 60 min each. At the start of sessions 1, 3, and 5, 20 grams of standard lab chow were placed in a dish on the operant chamber floor. Subjects were then placed into their assigned chamber and chose between consuming the freely available chow and responding on a progressive ratio schedule for sugar pellet reinforcement. The amount of standard chow consumed was recorded along with the breakpoint for each rat. During sessions 2, 4, and 6, the standard chow was not available, and subjects could respond on a progressive ratio schedule for sugar pellet reinforcement. Breakpoints for each rat were recorded. Free food and no-free food conditions were interleaved to examine if the subjects' baseline breakpoint changed after sessions with free food available.

3.3.4. Data Analysis

All statistical analyses were conducted using IBM SPSS Statistics version 27 for W. The alpha level for statistical significance was set at $p < 0.05$. Greenhouse-Geisser corrections were used in the event of Mauchly's test of sphericity being violated.

Three measures of performance were obtained throughout different phases: number of lever presses, breakpoint, and standard chow consumed. For autoshaping and fixed ratio schedules (FR-1, FR-3, FR-5), the number of lever presses per session was obtained. For progressive ratio, the breakpoint was obtained for each session. Finally, for operant effort-choice, the breakpoint and amount of standard chow (g) consumed was obtained per session.

3.4 Results

3.4.1 Acquisition of Lever Pressing via Autoshaping

A repeated measures ANOVA was conducted to examine the effect of sex and genotype on lever pressing across the nine autoshaping sessions. There was a significant effect of genotype $F_{1,35} = 16.12, p < 0.001$: DAD1^{-/-} mutants pressed the lever significantly less than DAD1^{+/+} rats across the nine sessions (see figure 3.1A). There was a significant effect of sex $F_{1,35} = 4.96, p < 0.05$: females pressed the lever significantly more than males across the nine sessions. There was no interaction between sex and genotype across all nine sessions $F_{1,35} = 0.31, p = 0.58$.

There was a significant effect of session $F_{1,35} = 54.35, p < 0.001$. Post hoc comparisons indicated that all subjects responded significantly more during sessions 4 – 9 than sessions 1 -3, which is indicative of lever pressing acquisition. There was a significant interaction between session and genotype $F_{1,35} = 14.21, p < 0.001$. Post hoc comparisons indicated that DAD1^{-/-} mutants pressed the lever significantly less than DAD1^{+/+} rats during sessions 5, 6, 7, 8, and 9 (see figure 3.1A). There was no significant interaction between session and sex $F_{1,35} = 1.65, p = 0.19$, and no significant interaction between session, sex, and genotype $F_{1,35} = 0.58, p = 0.62$.

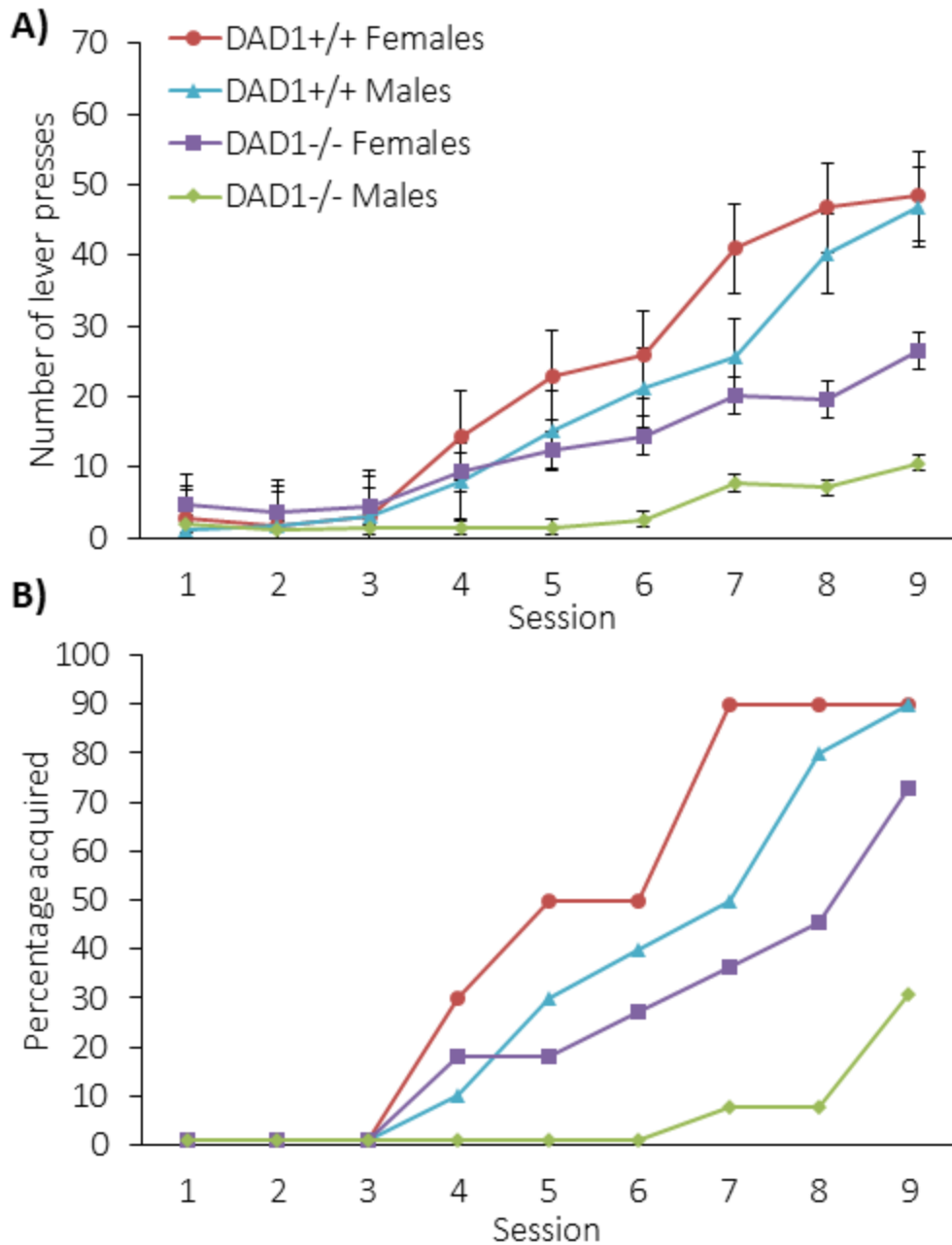


Figure 3.1. Acquisition of lever pressing via autoshaping. (A) Number of lever presses per autoshaping session. (B) Acquisition curve depicting percentage of subjects per group that acquired lever pressing per session. Error bars represent $\pm$ SEM.

A log rank test (Mantel-Cox) was conducted to determine if there were differences in the acquisition rates for the different groups: wildtype males, wildtype females, DAD1^{-/-} males, and DAD1^{-/-} females. The acquisition rates were significantly different, $\chi^2(2) = 20.25$, $p < .001$.

Pairwise comparisons indicate that DAD1^{-/-} mutant males had a significantly lower acquisition rates than DAD1^{-/-} mutant females ($\chi^2(2) = 6.37, p < .01$), wildtype females ($\chi^2(2) = 15.94, p < .001$), and wildtype males ($\chi^2(2) = 14.88, p < .001$; see Table 3.2, see figure 3.1B). Subjects that did not acquire lever pressing were removed for the following experiments.

Table 3.2

Acquisition rate for lever pressing

Subjects	N total	N acquired	% acquired
DAD1 ^{+/+} males	10	9	90%
DAD1 ^{+/+} females	10	9	90%
DAD1 ^{-/-} males	13	4	31%
DAD1 ^{-/-} females	11	8	73%

3.4.2 Fixed Ratio Schedules of Reinforcement

3.4.2.1 FR-1

A two-way ANOVA was conducted to examine the effect of sex and genotype on lever pressing on an FR-1 schedule of reinforcement. There was a significant effect of genotype, $F_{1,26} = 12.76, p < 0.001$; DAD1^{-/-} mutants pressed the lever significantly less than DAD1^{+/+} rats (see figure 3.2). There was no significant effect of sex $F_{1,26} = 0.017, p = 0.90$, and no significant interaction between sex and genotype $F_{1,26} = 0.12, p = 0.73$.

3.4.2.2 FR-2

A two-way ANOVA was conducted to examine the effect of sex and genotype on lever pressing on an FR-2 schedule of reinforcement. There was a significant effect of genotype, $F_{1,26} = 45.36, p < 0.001$; DAD1^{-/-} mutants pressed the lever significantly less than DAD1^{+/+} rats (see figure 3.2). There was no significant effect of sex, $F_{1,26} = 0.0004, p = 0.98$, and no significant interaction between sex and genotype, $F_{1,26} = 0.73, p = 0.40$.

3.4.2.3 FR-5

A two-way ANOVA was conducted to examine the effect of sex and genotype on lever pressing on an FR-5 schedule of reinforcement. There was a significant effect of genotype, $F_{1,26} = 51.45$, $p < 0.001$; DAD1^{-/-} mutants pressed the lever significantly less than DAD1^{+/+} (see figure 3.2). There was no significant effect of sex $F_{1,26} = 0.72$, $p = 0.40$, and no significant interaction between sex and genotype, $F_{1,26} = 0.23$, $p = 0.63$.

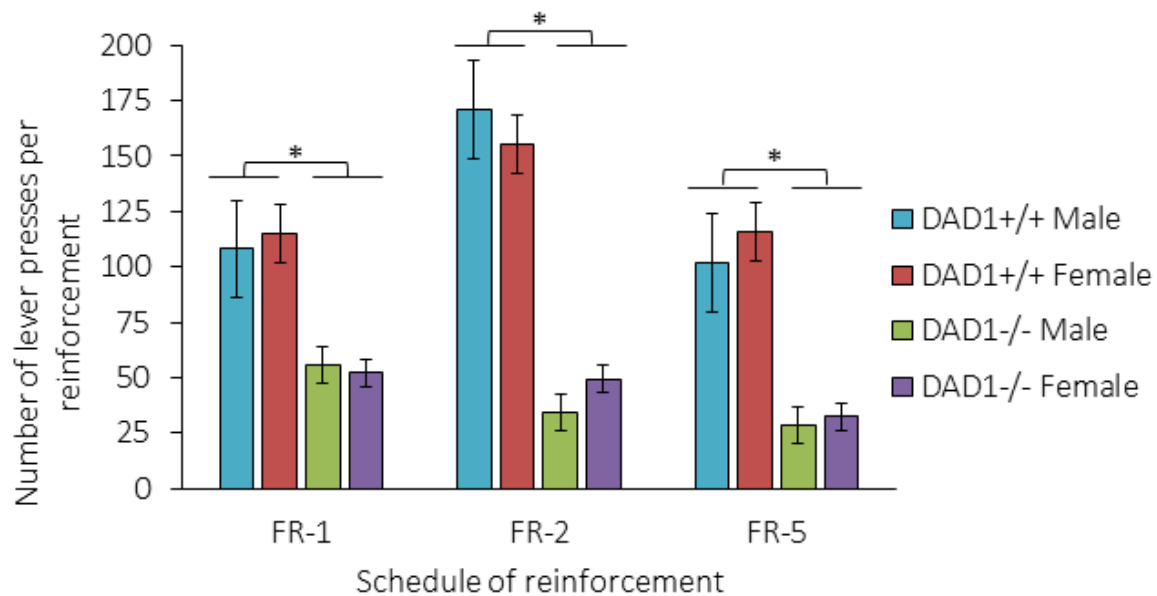


Figure 3.2. Average number of lever presses per group across fixed ratio schedules. *denotes $p < 0.05$. Error bars represent $\pm$ SEM.

3.4.3 Progressive Ratio

A repeated measures ANOVA was conducted to examine the effect of sex and genotype on breakpoint values across the three sessions. There was not a significant effect of session, $F_{1,35} = 1.63$, $p = 0.21$; subjects had consistent breakpoints across all three sessions. There was a significant genotype effect, $F_{1,35} = 16.44$, $p < 0.001$; DAD1^{-/-} mutants had significantly lower breakpoints than DAD1^{+/+} rats across the sessions. There was no significant sex effect, $F_{1,35} = 0.22$, $p = 0.65$, and no significant interaction between sex and genotype, $F_{1,35} = 0.22$, $p = 0.65$. There

was not a significant interaction between session and sex, $F_{1,35} = 0.21$, $p = 0.81$ or session and genotype, $F_{1,35} = 0.31$, $p = 0.74$ (see figure 3.3).

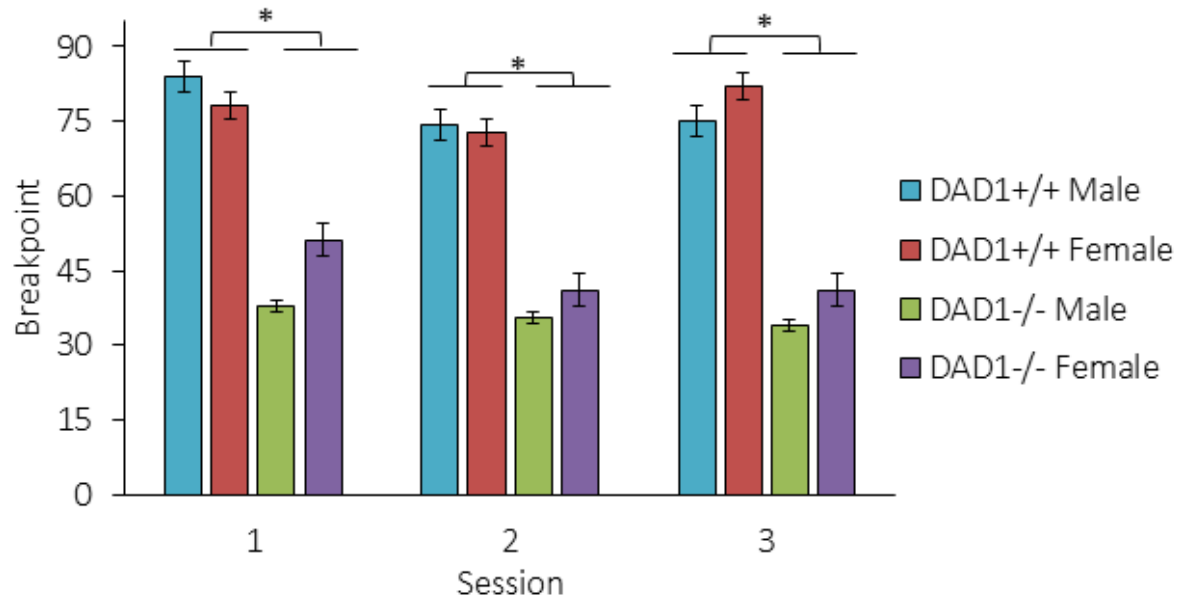


Figure 3.3. Average breakpoint ratio for each group across the three sessions. *denotes $p < 0.05$. Error bars represent $\pm$ SEM.

3.4.4 Operant Effort-Choice

3.4.4.1 Breakpoint Free Food Condition

A repeated measures ANOVA was conducted to examine the effect of sex and genotype on breakpoint values across the three food sessions. There was not a significant effect of session, $F_{1,35} = 0.41$, $p = 0.67$; subjects had consistent breakpoints across all three sessions. There was a significant genotype effect, $F_{1,35} = 20.05$, $p < 0.001$; DAD1^{-/-} mutants had significantly lower breakpoints than DAD1^{+/+} rats across the sessions (see figure 3.4). There was a significant interaction between session and genotype, $F_{1,35} = 4.25$, $p < 0.05$; DAD1^{-/-} mutants had significantly lower breakpoints than DAD1^{+/+} rats in all three sessions. There was no significant sex effect, $F_{1,35} = 3.31$, $p = 0.08$, no significant interaction between sex and genotype, $F_{1,35} = 0.18$, $p = 0.67$, and no significant interaction between session and sex, $F_{1,35} = 2.10$, $p = 0.13$.

3.4.4.2 Breakpoint No Free-Food Condition

A repeated measures ANOVA was conducted to examine the effect of sex and genotype on breakpoint values across the three non-food sessions. There was a significant effect of session, $F_{1,35} = 4.77$, $p < 0.05$. Post hoc comparisons indicated that all subjects had lower breakpoints in sessions 2 and 3 than the first session. There was a significant genotype effect, $F_{1,35} = 10.99$, $p < 0.01$; DAD1^{-/-} mutants had significantly lower breakpoints than DAD1^{+/+} rats across the sessions (see figure 3.4). There was no significant sex effect, $F_{1,35} = 0.49$, $p = 0.49$, no significant interaction between sex and genotype, $F_{1,35} = 0.05$, $p = 0.83$, no significant interaction between session and sex, $F_{1,35} = 2.04$, $p = 0.14$, and no significant interaction between session and genotype, $F_{1,35} = 2.43$, $p = 0.09$.

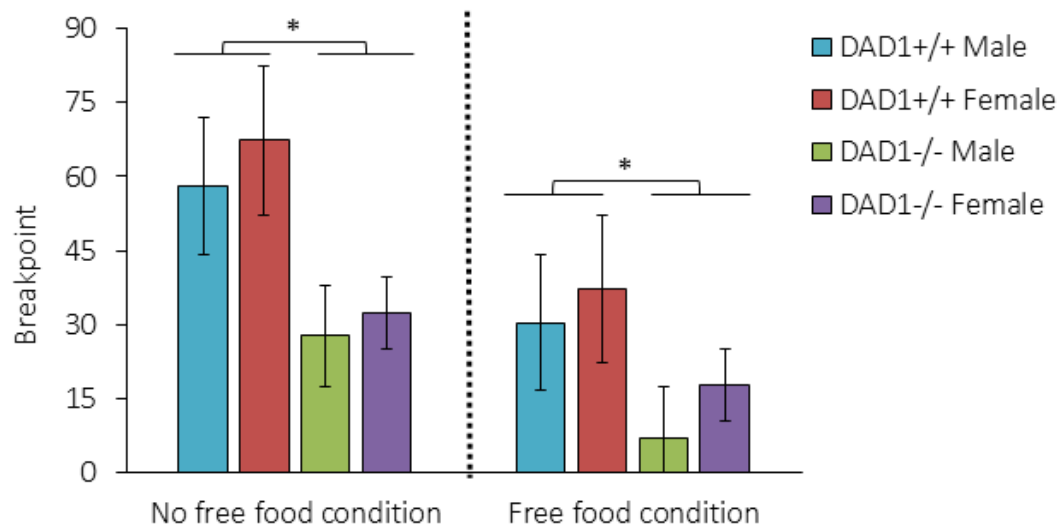


Figure 3.4. Average breakpoint ratio for each group between conditions. In the “no free food condition” subjects could press the lever for sugar pellets. During the “free food condition” subjects chose between pressing the lever for sugar pellets or consuming freely available chow. Error bars represent $\pm$ SEM. * denotes $p < 0.05$.

3.4.4.3 Change in Breakpoints Between Food Conditions

To examine the effect freely available food had on breakpoints, we calculated the difference in breakpoints between no food and food conditions. The difference was divided by the no food condition to determine the percentage change in breakpoints between conditions.

A two-way ANOVA was conducted to examine the effect of sex and genotype on percentage change in breakpoints between conditions. There was a significant effect of genotype, $F_{1,26} = 11.36$, $p < 0.01$; DAD1^{-/-} mutants had significantly larger decrease in breakpoints than DAD1^{+/+} rats. There was a significant effect of sex, $F_{1,26} = 8.31$, $p < 0.01$; males had significantly larger decrease in breakpoints than females. There was a significant interaction between sex and genotype, $F_{1,26} = 7.92$, $p < 0.01$. Tukey post hoc comparisons indicated that DAD1^{-/-} mutant males had a significantly larger decrease in breakpoints than DAD1^{-/-} mutant females, DAD1^{+/+} males, and DAD1^{+/+} females (see figure 3.5).

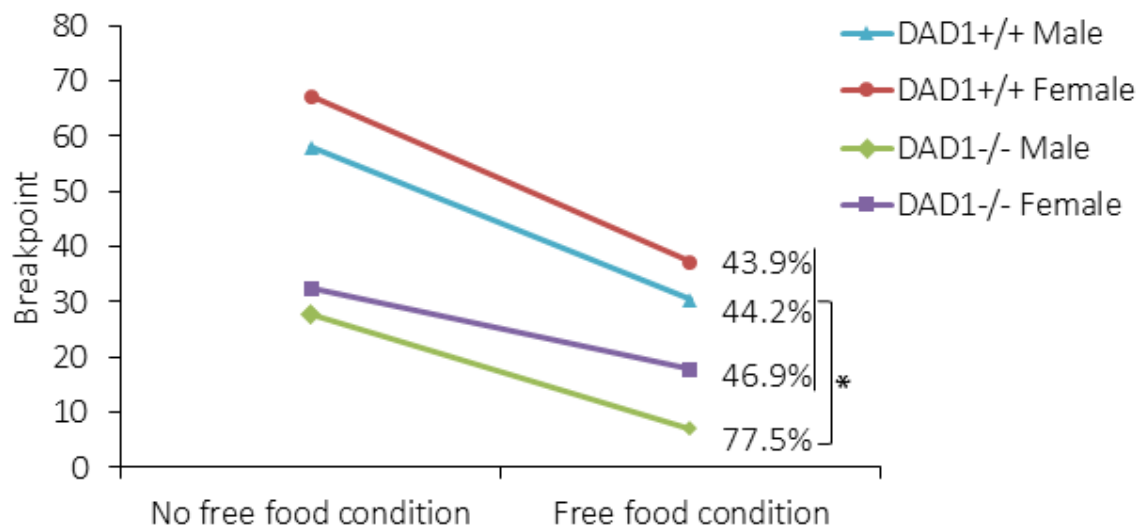


Figure 3.5. Change in breakpoint between food conditions. There was a significant genotype * sex interaction in change in breakpoints between food conditions. DAD1^{-/-} mutant males had the largest change in lever pressing between conditions, with a 77.5% decrease in lever pressing when free food was available.

3.4.4.4 Freely Available Food Consumed

Freely available food consumed was measured in grams and divided by each subject's body weight (in grams). A repeated measures ANOVA was conducted to examine the effect of sex and genotype on freely available food consumed across sessions. There was not a significant effect of session, $F_{1,35} = 0.76$, $p = 0.47$; subjects had consistent food consumption across all three sessions. There was an almost significant effect of genotype, $F_{1,35} = 3.66$, $p = 0.06$; DAD1^{-/-} mutants consumed more freely available food than DAD1^{+/+} rats. There was an almost significant interaction between sex and genotype, $F_{1,35} = 3.59$, $p = 0.07$; DAD1^{-/-} mutant males consumed more freely available food than DAD1^{-/-} mutant females, DAD1^{+/+} males, and DAD1^{+/+} females (see figure 3.6). There was no significant effect of sex, $F_{1,35} = 0.002$, $p = 0.96$, no significant interaction between session and sex, $F_{1,35} = 0.55$, $p = 0.46$, and no significant interaction between session and genotype, $F_{1,35} = 0.06$, $p = 0.81$.

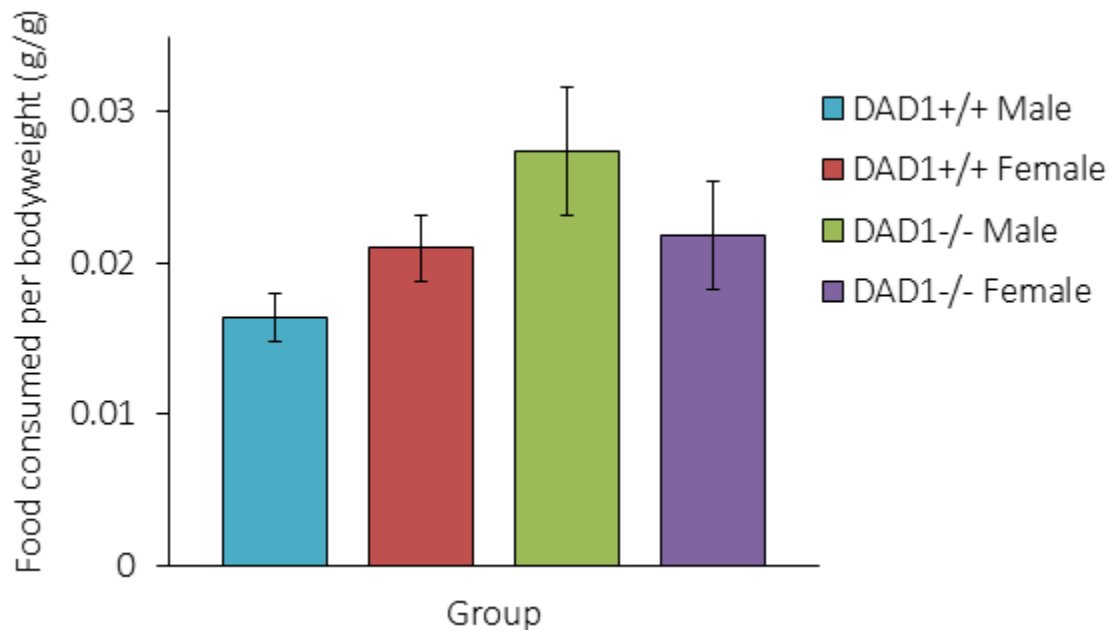


Figure 3.6. Average amount of standard lab chow consumed per group during "free food available" condition. Error bars represent $\pm$ SEM.

3.5 Discussion

This study explored the hypothesis that the dopamine D1 receptor mediates effort-based decision-making. In addition, we evaluated whether the effects of D1 receptor reductions were moderated by the subjects' sex. To test this, we assessed the exertion of effort in male and female DAD1^{-/-} mutant rats, which have a genetic reduction in D1 receptor structure and function (Homberg et al., 2016), and wildtypes in a variety of effort-based decision-making procedures. We predicted that the DAD1^{-/-} mutant rats would exert less effort in all procedures, and as predicted, DAD1^{-/-} mutant rats exerted significantly less effort than wildtypes—DAD1^{-/-} mutant rats pressed the lever significantly less on FR-1, FR-2, and FR-5 schedules and had significantly lower breakpoints on a progressive ratio schedule. Interestingly, we found that the effects of a genetic D1R reduction were significantly larger in male than female rats.

Prior to this study, our lab was unsuccessful in training male DAD1^{-/-} mutants to lever press (Squire, 2016). In the present study, we again found that only 31% of male DAD1^{-/-} mutants acquired lever pressing, however, interestingly, 73% of female DAD1^{-/-} mutants acquired lever pressing. Along with minimal acquisition of lever pressing, DAD1^{-/-} mutant males had a significantly larger decrease in breakpoints between food conditions in the operant effort-choice tasks and consumed on average more free food than controls and DAD1^{-/-} mutant females (approaching significance). There was no sex difference for DAD1^{-/-} mutants in responding on FR-1, FR-2, FR-5, and progressive ratio schedules. However, these experiments were only performed in subjects that had successfully acquired FR-1 lever pressing. Thus, this was not only the best performing cohort, but also (particularly for the male DAD1^{-/-} rats) a relatively small group of subjects. As a result, the statistical power may have been too small to detect statistical sex

differences across all experiments. Future studies could include more DAD1^{-/-} mutant males subjects to examine potential sex differences in fixed and progressive schedules.

There are several possible explanations to account for low acquisition rates in DAD1^{-/-} mutant male rats. Firstly, due to dopamine's role in motor activity, the DAD1^{-/-} mutants might exhibit locomotor impairments leading to a reduction in lever pressing. However, male DAD1^{-/-} mutants showed normal motor, sensory, and neurological characteristics along with normal exploratory behaviour in previous studies (Homberg et al., 2016; Squire et al., 2020). In the current study we did not measure locomotor activity during operant procedures, so it is possible that the DAD1^{-/-} mutants have a locomotor impairment not seen in previous experiments. Secondly, male DAD1^{-/-} mutants might exhibit a sucrose indifference and not find sucrose as rewarding as controls. However, data from our lab found that in a sucrose preference test, male DAD1^{-/-} mutants did not differ from controls in that they drank more sucrose solution than water, suggesting that they found the sucrose solution just as rewarding as controls (VandenBroeke, 2016). Additionally, when examining consumption of freely available sugar pellets (same pellets used in the present study), male DAD1^{-/-} mutants did not differ in amount of sugar pellets consumed when accounting for body weight differences (Squire, 2016). In previous studies, females were not tested for sucrose preference, so it is possible that a difference in sucrose preference for females influenced the results of the present study. A study is underway to examine if this is the case. Third, it is possible that general sex differences in operant responding can account for the lever pressing acquisition rates, however findings are largely mixed on whether sex influences operant responding. Some studies found that males pressed the lever more than females (Errante et al., 2021; Orsini et al., 2016; Presby et al., 2021; van Haaren et al., 1990), while other studies find that females responded more than males (Grimm et al., 2022; van Hest et al.,

1987), and some find no sex differences at all (Heinsbroek et al., 1987; Stringfield et al., 2019). In the present study, we also found no sex differences in wildtype controls, suggesting that the findings are not due to general sex differences in effort-based lever pressing.

A genotype * sex interaction is the most likely explanation for limited acquisition of lever pressing in the present study. Research on the interaction between sex and the dopamine D1 receptor is limited, however, these studies do suggest an interaction in specific behaviours. Latent inhibition is enhanced in female D1 knockout mice but not males, which suggests that the dopaminergic mechanism underlying latent inhibition potentiation may differ between sexes, specifically in the mediation by the D1 receptor (Bay-Ritcher et al., 2009). Michalik et al. (2015) found that both male and female D1 knockout mice showed attenuation of food-anticipatory activity, but the magnitude of attenuation was greater in females. These sex differences do not extend to all behaviours as female and male D1 knockout mice did not differ in locomotor function or food consumed (Bay-Ritcher et al., 2009; Michalik et al., 2015). In terms of operant responding, D1 knockout mice failed to acquire lever pressing (Young & Geyer, 2010) and failed to acquire cocaine self-administration (Caine et al., 2007) but there was no difference between males and females in both studies. El-Ghundi et al. (2003) found that D1 knockout mice pressed the lever minimally, however, not enough for successful acquisition, and only males were used. A possible explanation for the lack of sex differences in these operant studies is the authors used homozygous D1 knockout mice, which completely lack functional D1 receptors. In the present study, the homozygous *DAD1*^{-/-} mutants used are not complete knockouts and have only a 20-50% reduced functionality in the D1 receptor (Homberg et al., 2016). Alternatively, the differences between the studies may point to species difference which has been observed in other genetic studies as well (Ellenbroek & Yoon 2016).

The interaction between sex and the D1 receptor in the present study could be due to general sex differences in spontaneous and evoked dopaminergic transmission. Brundage et al. (2021) found sex differences in spontaneous striatal dopamine release in mice with male mice exhibiting regional differences in dopamine transmission in the dorsal striatum and Nacc regions, while female mice showed no regional differences. The mechanisms underlying these sex differences in dopamine release are largely unknown, but they hypothesised that it could be due to voltage-gated potassium channel functional differences (Brundage et al., 2021). Along with spontaneous dopamine release, there could be sex differences in evoked dopamine release causing subsequent differences in effort-based decision-making. Lefner et al. (2022) found that female rats had higher conditioned responding and faster response latencies during Pavlovian conditioning, but interestingly displayed lower reward-evoked dopamine responses compared to males. Future research could examine spontaneous and evoked dopamine release, specifically during effort-based decision-making, in $DAD1^{-/-}$ mutants and wildtypes to examine if this accounts for the sex differences.

In this respect, it is important to note that estradiol has an overall facilitating effect on dopaminergic transmission by increasing synthesis, release, and reuptake of dopamine via estradiol membrane receptors (Barth et al., 2015; Hidalgo-Lopez & Pletzer, 2017; Jacobs & D'Esposito, 2011). In effort-based decision-making, elevated concentrations of estradiol are associated with increased motivation and reward sensitivity, and there is greater goal-oriented behaviour when dopamine is stimulated in the presence of estradiol (Becker et al., 2016; Yoest et al., 2014). Agonism of alpha and beta estrogen receptors, increases high effort/high reward choices over low effort/low reward choices (Uban et al., 2012). Estradiol's enhancement of dopamine might account for the relatively higher acquisition rates seen in $DAD1^{-/-}$ mutant females as compared to $DAD1^{-/-}$

males. This enhanced dopaminergic system, might have boosted DAD1^{-/-} mutant females' motivation enough to acquire lever pressing while the DAD1^{-/-} mutant males were unable to acquire without this enhancement from estradiol. In line with this, wildtype females pressed the lever more during autoshaping and acquired lever pressing quicker than DAD1^{+/+} wildtype males (see figure 3.1), however this was not statistically significant. This suggests that if estradiol promotes dopamine release, it might be more enhanced in DAD1^{-/-} mutants, but further research needs to measure estradiol and dopamine levels in wildtype and DAD1^{-/-} mutant rodents and examine if this is correlated with their exertion of effort.

Clinical research further underlines estradiol's interaction with dopamine as a possible explanation for sex differences in incidences and severity of several neuropsychiatric disorders such as schizophrenia (Brzezinski-Sinai & Brzezinski, 2020). For example, age of schizophrenia onset differs between sexes with males being diagnosed in their late teens to early 20s while females are diagnosed in their late 20s to early 30s with an additional peak onset only seen in females around ages 40-50 around menopause (Brzezinski-Sinai & Brzezinski, 2020; Gogos et al., 2015). Estradiol levels are increased during puberty which could explain later diagnosis in women, and the decrease in estradiol during menopause could account for the second age of onset in females (Koebele et al., 2015; Li et al., 2016). Females also show fluctuations in severity of schizophrenia symptoms as their estradiol levels fluctuate throughout their reproductive cycle (Kokras & Dalla, 2014).

The unexpected interaction between sex and genotype in acquisition rates of lever pressing and decrease in breakpoints between food conditions emphasises the importance of including both male and female rodents in preclinical research. There has been a bias in preclinical research to only include male rodents due to hormonal fluctuations in females, however, meta-analyses show

that female rodents are no more variable than males in a multitude of neurological and behavioural tests (Becker et al., 2016; Beery, 2018). This bias in animal research and preclinical trials has caused poorer treatment outcomes for females (Beery, 2018; Thorne et al., 2022). Additionally, as mentioned previously, there are sex differences in incidences and severity of neuropsychiatric disorders which warrants the use of both males and females in animal research and preclinical trials (Bao & Swaab, 2010; Thibaut, 2016).

3.5.1 Conclusion

The present study's findings that DAD1^{-/-} mutant rats exert significantly less effort across all procedures supports and extends previous pharmacological research which showed that D1-like antagonist administration led rodents to consume freely available chow over lever pressing for sugar pellets (Yohn et al., 2015) and biased rodents towards the low-effort, smaller food reward in T-maze procedures (Schweimer & Hauber, 2006). This is the first study to show that a selective genetic reduction in the dopamine D1 receptor leads to blunted motivation and exertion of effort. Although, available pharmacological manipulations of the D1 receptor are not selective for only the D1 receptor, there is ongoing research to develop more selective D1 drugs using non-catechol agonists (Hall et al., 2019; Ye et al., 2013), which, from the present study's findings, could be vital for improving amotivation and blunted exertion of effort found in neuropsychiatric disorders, including schizophrenia and major depressive disorder. The unexpected finding that DAD1^{-/-} mutant females can acquire lever pressing at a rate over twice as much as mutant males warrants further investigation into possible sex differences in spontaneous and evoked dopamine release and a potential influence of estradiol on dopaminergic transmission. Additionally, these results highlight the importance of including both sexes in research, especially when there are sex

differences in incidences and severity of neuropsychiatric disorders (Brzezinski-Sinai & Brzezinski, 2020).

3.6 Extended Discussion (not included in publication)

Since this chapter was published as an individual paper, the discussion does not mention the “runt” observation in Chapter 2 that prompted the effort-based decision-making experiments. The impaired motivation in DAD^{-/-} mutants, specifically the reduction in exertion of effort, could explain why DAD1^{-/-} mutants had a higher likelihood of “runt” classification in the neonatal period. Although the present chapter generally support this explanation, future research could examine pups’ exertion of effort in relation to seeking milk to determine if this impaired motivation in operant-chambers applies to milk-seeking behaviour.

The following chapter builds upon the present study to investigate whether deficits in individual components of autoshaping (e.g., Pavlovian incentive learning, instrumental responding) can account for the DAD1^{-/-} mutants’ reduced acquisition of lever pressing in the present study.

Chapter 4

Pavlovian

Instrumental

Transfer

Chapter 4

Pavlovian Instrumental Transfer

4.1 Background

In the previous chapter (Chapter 3), $DAD1^{-/-}$ mutants demonstrated reduced motivation across fixed and progressive ratio schedules, specifically with less effort exerted. There was an unexpected sex * genotype interaction in acquisition of lever pressing via autoshaping with only 31% of $DAD1^{-/-}$ mutants males acquiring compared to 73% of $DAD1^{-/-}$ mutant females and 90% of wildtype males and females. During autoshaping, a light cue was paired with the delivery of sucrose pellets so that subjects learned that pressing the lever resulted in sucrose rewards. Autoshaping involves classical (Pavlovian) conditioning and operant conditioning; the sugar pellets are delivered when subjects press the lever (operant conditioning) or if subjects do not press the lever, pellets are delivered following cue-presentation (classical conditioning) (Kearns & Gomez Serrano, 2010). Given the sex * genotype interaction in acquisition of lever pressing via autoshaping, the current chapter investigates components of classical and operant conditioning independently to examine if impairments in classical and/or operant conditioning can account for the autoshaping acquisition results.

In classical conditioning, an unconditioned stimulus (US) is paired with a conditioned stimulus (CS) so that the CS eventually produces a conditioned response (CR) (Kearns & Gomez-Serrano, 2010). An important feature of classical conditioning is that the CS and US are presented independent of the subject's behaviour (Kearns & Gomez-Serrano, 2010; Maiorov, 2019). For example, in a classical lever-pressing experiment, food rewards would be given after presentation of the CS, whether the subject presses the lever or not. Performance in classical conditioning is modulated by Pavlovian incentive learning in which the cue acquires motivational properties to

enhance reward seeking behaviours, such as approaching the food hopper after learning a cue is associated with food delivery (Wassum et al., 2011).

It is well-documented that dopamine is involved in classical conditioning. Dopaminergic cells respond to sensory cues that predict rewards more than responding to the predicted reward itself, and these dopaminergic responses are dependent on cue-reward associations that develop during classical conditioning (Pan et al., 2005). Dopaminergic release varies depending on the expected value and probability of the reward (Schultz, 2007). There are no increases in dopamine release after the reward approach/seeking or during reward consumption, implicating dopamine in reward seeking behaviour involving cue-reward associations, but not reward consumption (Roitman et al., 2004).

Accumulating evidence suggests that the dopamine D1 receptor likely plays a role in dopamine's modulation of cue-reward associations. The dopamine D1 receptor antagonist, SCH-23390, impaired the formation of drug-cue associations and reduced the reinstatement of cue-induced reward-seeking behaviour in male rats (Berglind et al., 2006). Administration of SCH-23390 into the lateral PFC impaired the *formation* of cue-reward associations in monkeys but did not impair already established cue-reward associations (Puig & Miller, 2012). In *Drosophila*, compared to control subjects, those with a mutation in the D1 receptor had greatly impaired formation of cue-reward associations in Pavlovian olfactory conditioning (Kim et al., 2007). Across multiple species, antagonism of and mutations in the D1 receptor impaired formation of cue-reward associations and subsequent classical conditioning, which suggests that the D1 receptor might mediate Pavlovian incentive learning.

Opposite to classical conditioning, in operant conditioning, the presentation of the reward is *dependent* on the subject's behaviour (Kearns & Gomez-Serrano, 2010). For example, in an

operant lever-pressing experiment, food rewards would be given only if the subject presses the lever. Performance in operant conditioning is modulated by instrumental incentive learning in which the relationship between actions and their rewards is encoded and updated with the value of the reward which in turn enhances goal-directed behaviour, such as pressing the lever (Corbit & Balleine, 2015; Wassum et al., 2011).

Research suggests that dopamine modulates the exertion of effort in instrumental responding for rewards, but dopamine does not appear to modulate the incentive learning process in instrumental conditioning like it does with classical conditioning. Dopamine antagonism reduces instrumental responding for tasks with high response requirements (or high exertion of effort) but does not impact instrumental responding for tasks with low response requirement (Corbit & Balleine, 2015; Salamone et al., 2007; Salamone et al., 2022). Low doses of dopamine antagonists and depletion of Nacc dopamine biases instrumental responding towards low effort responding (Markou et al., 2013). Dopamine transporter knockdown mice (DAT KD) who have high extracellular levels of dopamine display intact instrumental incentive learning and increased exertion of effort for rewards (Cagniard et al., 2006; Salamone et al., 2022). Inhibition of dopamine synthesis in the Nacc did not alter incentive instrumental learning instrumental but did reduce the exertion of effort (Smith et al., 2021; Salamone et al, 2022). Taken together, these results suggest that dopamine mediates the exertion of effort but not instrumental incentive learning.

Evidence suggests that the dopamine D1 receptor likely plays a role in dopamine's modulation of effort exertion in instrumental responding. Administration of the D1-like antagonist, SCH-23390, into the Nacc, dose-dependently reduced instrumental responding in a lever-pressing paradigm while leaving the consumption of the food reward intact (Koch et al., 2000). Administration of D1/D2 antagonist, flupenthixol, did not influence instrumental incentive

learning but did reduce exertion of effort (Wassum et al., 2011). On the contrary, Smith-Roe and Kelley (2000) found that low doses of D1-like antagonist, SCH-23390, impaired the acquisition of instrumental learning and Beninger and Miller's (1998) review found that the D1 receptor was implicated in instrumental incentive learning. Together these findings suggest that the dopamine D1 receptor is implicated in the exertion of effort and possibly in instrumental incentive learning, but more research is needed to parse out the D1 receptor's role in instrumental incentive learning.

As mentioned above, this chapter investigates classical and operant conditioning independently to examine if impairments in these processes can account for the autoshaping acquisition results found in Chapter 3. Classical conditioning can influence instrumental responding, even if the instrumental responding is acquired independently of the classical conditioning. This process of transfer is called Pavlovian-to-Instrumental transfer (PIT) (Cartoni et al., 2013; Holland, 2004; Holmes et al., 2010). Positively valued conditioned stimuli promote instrumental responding and approach while negatively valued conditioned stimuli decrease responding and approach (Cartoni et al., 2016). The PIT paradigm examines this influence of classical conditioning on instrumental responding. The paradigm consists of classical conditioning independent of instrumental training before the combination of both during the test. During the test, the reward is not available and an increase in instrumental responding during the conditioned stimuli presentation is a measure of the transfer effect (Cartoni et al., 2013; Cartoni et al., 2016; Holland, 2004; Holmes et al., 2010).

Not surprising, due to the role of dopamine and the D1 receptor in Pavlovian incentive learning, a multitude of studies show that dopamine antagonism and lesions in the VTA impair the transfer effect with classical conditioning not enhancing instrumental responding (Corbit & Balleine, 2016; Holmes et al., 2010). Similarly, microinjections of the D1 antagonist, SCH-23390,

into the Nacc greatly reduced the transfer effect compared to vehicle and D2 antagonism (Lex & Hauber, 2008). Laurent et al. (2014) found that increased expression of delta-opioid receptors on cholinergic interneurons led to increased activity of D1 neurons and resulted in larger transfer effects.

In summary, it is clear that dopamine modulates Pavlovian incentive learning in classical conditioning and exertion of effort in instrumental responding, which is likely, partly mediated by the dopamine D1 receptor. In addition, dopamine and its D1 receptor have been implicated in instrumental incentive learning and the transfer of Pavlovian conditioning to instrumental responding.

4.2 Overall Aims of Chapter Four

This chapter investigates the role of the dopamine D1 receptor in transferring Pavlovian conditioning to instrumental responding in a PIT paradigm. Subjects, DAD1^{-/-} mutants and wildtype controls, undergo classical conditioning independently of operant conditioning before combining the two in the test phase to examine the impact classical conditioning has on instrumental responding (Takahashi et al., 2019).

It is predicted that:

1. DAD1^{-/-} mutants will display reduced exertion of effort in instrumental responding relative to wildtypes, with less lever presses across all schedules of reinforcement.
2. DAD1^{-/-} mutants will likely exhibit an impairment in instrumental incentive learning seen by a smaller difference between active and inactive levers across all schedules of reinforcement relative to wildtypes.
3. DAD1^{-/-} mutants will have a smaller transfer score during the PIT test, with classical conditioning not enhancing instrumental responding as much as in the wildtypes.

4. Males and females of both genotypes are included in this experiment. Due to the sex*genotype interaction in autoshaping in the previous chapter, it is predicted that DAD1^{-/-} mutant males will have the smallest transfer score.

4.3 Methods

4.3.1 Experimental Animals

Thirty-seven experimentally naïve adult Wistar (9 wildtype male, 11 wildtype female, 9 DAD1^{-/-} mutant male and 8 DAD1^{-/-} female) rats were used in this experiment. The rats were housed in groups of two to three based on matched sex and genotype in standard polycarbonate cages. Their weights were maintained at 80–90% of their free feeding weights (see Table 4.1 for descriptives) with food pellets delivered to their home cages once a day, after experimentation was completed for the day. Water was available ad libitum.

Table 4.1

Free feeding and food restricted weights per group

Subjects	Free feeding weight (g)*	Food restricted weight (g)*
DAD1 ^{+/+} males	461.67 (35.04)	412.65 (36.23)
DAD1 ^{+/+} females	282.67 (28.98)	254.86 (22.39)
DAD1 ^{-/-} males	353.11 (58.21)	319.31 (51.48)
DAD1 ^{-/-} females	207.00 (28.37)	186.53 (23.37)

*Mean weight per group with standard deviation presented in parenthesis

Their individually ventilated polycarbonate home cages were lined with saw dust and contained a red enrichment tube. The housing room was on a reverse 12-h light/dark cycle (lights off between 7 am and 7 pm). All experimental procedures took place during the rats' dark cycle, between 8 am and 6 pm. The housing room was kept at controlled temperature (19–21 °C) and humidity (55%).

4.3.2 Apparatus

All experiments were conducted in standard operant chambers (Med Associates Inc., St. Albans, VT, USA) measuring approximately 31 cm × 31 cm × 25 cm. During operant conditioning sessions and the PIT test, there were two non-retractable levers (removable) on the front interface panel, 5 cm above the floor. A sugar pellet dispenser was located at floor level in the centre of the front panel between the two levers. Standard 100 mA white lights were situated above every lever and were turned on during specific parts of the experiments (see Procedure below). MED-PC® V instrumentation and software were used to run experiments and for data collection. White odourless sugar pellets (45 mg Bio-serv® Flemington, NJ, USA) were used as reinforcement.

4.3.3 Procedure

The Pavlovian Instrumental Transfer (PIT) protocol followed several steps: habituation, Pavlovian conditioning, instrumental training, and Pavlovian reconditioning with a PIT test conducted after the last training session (see Figure 4.1) (Ostlund & Marshall, 2021; Takahashi et al., 2019).

4.3.3.1 Habituation

Animals were placed in their assigned operant chambers for 60 min for one habituation session. During the 60 min, sucrose pellets were delivered at varying and unpredictable time intervals with an average interval of two minutes. Neither levers nor cue-light were presented during the habituation session.

4.3.3.2 Pavlovian Conditioning

Pavlovian conditioning consisted of a 60 min session for five consecutive days. Levers were not available during these sessions. During a session, a constant cue-light, situated above where the right lever would be, was presented for 2 min at random time intervals with an average

intertrial interval of 2 – 3 minutes. Presentation of the cue-light (conditioned stimulus, CS) was accompanied by an immediate delivery of four sucrose pellets at the start of the cue. There were a total of nine CS and reward pairings per session.

4.3.3.3 Instrumental training

Instrumental training sessions began on the next day following Pavlovian conditioning. Each session lasted for 30 min, and the variable interval (VI) was increased across days. For the first three sessions, animals were trained on an FR-1 schedule of reinforcement in which every lever press resulted in immediate delivery of one sucrose pellet. On the fourth session, animals were trained on a VI-10 (5s, 10s, 15s) schedule of reinforcement in which a variable interval of time (average of 10s) passed and the first response after that time was reinforced with one sucrose pellet. On the fifth session, the VI increased to VI-20 (10s, 20s, 30s), and then eight sessions on VI-30 (10s, 20s, 30s, 40s, 50s). During instrumental training, the cue-light (CS) was not presented, both levers were available, and animals were rewarded for pressing the active (right) lever.

4.3.3.4 Pavlovian Reconditioning

The day after instrumental training finished, animals were subjected to one Pavlovian conditioning session. This session followed the same protocol described above, except the session only lasted for 30 min.

4.3.3.5 Pavlovian Instrumental Transfer Test

The day after Pavlovian reconditioning, all animals were subjected to a 36 min PIT test in which the cue-light was presented randomly four times (ITI of 4 – 10 minutes) for two minutes in length. Both levers were present throughout the entire session, but lever-responding was not reinforced with pellet delivery at any point. A two-minute period prior to the cue-light presentation was considered the “pre-CS period”. Responding on both levers was recorded throughout the entire

session. The number of active (right) lever-presses prior to cue-light (pre-CS) and during the cue-light (CS) presentation was used to quantify the transfer effect.

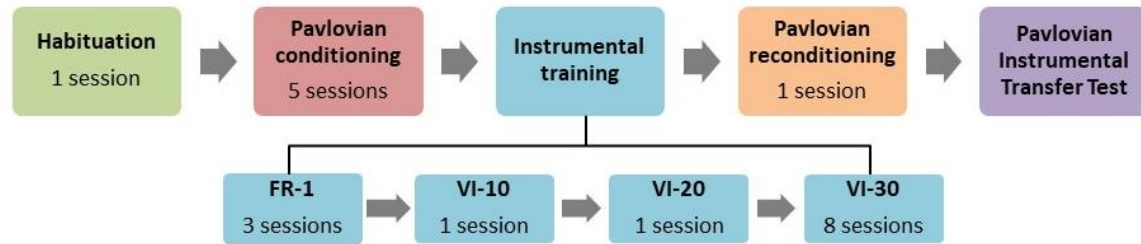


Figure 4.1. Timeline of the Pavlovian Instrumental Transfer protocol.

4.3.4 Data Analysis

The variables of interest for instrumental training were the number of lever presses for active and inactive levers. The variables of interest for the PIT test were the number of lever presses during the pre-CS and cue blocks. There were no measurements recorded during Pavlovian conditioning since the levers were removed during this phase.

For instrumental training, the variables were examined with repeated measures ANOVAs with genotype ($DAD1^{-/-}$ mutants, wildtypes) and sex (male, female) as the between-subject factor and session as the within-subjects factor. Mauchly's test of sphericity indicates that the assumption of sphericity was violated for FR-1 and VI-30 instrumental responding ($p < 0.01$), therefore Greenhouse-Geisser corrections were used.

For the PIT test, the lever pressing was examined with two-way (lever: active, inactive; condition: pre-CS, CS) repeated measures ANOVAs with genotype ($DAD1^{-/-}$ mutants, wildtypes) and sex (male, female) as the between-subject factors. Transfer scores were calculated by taking the absolute value of the number of *cue* lever presses minus the number of *pre-cue* lever presses for each time point (Holland, 2004). Transfer scores were examined with repeated measures ANOVAs with genotype ($DAD1^{-/-}$ mutants, wildtypes) and sex (male, female) as the between-subject factor and time point (1-4) as the within-subjects factor. Mauchly's test of sphericity

indicates that the assumption of sphericity was not violated for lever pressing ($p = 0.66$) nor transfer scores ($p = 0.82$), therefore no corrections were used.

All data were analysed for statistical significance using Jamovi (version 2.3) and SPSS (version 27). The alpha level for statistical significance was $p < 0.05$ and partial eta squared (η^2) was used for the effect size. All significant interactions were followed up with post hoc Student's t-tests.

4.4 Results

4.4.1 Instrumental Training

4.4.1.1 FR-1 Schedule of Reinforcement

DAD1^{-/-} mutants pressed the active lever ($p < 0.05$) significantly less than wildtypes, indicating reduced exertion of effort and/or lack of instrumental learning in DAD1^{-/-} mutants (lever * genotype: $F_{1,34} = 8.43$, $p < 0.01$, $\eta^2 = 0.20$). Wildtypes pressed the active lever during sessions 2 and 3 ($p < 0.01$ for both sessions) significantly more than the inactive lever (session * lever * genotype: $F_{1,34, 45.48} = 6.79$, $p < 0.01$, $\eta^2 = 0.10$), which is indicative of instrumental responding acquisition (see Figure 4.2). DAD1^{-/-} mutants, on the other hand, did not significantly differ between levers or sessions, however lever pressing was relatively minimal across both levers possibly resulting in a floor effect ($p = 1.00$ for all post hoc tests).

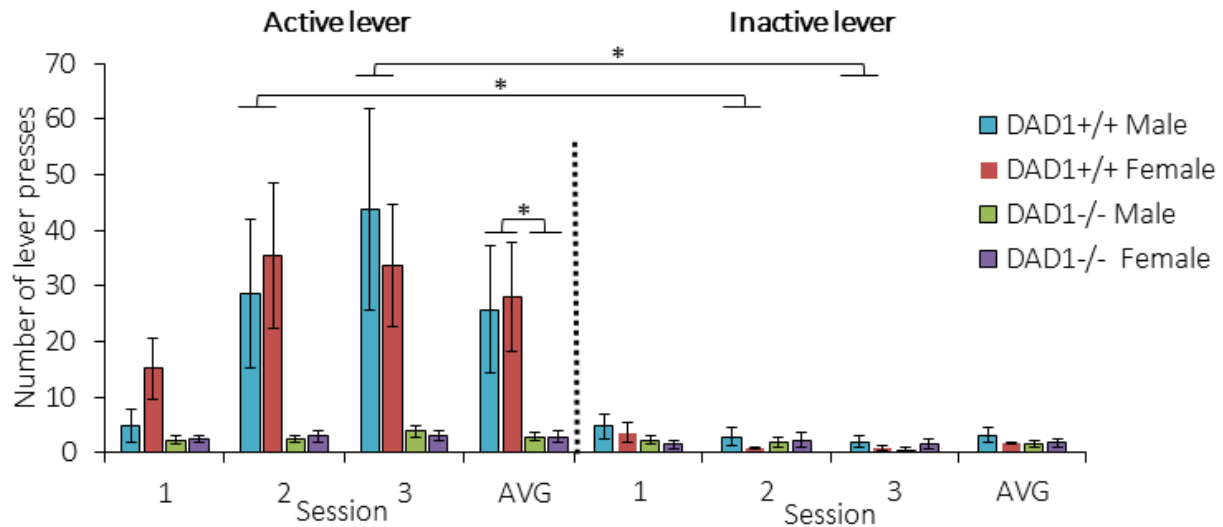


Figure 4.2. Active and inactive lever presses on a FR-1 schedule of reinforcement. DAD1^{-/-} mutants, on average, pressed the active lever less than wildtypes. Wildtypes pressed the active lever more than inactive during sessions 2 and 3. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

4.4.1.2 VI-10 and VI-20 Schedules of Reinforcement

During VI-10 and VI-20, wildtypes pressed the active lever significantly ($p < 0.001$) more than DAD1^{-/-} mutants (VI-10 lever * genotype: $F_{1,34} = 19.75$, $p < 0.001$, $\eta_p^2 = 0.37$; VI-20 lever * genotype: $F_{1,34} = 22.30$, $p < 0.001$). Within genotype, wildtypes pressed the active lever significantly more than the inactive lever ($p < 0.001$ for both schedules), which is indicative of instrumental responding acquisition (see Figure 4.3). DAD1^{-/-} mutants did not significantly differ between levers in VI-10 ($p = 0.99$) nor VI-20 ($p = 0.97$), however lever pressing was relatively low across both levers possibly resulting in a floor effect.

During VI-20, wildtype males pressed the active lever significantly more than DAD1^{-/-} mutant males and females ($p < 0.001$ for both) (lever*sex*genotype: $F_{1,34} = 4.33$, $p < 0.05$, $\eta_p^2 = 0.11$). Within genotype, wildtype males pressed the active lever significantly more than the inactive lever ($p < 0.001$), while wildtype females had a trending significance towards pressing the active lever more than the inactive ($p = .06$), which is indicative of instrumental responding acquisition (see Figure 4.3). DAD1^{-/-} mutant males ($p = 0.99$) and females ($p = 1.00$) did not

significantly differ between levers, likely due to a floor effect as described above. For VI-10, there were no interactions between lever, sex, and genotype ($F_{1,34} = 1.86$, $p = 0.18$, $\eta_p^2 = 0.05$).

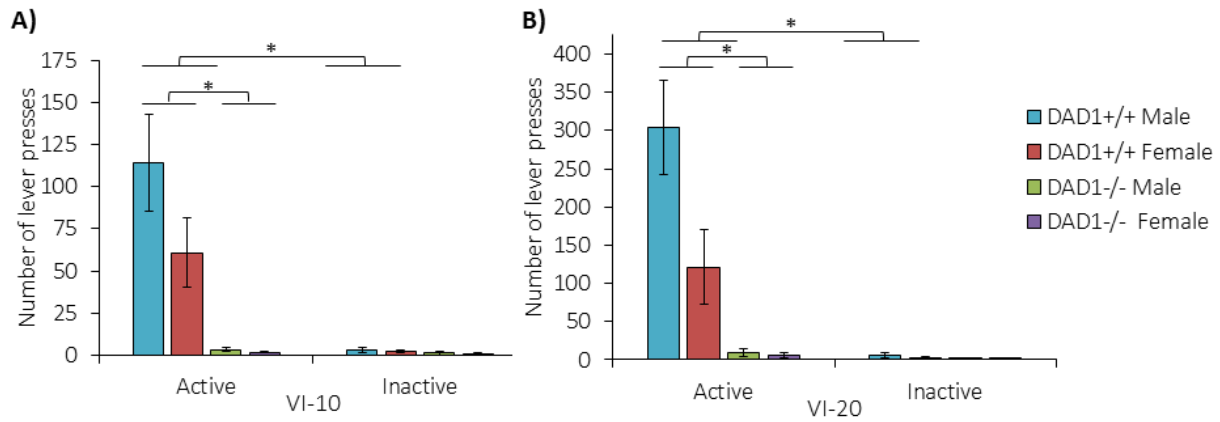


Figure 4.3. Active and inactive lever presses during VI-10 (A) and VI-20 (B). Wildtypes pressed the active lever significantly more than DAD1^{-/-} mutants on both schedules. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

4.4.1.3 VI-30 Schedule of Reinforcement

Wildtypes pressed the active lever during sessions 1 – 8 significantly more than DAD1^{-/-} mutants pressed the active and inactive levers ($p < 0.01$ across all post hoc tests) (lever effect: $F_{1,34} = 36.25$, $p < 0.001$, $\eta_p^2 = 0.52$; session * genotype: $F_{2.98, 101.19} = 6.08$, $p < 0.001$, $\eta_p^2 = 0.10$; see figure 4.4). Within genotype, wildtypes pressed the active lever during sessions 1 – 8 more than they pressed the inactive lever ($p < 0.01$ across all post hoc tests), which is indicative of instrumental responding acquisition (session * lever * genotype: $F_{2.85, 97.03} = 6.15$, $p < 0.001$, $\eta_p^2 = 0.15$; see Figure 4.4). DAD1^{-/-} mutants did not significantly differ between levers or sessions, which is likely caused by low responding rates overall, especially in the mutant females ($p = 1.00$ for all post hoc tests).

There was a sex effect ($F_{1,34} = 6.46$, $p < 0.05$, $\eta_p^2 = 0.16$) with males pressing the levers more than females. Within sex, males pressed the active lever significantly more than the inactive lever ($p < 0.001$) while females had a trending significance towards pressing the active lever more than the inactive ($p = 0.07$), which is overall indicative of instrumental responding acquisition

(lever*sex: $F_{1,34} = 6.36$, $p < 0.05$, $\eta_p^2 = 0.16$). DAD1^{-/-} mutant *males* increase lever pressing across sessions (see figure 4.4) which suggests they do acquire *some* level of instrumental responding; however, this does not reach statistical significance (lever*sex*genotype interaction: $F_{1,34} = 2.11$, $p = 0.16$, $\eta_p^2 = 0.06$), likely due to the large individual variance and low responding rates in DAD1^{-/-} mutant *females*.

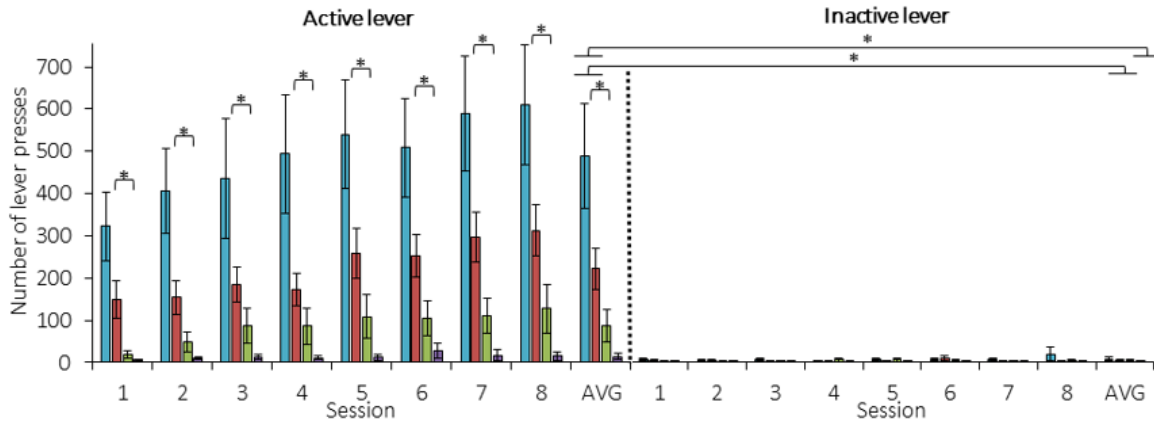


Figure 4.4. Average lever presses on a VI-30 schedule of reinforcement during eight sessions. Wildtypes pressed the active lever significantly more than DAD1^{-/-} mutants pressed the active lever during all eight sessions. Across the sessions, wildtypes pressed the active lever more than DAD1^{-/-} mutants pressed both levers. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

4.4.2 Pavlovian Instrumental Transfer Test

4.4.2.1 PIT Conditions

Wildtypes pressed the active lever during pre-CS significantly more than the active ($p < 0.001$) lever during CS (condition * lever * genotype: $F_{1,30} = 7.56$, $p < 0.01$, $\eta_p^2 = 0.20$), which is indicative of a negative transfer effect likely due to a response competition between lever pressing and pellet dispenser approach (condition * genotype: $F_{1,30} = 9.29$, $p < 0.01$, $\eta_p^2 = 0.24$). DAD1^{-/-} mutants did not differ between pre-CS and CS conditions, indicating a lack of transfer effect ($p = 0.99$) (see figure 4.5).

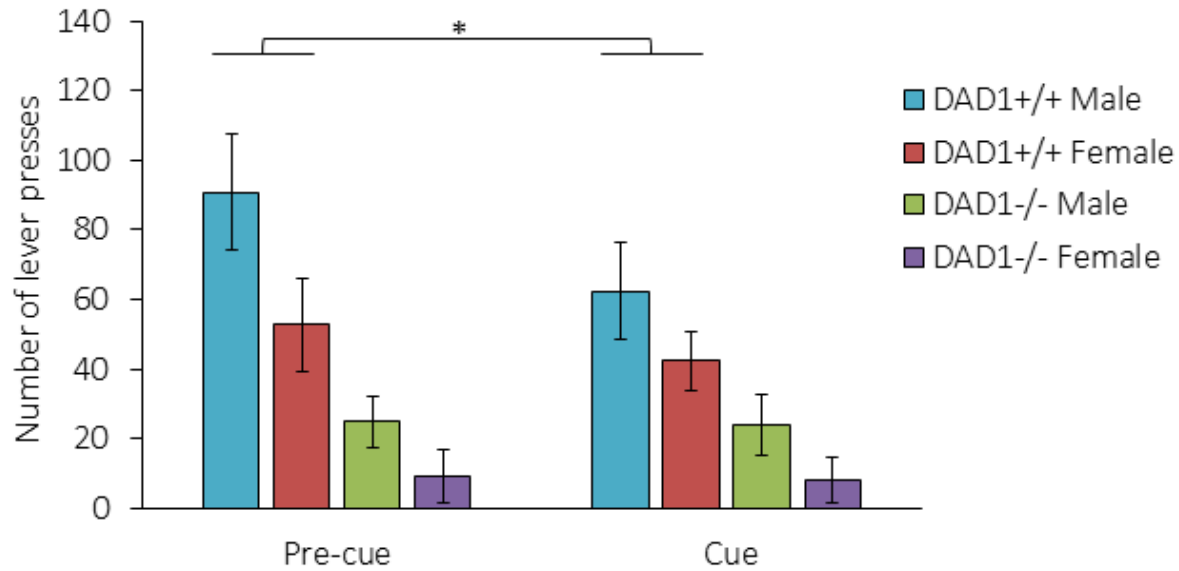


Figure 4.5. Average active lever presses during the PIT test. Wildtypes pressed the active lever during the pre-cue significantly more than during the cue across all times points. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

The cue-light was presented four times during the PIT test, resulting in four blocks throughout the session. Wildtypes pressed the lever significantly less during times 3 and 4 than times 1 and 2, which is expected with extinction (time * genotype: $F_{3,90} = 9.03$, $p < 0.001$, $\eta_p^2 = 0.23$) (see figure 4.6).

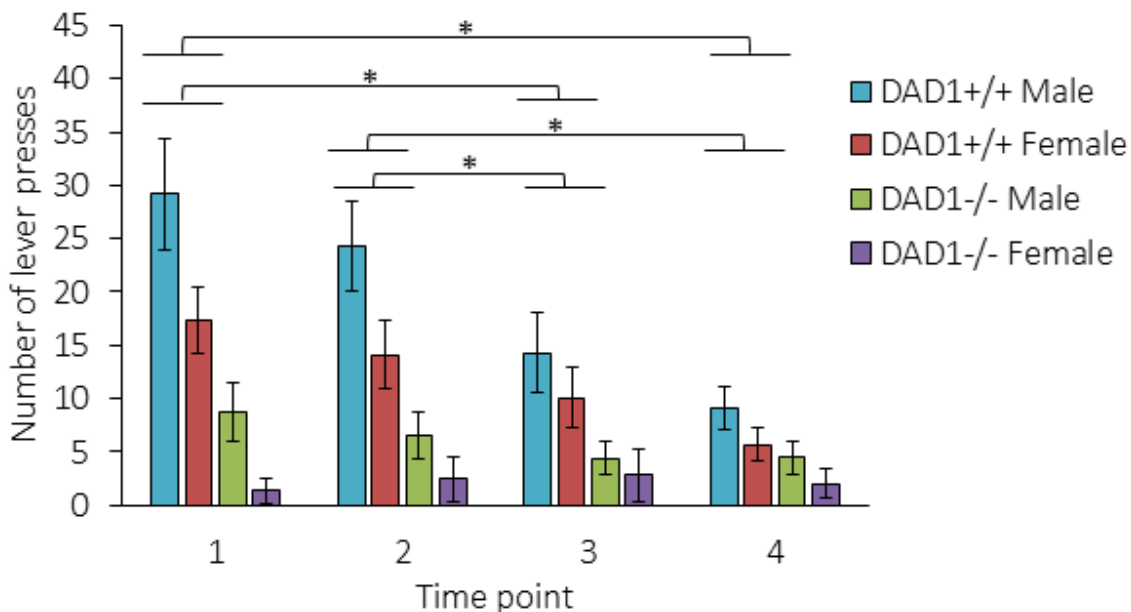


Figure 4.6. Average active lever presses per time block. Wildtypes decreased lever pressing across sessions. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

4.4.2.2 Transfer Scores

DAD1^{-/-} mutants had a significantly smaller absolute transfer score than wildtypes across the time points (genotype effect: $F_{1,30} = 12.97$, $p < 0.01$, $\eta_p^2 = 0.30$), which suggests that Pavlovian conditioning had a greater effect on instrumental responding for wildtypes. Males had a significantly larger absolute transfer score than females across the time points (sex effect: $F_{1,30} = 6.24$, $p < 0.05$, $\eta_p^2 = 0.17$; see figure 4.7). Absolute transfer scores did not differ across time points ($F_{3,90} = 0.75$, $p = 0.53$, $\eta_p^2 = 0.02$).

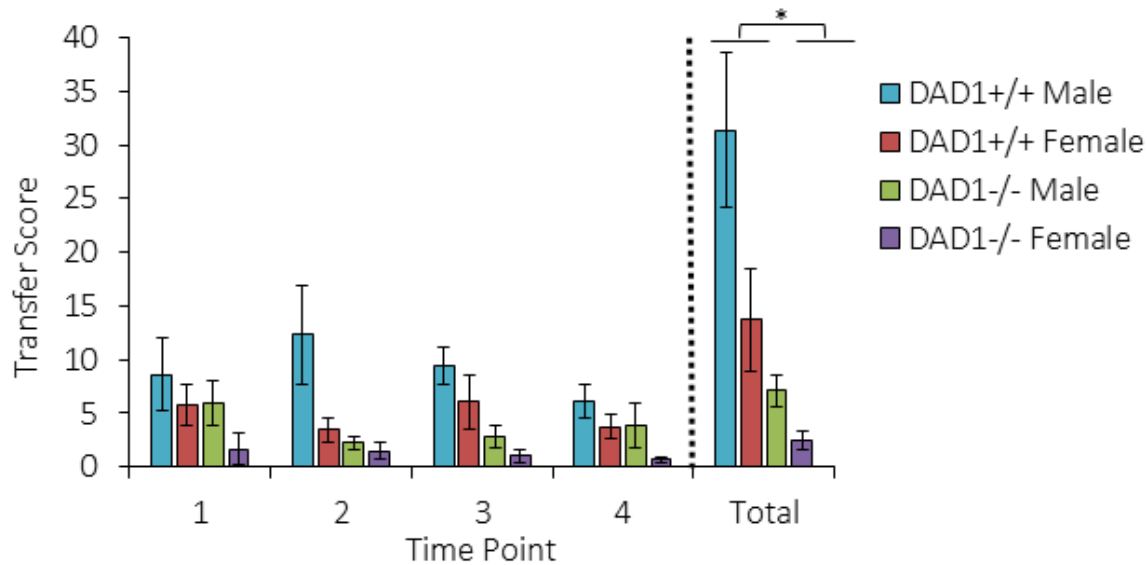


Figure 4.7. Average transfer scores (absolute value) across time points during the PIT test. DAD1^{-/-} mutants had significantly smaller transfer scores than wildtypes. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

4.5 Discussion

The aim of this experiment was to examine if impairments in classical or operant conditioning could account for the sex * genotype interaction in autoshaping in the previous chapter. Classical and operant conditioning were examined using the Pavlovian Instrumental Transfer (PIT) paradigm. It was hypothesised that DAD1^{-/-} mutants would exert less effort than wildtypes, demonstrated by pressing the active lever less during instrumental responding phases.

As predicted, DAD1^{-/-} mutants, regardless of sex, pressed the levers significantly less than wildtypes across all schedules of reinforcement, which was also found in the previous chapter. These results support the postulation that dopamine modulates exertion of effort in instrumental responding (Cagniard et al., 2006; Corbit & Balleine, 2016; Salamone et al., 2007; Salamone et al., 2022), and further support the assertion that dopamine's modulation of effort-exertion occurs via the dopamine D1 receptor (Koch et al., 2000; Wassum et al., 2011).

It was hypothesised that DAD1^{-/-} mutants would exhibit impaired instrumental incentive learning with a smaller difference between active and inactive levers compared to wildtypes. As expected, wildtypes learned that the active lever resulted in sucrose pellet delivery with significantly more presses on the active than inactive lever across all schedules of reinforcement. On the other hand, DAD1^{-/-} mutants did not significantly differ between the active and inactive levers during all schedules of reinforcement which supports the findings that D1-antagonism impaired instrumental incentive learning (Beninger & Miller, 1998; Smith-Roe & Kelly, 2000). On the other hand, although not statistically significant, DAD1^{-/-} mutant males increased lever pressing across the eight VI-30 sessions indicating that they do acquire *some* level of instrumental responding. Overall, interpretation of these results should be done with caution because instrumental responding was overall low for DAD1^{-/-} mutants, so it is possible that there are no significant differences between levers due to a floor effect.

It was hypothesised that DAD1^{-/-} mutants would exhibit a reduced transfer effect during the PIT test with a smaller transfer score than wildtypes. In line with this, DAD1^{-/-} mutants had a significantly smaller absolute transfer score than wildtypes across the four time points during the PIT test. Interestingly, DAD1^{-/-} mutants did not show any significant difference in lever pressing between the cue conditions, indicating an overall lack of transfer effect. However, it is difficult to

conclude whether the lack of transfer effect is due to an impairment in transfer of Pavlovian conditioning to instrumental responding, or whether overall instrumental responding was too low in DAD1^{-/-} mutants to see a difference between cue conditions. Additionally, it is possible that the DAD1^{-/-} mutants have impaired Pavlovian incentive learning and did not learn to associate the light cue with sucrose pellet delivery during the initial Pavlovian conditioning, therefore, there was no cue-reward associations to transfer to instrumental responding. To address this, future research could measure pellet dispenser entries, latency to the dispenser and percentage of time spent at the dispenser to examine the animals' approach behaviour during cue presentation. If DAD1^{-/-} mutants have impaired formation of cue-reward associations, we would expect *reduced* approach behaviour during the cue presentation. Impaired formation of cue-reward associations would mean that in the PIT test, there is no cue-reward associations for subjects to transfer to instrumental responding. On the other hand, if DAD1^{-/-} mutants display intact cue-reward associations, then the lack of transfer effect in the present study would likely be due to an impairment in the *transfer* of Pavlovian conditioning not the initial learning of cue-reward associations.

Due to their low autoshaping acquisition in the previous chapter, it was hypothesised that DAD1^{-/-} mutant males would have the smallest transfer score. Contrary to predictions, DAD1^{-/-} mutant females had the smallest transfer score. When looking at the total transfer score, the order from largest to smallest transfer score was as follows: wildtype males, wildtype females, DAD1^{-/-} mutant males and DAD1^{-/-} mutant females. This order is also present in the instrumental responding phase with males pressing the lever significantly more than females on variable schedules of reinforcement. A review paper by Dalla and Shors (2009) found that male rats perform better than female rats in Pavlovian conditioning paradigms, whereas females tend to outperform males in operant conditioning. For example, in an operant-based procedure, females

responded at higher rates than males for a sucrose solution (Grimm et al., 2022) and females responded quicker than males in an operant-avoidance paradigm (Simpson & Kelly, 2012). The present study partially contradicts the literature with males outperforming females in operant responding on variable schedules while females outperformed males during fixed ratio schedules. In the previous chapter, females outperformed males in acquiring lever pressing but this clear sex difference disappeared when subjects transitioned from autoshaping to fixed and progressive ratio schedules. Future research could incorporate variable, fixed and progressive ratios of reinforcement within the same paradigm to clarify if sex differences seen in DAD1 animals vary based on schedule of reinforcement.

Wildtypes significantly differed between pre-CS and CS conditions during the PIT test indicating that Pavlovian incentive learning was transferred to instrumental responding as expected. Contrary to the expected *positive* transfer effect, wildtypes pressed the levers significantly less during CS than during pre-CS, which is indicative of a *negative* transfer effect. The plausible explanation for this is a competition between Pavlovian conditioning and instrumental responding. Pavlovian conditioning typically enhances instrumental responding; however, cue-reward associations from Pavlovian conditioning can elicit conditioned approaches to the CS (i.e., sign-tracking) along with conditioned approaches to the reward delivery site, pellet dispenser (i.e., goal-tracking) (Cartoni et al., 2016; Holmes et al., 2010). Consequently, during the PIT test, these two conditioned behaviours cause a competition between the targeted instrumental response (e.g., lever pressing) and the conditioned Pavlovian response (e.g., approaching the pellet dispenser). For the wildtypes, their reward-cue association might have enhanced approaching the pellet dispenser in search for sucrose pellets instead of pressing the lever. Although the wildtypes

had a negative transfer effect in a different direction than hypothesised, Pavlovian conditioning had a clear influence on instrumental responding as expected.

The present study's results support a prominent role of the dopamine D1 receptor in exertion of effort and possibly in instrumental incentive learning with overall low responding rates. Regarding Pavlovian conditioning, the lack of transfer effect seen in $DAD1^{-/-}$ mutants could be due to an impairment in the transfer of cue-reward associations to instrumental responding and/or an impairment in the actual formation of cue-reward associations during initial Pavlovian conditioning. Unfortunately, the results of the present study do not indicate how the D1R impacts the formation of cue-reward associations and the transfer of these associations, therefore more research is needed for concrete conclusions.

Continuing with the investigation of the dopamine D1 receptor in a variety of functions, the following two chapters investigate how a reduction in the D1R impacts hedonic processing.

Chapter 5

Anticipation and Consumption of Sucrose Solution

Chapter 5

Anticipation and Consumption of Sucrose Solution

5.1 Background

As discussed in Chapter 1, dysfunction in hedonic processing is seen across a variety of neuropsychiatric disorders, highlighting the need for research into the underlying mechanisms of hedonic processing (American Psychiatric Association, 2013; Prabhakar et al., 2022). This chapter explores the effects of a genetic reduction in D1R on both anticipatory and consummatory phases of hedonic processing. The following components of the two phases will be examined: anticipatory behaviour, anticipatory pleasure, consummatory behaviour, and consummatory pleasure.

Several different paradigms have been reported to measure anticipatory and consummatory behaviour, and subsequent pleasure, in animal models, specifically rodents. The following paradigms are commonly used to measure consummatory components: (1) affective taste reactivity, (2) sucrose preference test, (3) conditioned place preference, and (4) social interaction test. These paradigms use stimuli that are primary reinforcers with innate properties (e.g., food, social interaction, sexual behaviour, etc) (Markov, 2022). The consummatory behaviour during these paradigms is suggestive of the subject's affective state or consummatory pleasure.

In the (1) affective taste reactivity paradigm, tongue protrusions, elevation of mouth corners (e.g., 'smiling'), and paw licking are indicative of the rodent's consummatory pleasure. The quantity and timing of these reactions are measured to determine the level of pleasure experienced (Berridge, 2000). In the (2) sucrose preference test, subjects can choose between consuming a sucrose solution or water, and their preference for the sucrose solution over water (e.g., ratio of sucrose solution to water consumed) is indicative of their consummatory pleasure (Markov, 2022; Scheggi et al., 2018). In (3) conditioned place preference, animals are exposed to

two compartments with one compartment containing a rewarding stimulus such as drug administration, social interaction, novel object, brain stimulation, or food/sweets. The animal learns to associate the compartment with the rewarding stimulus and when given access to both compartments, they prefer the compartment previously paired with the reward. The amount of time spent in the reward-associated compartment is an indicator of consummatory pleasure (Bevins & Besheer, 2005; Huston et al., 2013). Lastly, since rats are highly social creatures, (4) social interaction is inherently pleasurable; measuring the amount of time spent interacting with a conspecific versus a non-social stimulus is a measure of consummatory pleasure (Kondrakiewicz et al., 2019; Scheggi et al., 2018; Van den Berg et al., 1999).

When a neutral stimulus (e.g., light, compartment) is repeatedly paired with a reward, the stimulus becomes a predictive cue indicative of reward delivery. Rewards can include standard chow, sweets, sucrose solutions, sexual encounters, social interaction, brain stimulation and enriched housing to name a few (van den Bos, 2019; van den Bos et al., 2003; van der Harst et al., 2003). This predictive cue can produce anticipatory behaviour in which the animal predicts and experiences pleasure in anticipation of the reward delivery after cue presentation. Anticipatory behaviour in rats is comprised of an exponential increase in locomotor activity (rearing, walking, running) and anticipatory pleasure is measured via 50kHz affective vocalisations (van den Bos, 2019; van den Bos et al., 2003; van der Harst et al., 2003).

Importantly, consummatory and anticipatory paradigms can be combined to examine both components of hedonic processing. Rats can be trained to anticipate the arrival of a reward in a specific setting, such as a testing arena, and their behaviour prior to reward delivery reflects their anticipatory behaviour and pleasure (Burke et al., 2021; Cuttoli et al., 2022; Goedhoop et al., 2023). For example, an animal can be placed into a compartment and then presented with the

opportunity to engage in social interaction or consume a sucrose solution. Following repeated presentations of the compartment followed by reward, the animal learns to associate the compartment with a future rewarding experience, therefore, their locomotor movement, rearing and 50kHz vocalisations when waiting for the reward can be measured (Burke et al., 2021; Cuttoli et al., 2022; Goedhoop et al., 2023). These anticipatory and consummatory paradigms can be modified to examine motivation. For example, a demand requirement could be added which requires subjects to press the lever or climb over a barrier to obtain a reward after presentation of a cue (Gill et al., 2020; Lovic et al., 2011).

Consummatory behaviour is, in part, mediated by serotonergic, opioid and endocannabinoid systems while anticipatory behaviour is, in part, mediated by midbrain dopaminergic pathways, likely regulated by the D1 receptor (Berridge & Robinson, 1998; 2003; Höflich et al., 2019). In previous studies, $DAD1^{-/-}$ mutants did not differ from control animals in their preference for a sucrose solution over water, nor in their consumption of freely available sugar pellets (Squire, 2016), suggesting that the $DAD1^{-/-}$ mutants have intact consummatory behaviour. On the other hand, in an anticipatory locomotor experiment, $DAD1^{-/-}$ mutants failed to display anticipatory behaviours with no exponential increase in rearing and locomotion for standard lab chow (Vanden Broeke, 2016). The consumption of chow was not recorded, so it is not clear if $DAD1^{-/-}$ mutants had impaired anticipatory behaviour and normal consummatory behaviour, or if consummatory behaviour was also decreased. Additionally, 50 kHz vocalisations are a reliable measure of anticipatory pleasure, but they were not measured in previous studies. Importantly, previous research on hedonic processing in $DAD1^{-/-}$ mutants used male subjects and since sex hormones in females can alter dopaminergic signalling, it is vital to examine hedonic processing in both $DAD1^{-/-}$ mutant males and females. Therefore, the present study builds upon

previous research to examine how DAD1^{-/-} mutants and wildtype controls of both sexes consume *and* anticipate a sucrose solution with a variety of anticipatory measures (rearing, locomotion, 50 kHz vocalisations).

5.2 Overall Aims of Chapter Five

Subjects, DAD1^{-/-} mutants and wildtype controls, are placed into anticipation boxes for ten minutes and then given access to an adjacent chamber with a 2.5% sucrose solution for ten additional minutes. This procedure is repeated for a total of seven days so that the subjects learn that the anticipation box is followed by access to a sweetened solution, thus inducing anticipatory behaviour and pleasure (increased rearing, locomotion, 50kHz ultrasonic vocalisations) (Buck et al., 2014; Barbano & Cador, 2006; Van den Berg et al., 1999).

It is predicted that:

1. DAD1^{-/-} mutants will display normal consummatory behaviour and pleasure by consuming a similar amount of sucrose solution and emitting a similar number of 50 kHz ultrasonic vocalisations during the consummatory phase as DAD1^{+/+} wildtypes.
2. DAD1^{-/-} mutants will have deficits in anticipatory behaviour and pleasure seen in reduced exponential increase in locomotor activity, rearing and 50 kHz ultrasonic vocalisations during the anticipatory phase.
3. Males and females of both genotypes are included in this experiment. Sex differences in this experiment are exploratory.

5.3 Methods

5.3.1 Experimental Animals

Thirty-six adult Wistar (10 wildtype male, 9 wildtype female, 10 DAD1^{-/-} mutant male, 7 DAD1^{-/-} mutant female) rats were used in this experiment. To minimise a potential extraneous

order effect (Shoji & Miyakawa, 2021), some subjects were used in previous experiments (randomly assigned) while others were experimentally naïve at the start of the present study. The rats were housed in groups of two to three based on matched sex and genotype in standard polycarbonate cages. Food and water were available *ad libitum* at all times except during the experiment. Their individually ventilated polycarbonate home cages were lined with saw dust and contained a red enrichment tube. The housing room was on a reverse 12-h light/dark cycle (lights off between 7 am and 7 pm). All experimental procedures took place during the rats' dark cycle, between 8 am and 6 pm. The housing room was kept at controlled temperature (19–21 °C) and humidity (55%).

5.3.2 Apparatus

This experiment took place in a black, polycarbonate, rectangular open-air chamber (67.5 x 45 x 45 cm). The chamber was divided into a square arena for the anticipation phase (45 x 45 x 45 cm; “anticipation box”) and a rectangular arena for the consumption phase (45 x 22.5 x 45 cm; “sucrose chamber”). A portion of the divider was removable to allow the animal to explore both chambers during the consumption phase (see Figure 5.1 for the experimental setup). The chamber was cleaned with 70% ethanol and paper towels between each subject.

The sucrose solution was available in the smaller chamber in a clear plastic drink bottle with a non-leak stopper in the drinking tube. The 2.5% sucrose solution was prepared by mixing 2.5g of commercial sugar with 97.5g of distilled water. Each subject was assigned an individual bottle that remained the same across the sessions.

An UltraMic (250K 16 bi; Java Sound) microphone was suspended over each chamber to record the vocalisations during the anticipatory phase and consummatory phase using Audacity recording software (Audacity Team, 2020). One Dell processor laptop and one Dell desktop

computer were used to operate Audacity (one per microphone). All experimental testing occurred during the animals' dark cycle under red light and was recorded using EthoVision XT software (Noldus, Netherlands). EthoVision XT (Nolus, Netherlands) and BORIS (Friard & Gamba, 2016) were both used to score and analyse the animals' behaviour.

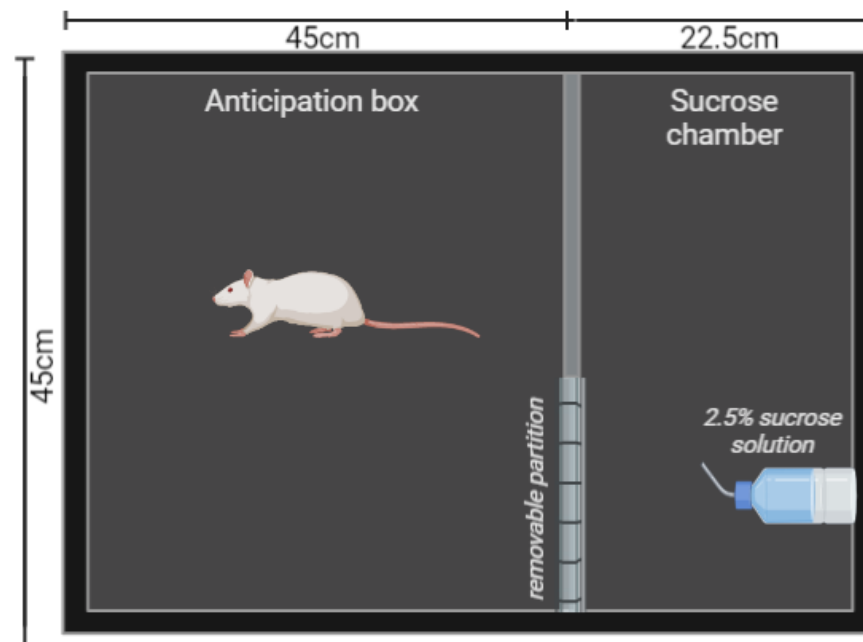


Figure 5.1. Experimental setup. The chamber was divided into an “anticipation box” for the anticipation phase and a “sucrose chamber” for sucrose consumption.

5.3.3 Procedure

On days 1 to 3, animals were transported to the testing room and habituated to both chambers (partition removed) for 10 minutes daily. Following habitation on day 1, each animal's weight was recorded. From day 4 on, subjects were placed in the anticipation box for 10 minutes with the partition closed to record anticipatory behaviour. After 10 minutes, the partition was removed, and the rat was given access to both chambers. The sucrose solution bottle was weighed (in grams) before and after the session to measure sucrose consumption (see figure 5.2). This procedure was repeated for a total of seven days (sessions 1 – 7) to allow the rat to associate the placement in the anticipation box with subsequent access to a sucrose solution, thereby inducing

anticipatory behaviour and pleasure (Barbano & Cador, 2006; Knutson et al., 2002; Makowska & Weary, 2016). Following the experiment on day 10, each animal's weight was recorded.



Figure 5.2. An image of one subject consuming sucrose solution.

The frequency and duration of rearing (unsupported and supported) were scored using BORIS software offline. Supported rearing was defined as an animal standing on its hind legs using the chamber wall as support while unsupported rearing was defined as an animal standing on its hind legs without using support (Sturman et al., 2018). As discussed in Chapter 2 (section 2.3.4), audio files underwent the following steps: cleaned using Squeaky Clean, processed with DeepSqueak v3, manually reviewed, and variables were exported (number of calls, duration of calls, principle frequency).

5.3.4 Data Analysis

The variables of interest were divided into anticipation and consumption phases. The variables in the anticipation phase were frequency (number of times behaviour occurs) and duration (total length of time behaviour occurs for) of supported rearing, unsupported rearing, total rearing, distance travelled and 50kHz vocalisations (number of calls, duration, frequency). The variables in the consummatory phase were the amount of sucrose solution consumed and 50kHz vocalisations (number of calls, duration, frequency). Subjects' weight was measured on the first and last days of the experiment as an additional variable.

All variables were examined with repeated measures ANOVAs with genotype (DAD1^{-/-} mutants, DAD1^{+/+} wildtypes) and sex (male, female) as the between-subject factor and session as the within-subjects factor (1 – 7). Mauchly's test of sphericity indicated that the assumption of sphericity was violated for all variables ($p < 0.01$), except for sucrose consumed, therefore Greenhouse-Geisser corrections were used. All data were analysed for statistical significance using Jamovi (version 2.3) and SPSS (version 27). The alpha level for statistical significance was $p < 0.05$ and partial eta squared (η^2) was used for the effect size. All significant interactions were followed up with post hoc Student's t-tests.

5.4 Results

5.4.1 Weight

Across the two time points, there was a significant sex * genotype interaction ($F_{1,32} = 17.20$, $p < 0.001$, $\eta^2 = 0.35$). Inspection of figure 5.3 showed that wildtype males weighed significantly more than other groups, while wildtype females did not differ from DAD1^{-/-} mutant females. These results replicate Homberg and colleagues' (2016) finding that DAD1^{-/-} mutant males have a reduced bodyweight compared to wildtypes. Previous studies did not include females; therefore, the sex * genotype interaction adds upon previous literature.

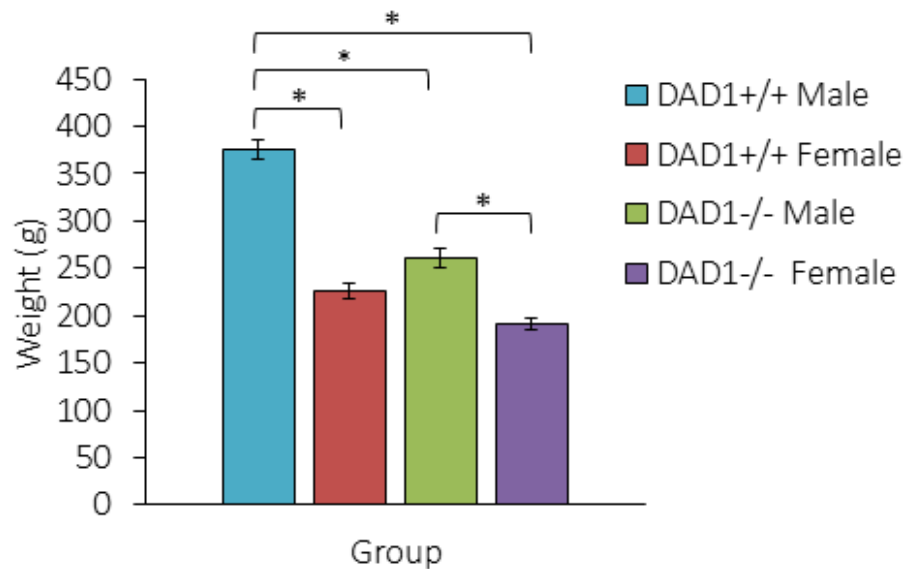


Figure 5.3. Average weight across time points. Wildtype males weighed significantly more than other groups, while wildtype females did not differ from DAD1^{-/-} mutant females. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

5.4.2 Consumption Phase

5.4.2.1 Sucrose Solution Consumption

Due to the sex and genotype effects on body weight, sucrose consumption was examined via the amount of sucrose consumed (mg) per body weight (kg). Across the sessions, DAD1^{-/-} mutants consumed more sucrose solution than wildtypes (genotype effect: $F_{1,32} = 4.14$, $p < 0.05$, $\eta_p^2 = 0.12$), and females consumed more sucrose solution than males (sex effect: $F_{1,32} = 6.55$, $p < 0.05$, $\eta_p^2 = 0.17$), but there was not a significant sex * genotype interaction ($F_{1,32} = 0.84$, $p = 0.37$, $\eta_p^2 = 0.03$) (see figure 5.4). DAD1^{-/-} mutants sucrose consumption significantly varied across the sessions (session * genotype: $F_{6,192} = 2.56$, $p < 0.05$, $\eta_p^2 = 0.07$), which is likely due to the DAD1^{-/-} mutant females increasing their consumption across sessions while the other groups' consumption remained stable across sessions (see figure 5.5).

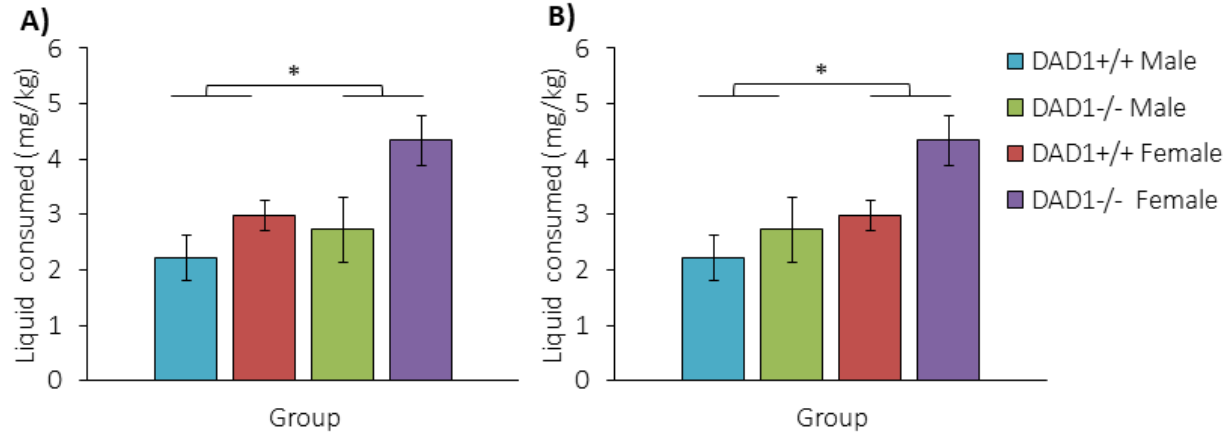


Figure 5.4. Average sucrose consumption (mg/kg) across sessions. (A) DAD1^{-/-} mutants consumed more sucrose solution than wildtypes. (B) Females consumed more sucrose solution than males. Error bars represent $\pm$ SEM. *denotes significance to $p < .05$.

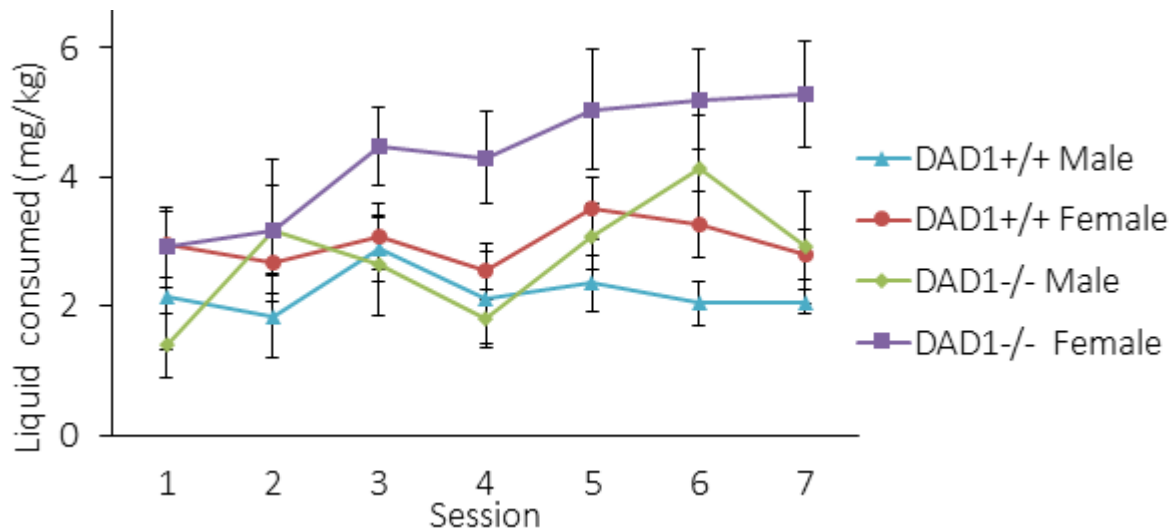


Figure 5.5. Average sucrose solution consumed (mg/kg) per session. DAD1^{-/-} mutants' consumption varied across sessions. Error bars represent $\pm$ SEM.

5.4.2.2 Number of 50 kHz Ultrasonic Vocalisations

There were no sex ($F_{1,27} = 0.50$, $p = 0.49$, $\eta_p^2 = 0.02$) nor genotype ($F_{1,27} = 1.76$, $p = 0.20$, $\eta_p^2 = 0.06$) effects on the number of 50 kHz vocalisations emitted across sessions, suggesting that subjects had similar levels of consummatory pleasure. The number of vocalisations remained stable across sessions, although the individual variance was relatively large (session effect: $F_{5,22, 140.93} = 2.38$, $p = 0.06$, $\eta_p^2 = 0.08$; see figure 5.6).

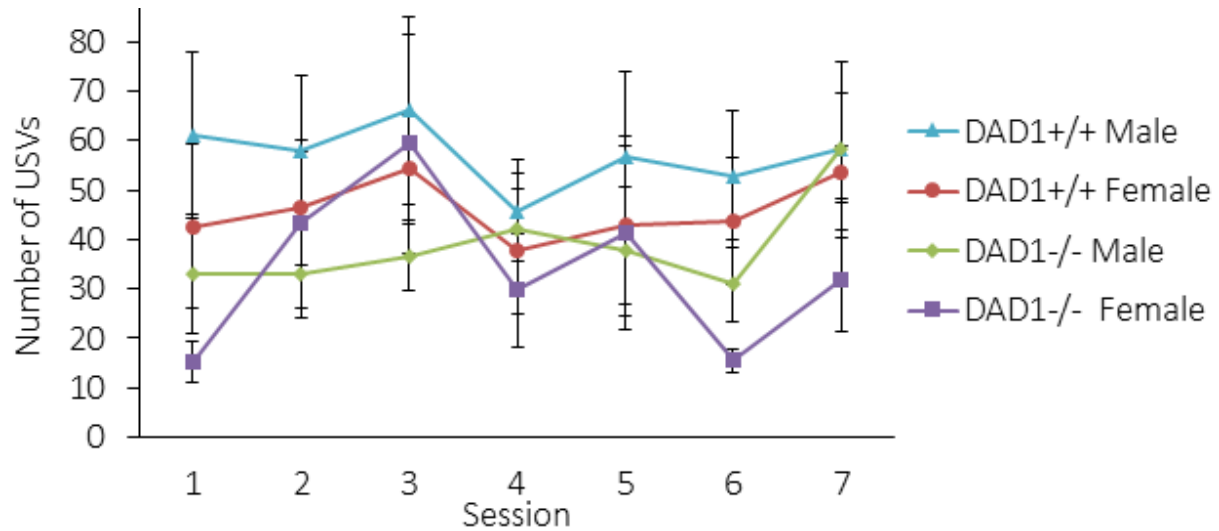


Figure 5.6. Average number of 50 kHz vocalisations emitted per consummatory session. Average number of vocalisations remained stable across sessions. Error bars represent $\pm$ SEM.

5.4.2.3 Average Frequency and Duration of 50 kHz Ultrasonic Vocalisations

There were no genotype differences in the average frequency ($F_{1,27} = 0.55$, $p = 0.46$, $\eta_p^2 = 0.02$) and duration ($F_{1,27} = 0.68$, $p = 0.42$, $\eta_p^2 = 0.02$) of 50 kHz vocalisations emitted across the consummatory sessions. There was no sex effect for *frequency* ($F_{1,27} = 1.88$, $p = 0.18$, $\eta_p^2 = 0.07$) of vocalisations, but males had a significantly longer *duration* ($F_{1,27} = 5.59$, $p < 0.05$, $\eta_p^2 = 0.17$) of 50 kHz vocalisations than females (see figure 5.7).

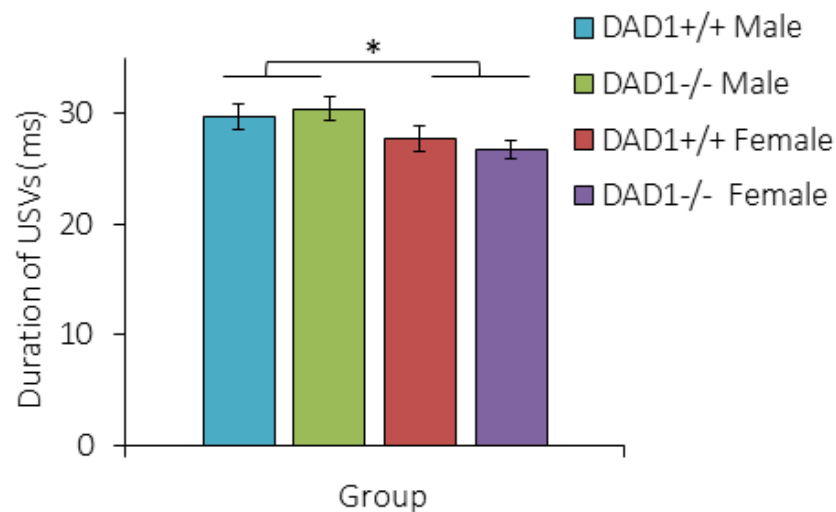


Figure 5.7. Sex difference in the average duration of 50 kHz vocalisations emitted during consumption sessions. Males emitted significantly longer calls than females. Error bars represent $\pm$ SEM. *denotes significance to $p < .05$.

The average frequency ($F_{5.94, 160.28} = 0.56$, $p = 0.47$, $\eta_p^2 = 0.02$) and duration ($F_{2.24, 60.57} = 1.85$, $p = 0.13$, $\eta_p^2 = 0.06$) of vocalisations remained stable across sessions (see figure 5.8).

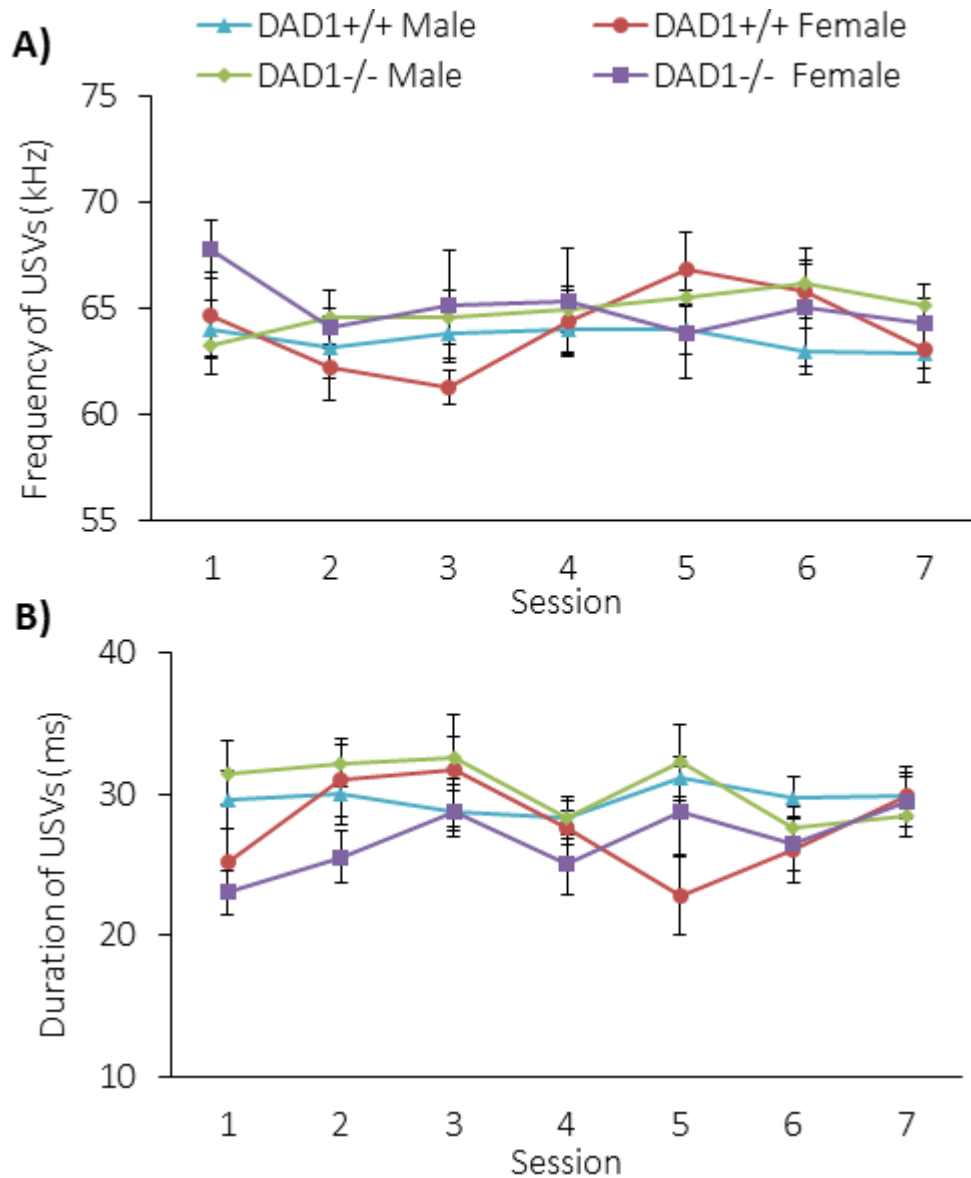


Figure 5.8. Average frequency and duration of 50 kHz vocalisations emitted per consummatory session. The frequency (A) and duration (B) of vocalisations remained stable across sessions. Error bars represent $\pm$ SEM.

5.4.3 Anticipation Phase

5.4.3.1 Unsupported Rearing

DAD1^{-/-} mutants had significantly lower frequency ($F_{1,32} = 32.51$, $p < 0.001$, $\eta_p^2 = 0.50$) and duration ($F_{1,32} = 20.64$, $p < 0.001$, $\eta_p^2 = 0.39$) of unsupported rears than wildtypes across sessions (see figure 5.9). There was not a significant effect of sex in frequency ($F_{1,32} = 0.18$, $p = 0.68$, $\eta_p^2 = 0.006$) nor duration ($F_{1,32} = 2.44$, $p = 0.13$, $\eta_p^2 = 0.07$) of unsupported rears (see figure 5.9). There was not a significant difference in frequency ($F_{3,84, 122.97} = 2.09$, $p = 0.08$, $\eta_p^2 = 0.06$) nor duration ($F_{3,84, 122.97} = 2.39$, $p = 0.06$, $\eta_p^2 = 0.07$) of unsupported rears across sessions, suggesting a lack of anticipatory behaviour across genotype groups (see figure 5.10).

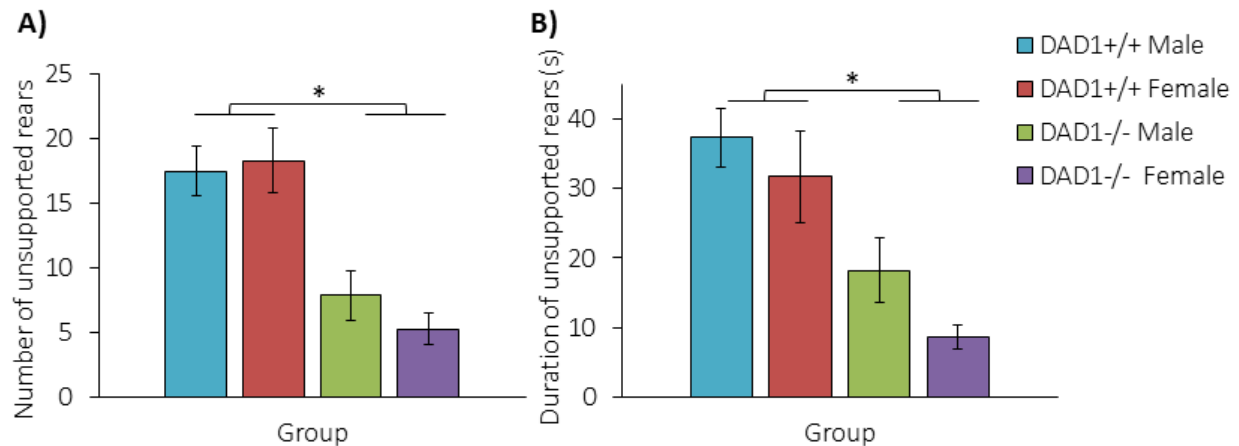


Figure 5.9. Genotype differences in average frequency and duration of unsupported rears during anticipatory sessions. DAD1^{-/-} mutants had significantly lower frequency (A) and duration (B) of unsupported rears than wildtypes. Error bars represent ± SEM. *denotes significance to $p < .05$.

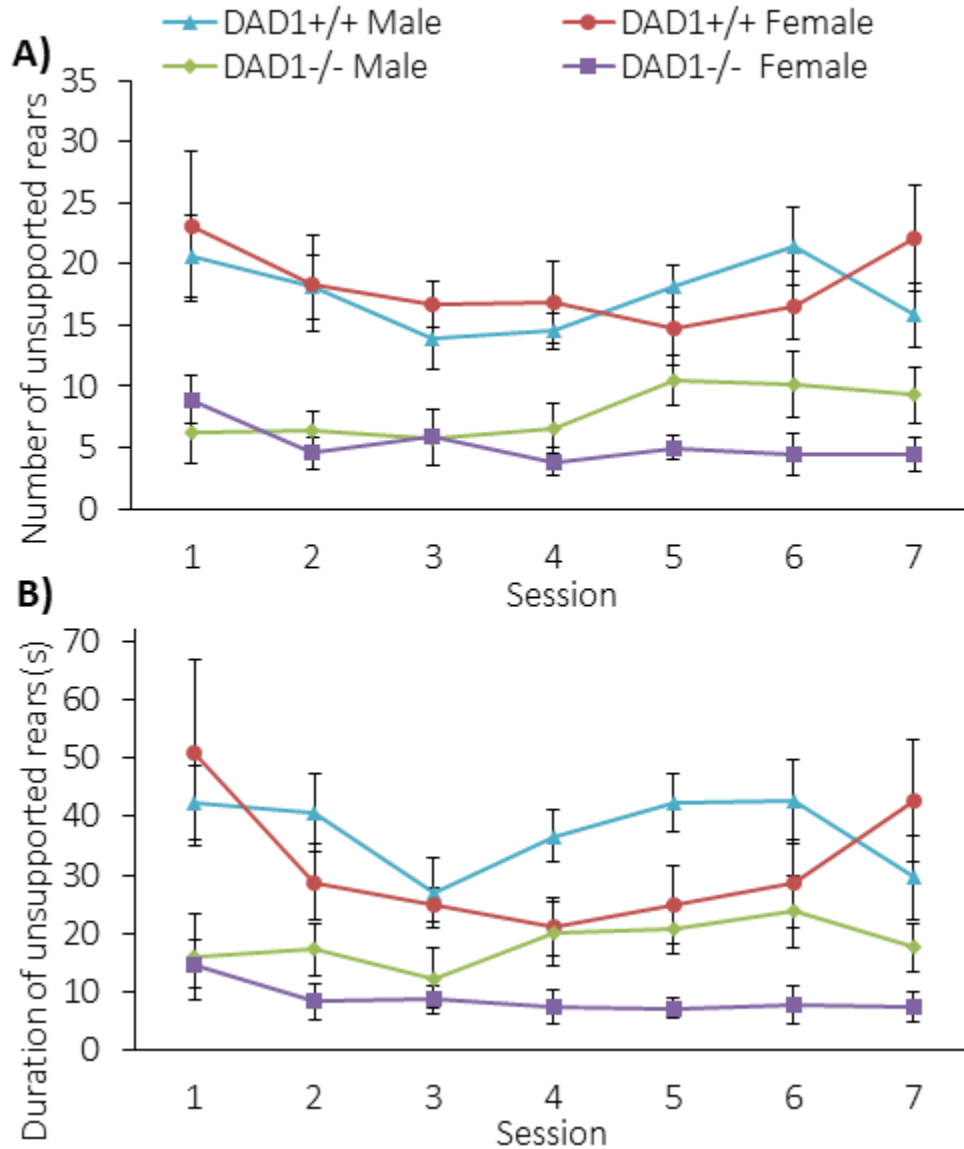


Figure 5.10. Average frequency and duration of unsupported rears per anticipatory session. Frequency (A) and duration (B) remained stable across anticipatory sessions. Error bars represent $\pm$ SEM.

5.4.3.2 Supported Rearing

Contrary to unsupported rears, there was not a significant genotype effect for *frequency of supported rears* ($F_{1,32} = 0.31$, $p = 0.58$, $\eta_p^2 = 0.009$), however, $DAD1^{-/-}$ mutants had significantly smaller *duration of supported rears* ($F_{1,32} = 7.49$, $p < 0.05$, $\eta_p^2 = 0.19$) than wildtypes across sessions (see figure 5.11A). Females had significantly larger frequency ($F_{1,32} = 20.51$, $p < 0.001$,

$\eta_p^2 = 0.39$) and duration ($F_{1,32} = 6.33$, $p < 0.05$, $\eta_p^2 = 0.19$) of supported rears than males across sessions (see figure 5.11B)

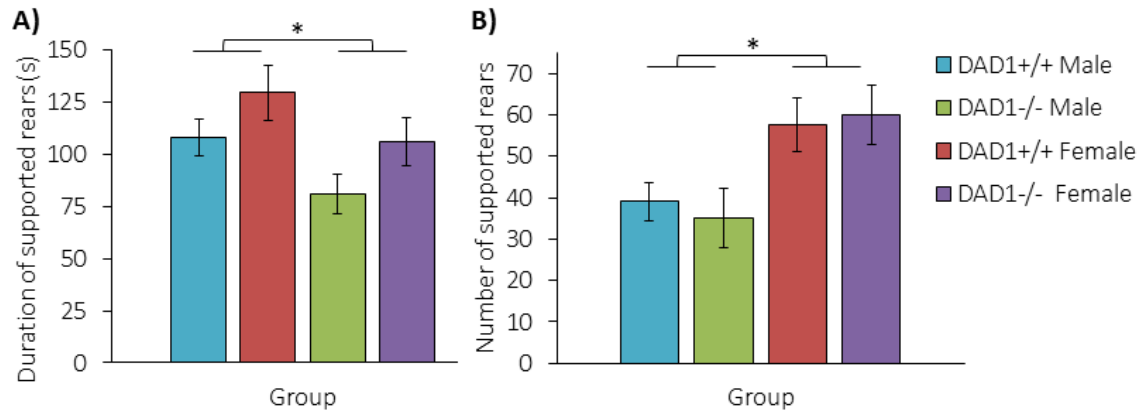


Figure 5.11. Average duration and number of supported rears across anticipatory phases. (A) DAD1^{-/-} mutants and males had significantly smaller durations of supported rears. (B) Males had significantly smaller durations of supported rears. Error bars represent $\pm$ SEM. *denotes significance to $p < .05$.

Subjects had significantly greater frequency ($F_{4.30, 137.67} = 3.48$, $p < 0.01$, $\eta_p^2 = 0.10$) and duration ($F_{4.88, 156.12} = 3.39$, $p < 0.01$, $\eta_p^2 = 0.10$) of supported rears during session 5 and 6 than during session 1, which is indicative of *minor* anticipatory behaviour (see figure 5.12).

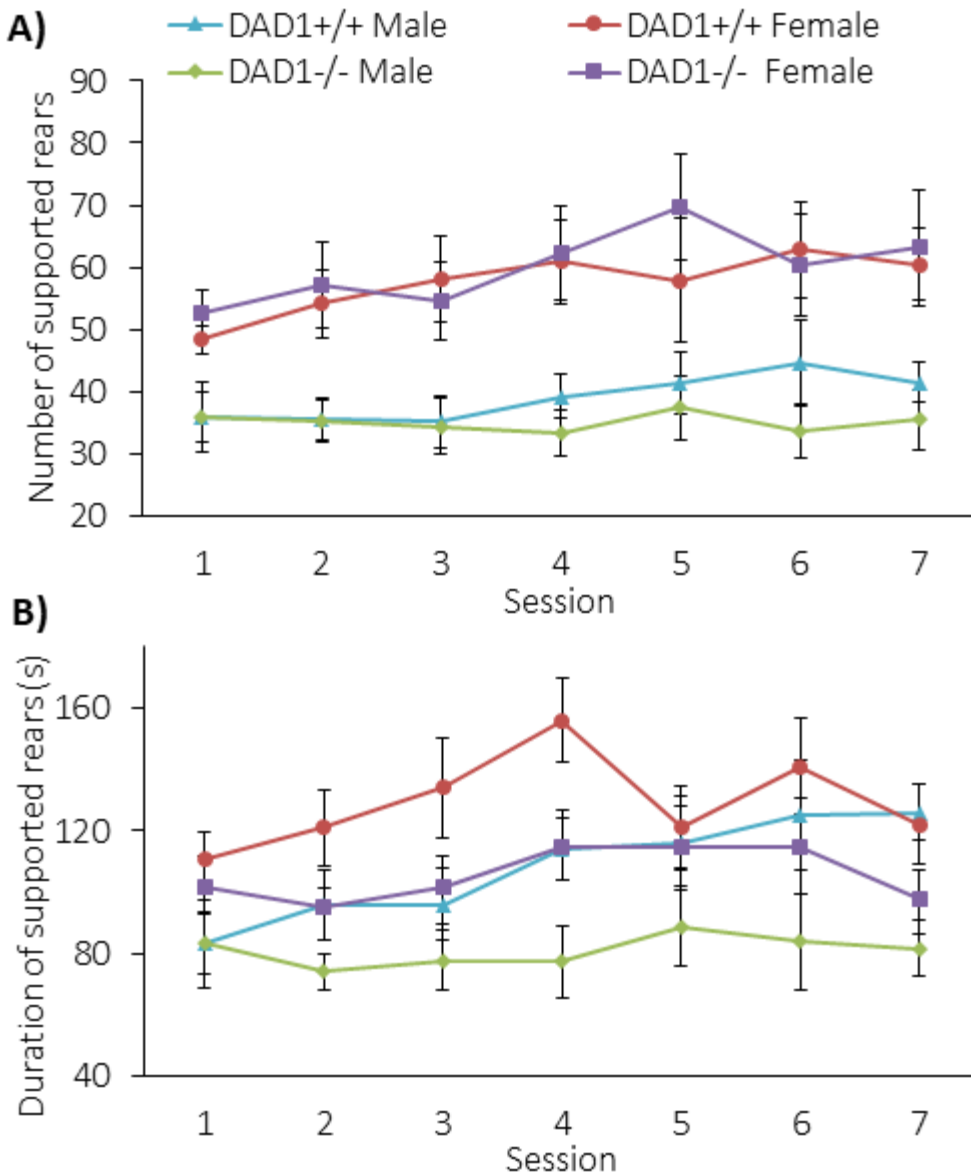


Figure 5.12. Frequency and duration of supported rears per anticipatory session. Subjects had significantly greater frequency (A) and duration (B) of supported rears during sessions 5 and 6 than during session 1. Error bars represent $\pm$ SEM.

5.4.2.3 Total Rearing

Across the sessions, DAD1^{-/-} mutants had significantly smaller frequency ($F_{1,32} = 4.90$, $p < 0.05$, $\eta_p^2 = 0.13$) and duration ($F_{1,32} = 16.16$, $p < 0.001$, $\eta_p^2 = 0.34$) of total rears than wildtype rats (see figure 5.14).

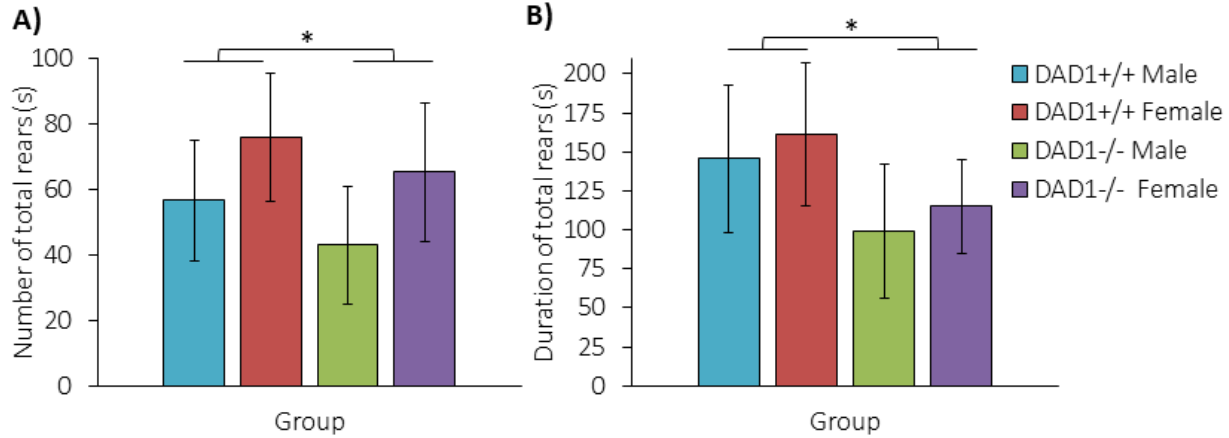


Figure 5.14. Genotype differences in average frequency and duration of total rears during anticipatory sessions. DAD1^{-/-} mutants had significantly smaller frequency (A) and duration (B) of total rears than wildtypes. Error bars represent $\pm$ SEM. *denotes significance to $p < .05$.

Females had significantly more total rears than males across sessions ($F_{1,32} = 14.57$, $p < 0.001$, $\eta_p^2 = 0.31$), but did not have significantly greater *duration* of total rears ($F_{1,32} = 1.85$, $p = 0.18$, $\eta_p^2 = 0.06$). There were no significant interactions between sex and genotype for frequency ($F_{1,32} = 0.08$, $p = 0.78$, $\eta_p^2 = 0.002$) nor duration ($F_{1,32} = 0.09$, $p = 0.99$, $\eta_p^2 = 0.001$).

There was a significant effect of session for frequency of total rears ($F_{4,15, 132.67} = 2.70$, $p < 0.05$, $\eta_p^2 = 0.08$), however post hoc comparisons indicate that frequency of total rears did not significantly differ between sessions ($p > 0.05$ for all post hoc tests). The lack of session effect suggests a lack of anticipatory behaviour with no significant increase in frequency of total rears (see figure 5.15A). For total duration, subjects had a significantly larger duration ($F_{4,39, 140.39} = 2.68$, $p < 0.05$, $\eta_p^2 = 0.08$) of total rears during session 6 than sessions 2 ($p < 0.05$) and 3 ($p < 0.05$), which suggests minimal anticipatory behaviour (see figure 5.15B).

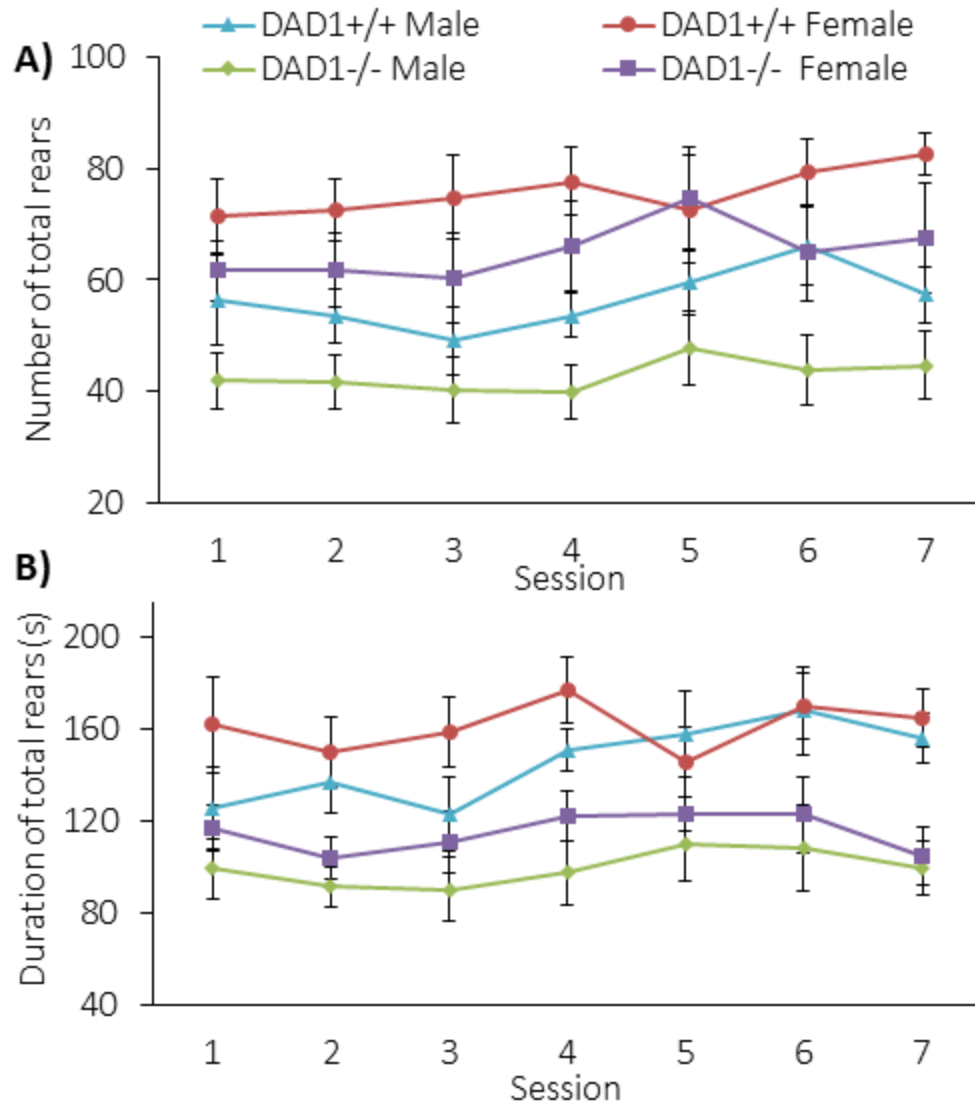


Figure 5.15. Frequency and duration of total rears per anticipatory session. Frequency of rears (A) remained stable across sessions while subjects had significantly larger durations (B) of total rears in session 6 than session 2. Error bars represent $\pm$ SEM.

5.4.2.4 Distance Travelled

Females travelled significantly more than males across the sessions ($F_{1,32} = 5.84$, $p < 0.05$, $\eta_p^2 = 0.15$; see figure 5.16). There was an approaching significant effect of genotype ($F_{1,32} = 3.94$, $p = 0.06$, $\eta_p^2 = 0.10$); DAD1^{-/-} mutants travelled more than wildtypes.

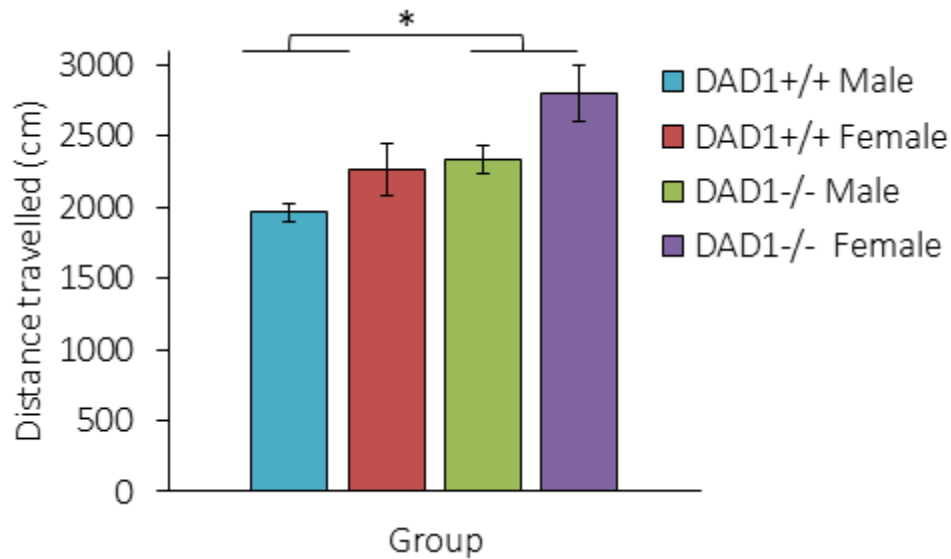


Figure 5.16. Average distance travelled during anticipatory sessions. Females travelled more than males. Error bars represent $\pm$ SEM. *denotes significance to $p < .05$.

Contrary to our predictions, subjects did not increase their distance travelled across sessions ($F_{4.34, 138.90} = 1.21$, $p = 0.31$, $\eta_p^2 = 0.04$); in fact, females travelled significantly less during session 7 than session 5 ($p < 0.05$), suggesting a lack of anticipatory behaviour across subjects (session * sex: $F_{4.34, 138.90} = 2.97$, $p < 0.05$, $\eta_p^2 = 0.09$; see figure 5.17).

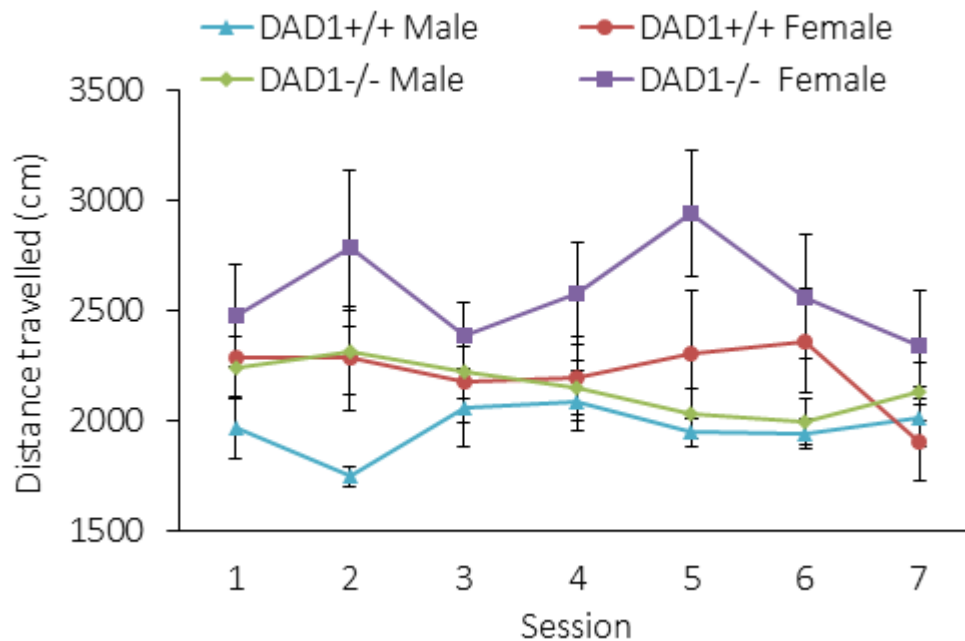


Figure 5.17. Average distance travelled per anticipatory session. Females travelled less during session 7 than 5. Error bars represent $\pm$ SEM.

5.4.2.5 50 kHz Ultrasonic Vocalisations

There was no sex ($F_{1,27} = 3.08$, $p = 0.09$, $\eta_p^2 = 0.10$) nor genotype ($F_{1,27} = 1.397$, $p = 0.25$, $\eta_p^2 = 0.05$) differences in the number of 50 kHz vocalisations subjects emitted. Contrary to our predictions, subjects did not increase the number of 50 kHz vocalisations across sessions ($F_{4,05,109.27} = 0.76$, $p = 0.59$, $\eta_p^2 = 0.03$), suggesting a lack of anticipatory pleasure across subjects (see figure 5.18).

Although most studies only include the number of 50 kHz vocalisations for anticipatory pleasure, we examined the average frequency (kHz) and duration of vocalisations for additional measures (Brenes & Schwarting et al., 2015). Across sessions, there was no sex or genotype differences in the average frequency nor duration of 50 kHz vocalisations emitted, and the average frequency and duration of vocalisations remained stable across sessions (see Appendix C, Table A1.1 for statistics and figure A1.1).

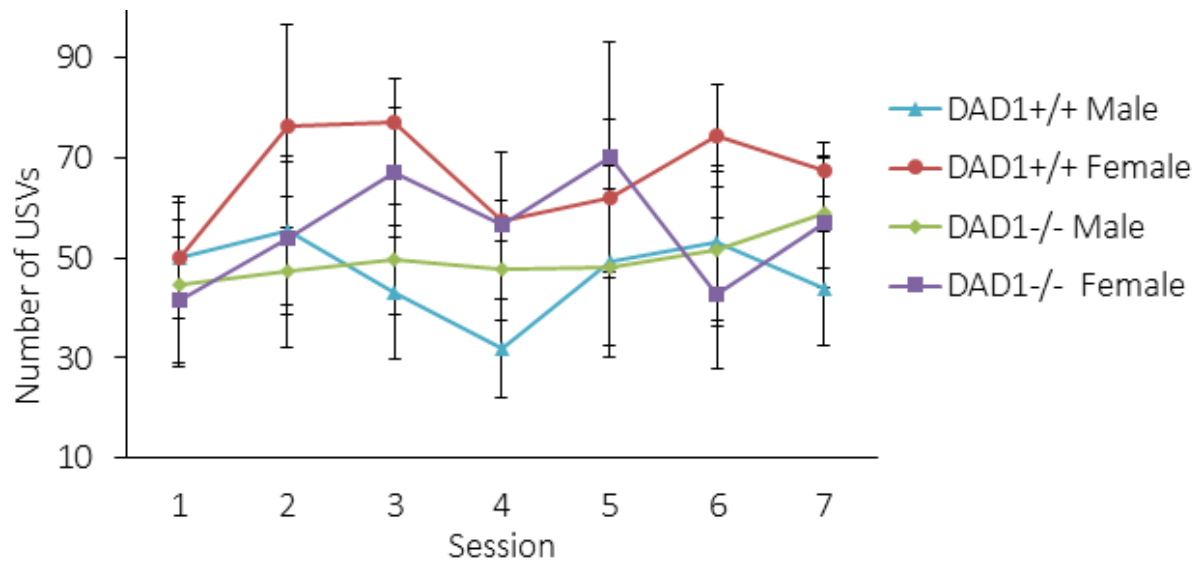


Figure 5.18. Average number of 50 kHz vocalisations emitted per anticipatory session. The average number of vocalisations remained stable across sessions. Error bars represent $\pm$ SEM.

5.5 Discussion

The aim of this experiment was to examine how DAD1^{-/-} mutants consume *and* anticipate a 2.5% sucrose solution. It was hypothesised that DAD1^{-/-} mutants would exhibit normal consummatory behaviour and pleasure by consuming a similar amount of sucrose solution and emitting a similar number of 50 kHz vocalisations as wildtypes. As predicted, during the consummatory phase, DAD1^{-/-} mutants did not differ from wildtypes in vocalisations, suggesting intact pleasure while having access to and consuming a sucrose solution. Surprisingly, DAD1^{-/-} mutants consumed significantly *more* sucrose solution than wildtypes, suggesting that DAD1^{-/-} mutants have intact or even enhanced consummatory behaviour. Intact consummatory behaviour and pleasure supports genetic (Squire, 2016; Vanden Broeke, 2016) and pharmacological research (Baldo et al., 2002; Buck et al., 2014) that indicates the dopamine D1 receptor is *not* implicated in consummatory processes while on the other hand, enhanced consummatory behaviour would implicate the D1 receptor.

Previous experiments with DAD1^{-/-} mutants used only males, therefore, sex differences in the present study were exploratory. We found that females consumed more sucrose solution than males, which was likely driven by the DAD1^{-/-} mutant females drinking more sucrose solution than other subjects. Since previous research with DAD1^{-/-} mutants did not include females, we cannot confirm if this sex difference in consumption of sucrose is typical for DAD1^{-/-} mutants. Beyond D1R research, there are contradictory findings in terms of sex differences in sucrose preference and consumption. Some studies found that female rats consumed less sucrose solution and displayed lower licking rates than males (Curtis et al., 2004), while other studies found that females consumed more sucrose solution in bottle-choice and binge intake paradigms (Grimm et al., 2022). Additionally, several studies found no sex differences in sucrose consumption (Sclafani

et al., 1987) or preference (Fonseca-Rodrigues et al., 2022). To further investigate the consummatory sex differences, future research could conduct sucrose preference (Markov, 2022) and successive negative contrast tests (Jiménez-García et al., 2019) in female and male DAD1^{-/-} mutants.

It was hypothesised that DAD1^{-/-} mutants would exhibit deficits in anticipatory behaviour and pleasure shown by a lack of exponential increase in locomotor activity, rearing and 50 kHz vocalisations. Unexpectedly, and contrary to this hypothesis, both DAD1^{-/-} mutants and wildtypes failed to show anticipatory behaviour and pleasure across most measures; there was no consistent increase in unsupported and total rearing, locomotor activity, nor 50 kHz vocalisations across the seven sessions. The only anticipatory behaviour was the increase in frequency of supported rearing; however, it is unclear whether supported and unsupported rearing both reflect anticipatory behaviour. Selective breeding for higher unsupported rearing does not increase supported rearing (Layfield et al., 2023) and testing conditions that impact unsupported rearing (e.g., acute stress, aversive stimuli) do not impact supported rearing (Sturman et al., 2018). Sturman and colleagues (2018) proposed that unsupported rearing is reflective of an animal's affective state while supported rearing is reflective of general activity, therefore, we cannot conclude that the minor increase in supported rearing in the present study reflects anticipatory behaviour nor pleasure. Additionally, the increase in rearing seen in the present study's wildtypes does not reflect increases seen in previous studies (Vanden Broeke, 2016). In the present study, wildtypes' supported rearing increased by only 21% across sessions, whereas, in a previous anticipatory locomotor study rearing increased by 263% (Vanden Broeke, 2016). Although DAD1^{-/-} mutants failed to show anticipatory behaviour, we cannot conclude that this is due to the dopamine D1 receptor because the wildtypes also failed to show concrete anticipatory behaviour.

One potential reason for the lack of anticipatory behaviour across subjects is that the subjects were not motivated enough to consume the sucrose solution without food and water restriction. However, this is unlikely because food and water deprivation either does not alter sucrose solution consumption (Forbes et al., 1996) or it can even reduce consumption (Hatcher et al., 1997). Therefore, it is not recommended to food or water restrict rats prior to sucrose consumption tests (Markov, 2022). To eliminate motivational factors as an explanation, future studies could food and water restrict half of the cohort and examine if they consume the solution differently than non-restricted subjects.

Another potential reason for the lack of anticipatory behaviour across subjects is that the sucrose solution had a relatively low reward value, so it did not cause enough motivation to induce anticipatory behaviour. Although *DAD1^{-/-}* mutants consumed more sucrose solution than wildtypes, the consumption of sucrose solution was relatively small overall. Excluding body weight, in the present study, subjects averaged 3mg of 2.5% sucrose solution; whereas, in a previous study in males, subjects consumed an average of 80mg of 10% sucrose solution (Squire, 2016), suggesting that a 2.5% solution might not be rewarding enough to induce anticipatory pleasure. We chose a 2.5% sucrose solution because the literature suggests that sucrose concentrations need to be minimal to avoid satiation but sufficient enough to provoke preference over water (Markov, 2022). Moreover, we wanted to avoid a potential ceiling effect, where the incentive value is so strong it would mask a potential genotype effect. Consumption of sucrose solution follows an inverted U-shaped curve from 0.7% to 30% with lowest consumptions at 0.7% and 30% concentrations, and peak consumption at 8%, which is why sucrose preference tests typically use concentrations between 1 - 10% (Markov, 2022; Matthews et al., 1995). However, there is no standard concentration across studies and recent research found that rats were

indifferent between 2% and 8% concentrations (Fam et al., 2022; Markov, 2022). As mentioned in this chapter's background, sucrose preference tests are a standard paradigm for consummatory pleasure by examining preference for sucrose solutions over water, but this paradigm does not incorporate anticipatory behaviour. In the present study, contrary to sucrose preference tests, subjects were only given sucrose solution, so it is possible that 2.5% was not rewarding enough since there was no comparison to water. To address the reward value of the 2.5% sucrose solution, future research could test varying sucrose solution concentrations to determine which concentration is sufficient to produce anticipatory behaviour and pleasure.

In summary, $DAD1^{-/-}$ mutants consumed more sucrose solution than wildtypes showing that a genetic reduction in the D1 receptor does not reduce consummatory behaviour nor pleasure, supporting previous evidence that D1 receptor antagonism or blockage does not impair consummatory processes (Baldo et al., 2002; Buck et al., 2014; El-Ghundi et al., 2003; Soares-Cunha et al., 2016). Subjects, regardless of genotype, failed to show anticipatory behaviour across measures which is likely due to the sucrose solution not having a high enough reward value, causing overall low consumption levels. Due to the lack of anticipatory behaviour in wildtypes in the present study, we cannot determine whether the dopamine D1 receptor plays a role in anticipatory behaviour and pleasure. The following chapter continues to investigate the role of the D1 receptor in anticipatory processes by examining how $DAD1^{-/-}$ mutants anticipate social play which is one of the most rewarding behaviours of a rat (Vanderschuren et al., 2008). Seidisarouei et al. (2021) found that juvenile rats preferred social interaction over sucrose consumption, therefore, social play in the next chapter may be rewarding enough to induce anticipatory behaviour in wildtypes which we did not see in the present study.

Chapter 6

Anticipation of and Engagement in Social Play

Chapter 6

Anticipation of and Engagement in Social Play

6.1 Background

Building upon the previous chapter, the present chapter explores the effects of a genetic reduction in the dopamine D1R on social cognition, social behaviour and anticipation, specifically for the opportunity for juvenile rats to engage in social play.

Rats are highly social animals and engage in social play behaviour soon after locomotor development, peaking around postnatal days 30 - 40 and declining with sexual maturation (van Kerkhof et al., 2013; Vanderschuren et al., 1997). Social play is of utmost importance for social, cognitive, emotional and sensorimotor development; rats who have the opportunity to engage in social play have better socio-cognitive and emotional skills than rats raised without the opportunity to engage in social play (Bijlsma et al., 2022; Pellis et al., 2019; Vanderschuren & Trezza, 2014).

Not only is social play vital for development, but it is also a highly pleasurable and rewarding activity to engage in. Social play's rewarding value can be seen in its use in T-maze learning, operant conditioning and conditioned place preference (Trezza et al., 2010;; Vanderschuren et al., 2016). Due to its rewarding value, researchers can examine the motivational aspects of play by measuring the number of play initiations and examine anticipation by training subjects to anticipate the opportunity to play with their cage mate and examine their behaviour prior to this opportunity (Burke et al., 2021).

To examine social play, two rats of matching sex and weight are placed together in an arena for a set amount of time (5 to 15 minutes), and researchers record the occurrence of play behaviour. There are typically two approaches for quantifying play behaviour. In the first and most frequently used approach, the most characteristic and easily recognisable parameters are scored which are

pouncing and pinning (Pellis et al., 2022; van Kerkhof et al., 2013). Pouncing is when one rat noses or rubs the nape of the other subject's neck to initiate play which reflects a subject's motivation to engage in play. Pinning, on the other hand, is when one subject lays on its dorsal side while the partner holds them down (Pellis et al., 2022; Trezza et al., 2010; van Kerkhof et al., 2013; (Sivy et al., 2011; Vanderschuren et al., 2016). In the second approach, the microstructure of play is examined by scoring multiple play behaviours such as chasing, pouncing, pinning and wrestling. Social play is highly ritualized, and typically starts with one animal pouncing another animal (Pellis et al., 2022; Trezza et al., 2010; van Kerkhof et al., 2013; Vanderschuren et al., 2016). The animal that is pounced upon can respond by either evading the partner (typically associated with chasing), wrestling (pushing, pawing, and grabbing) the partner or rotating to its dorsal surface while the partner pins them down.

The first approach is sufficient for determining the amount of play and subsequently the subject's motivation to play while the second approach can account for differences in *how* subjects engage in play (e.g., how genetic, pharmacological, environment, etc manipulations impact the structure of play). In addition to measuring types of play, socio-emotional communication, 50kHz ultrasonic vocalisations, during play reflect the positive affective state associated with play (Burgdorf et al., 2008; Knutson et al., 2002; Pellis et al., 2022). Recently, Pellis and colleagues (2022) called for a comprehensive method of measuring play behaviour as studies vary in which of the two approaches they use (Pellis et al., 2022). They recommend combining the two approaches to analyse the quantity and structure of play, along with recording 50kHz ultrasonic vocalisations.

Various neurotransmitters have been implicated in different aspects of social play: serotonin likely modulates social cues during play, opioids likely modulate the hedonic experience

of play (consummatory pleasure), and dopamine likely modulates the incentive motivation to play (Kellman & Radwan, 2022). Non-selective dopamine antagonists, such as haloperidol, chlorpromazine and alpha-flupentixol, inhibit social play, and the enhancement of social play by cannabinoid agonists is blocked by dopamine antagonists (Vanderschuren et al., 2016). Chronic treatment with modafinil, which reduces dopamine release in the Nacc, reduces social behaviour (Cid-Jofré et al., 2021). It is clear that dopamine antagonism impairs social play behaviour, but dopamine agonism does not reliably increase social play behaviour; it has been shown to both increase and decrease social play (Kellman & Radwan, 2022). These conflicting results suggest that dopamine is likely involved in the incentive motivation of social play but not the expression of play itself (Achterberg et al., 2016).

However, role of the dopamine D1 receptor in social play is less unequivocal. Dopamine D1 receptor *agonism* with SKF-38393, has been found to both suppress social play behaviour (Siviy et al., 1996) and increase play behaviour (Olesen et al., 2005). On the other hand, the dopamine D1 receptor *antagonist* SCH-23390 inhibited social play (Vanderschuren et al., 2016) and reduced initiation of play which suggests a role for the D1-receptor in the motivation for social play (Manduca et al., 2016). The contradictory findings above could be due to the pharmacological manipulation not being selective solely for the D1 receptor (Homberg et al., 2016) and/or that the D1 receptor is involved mainly with the motivational aspects of social play, rather than in the ongoing engagement in play (Achterberg et al., 2016; Manduca et al., 2021).

There is no previous research examining how a genetic manipulation of the D1 receptor impacts social play and no pharmacological or genetic research on how the D1 receptor effects the anticipation of social play specifically. *DAD1^{-/-}* mutant *adults* display reduced social cognition and behaviour (Homberg et al., 2016), therefore the present study examines if this deficit in adult social

functioning occurs in the juvenile period. The present study builds upon the literature to examine how DAD1^{-/-} mutants and wildtype controls anticipate *and* engage in social play, using a comprehensive method (Pellis et al., 2022).

6.2 Overall Aims of Chapter Six

Subjects, DAD1^{-/-} mutants and wildtype controls, are isolated from their cage mates for 3.5 hours to induce an optimal degree of motivation for social play (Niesink & Van Ree, 1989; Van Hasselt et al., 2012). After isolation, subjects are placed into individual anticipation boxes for ten minutes and then both animals are placed in the play arena for 10 minutes to examine their engagement in play behaviour. This procedure is repeated for a total of seven days so that subjects learn that the anticipation box is followed by the opportunity to engage in social play, thus inducing anticipatory behaviour and pleasure (increased rearing, locomotion, 50kHz ultrasonic vocalisations).

It is predicted that:

1. DAD1^{-/-} mutants will display deficits in anticipatory behaviour and pleasure seen as a less pronounced increase in locomotor activity, rearing and 50 kHz ultrasonic vocalisations.
2. Pharmacological manipulations of the D1R show conflicting effects on social play but suggest that the D1 receptor is implicated in the motivational aspect of play.
 - a. DAD1^{-/-} mutants will display diminished motivation to engage in social play with fewer initiations of play (less pouncing). Consequently, DAD1^{-/-} will have fewer engagements in social play (less chasing, wrestling, pinning, etc).
 - b. There is limited research on the microstructure (or sequence) of play, therefore the sequence of social play is exploratory.

3. Previous DAD1^{-/-} mutant research on social cognition used only males, therefore, sex differences in this experiment are exploratory.

6.3 Methods

6.3.1 Experimental Animals

Seventy juvenile (20 wildtype male, 14 wildtype female, 20 DAD1^{-/-} mutant male, 16 DAD1^{-/-} mutant female) rats were used in this experiment (see table 6.1). Some subjects were used in the isolation induced vocalisation experiment prior to weaning (see Chapter 2) while others were experimentally naïve (randomly assigned) at the start of the present study. Animals were weaned on PND 21 and housed in standard polycarbonate cages in pairs based on matching genotype, sex and weight. Food and water were available *ad libitum* at all times except during the experiment. Their individually ventilated polycarbonate home cages were lined with saw dust and contained a red enrichment tube. The housing room was on a reverse 12-h light/dark cycle (lights off between 7 am and 7 pm). All experimental procedures took place during the rats' dark cycle, between 8 am and 6 pm. The housing room was kept at controlled temperature (19–21 °C) and humidity (55%).

Table 6.1

Number of subjects and pairs per group

Subjects	N subjects	N pairs
DAD1 ^{+/+} males	20	10
DAD1 ^{+/+} females	14	7
DAD1 ^{-/-} males	20	10
DAD1 ^{-/-} females	16	8

6.3.2 Apparatus

During the isolation phase, animals were isolated from their cage mate for 3.5 hours in standard polycarbonate cages (identical to their home cage) with the same items as their home

cage: bedding, a tunnel, one drink bottle, food chow and a chew block. One animal from each pair was assigned an isolation cage that remained the same over the experiment.

The anticipation phase took place in two black polycarbonate, rectangular open-air chambers (30 x 18 x 38 cm; “anticipation box”). Each animal from a pair was assigned one of the two chambers which remained the same across the experiment. The social play phase took place in a black, polycarbonate, rectangular open-air chamber (40 x 25 x 30 cm; “play arena”) with a thin layer of bedding spread across the bottom.

An UltraMic (250K 16 bi; Java Sound) microphone was suspended over each anticipation box and the play arena to record the vocalisations using Audacity software (Audacity Team, 2020). Two Dell processor laptops and one Dell desktop computer were used to operate Audacity (one per microphone). Acoustic foam was affixed to polycarbonate sheets surrounding the chambers and arena to reduce noise in USV recordings (see figure 6.1 for the layout of the chambers). The chambers and arena were cleaned with 70% ethanol and paper towels between each pair. All experimental testing occurred during the animals’ dark cycle under red light and was recorded using EthoVision XT software (Noldus, Netherlands). EthoVision XT (Noldus, Netherlands) and BORIS (Friard & Gamba, 2016) were both used to score and analyse the animals’ behaviour.



Figure 6.1. Set up for the anticipation of social play experiment. The grey rectangles represent acoustic foam taped onto polycarbonate sheets to minimise noise during recordings.

6.3.3 Procedure

The following method was derived from Vanderschuren et al. (2008), Burke et al. (2017) and Pettie (2020). On PND 25, animals were transported to the testing room and individually habituated to their assigned anticipation box for 10 minutes and then immediately habituated to the play arena for an additional 10 minutes. This habituation procedure was repeated on PND 26 and PND 27.

On testing days (PND 28 – 34), animals were isolated from their cage mate for 3.5 hours in their designated isolation cages in the housing room. After three hours and 20 minutes of isolation, the two cages were transported to the testing room for the last 10 minutes of isolation. The animals were then placed in their designated anticipation box for 10 minutes to examine anticipatory behaviour. After 10 minutes of anticipation, both animals were placed in the play arena for 10 minutes to examine their play behaviour. Following play, animals were weighed and returned to their home cage in the housing room. This isolation, anticipation and play procedure was repeated for a total of seven days (PND 28 – 34) to allow subjects to learn that the placement

in the anticipation box was followed by the opportunity to play and therefore induces anticipatory behaviour (Barbano & Cador, 2006; Buck et al., 2014; Burke et al., 2017; Knutson et al., 1998; Knutson et al., 2002; Makowska & Weary, 2016).

Later, the anticipation (rearing) and play (pouncing, chasing, pinning, wrestling and passive) behaviours were scored using BORIS software. Anticipation behaviour was scored per subject and play behaviour was scored per pair. Supported rearing was defined as an animal standing on its hind legs using the chamber wall as support while unsupported rearing was defined as an animal standing on its hind legs without using support (Sturman et al., 2018). Pouncing was defined as one animal soliciting another animal by nosing or rubbing the nape of its neck (Manduca et al., 2014; van Kerkhof et al., 2013; Vanderschuren et al., 2016). Chasing was defined as one rat following the other to initiate play (Trezza et al., 2010). Pinning was defined as one animal lying with its dorsal surface on the ground with the other animal standing over it (Manduca et al., 2014; Vanderschuren et al., 2016). Wrestling was defined as both subjects standing upright on their hind legs pawing at each other (boxing) (Vanderschuren et al., 1997). Passive behaviour was defined as the animals being in close contact (within a body length of each other) without active social play (e.g. sitting next to each other grooming) (Vanderschuren et al., 1997).

6.3.4 Data Analysis

The variables of interest were divided into two phases: anticipation and consumption. The variables of interest in the anticipation phase were frequency (number of times this behaviour occurs) and duration (total length of time this behaviour occurs for) of supported rearing, unsupported rearing and total rearing (unsupported + supported), distance travelled and 50kHz ultrasonic vocalisations (average frequency, call length, number of calls).

The variables of interest in the play phase were the frequency of pouncing, chasing, pinning, wrestling and passive social behaviour, the duration of passive social behaviour and 50kHz ultrasonic vocalisations (frequency, call length, number of calls). We did not measure the duration of pouncing, chasing, pinning and wrestling as these behaviours are very short in duration and occur in rapid succession, therefore scoring their frequency is more informative than their duration (van Kerkhof et al., 2013).

In addition to analysing the frequency of social behaviours, we examined the structure of the play by comparing the transitions between the individual components of play behaviour. The sequences of all behaviours were transformed into first-order Markov chain transition matrices in which the preceding behaviour is displayed in the columns (six states: chasing, pinning, wrestling, pouncing and passive) and the successive behaviour in the rows (six states: chasing, pinning, wrestling, pouncing and passive). The probability of each transition was computed by dividing each transition (e.g. pinning followed by pouncing) by the total number of transitions for that state (e.g. all pinning) (Casarrubea et al., 2022; Tejada et al., 2010; Van den Berg et al., 1999; churen et al., 1996). Transition matrices were computed for each play pair across the seven play sessions using a script in R Studio for first-order Markov chains, resulting in 35 transition matrices (one per pair with sessions combined).

The variables of interest were examined with repeated measures ANOVAs with genotype (DAD1^{-/-} mutants, wildtypes) and sex (male, female) as the between-subject factor and session (1 – 7) as the within-subjects factor. Mauchly's test of sphericity indicated that the assumption of sphericity was violated for all anticipation variables of interest, ($p < 0.05$), therefore Greenhouse-Geisser corrections were used for anticipation phase variables. A Greenhouse-Geisser correction was used for some social play measures (see specifics below). The repeated measures ANOVAs

were used to analyse the frequency and duration of rears (unsupported, supported, total), distance travelled, 50 kHz vocalisations, frequency of social behaviour (pinning, chasing, wrestling, pouncing and passive), duration of passive behaviour and probabilities of transitions between behaviours.

All data were analysed for statistical significance using Jamovi (version 2.3) and SPSS (version 27). The alpha level for statistical significance was $p < 0.05$ and partial eta squared (η_p^2) was used for the effect size. All significant analyses were followed up with Tukey HSD post hoc tests, and/or planned comparisons were used to follow up potential effects using Student's t-test.

6.4 Results

6.4.1 Play Behaviour

6.4.1.1 Chasing

Mauchly's test of sphericity indicated that the assumption of sphericity was not violated ($p = 0.81$). Across the sessions, DAD1^{-/-} mutants had significantly more chasing events than wildtypes ($F_{1,28} = 5.18, p < 0.05, \eta_p^2 = 0.16$) and males had significantly more chasing events than females ($F_{1,28} = 4.61, p < 0.05, \eta_p^2 = 0.14$; see figure 6.2). However, there was no significant interaction between session and genotype ($F_{1,28} = 1.40, p = 0.22, \eta_p^2 = 0.05$). Females' engagement in chasing significantly varied across the sessions with females displaying an inverse U-shaped curve (session * sex: $F_{6,168} = 4.43, p < 0.001, \eta_p^2 = 0.14$; see Appendix D, figure A1.1). Males did not significantly differ across the sessions ($p > 0.05$ for all post hoc tests).

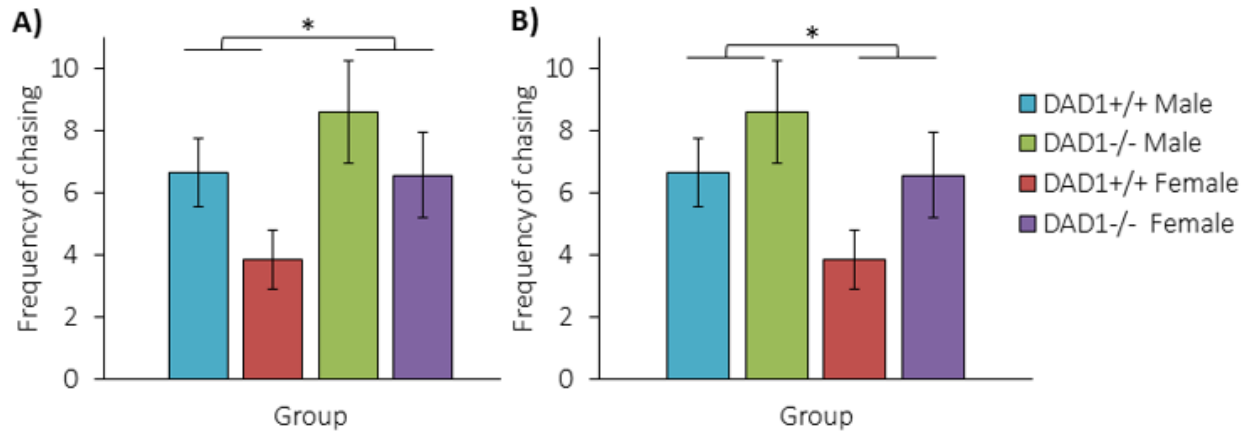


Figure 6.2. Average frequency of chasing across sessions. (A) DAD1^{-/-} mutants chased more than wildtypes. (B) Males chased more than females. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

6.4.1.2 Pinning

Mauchly's test of sphericity indicated that the assumption of sphericity was not violated ($p = 0.10$). Contrary to chasing, across the sessions, DAD1^{-/-} mutants did not significantly differ from wildtypes in frequency of pins ($F_{1,28} = 0.23$, $p = 0.64$, $\eta_p^2 = 0.008$), and males did not significantly differ from females ($F_{1,28} = 0.82$, $p = 0.37$, $\eta_p^2 = 0.003$). Females' engagement in pinning significantly varied across the sessions with females displaying an inverse U-shaped curve across sessions (session $\times$ sex: $F_{6,168} = 3.02$, $p < 0.01$, $\eta_p^2 = 0.10$; see figure 6.3). Males did not significantly differ across the sessions ($p > 0.05$ for all post hoc tests).

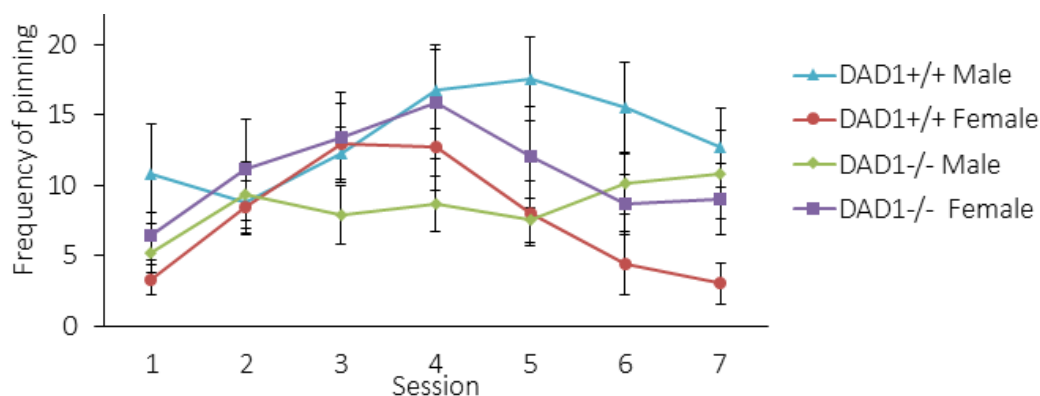


Figure 6.3. Average frequency of pinning events per session. Females' frequency of pinning varied across sessions. Error bars represent $\pm$ SEM.

6.4.1.3 Wrestling

Mauchly's test of sphericity indicates that the assumption of sphericity was violated, $p < 0.05$, therefore Greenhouse-Geisser corrections were used for within subjects' analyses. Across the sessions, DAD1^{-/-} mutants had significantly more wrestling events than wildtypes ($F_{1,28} = 5.67$, $p < 0.05$, $\eta_p^2 = 0.17$; see figure 6.4). Males did not significantly differ from females in frequency of wrestling ($F_{1,28} = 2.60$, $p = 0.12$, $\eta_p^2 = 0.09$). Subjects' engagement in wrestling decreased across the sessions with subjects wrestling significantly less during sessions 6 than sessions 1 ($p < 0.01$) and 5 ($p < 0.05$) (session * sex: $F_{4.28, 119.85} = 4.82$, $p < 0.01$, $\eta_p^2 = 0.11$; see Appendix D, figure A1.2).

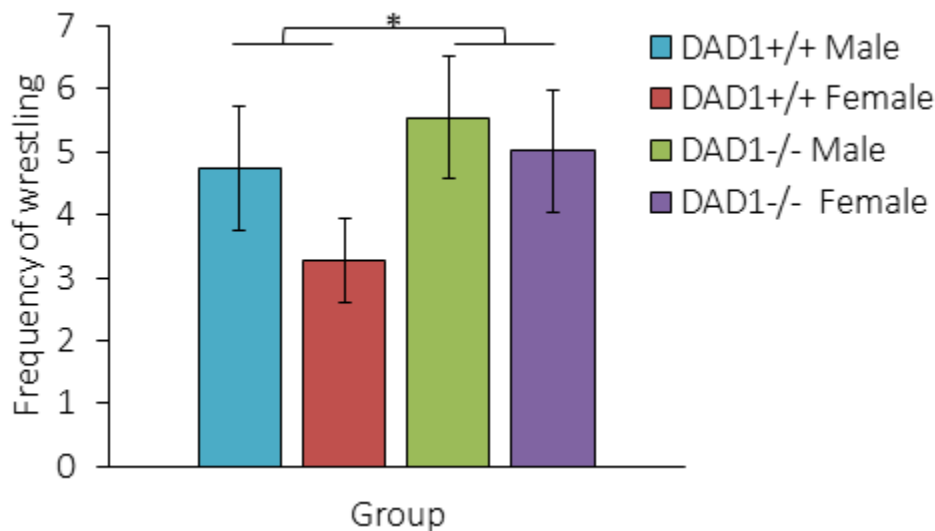


Figure 6.4. Average frequency of wrestling across sessions. DAD1^{-/-} mutants wrestled more than wildtypes. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

6.4.1.4 Pouncing

Mauchly's test of sphericity indicates that the assumption of sphericity was violated, $p < 0.05$, therefore Greenhouse-Geisser corrections were used for within subjects' analyses. Across the sessions, DAD1^{-/-} mutants did not significantly differ from wildtypes in frequency of pounces ($F_{1,28} = 1.66$, $p = 0.17$, $\eta_p^2 = 0.06$), and males did not significantly differ from females ($F_{1,28} = 0.81$, $p = 0.52$, $\eta_p^2 = 0.03$). Females' engagement in pouncing significantly varied across the

sessions with females displaying an inverse U-shaped curve across sessions (session * sex: $F_{3,94, 110.19} = 5.43$, $p < 0.001$, $\eta_p^2 = 0.16$; see figure 6.5). Males did not significantly differ across the sessions ($p > 0.05$ for post hoc tests).

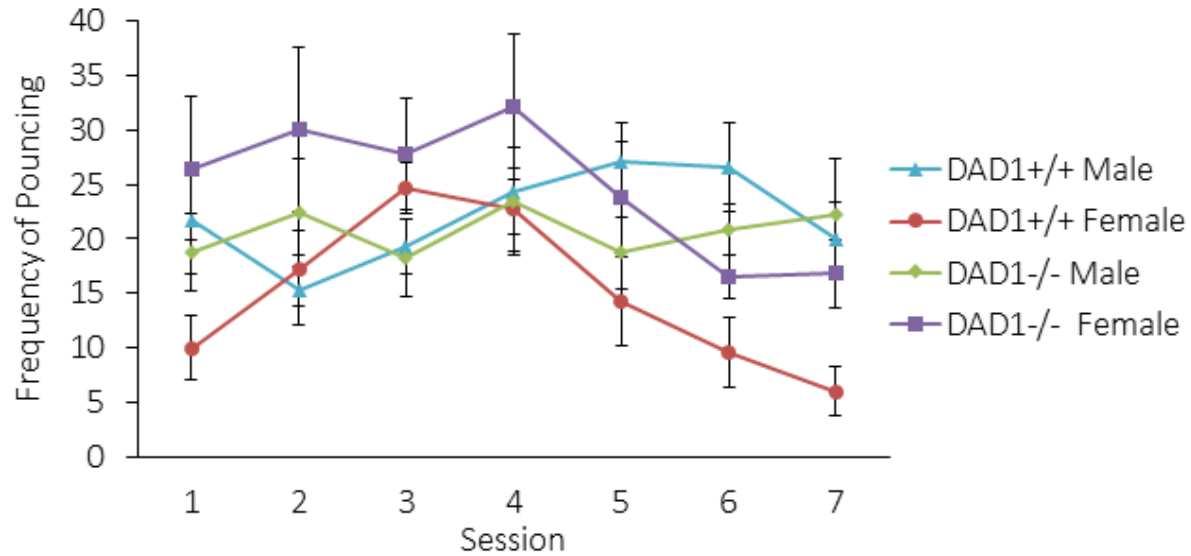


Figure 6.5. Average frequency of pouncing events per session. Females' frequency of pouncing varied across sessions. Error bars represent $\pm$ SEM.

6.4.1.5 Total Play Behaviour

Frequencies of all active play behaviours (chasing, wrestling, pinning, pouncing) were combined to determine the frequency of total play behaviour. Mauchly's test of sphericity indicates that the assumption of sphericity was not violated ($p = 0.24$). Although there were genotype and sex differences within certain play behaviours, DAD1^{-/-} mutants and wildtypes did not significantly differ in total play behaviour ($F_{1,28} = 1.40$, $p = 0.25$, $\eta_p^2 = 0.05$), and males did not significantly differ from females ($F_{1,28} = 1.08$, $p = 0.31$, $\eta_p^2 = 0.04$). As seen with some play behaviours, females' engagement in total play behaviour significantly varied across the sessions with females displaying an inverse U-shaped curve (session * sex: $F_{6, 168} = 5.11$, $p < 0.001$, $\eta_p^2 = 0.15$; see figure 6.6). Males did not significantly differ across the sessions ($p > 0.05$ for all post hoc tests).

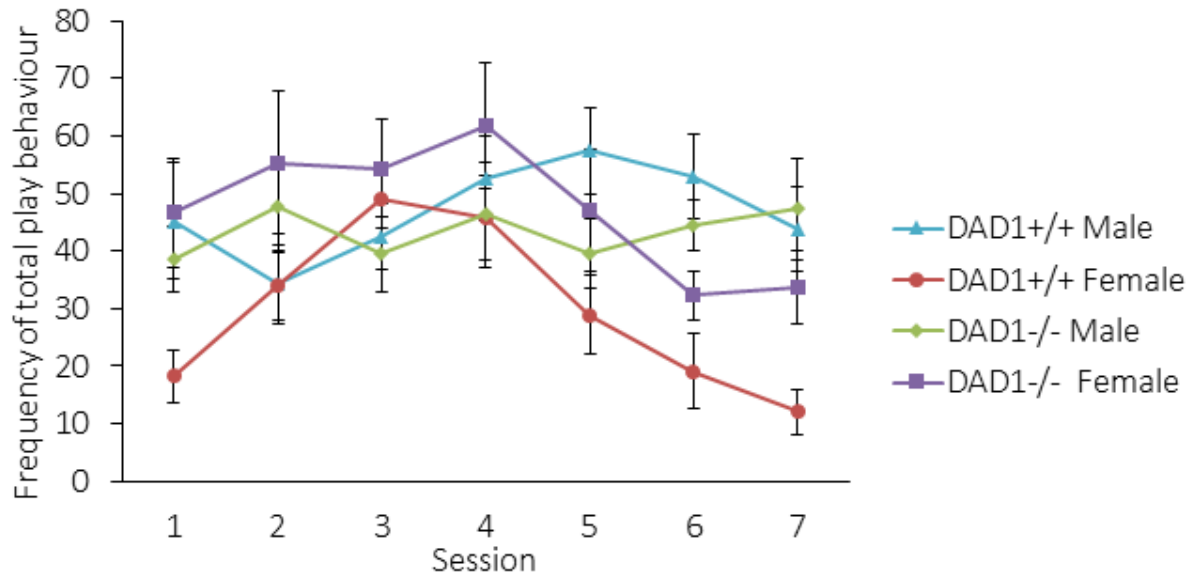


Figure 6.6. Average frequency of total play behaviour per session. Females' frequency of play behaviour varied across sessions. Error bars represent $\pm$ SEM.

6.4.1.6 Passive Social behaviour

Mauchly's test of sphericity indicates that the assumption of sphericity was violated for passive behaviour, $p < 0.05$, therefore Greenhouse-Geisser corrections were used. DAD1^{-/-} mutants engaged in more passive social *events* than wildtypes across sessions ($F_{1,28} = 5.43$, $p < 0.05$, $\eta_p^2 = 0.16$), but spent significantly less *time* engaged in passive social events ($F_{1,28} = 7.83$, $p < 0.05$, $\eta_p^2 = 0.07$; see figure 6.7). Males and females did not significantly differ in frequency ($F_{1,28} = 2.40$, $p = 0.13$, $\eta_p^2 = 0.08$) nor duration ($F_{3,31} = 0.67$, $p = 0.42$, $\eta_p^2 = 0.006$) of passive social behaviour. Frequency of passive social behaviour did not significantly differ across sessions ($F_{4,22,118.22} = 1.726$, $p = 0.15$, $\eta_p^2 = 0.05$). However, duration of passive social behaviour did differ across sessions ($F_{3.76,105.19} = 2.72$, $p < 0.05$, $\eta_p^2 = 0.013$; see Appendix D, figure A1.3).

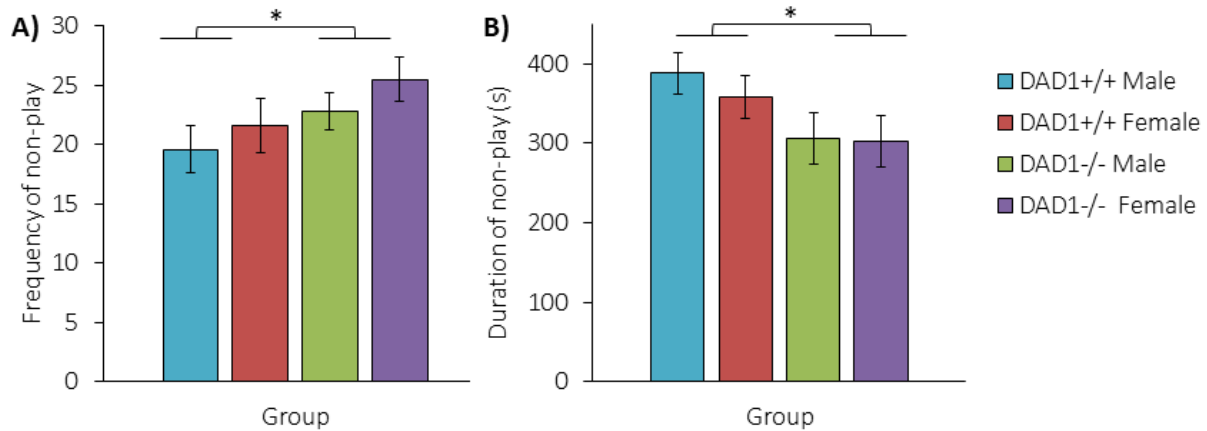


Figure 6.7. Average frequency and duration of non-play social behaviour across sessions. (A) DAD1^{-/-} mutants engaged in more non-play behaviour, but (B) wildtypes had a longer duration of non-play behaviour. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

6.4.1.7 Social Play Transitions

Mauchly's test of sphericity indicates that the assumption of sphericity was violated, $p < 0.001$, therefore Greenhouse-Geisser corrections were used. Subjects did not equally engage in behaviour across play types ($F_{2.37, 73.36} = 102.88$, $p < 0.001$, $\eta_p^2 = 0.77$). Subjects, regardless of sex or genotype, engaged in more pouncing and passive behaviour than chasing, wrestling and pinning ($p < 0.001$ for all post hoc tests) with wrestling having the lowest frequency of all behaviour. Subjects did not significantly differ in engagement between pouncing and passive behaviour ($p = 0.84$). Although there were genotype differences in frequency within certain play behaviours (e.g., DAD1^{-/-} mutants wrestling and chasing more), subjects did not significantly differ in their engagement between play behaviours (genotype effect: $F_{1,31} = 3.03$, $p = 0.09$, $\eta_p^2 = 0.09$; sex effect: $F_{1,31} = 0.57$, $p = 0.46$, $\eta_p^2 = 0.02$; see figure 6.8). For example, DAD1^{-/-} mutants wrestled more than wildtypes, but all subjects engaged in wrestling the least out of all behaviours.

Overall, there was only one significant difference in the sequence of play between genotypes and one between sexes. DAD1^{-/-} mutants had a significantly larger probability of transitioning from pouncing to wrestling ($p < 0.05$) than wildtypes (transition * genotype: $F_{1,69}$,

$_{52.31} = 4.97$, $p < 0.05$, $\eta_p^2 = 0.14$). Females had a significantly larger probability of transitioning from passive back into passive behaviour ($p < 0.05$) than males (transition * sex: $F_{1.69, 52.31} = 7.14$, $p < 0.01$, $\eta_p^2 = 0.19$; see figure 6.8). It is interesting to note that subjects, regardless of genotype and sex, did not differ between their strongest transition probability within each behaviour, expect for transitions from pinning. Although not significantly different, the largest transition probability from pinning for DAD1^{-/-} mutants was to passive behaviour with pouncing following closely behind. Whereas the largest transition probability from pinning for wildtypes was to pouncing with passive behaviour following closely behind (see figures 6.8 and 6.9).

Most significant differences in the structure of play were between transitions. There was not an equal probability of transitioning between behaviours; subjects, regardless of genotype or sex, engaged in social play in a specific sequence. For pinning, subjects had a significantly higher probability of transitioning from pinning to pouncing and passive than transitioning to chasing, wrestling and pinning (transition effect: $F_{1.28, 39.68} = 80.97$, $p < 0.001$, $\eta_p^2 = 0.72$; $p < 0.05$ for all post hoc tests). For chasing, subjects had a significantly higher probability of transitioning from chasing to pouncing than transitioning to pinning, chasing, wrestling and pouncing (transition effect: $F_{1.76, 54.51} = 354.89$, $p < 0.001$, $\eta_p^2 = 0.72$; $p < 0.01$ for all post hoc tests). For wrestling, subjects had a significantly higher probability of transitioning from wrestling to pouncing and passive behaviour than transitioning to pinning, chasing and wrestling (transition effect: $F_{1.20, 37.09} = 111.40$, $p < 0.001$, $\eta_p^2 = 0.78$; $p < 0.001$ for all post hoc tests). For pouncing, subjects had a significantly higher probability of transitioning from pouncing to pinning than transitioning to chasing ($p < 0.001$), wrestling ($p < 0.001$), pouncing ($p < 0.001$) and passive behaviour ($p < 0.05$) (transition effect: $F_{1.69, 52.31} = 111.72$, $p < 0.001$, $\eta_p^2 = 0.79$). For passive behaviour, subjects had a significantly higher probability of transitioning from passive behaviour to pouncing and passive

behaviour than transitioning to pinning, chasing and wrestling (transition effect: $F_{2.02, 62.49} = 205.84$, $p < 0.001$, $\eta_p^2 = 0.87$; $p < 0.001$ for all post hoc tests; see figure 6.8).

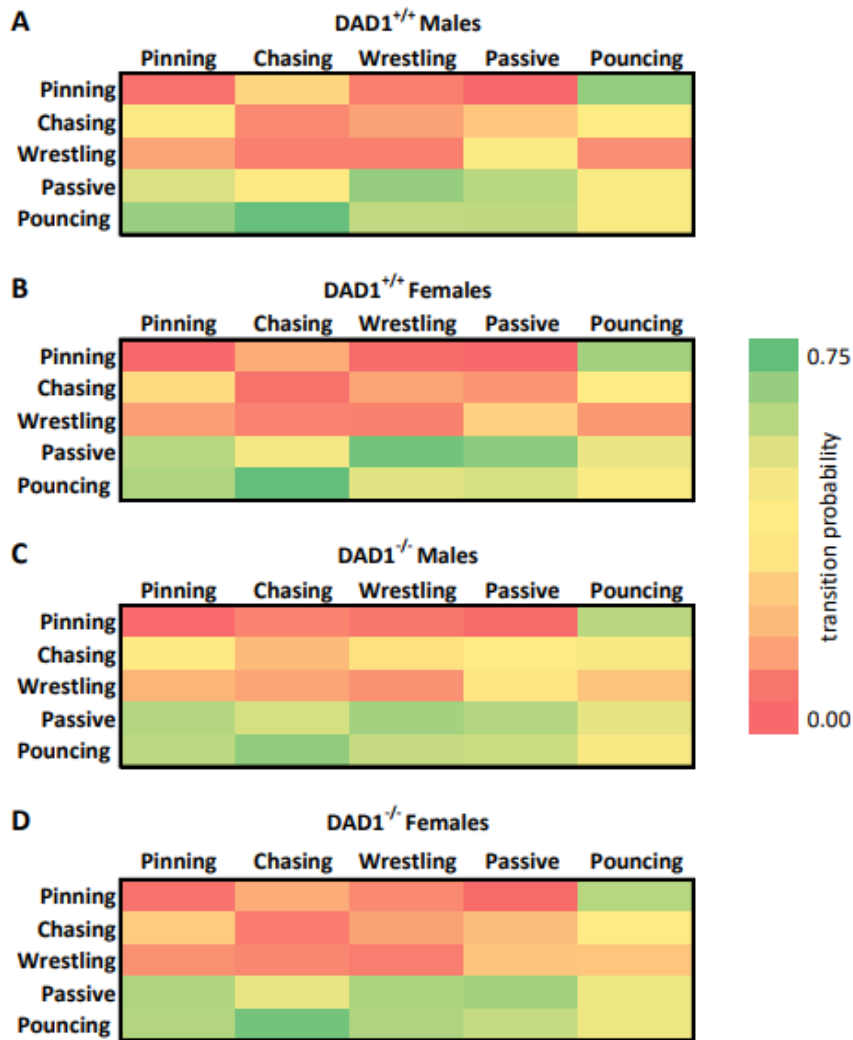
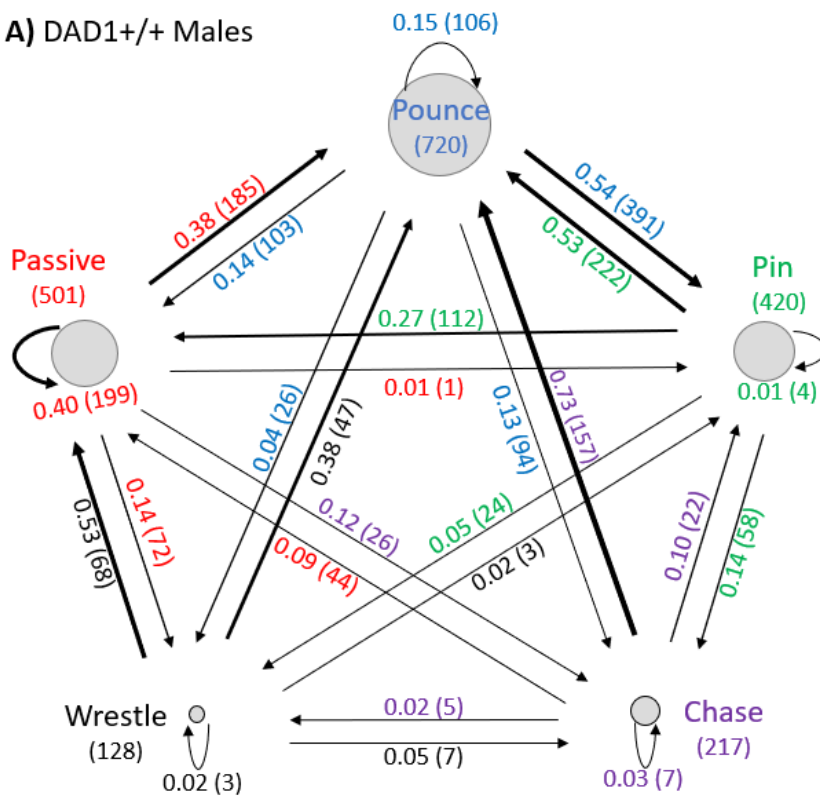
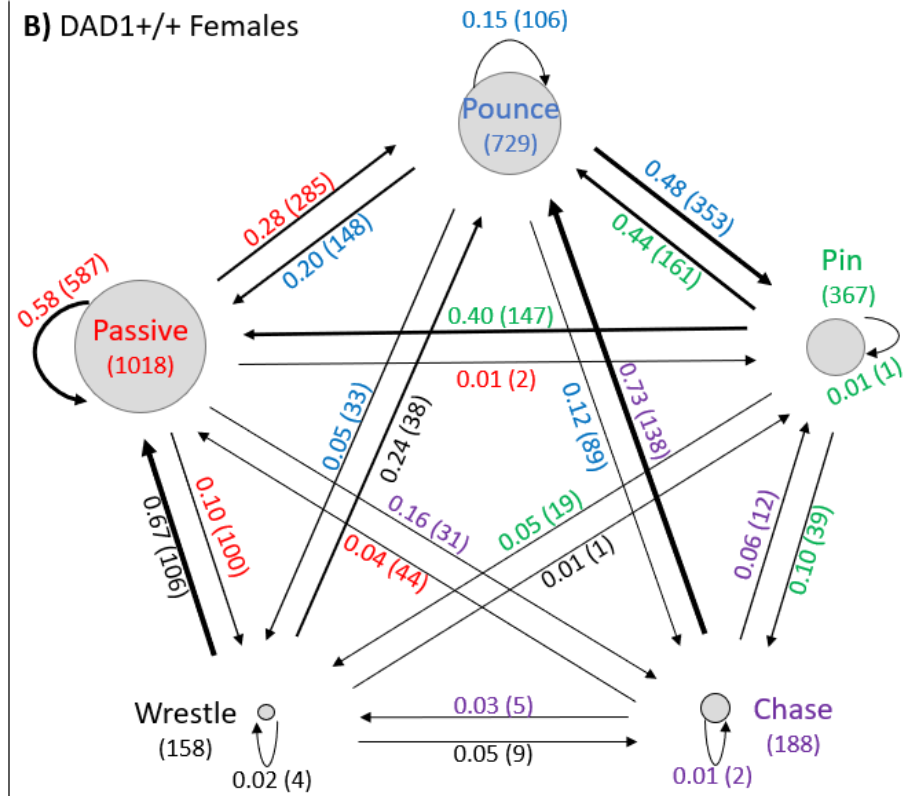


Figure 6.8. Heatmap transition matrices for each group: wildtype males (A), wildtype females (B), DAD1^{-/-} mutant males (C) and DAD1^{-/-} mutant females (D). Heat maps were computed to illustrate the transitions with the highest and lowest probabilities. Red colours indicate transitions with the *lowest* probabilities and green colours indicate the *highest* probabilities.

A) DAD1+/+ Males



B) DAD1+/+ Females



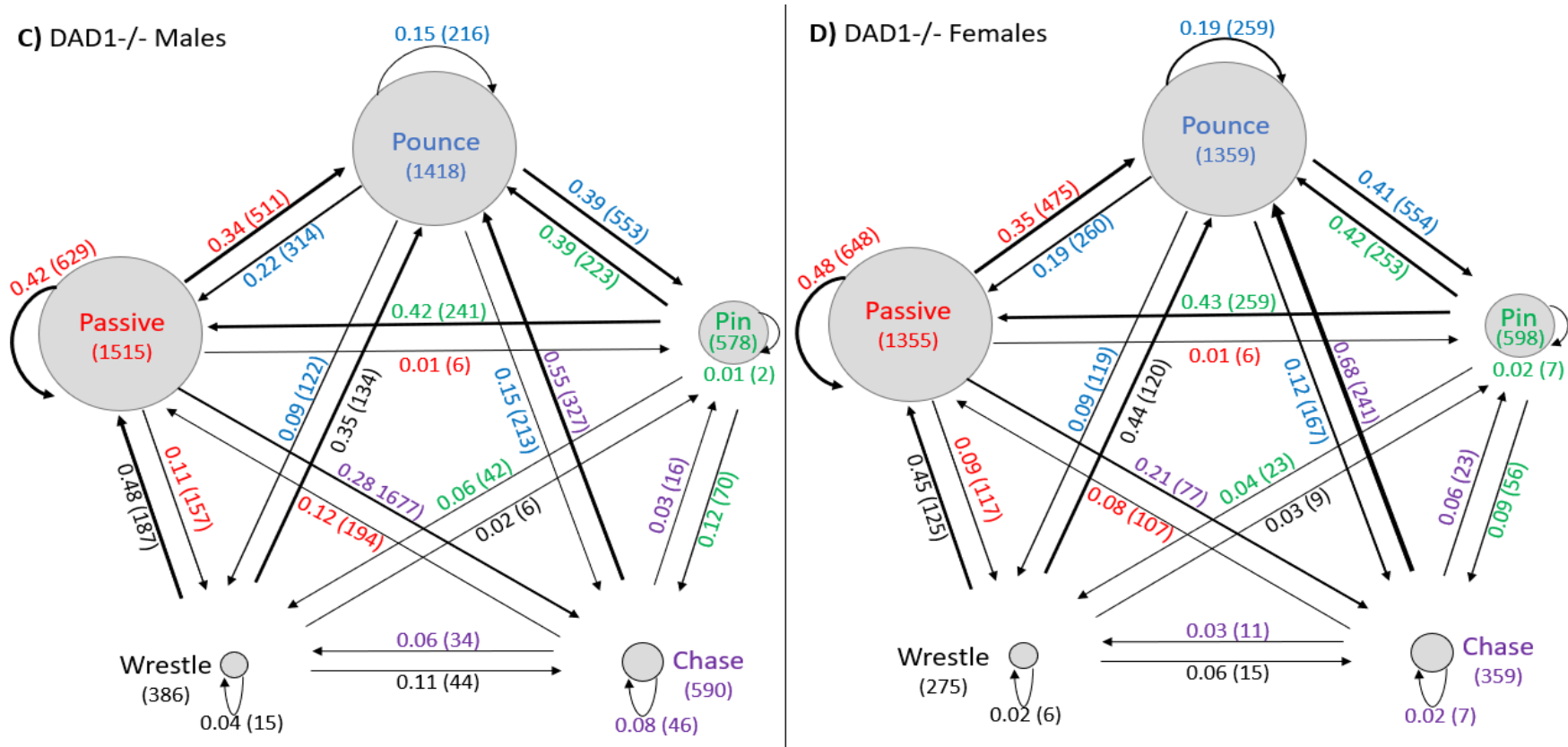


Figure 6.9. Behavioural kinetograms for each group: wildtype males (A), wildtype females (B), DAD1^{-/-} mutant males (C) and DAD1^{-/-} mutant females (D). The transition matrices were transformed into behavioural kinetograms in which the size of the circles corresponds to the total number of transitions for each behaviour and the thickness of arrows is proportional to the probability of each transition. Thick arrows represent the highest probability. Thin arrows represent the lowest probability. Frequencies for transitions are shown in brackets. *Note: the kinetograms are a sum of behaviour across sessions, not an average.*

6.4.1.8 50 kHz Ultrasonic Vocalisations

There was no sex ($F_{1,30} = 1.37, p = 0.25, \eta_p^2 = 0.04$) or genotype ($F_{1,30} = 0.41, p = 0.52, \eta_p^2 = 0.01$) effects on the number of 50 kHz vocalisations emitted across, suggesting that subjects had similar levels of consummatory pleasure during play. The number of vocalisations remained stable across sessions, although the individual variance was relatively large (session effect: $F_{4,12, 123.58} = 1.84, p = 0.12, \eta_p^2 = 0.05$; see figure 6.10).

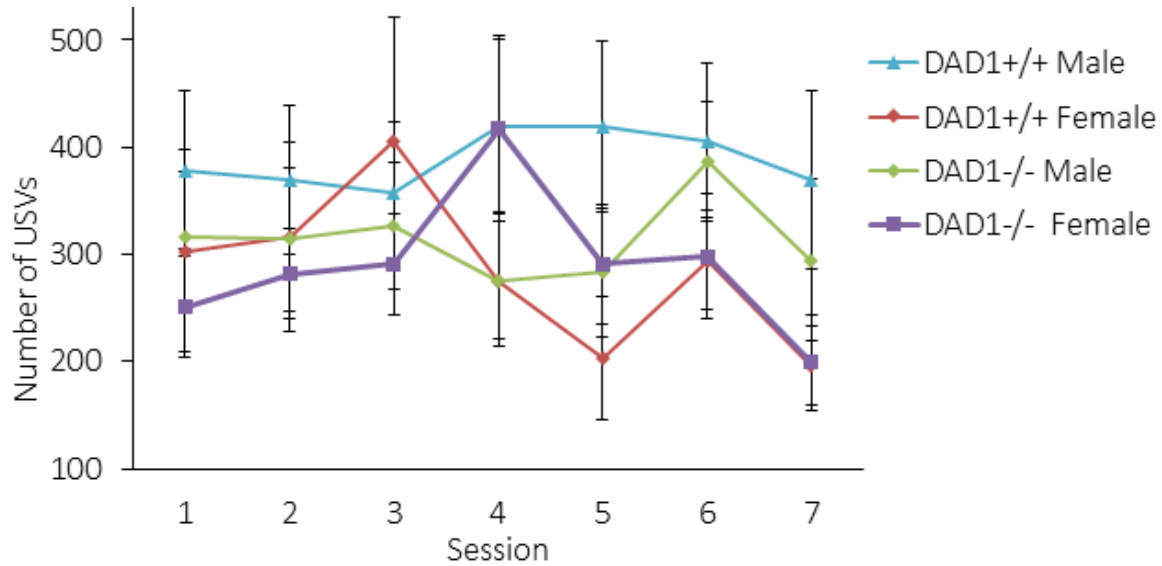


Figure 6.10. Average number of 50 kHz vocalisations emitted per play session. The average number of vocalisations remained stable across sessions. Error bars represent $\pm$ SEM.

There were no genotype or sex differences in the average frequency (kHz) or duration of 50 kHz vocalisations subjects emitted across play sessions (frequency genotype effect: $F_{1,30} = 0.93, p = 0.34, \eta_p^2 = 0.03$; frequency sex effect: $F_{1,30} = 0.04, p = 0.95, \eta_p^2 = 0.001$; duration genotype effect: $F_{1,30} = 0.30, p = 0.59, \eta_p^2 = 0.01$; duration sex effect: $F_{1,30} = 0.02, p = 0.92, \eta_p^2 = 0.001$; see Appendix D, A1.4 for additional statistics and figures).

6.4.2 Anticipatory Phase

6.4.2.1 Rearing

DAD1^{-/-} mutants had significantly lower frequency ($F_{1,65} = 22.54, p < 0.001, \eta_p^2 = 0.26$) and duration ($F_{1,65} = 28.04, p < 0.001, \eta_p^2 = 0.30$) of unsupported rears than wildtypes across sessions (see figure 6.11). Between groups, DAD1^{-/-} mutant males had significantly fewer unsupported rears than DAD1^{-/-} mutant females and significantly lower frequency and duration of unsupported rears than wildtype males and females (frequency genotype * sex: $F_{1,65} = 4.22, p < 0.05, \eta_p^2 = 0.06$; duration genotype * sex: $F_{1,65} = 4.12, p < 0.05, \eta_p^2 = 0.06$). Although there was a significant genotype effect for frequency and duration of unsupported rears, there were no genotype differences in percentage change (frequency genotype effect: $F_{1,66} = 3.57, p = 0.06, \eta_p^2 = 0.05$; duration genotype effect: $F_{1,66} = 3.63, p = 0.06, \eta_p^2 = 0.05$) (see Appendix D, figure A2.1).

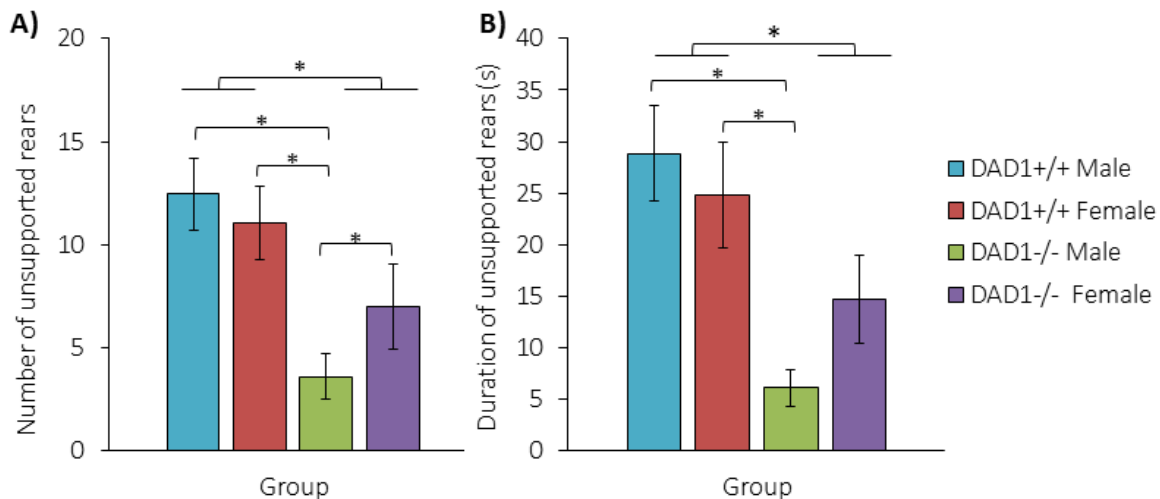


Figure 6.11. Average frequency and duration unsupported rears across anticipatory sessions. (A) DAD1^{-/-} mutants had fewer unsupported rears than wildtypes. DAD1^{-/-} males had fewer rears than other groups. (B) DAD1^{-/-} mutants had smaller duration of unsupported rears. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

Wildtypes significantly increased in frequency ($F_{4.75, 308.80} = 2.47, p < 0.05, \eta_p^2 = 0.04$) and duration ($F_{5.50, 357.24} = 4.22, p < 0.001, \eta_p^2 = 0.06$) of unsupported rears across the sessions, which

is indicative of anticipatory behaviour. DAD1^{-/-} mutants, on the other hand, did not significantly differ in frequency nor duration of unsupported rears across sessions, showing a lack of anticipatory behaviour (see figure 6.12). As mentioned in the previous chapter (see Chapter 5.5), it is unclear whether supported and unsupported rearing both reflect anticipatory behaviour, therefore the supported rearing statistics and figures are in section A2.2 and A2.3 of Appendix D.

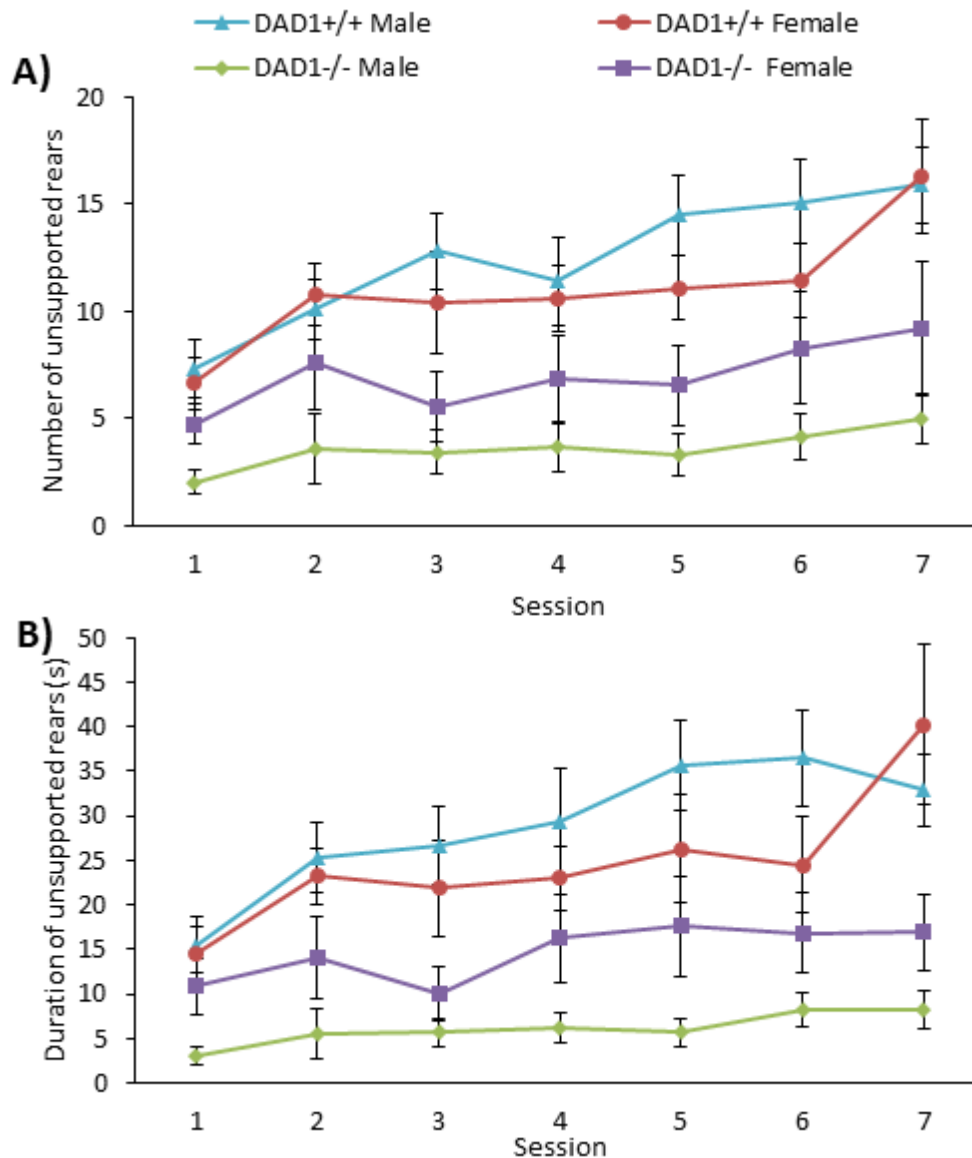


Figure 6.12. Frequency and duration of unsupported rears per anticipatory session. Wildtypes significantly increased in frequency (A) and duration (B) of unsupported rears across the seven sessions. Error bars represent $\pm$ SEM.

6.4.2.2 Distance Travelled

Females covered significantly more distance than males across sessions ($F_{1,66} = 4.95$, $p < 0.05$, $\eta_p^2 = 0.07$; see figure 6.13), but there was no significant difference between genotypes ($F_{1,66} = 0.24$, $p = 0.63$, $\eta_p^2 = 0.01$). Subjects significantly increased in distance moved across the seven sessions, which is indicative of anticipatory behaviour (session effect: $F_{3.11, 205.36} = 4.94$, $p < 0.01$, $\eta_p^2 = 0.07$; see figure 6.14).

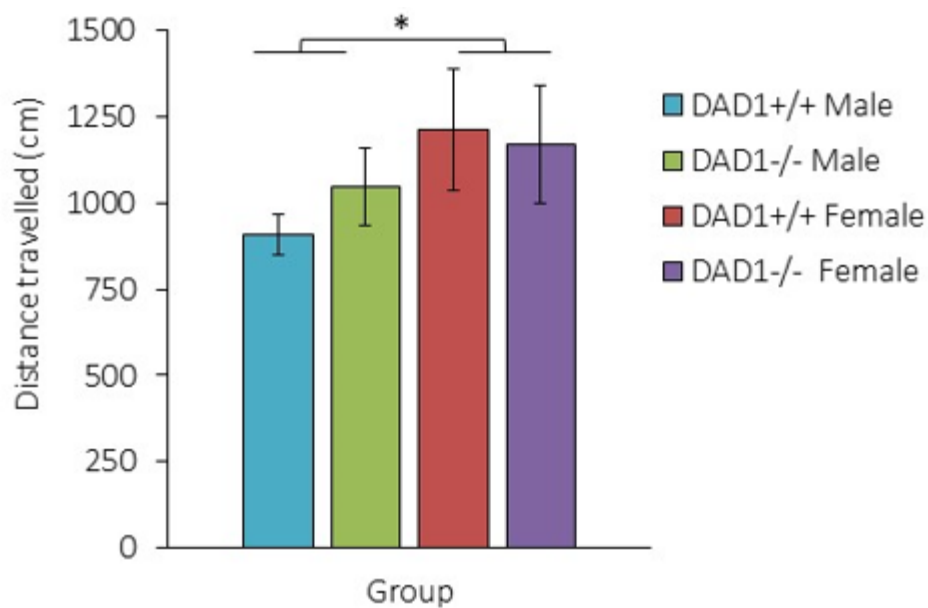


Figure 6.13. Average distance travelled during anticipatory sessions. Females covered more distance than males. Error bars represent $\pm$ SEM. *denotes significance to $p < .05$.

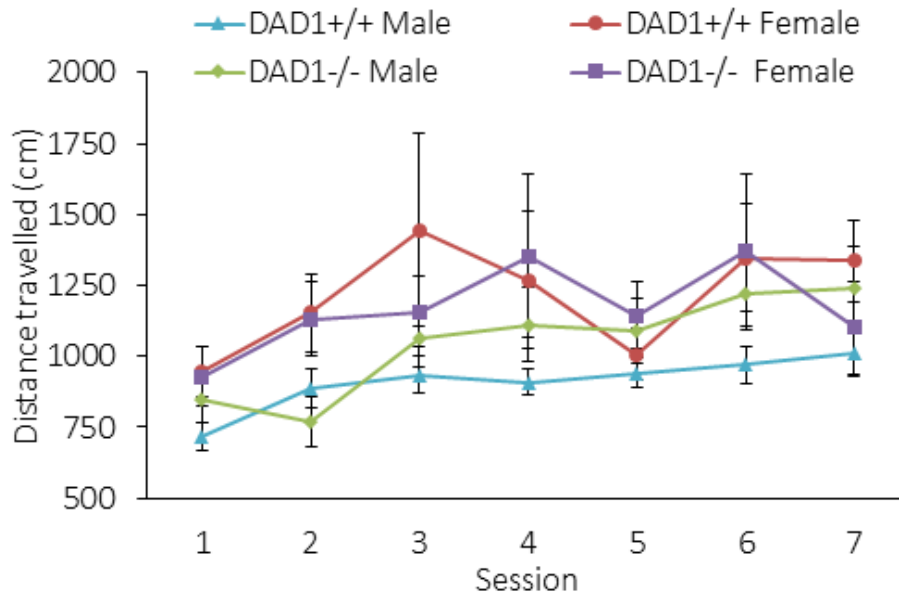


Figure 6.14. Average distance travelled per anticipatory session. Movement increased across sessions. Error bars represent $\pm$ SEM.

6.4.2.3 50 kHz Ultrasonic Vocalisations

DAD1^{-/-} mutants emitted significantly fewer vocalisations than wildtypes across anticipation sessions ($F_{1,64} = 8.21$, $p < 0.05$, $\eta_p^2 = 0.34$; see Appendix D, Figure A2.7). There was no sex effect in the number of vocalisations subjects emitted ($F_{1,64} = 0.11$, $p = 0.74$, $\eta_p^2 = 0.007$). Wildtypes significantly increased the number of vocalisations emitted across sessions, which is indicative of anticipatory pleasure (session * genotype: $F_{1.38, 22.01} = 9.74$, $p < 0.01$, $\eta_p^2 = 0.38$; $p < 0.05$ for all post hoc tests). DAD1^{-/-} mutants, on the other hand, did not significantly increase the number of vocalisations emitted across sessions, showing a lack of anticipatory pleasure (see figure 6.15).

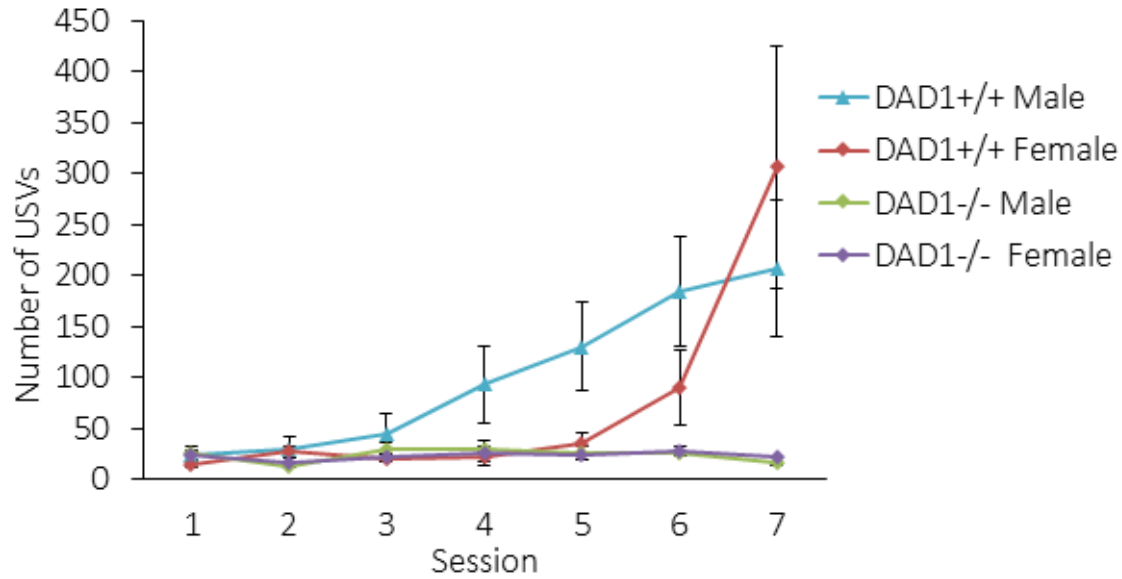


Figure 6.15. Average number of vocalisations emitted per anticipatory sessions. Wildtypes emitted more vocalisations across sessions showing anticipatory pleasure; whereas, DAD1^{-/-} mutants vocalisations remained stable. Error bars represent $\pm$ SEM.

Across sessions, there were no sex or genotype differences in the average frequency (kHz) or duration of vocalisations emitted across sessions (frequency genotype effect: $F_{1,64} = 0.17$, $p = 0.69$, $\eta_p^2 = 0.01$; frequency sex effect: $F_{1,64} = 0.07$, $p = 0.80$, $\eta_p^2 = 0.004$; duration genotype effect: $F_{1,64} = 0.04$, $p = 0.95$, $\eta_p^2 = 0.001$; duration sex effect: $F_{1,64} = 0.34$, $p = 0.57$, $\eta_p^2 = 0.02$; see Appendix D for additional statistics and figure A2.8).

6.5 Discussion

The aim of this study was to investigate if social cognitive and behavioural deficits seen in adult DAD1^{-/-} mutants (Homberg et al., 2016) are seen in the juvenile period by examining how DAD1^{-/-} mutants anticipate and engage in social play behaviour. It was hypothesised that DAD1^{-/-} mutants would display diminished motivation to engage in social play with fewer initiations of play. Contrary to this hypothesis, DAD1^{-/-} mutants did not differ from wildtypes in frequency of pouncing, which contradicts literature that found that D1 antagonism with SCH-23390, reduced pouncing (Manduca et al., 2021). It is possible that in other studies, SCH-23390 reduced initiation

of play due to a dosing effect; SCH-23390 might block more D1 receptors than a genetic reduction does. The 20-50% reduction seen in DAD1^{-/-} mutants might not be enough of a reduction to cause changes in initiation and engagement in social behaviour (Siviy et al., 2011; Undieh, 2010). Additionally, it is possible that the isolation period induced motivation too strong to detect subtle differences in play. Future research could decrease the isolation period to examine if lower motivation in DAD1^{-/-} mutants leads to marked impairments in play behaviour.

Although we did not find a genotype effect for pouncing, there were subtle, nonsignificant differences in transitions from pinning behaviour. From pinning, wildtypes' highest probability was transitioning to pouncing which suggests they initiate another round of play directly after finishing a bout with pinning. DAD1^{-/-} mutants highest transition from pinning was to passive behaviour which suggests they might take a “break” before initiating another bout of play. These differences in transitions from pinning were not significant, so we cannot conclude that the dopamine D1 receptor modulates initiation of social play. To further examine the role of the dopamine D1 receptor in initiation of social play, future research could pair DAD1^{-/-} mutants with wildtypes and measure if the wildtypes initiate play bouts more than mutants.

It was hypothesised that DAD1^{-/-} mutants would display diminished motivation to initiate play consequently displaying reduced engagement in social play relative to wildtypes. Contrary to predictions, DAD1^{-/-} mutants did not significantly differ in pouncing, pinning and total social play behaviour, which are the most characteristic parameters of social play (Pellis et al., 2022; van Kerkhof et al., 2013). Unexpectedly, DAD1^{-/-} mutants engaged in *more* chasing, wrestling and passive behaviour than wildtypes, but had a shorter duration of passive behaviour. Passive behaviour is indicative of the subjects engaging in social interaction without actively playing, so it is possible that passive behaviour serves as a “break” or recovery from active play bouts.

DAD1^{-/-} mutants engaged in more chasing and wrestling, therefore, they might need more frequent, but short “breaks” during play, possibly due to the D1 receptor’s role in exertion of effort seen in previous chapters (3 and 4) and the literature (Schweimer & Hauber, 2006; Yohn et al., 2015). Even though DAD1^{-/-} mutants engaged in more chasing and wrestling, subjects, regardless of genotype, engaged in chasing and wrestling the least out of all behaviours, which supports van Kerkhof et al.’s (2013) finding that the occurrence of wrestling and chasing are highly variable; whereas, pouncing and pinning are highly characteristic and occur frequently in social play.

The sequence of play was exploratory, and we found that DAD1^{-/-} mutants had an overall similar sequence of play as wildtypes. Out of the 25 transitions between the five behaviours, DAD1^{-/-} mutants only differed from wildtypes in one transition: DAD1^{-/-} mutants had a significantly larger probability of transitioning from pouncing to wrestling. As discussed earlier, DAD1^{-/-} mutants engaged in more wrestling which might be reflected in their transition from pouncing to wrestling. The sequence of play results supports the literature that play behaviour follows a specific structure. Decades of research has shown that a bout of social play begins with one rat pouncing another rat by nosing or rubbing the nape. The rat that is pounced upon can respond by either evading the partner (which typically causes the partner to chase after), wrestling the partner or rotating to its dorsal surface while the partner pins them down (Pellis et al., 2022; Trezza et al., 2010; van Kerkhof et al., 2013; Vanderschuren et al., 2016). It is not clear from the literature if there is an equal probability of transitioning from pouncing to chasing, wrestling, or pinning since only a handful of studies have examined sequences of play (Melotti et al., 2014; Pellis et al., 1993; Vanderschuren, 1995). Our results suggest that there is not an equal probability between transitions; pouncing was most often followed by pinning, then chasing and least frequently followed by wrestling. Future social play research should incorporate sequential

analysis to gain more insight into the structure of play. Social play is a commonly used paradigm, especially for looking at the effects of neonatal conditions, environmental stressors and drugs, therefore it is important to have a comprehensive understanding of play behaviour to better understand how different manipulations alter play frequency and sequence (Pellis et al., 2022).

Previous experiments examining the D1R and social play only used males, therefore, sex differences in the present study were exploratory. We found that there was only one sex difference in frequency of play behaviour with males chasing more than females, and the sequence of play did not differ between sexes. Males generally exhibit more social play behaviour than females, specifically in pouncing, pinning and wrestling which contradicts our results that only found sex differences in chasing (Auger & Olesen, 2009; Marquardt et al., 2023; Meaney & Stewart, 1981). A potential explanation for the lack of sex differences in our study could be due to the heterozygous genotype, $DAD1^{+/-}$, of the mothers. Neonatal treatment with D1 agonists increases estrogen receptor-regulated gene expression and subsequently increases social play. This increase in social play can be blocked by treatment with an estrogen receptor antagonist (Olesen et al., 2005). The heterozygous genotype might cause altered estrogen levels in the mothers which subsequently altered play behaviour and disrupted the typical sex differences. To examine the lack of sex differences, future research could measure estrogen levels in pregnant heterozygous $DAD1^{+/-}$ rats and subsequently in their pups during the neonatal and juvenile period.

Interestingly, although sexes did not differ in most play behaviours, females' engagement in play varied more across sessions. Females had an inverted U-shaped curve in chasing, pinning, pouncing and total play behaviour; females engaged less in these display behaviours during the first few sessions, peaked in their engagement over the next few sessions, and then decreased in the last few sessions. Social play emerges in rats around postnatal day 18 – 20, reaches its peak

between days 25 - 40, and declines with sexual maturation. Fluctuations in female rodent behaviour, specifically social behaviour, are associated with the estrous cycle (Chari et al., 2020). However, the estrous cycle develops in rats around 6 - 8 weeks of age (Marcondes et al., 2002) and the juveniles in our study were aged 3 - 4 weeks during testing so estrous cycle fluctuations likely do not explain the inverted U-shaped curve we found. To address the unexpected fluctuations in females' social play behaviour, we could examine play behaviour and sex hormone levels from postnatal day 25 through sexual maturation and see if the fluctuations in play are associated with changing hormone levels pre-puberty and into maturation.

Regarding anticipatory processes, we hypothesised that DAD1^{-/-} mutants would have deficits in anticipatory behaviour and pleasure seen as a blunted increase in locomotor activity, rearing and 50 kHz ultrasonic vocalisations across the sessions. We found that wildtypes showed significant increases in unsupported rearing and 50kHz vocalisations, two reliable and prominent measures of anticipatory behaviour, which is indicative of anticipatory pleasure for social play. DAD1^{-/-} mutants, on the other hand, did not show an increase in unsupported rearing or 50kHz vocalisations, suggesting a lack of anticipatory pleasure even with intact motivation to engage in play. These results support the pharmacological and D1 knockout research implicating the D1 receptor in anticipatory behaviour and subsequent pleasure (Assali et al., 2021; Baldo et al., 2002; Gallardo et al., 2014; Liu et al., 2012; Zhang et al., 2009).

Overall, DAD1^{-/-} mutants mostly engaged in a similar amount and structure of social play, along with initiating play similarly to wildtypes. The pharmacological literature is contradictory on whether the dopamine D1 receptor is involved in social play and/or its motivational aspect (Achterberg et al., 2016; Manduca et al., 2016; Olesen et al., 2005; Siviyy et al., 1996; Vanderschuren et al., 2016). Our results with a genetic manipulation suggest that the dopamine

D1 receptor might have a minor role in social play but is likely not the main regulator of play behaviour. Even though $DAD1^{-/-}$ mutants only have minor differences in social play compared to wildtypes, it is possible that this contributes to the social deficits in adult $DAD1^{-/-}$ mutants (Homberg et al., 2016). Since social play is important for social, cognitive and emotional development, these (Bijlsma et al., 2022; Pellis et al., 2019; Vanderschuren & Trizz, 2013), alterations, although minor, could lead to greater impairments in adult social behaviour. In terms of anticipation, the $DAD1^{-/-}$ mutants' lack of anticipatory behaviour supports a prominent role of the D1 receptor in anticipatory pleasure (Assali et al., 2021; Baldo et al., 2002; Gallardo et al., 2014; Liu et al., 2012; Zhang et al., 2009).

Outside of the dopamine D1 receptor, the results from the present study support Pellis and colleagues (2022) call for a comprehensive method for measuring social play behaviour in rats. Pouncing and pinning are the most characteristic and easily recognisable parameters of social play, which means most studies only examined these two behaviours or they examined the amount of total play behaviour (Pellis et al., 2022; van Kerkhof et al., 2013). We did not find genotype differences in pouncing, pinning or total social play behaviour but did find differences in chasing, wrestling, passive behaviour, and minor differences in sequences of play. The comprehensive social play method allowed us to detect subtle differences specific play behaviours (chasing, wrestling, passive behaviour) that other methods might have missed. Future research examining social play behaviour should incorporate a comprehensive method that includes a variety of play behaviours (beyond pouncing and pinning), the sequence of play and 50kHz ultrasonic vocalisations for socio-emotional communication (Pellis et al., 2022).

Chapter 7

General Discussion

Chapter 7

General Discussion

7.1 Thesis Aims and Predictions

Neuropsychiatric disorders are highly heterogeneous; there is overlap in symptoms *across* disorders, variability in symptomology *within* disorders is common, diagnoses are minimally stable long-term, and there is polygenic overlap between disorders (AL-Asadi et al., 2015; Allsopp et al., 2019; American Psychiatric Association, 2013; Gandal et al., 2018; Goldberg, 2011; Schulze et al., 2011). Consequently, current pharmacological treatment, designed for independent disorders, does not treat all symptoms exhibited in disorders, and these symptoms are typically associated with poor functional outcomes in occupational and social domains (Hyman, 2013; Miller & Raison, 2022). To resolve the limitations of disorder-based research, the RDoC framework implements a domain-based approach that enables the examination of the neurobiological mechanisms underlying domains of human functioning to create targeted treatment for dysfunctions seen across disorders (National Institute of Mental Health, 2023).

This thesis investigated the neurobiological mechanisms underlying anhedonia, avolition and asociality, which are dysfunctions in hedonic processing, motivation and social processes respectively. These symptoms, or dysfunctions, are seen across multiple neuropsychiatric disorders and are not currently improved with pharmacological treatment (Gorwood, 2008; Hyman, 2012; Miller & Raison, 2022; Nestler & Hyman, 2010; Whitton & Pizzagalli, 2022). Pharmacological and D1R knockout studies implicate the D1R in anticipatory behaviour and pleasure, motivation and effort-based decision making, and social processes (Assali et al., 2021; Baldo et al., 2002; Homberg et al., 2016; Salamone & Correa, 2012; Smith-Roe & Kelley, 2000). As discussed in the general introduction, pharmacology is not selective for solely the D1 receptor

(Homberg et al., 2016; Jones-Tabah et al., 2022; Ramos et al., 2005) and knockout models have complete inactivation of the D1R which can lead to compensatory mechanisms and is less translational to humans (National Institute of Mental Health; Nelson, 1997).

To overcome these limitations, this thesis used DAD1^{-/-} mutants, which have a genetic reduction in the D1R, to build upon previous research implicating the D1R in anticipatory pleasure, motivation, and social processes (Homberg et al., 2016; Smits et al., 2006). The overarching hypothesis is that the D1R is implicated in these three functions, therefore the prediction was that DAD1^{-/-} mutants would display deficits in socio-emotional communication, anticipatory pleasure, motivation for and engagement in social play, and exertion of effort. Additionally, to expand previous research with male DAD1^{-/-} mutants, male *and* female rats were used in all studies in this thesis. An overall summary of the findings and their relationship to the hypotheses is discussed in the following section.

7.2 Summary of Findings

This section summarises the main findings from the five studies in this thesis (see Table 7.1 for a visual summary). Chapter 2 examined the role of the D1R in neonatal development and socio-emotional communication. In line with previous studies (Homberg et al., 2016), neonatal DAD1^{-/-} mutants displayed reduced vocalisations when isolated from their dam and littermates. DAD1^{-/-} pups weighed less than wildtype and heterozygous pups, and interestingly DAD1^{-/-} mutants had a significantly higher chance of “runt” classification (i.e., weighing less than 90% of the litter’s average weight) than wildtypes. This observation led me to hypothesise that DAD1^{-/-} mutants might have a greater chance of “runt classification” due to impairments in motivation, or willingness to exert effort, for milk especially in larger litters which ultimately prompted the effort-based decision-making experiments in Chapter 3.

Chapter 3 explored the role of the D1R in motivation and effort-based decision-making across a variety of operant-chamber experiments. As predicted, DAD1^{-/-} mutants displayed deficits in motivation with significantly less lever pressing across all schedules of reinforcement and lower breakpoints on a progressive ratio schedule relative to wildtypes. Interestingly, there were several sex*genotype interactions; DAD1^{-/-} mutant males had a significantly lower acquisition rate and males preferred consuming freely available food chow over pressing the lever for higher value rewards relative to DAD1^{-/-} mutant females and wildtype males and females. The implications of these sex differences are discussed in a later section. Chapter 4 explored whether impairments in individual components of autoshaping (e.g., Pavlovian incentive learning, instrumental responding) could account for the DAD1^{-/-} mutant males' reduced acquisition of lever pressing in Chapter 3. In a Pavlovian Instrumental Transfer paradigm, DAD1^{-/-} mutants displayed diminished exertion of effort with lower instrumental responding across all schedules of reinforcement and a lack of transfer effect, but additional research is needed to determine the cause of an impaired transfer effect.

Chapters 5 and 6 examined the role of the D1R in hedonic processing, specifically in anticipatory and consummatory pleasure. In Chapter 5, DAD1^{-/-} mutants exhibited intact consummatory pleasure for a sucrose solution as predicted. Neither DAD1^{-/-} mutants nor wildtypes showed anticipatory behaviour nor pleasure; rearing, locomotor activity and 50 kHz vocalisations did not increase across sessions. The lack of anticipatory behaviour in wildtypes suggests that the sucrose solution was not rewarding enough to induce anticipatory pleasure. Therefore, Chapter 6 continued to investigate anticipatory pleasure by examining how DAD1^{-/-} mutants anticipate social play which is one of the most rewarding behaviours of a rat (Vanderschuren et al., 2008). Deficits were seen in social cognition and behaviour in *adult* DAD1^{-/-} mutants (Homberg et al., 2016), but

contradictory to predictions, DAD1^{-/-} mutant *juveniles* displayed intact initiation of social play in Chapter 6. There were subtle differences between genotypes in some play behaviours, which might contribute to the deficits in adult DAD1^{-/-} mutants' social cognition and interaction (Homborg et al., 2016). As predicted, wildtypes displayed anticipatory behaviour and pleasure for the opportunity to engage in social play and although DAD1^{-/-} mutants displayed intact motivation to engage in play, they did not show anticipatory behaviour prior to play, suggesting a deficit in anticipatory pleasure.

Overall, DAD1^{-/-} mutants exhibited intact consummatory pleasure and initiation of social play, subtle differences in social play behaviours, and diminished anticipatory pleasure, reduced exertion of effort and deficits in instrumental responding. The implications of these findings are discussed in a later section.

Table 7.1*Summary of Main Findings*

Study	Main Findings	Hypothesis supported?
Pup USVs (Chapter 2)	1. DAD1 ^{-/-} emitted fewer vocalisations relative to wildtypes on PND 7 but did not differ on PND 14.	Partially
	2. DAD1 ^{-/-} mutants did not differ from wildtypes in number, average duration, or bandwidth, but emitted calls at a higher average frequency.	Mostly
	3. Males did not differ from females in number, average duration, or bandwidth, but emitted calls at a higher average frequency.	Mostly
	4. DAD1 ^{-/-} mutants weighed less than wildtypes and had a higher chance of being a runt than wildtypes. Offspring of large litters weighed less than offspring of small litters.	Yes
Effort-Based Decision- Making (Chapter 3)	1. DAD1 ^{-/-} mutants pressed the lever significantly less than wildtypes across all schedules of reinforcement.	Yes
	2. DAD1 ^{-/-} mutants had significantly lower breakpoint ratios than wildtypes.	Yes
	3. DAD1 ^{-/-} mutant males had a significantly larger decrease in breakpoints between conditions and consumed more freely available food (not significant).	Mostly
	4. DAD1 ^{-/-} mutant males had a significantly lower acquisition rate of lever pressing than DAD1 ^{-/-} mutant females and wildtype males and females.	Not hypothesised
Pavlovian Instrumental Transfer (Chapter 4)	1. DAD1 ^{-/-} mutants pressed the lever significantly less than wildtypes across all schedules of reinforcement.	Yes
	2. DAD1 ^{-/-} mutants displayed no difference between active and inactive levers (could be due to floor effect).	Partially
	3. DAD1 ^{-/-} mutants had significantly smaller absolute transfer scores than wildtypes with DAD1 ^{-/-} mutant females having the smallest transfer score.	Partially
Sucrose Solution (Chapter 5)	1. DAD1 ^{-/-} mutants showed intact and consistent consummatory pleasure for a sucrose solution.	Yes
	2. Neither DAD1 ^{-/-} mutants nor wildtypes displayed anticipatory behaviour or pleasure; there was not a significant increase in rearing, locomotion nor 50 kHz vocalisations.	Partially
Social Play (Chapter 6)	1. Wildtypes exhibited anticipatory pleasure with a significant increase in 50 kHz vocalisations and unsupported rearing. DAD1 ^{-/-} mutants did not show an increase in vocalisations nor unsupported rearing, suggesting a lack of anticipatory pleasure.	Yes
	2. DAD1 ^{-/-} mutants displayed similar motivation to engage in social play and similar structure of play relative to wildtypes. DAD1 ^{-/-} mutants engaged in chasing and wrestling more than wildtypes.	No

7.3 Limitations

There are several limitations in this thesis that warrant discussion. In Chapter 4, DAD1^{-/-} mutants failed to exhibit a transfer effect from classical conditioning to instrumental responding. Unfortunately, we were unable to record behaviour during classical conditioning, so it is unclear if the DAD1^{-/-} mutants even acquired cue-reward associations during classical conditioning. If the DAD1^{-/-} mutants did acquire light-sucrose pellet associations, then the DAD1^{-/-} mutants were likely unable to transfer this to instrumental responding. On the other hand, if the DAD1^{-/-} mutants did not acquire light-sucrose pellet associations, then there would not have been associations to transfer to instrumental responding. To address this limitation, future research could measure pellet dispenser entries, latency to the dispenser and percentage of time spent at the dispenser to examine the DAD1^{-/-} mutants' approach behaviour during cue presentation and determine whether they acquired the Pavlovian relationship between the cue and sucrose pellet delivery.

In Chapter 6, we did not find marked differences in social play behaviour between genotypes. As discussed in Chapter 1, social processes are mediated by a network of brain regions (i.e., the “social brain”) including, but not limited to, the amygdala, mPFC, temporoparietal junction (TPJ), and superior temporal sulcus/gyrus (STS/STG). The DAD1^{-/-} mutation does not reduce D1R insertion equally across brain regions (e.g., 20% in olfactory tubercle and 50% in PFC and subcortical regions), therefore it is possible that some “social brain” regions are less impacted by the D1R mutation resulting in normal play behaviour (Homberg et al., 2016). Quantification of “social brain” regions using autoradiography could be completed to determine the level of D1R reduction in regions implicated in social processes. Additionally, although a genetic reduction is beneficial with less compensatory consequences than complete knockouts (Nelson, 1997), a 20-50% reduction in D1R might be too small of a reduction to detect subtle

differences in functioning. For example, a larger reduction in D1R might be necessary to induce alterations in social play behaviour. Lastly, the Markov chain analysis of social play behaviours did not include time as a factor; the subtle differences seen in sequence of play might become more apparent with time factored in which future studies could include (Van den Berg et al., 1999; Vanderschuren et al., 1995).

We found sex differences in several experiments, such as acquisition of lever pressing in Chapter 3 and stability of play behaviour across sessions in Chapter 6, but we did not measure sex hormones or track the estrous cycle. Estradiol is of particular interest because it facilitates dopaminergic transmission with increased phasic dopamine firing during the estrus stage which increases motivated behaviour in rats and humans (Barth et al., 2015; Hidalgo-Lopez & Pletzer, 2017; Jacobs & D'Esposito, 2011; Perez et al., 2014). Without estrous cycle tracking and sex hormone measurement, it is unclear whether estradiol directly influenced the female's behaviour in this thesis. To address this limitation, future work could track the estrous cycles and measure sex hormone levels to determine if hormonal differences correlate with behaviour.

Lastly, it is important to note that neuropsychiatric disorders are complex with polygenic and environmental influences (Gandal et al., 2018; Schulze et al., 2011). Although it is clear that the D1R is implicated in anticipatory behaviour and motivation, this accounts for one mechanism underlying dysfunctions seen across neuropsychiatric disorders. The D1R likely interacts with other genes and environmental factors to influence dysfunctions, which future research could investigate to give a clearer picture of all factors involved.

7.4 Implications and Future Work

7.4.1 Sex Differences

As mentioned above, there were sex differences in motivated behaviour and social play engagement across sessions. Regarding motivation, DAD1^{-/-} mutant males had a lower lever-pressing acquisition rate and preferred the low effort/low reward option in a choice task relative to wildtypes and DAD1^{-/-} mutant females. DAD1^{-/-} mutant females, on the other hand, did not differ from wildtypes in acquisition of lever pressing or preference for low versus high effort rewards. Regarding social play, males and females did not differ in their overall amount of engagement in social play, but there was a difference in the stability of play over the sessions; males' engagement in all play behaviours remained stable across sessions while females displayed an inverted U-shaped curve in chasing, pinning, pouncing and total play behaviour. These results highlight the vital importance of including both sexes in research, especially with the sex differences in incidences and severity of several neuropsychiatric disorders such as schizophrenia, MDD, ASD, ADHD, and anxiety disorders (Brzezinski-Sinai & Brzezinski, 2020; Green et al., 2019).

It is common for animal research results from males to be extended to females (Lee, 2018), but as shown in this thesis (and other research), there can be sex differences in behaviour suggesting that we cannot always extrapolate findings to both sexes. With this, estradiol exerts an overall facilitating effect on dopaminergic transmission by increasing synthesis and release of dopamine via estradiol membrane receptors (Barth et al., 2015; Hidalgo-Lopez & Pletzer, 2017; Jacobs & D'Esposito, 2011). Elevated estradiol concentrations have been associated with increased motivation and goal-oriented behaviour which could account for the sex differences in motivated behaviour seen in Chapter 3 (Becker et al., 2016; Yoest et al., 2014). Overall, the inclusion of both sexes in this thesis showed sex differences in some behaviour, therefore, future

research exploring the role of the D1R, or any mechanism, should incorporate both sexes to ensure any differential influence of sex hormones is captured and accounted for, and is ultimately more translational to humans (Beery & Zucker, 2011; Holdcroft, 2007; Lee, 2018).

7.2.2 The Dopamine D1 Receptor

Social processes refer to how individuals perceive, recognise, interpret and respond to social cues and stimuli, and can be broken down into three interdependent components: social perception, social cognition and social behaviour (Beaudoin & Beauchamp, 2020). Social play in juvenile rats encompasses all components of social processes, with an emphasis on social behaviour. *Juvenile* DAD1^{-/-} mutants displayed intact motivation to engage in social play with minor differences in quantity of specific social play behaviours, which contrasts with the diminished social cognition seen in *adult* DAD1^{-/-} mutants (Homberg et al., 2016). Social processes begin developing in infancy with shifting social targets (e.g., parents, peers) and functions throughout development; peer play in childhood builds social and cognitive skills while relationships in adulthood likely reflect evolutionary pair-bonding (Burnett et al., 2011; Foulkes et al., 2018; Hammond & Brownell, 2015; Kilford et al., 2016; Nelson et al., 2016). It is possible that the D1R might not modulate social processes throughout the entire lifespan, rather it is involved in social cognition and behaviour in adulthood which serves a different function than childhood peer play. On the other hand, as discussed in Chapter 6, the isolation period in our study might have induced motivation too strong to detect differences in play behaviour. Altering the protocol with smaller isolation periods could potentially highlight marked genotype differences in social play, therefore, we cannot concretely conclude that the D1R *only* plays a role in adult social processes.

Motivated behaviour occurs when individuals engage in reward-seeking behaviour aimed at receiving a reward or achieving a goal which includes behavioural activation, exertion of effort and task engagement (Berridge & Robinson, 2003; Berridge et al., 2009). DAD1^{-/-} mutants displayed deficits in reward-seeking behaviour across studies; DAD1^{-/-} mutants displayed fewer neonatal isolation-induced vocalisations, pressed the lever significantly less, and preferred consuming freely available food chow over pressing the lever for higher value rewards. Ultimately, the diminished exertion of effort the DAD1^{-/-} mutants displayed across three studies supports studies that implicated the D1R in motivation, which suggest that the D1R likely plays a role in reduced willingness to exert effort across a variety of neuropsychiatric disorders (Berglind et al., 2006; Kim et al., 2007; Puig & Miller, 2012).

There are two distinct phases of hedonic processing: the anticipatory phase involves the increase in behaviour and the pleasure associated with anticipation of a predicted reward, whereas, consummatory pleasure encompasses the in-the-moment experience of pleasure in the presence of a reward or “liking” a reward (Berridge & Robinson, 2003). DAD1^{-/-} mutants displayed intact consummatory pleasure for sucrose and social rewards showing that the D1R does *not* mediate the consummatory phase of hedonic processing (Bressan & Crippa, 2005; Berridge & Robinson, 2003). Even though DAD1^{-/-} mutants engaged in play similarly to wildtypes, DAD1^{-/-} did not display anticipatory behaviour for the opportunity to engage in social play, showing a deficit in anticipatory pleasure. Ultimately, the diminished anticipatory pleasure seen in DAD1^{-/-} supports previous research implicating the D1R as a mechanism underlying anticipatory anhedonia seen in a variety of neuropsychiatric disorders.

This thesis suggests a clear role for the D1R in anticipatory pleasure and motivated behaviour which has implications for targeted treatment for anticipatory anhedonia and avolition

seen across neuropsychiatric disorders (Berglind et al., 2006; Kim et al., 2007; Puig & Miller, 2012). As discussed in Chapter 1, there are currently no drugs available that exhibit substantial selectivity for the D1 receptor over the D5 receptor, moreover some D1R-binding drugs also show affinity for non-dopamine receptors (Homberg et al., 2016; Jones-Tabah et al., 2022; Ramos et al., 2005). Consequently, there is no pharmacological manipulation clinically available to enhance D1R function which could improve anticipatory anhedonia and avolition (Kim et al., 2015). However, there are promising results with recent development of D1R positive allosteric modulators that can be specialised to target secondary messenger pathways of the D1R (Jones-Tabah et al., 2022; Kim et al., 2015; Svensson et al., 2019). These allosteric modulators are undergoing preclinical research and preliminary results show that these drugs could benefit symptoms seen across disorders including, but not limited to, anhedonia, asociality and avolition (Desai et al., 2021; Jones-Tabah et al., 2022; McCarthy et al., 2022).

In addition to pharmacological treatment, targeted gene therapy could improve dysfunctions seen across neuropsychiatric disorders. As mentioned previously, there are five SNPs in human D1 receptors identified to date (hD1R; T37P, T37R, R50S, S199A, and A229). These SNPs are associated with changes in receptor expression and likely associated with dysfunctions seen in neuropsychiatric disorders (Al-Fulaij et al., 2008). Targeted gene strategies introduce specific genetic material via vectors to alter cell functions in specific areas of the body. Strategies include, but are not limited to replacing non-functional genes, providing a repair template that enables cells to erase mutations, and inactivating mutated genes (Bulcha et al., 2021; Humbert et al., 2012; Zhao et al., 2022). In the central nervous system, viruses, specifically adeno-associated viruses (AAVs) and lentiviruses (LVs), are used as vectors to deliver the genetic material to neuronal targets in specific brain regions (Ntetsika et al., 2021). Preclinical trials with viral vectors

targeting dopaminergic neurons show promising results for treating Parkinson's Disease (Ntetsika et al., 2021), therefore gene strategies targeting D1R mutations could be developed to treat dysfunctions in hedonic processing and motivation. A multitude of brain regions are involved in these processes with key regions including the amygdala for anticipatory processes, and the Nacc and PFC for anticipatory and motivational processes (Beaudoin & Beauchamp, 2020; Berridge & Robinson, 2003; Höflich et al., 2019; Linnet, 2014; Tso et al., 2018). The D1R is abundant with high density in the Nacc, and with lesser density in the amygdala and PFC (Bhatia et al., 2022; Cadet et al., 2010; Martel & McArthur, 2020), therefore viral vectors could be used to replace or fix mutated D1R genes in one of these brain regions and examine what behavioural dysfunctions are improved (Bulcha et al., 2021; Humbert et al., 2012; Zhao et al., 2023).

In summary, this thesis supports a role of the D1 receptor in anticipatory pleasure and motivated behaviour. Dysfunctions in anticipatory pleasure (i.e., anticipatory anhedonia) and motivated behaviour (i.e., avolition) are common across multiple neuropsychiatric disorders and are associated with poor functional outcomes (Gorwood, 2008; Hyman, 2012; Miller & Raison, 2022; Nestler & Hyman, 2010; Whitton & Pizzagalli, 2022). Importantly, the D1R is not a prominent target of current pharmacology used to treat neuropsychiatric disorders which could explain why anticipatory anhedonia and avolition are not improved with available pharmacological interventions (Bruns et al., 2018). To improve functional outcomes in individuals experiencing these symptoms, D1-targeted pharmacology and -gene targeted therapy should be explored.

7.2.3 Research Domain Criteria (RDoC)

RDoC was created to improve pharmacological treatment of neuropsychiatric disorders by providing researchers with a multidimensional conceptualisation of disorders; disorders are not fully independent syndromes but rather represent clusters of symptoms that partially overlap

(National Institute of Mental Health, 2023; Newson et al., 2021). RDoC encourages researchers to examine the neurobiological mechanisms of human functioning to create targeted treatment for dysfunctions seen across disorders. Lifespan development is incorporated into the RDoC framework since behaviour and cognition develop throughout the lifespan (National Institute of Mental Health, 2023). This thesis used an RDoC framework to investigate the neurobiological mechanisms of anhedonia, asociality and avolition. The three symptoms were examined within the context of the domains they correspond to: positive valence systems domain (e.g., reward responsiveness, reward learning, reward valuation) and social processes domain (e.g., affiliation and attachment, social communication, etc) across the lifespan.

The results in this thesis support the use of the RDoC framework in future research for several reasons. Firstly, the components associated with these three symptoms were differentially impacted by a genetic reduction in the D1R. Results implicate the D1R in anticipatory anhedonia and motivational dysfunctions (i.e., positive valence systems domain) but did not find a clear role for juvenile social behaviour (i.e., the social processes domain), in other words, the D1R is involved in some aspects of neuropsychiatric disorders but not all. This supports researchers looking at individual components of functioning (i.e., domains) rather than mechanisms of entire disorders. Secondly, there were differences in social processes across the lifespan. *DAD1^{-/-}* mutants previously displayed impairments in social cognition and behaviour in adulthood (Homberg et al., 2016) but these deficits did not extend to social play behaviour in juvenile *DAD1^{-/-}* mutants in this thesis. Consequently, the D1R's role in social processes might depend on the developmental stage which supports researchers looking at functioning across the lifespan as RDoC encourages. Overall, this thesis suggests that impairments in the D1R is one mechanism leading to dysfunctions

in anticipatory anhedonia and motivated behaviour with room for future research to examine environmental or polygenic interactions with the D1R.

Ultimately, the RDoC approach in this thesis provided insights into one mechanism underlying constructs in the *positive valence systems* and *social processes* domains of human functioning and how these mechanisms might change throughout development. Examining symptoms of disorders and their associated components gives us greater insight into their underlying mechanisms which can lead to improvements in treatment of neuropsychiatric disorders. Future treatment of neuropsychiatric disorders could be based on presentation of symptoms, not disorder diagnosed (e.g., treating dysfunction in hedonic processing rather than MDD). It is important to note that the diagnosis of neuropsychiatric disorders using the DSM is the best available classification as of the present, but some researchers suggest that future diagnosis of neuropsychiatric disorders might be based on common neurobiological pathways, not symptoms (e.g., diagnosis based on a dysregulated dopaminergic pathway not symptoms), (Cuthbert, 2020; Hakak-Zargar et al., 2022; Morris et al., 2022).

7.4 Conclusion

This thesis used an RDoC approach to investigate the neurobiological mechanisms underlying dysfunctions in hedonic processing, social processes and motivated behaviour seen across a multitude of neuropsychiatric disorders. To this end, rats with a selective genetic reduction in the dopamine D1 receptor underwent a series of five studies that examined neonatal development, exertion of effort, anticipation for sucrose and social reward, consumption of sweet rewards and engagement in social play across their lifespan. This reduction in the dopamine D1 receptor led to blunted exertion of effort and diminished anticipatory pleasure while consummatory pleasure and social play were not impacted. Ultimately, this thesis furthered

pharmacological and knockout research, implicating a clear role for the D1R in motivation and anticipatory pleasure which could be targeted for treatment with the development of selective D1 pharmacology, such as the D1 positive allosteric modulators undergoing preclinical trials (Jones-Tabah et al., 2022; Kim et al., 2015; Svensson et al., 2019). This thesis also supports using the RDoC framework to examine mechanisms underlying symptoms rather than disorders and highlights the vital importance of including males *and* females in animal research as there can be sex differences in behaviour.

Future research should continue using selective genetic reductions in the D1R to examine other dysfunctions seen across disorders, such as sleep and memory impairments (Takahashi et al., 2012; Zhang et al., 2022). Additionally, since neuropsychiatric disorders are complex with polygenic and environmental influences, future work can build upon the current thesis to examine how other genetic (e.g., mutations in serotonin transporter) and/or environmental conditions (e.g., early life stress) might interact with reductions in the D1 receptor to influence cognition and behaviour. Additionally, D1-targeted pharmacology (e.g., positive allosteric modulators) and D1-gene targeted therapy using viral vectors should be explored for treating dysfunctions such as anticipatory anhedonia, asociality, and avolition.

Appendix

Appendix A – Chapter Two

A1. Litter Descriptive Statistics

A1.1 Average Weight and Growth Per Litter

Table A1.1

Descriptive Statistics for Each Litter

Litter ID	Litter Size	Sex Ratio	PND 7 Weight (g)	PND 14 Weight (g)	Weight Change (%)
A	8	0.63	24.00 ± 1.69	41.10 ± 1.96	71.70 ± 5.87
B	5	0.40	21.20 ± 0.84	42.80 ± 2.77	102.00 ± 10.50
C	17	0.42	14.30 ± 1.62	23.90 ± 4.50	67.80 ± 27.00
D	10	0.50	28.80 ± 1.93	38.60 ± 2.72	34.40 ± 11.20
E	17	0.65	14.30 ± 1.57	25.50 ± 3.73	77.80 ± 14.50
F	11	0.82	17.50 ± 1.29	34.50 ± 1.63	98.10 ± 12.80
G	9	0.78	21.60 ± 2.97	39.30 ± 2.52	70.90 ± 3.60
H	14	0.64	16.70 ± 1.37	29.80 ± 2.74	79.80 ± 14.10
I	11	0.64	21.70 ± 1.10	35.90 ± 1.76	65.60 ± 10.10
J	12	0.42	20.40 ± 1.62	34.00 ± 2.00	77.40 ± 3.68
K	16	0.44	17.90 ± 0.80	28.50 ± 4.32	59.20 ± 23.30
L	15	0.33	17.10 ± 2.57	28.40 ± 3.88	68.60 ± 34.30
M	17	0.47	17.70 ± 2.58	28.80 ± 3.83	71.40 ± 34.10
N	18	0.33	14.00 ± 1.22	23.00 ± 2.40	64.60 ± 15.00
O	17	0.47	14.80 ± 1.88	24.40 ± 4.47	65.20 ± 22.20
P	16	0.56	12.60 ± 3.27	28.20 ± 5.22	143.00 ± 22.20
All litters	13.31	0.53	17.63 ± 4.39	29.88 ± 6.65	74.30 ± 31.02

± indicates standard deviation.

A1.2 Runt Demographics Based on Litter Size

Table A1.2

Runt Demographics

Subjects	All litters	≤12 litters	>12 litters
Number of runts	2.38± 1.66	1.60 ± 0.55	2.88 ± 1.96
Runts per litter (%)	17.20 ± 10.20	16.10 ± 5.36	17.80 ± 12.70
DAD1 ^{-/-} mutant (%)	54.84 ± 36.40	43.00 ± 41.80	53.60 ± 34.90
DAD1 ^{+/+} wildtype (%)	12.90 ± 19.70	20.00 ± 27.40	5.95 ± 12.10
DAD1 ^{+/-} heterozygous (%)	32.26± 44.00	37.00 ± 54.80	40.50 ± 40.00

± indicates standard deviation.

A2. Sex Composition

The proportion of males relative to the number of total pups was calculated per litter to determine the sex ratio (Romano et al., 2012). A binomial one-proportion z test was used to determine if the sex ratios differed from the expected proportion of 0.5 (Cameron, 2004; Romano et al., 2012). The average sex ratio of the 16 litters was 0.53 ± 0.15 which did not significantly differ from the expected sex ratio ($z = 1.20, p = 0.23$). Twelve out of sixteen litters did not significantly differ from 0.50 ($p > 0.05$ for the 12 litters) while litters F, G, L and N significantly differed. Litter F ($z = 10.5, p < 0.001$) and litter G ($z = 8.31, p < 0.001$) had male biases with sex ratios of 0.82 and 0.78 respectively. Litter L ($z = -6.50, p < 0.001$) and litter N ($z = -7.09, p < 0.001$) had female biases with a sex ratio of 0.33 for both litters (see figure A.1). There was not a significant relationship between sex ratio and litter size ($r(14) = -0.33, p = 0.21$), suggesting that litter size did not impact the sex composition which supports the finding that prenatal factors that influence sex ratio (e.g., dam carbohydrate intake) do not impact litter size suggesting that the two variables are independent (Rosenfeld et al., 2003).

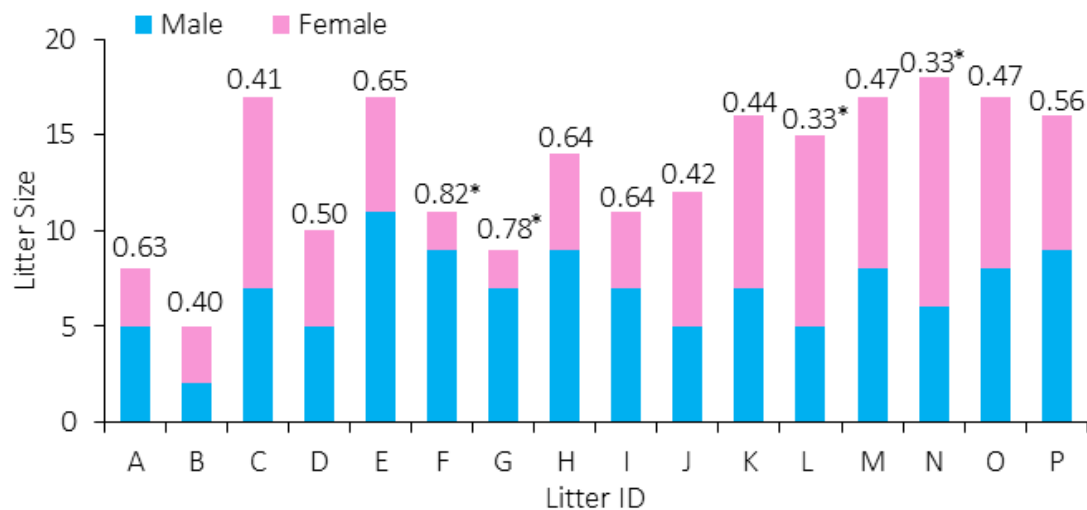


Figure A2.1. Litter size and sex composition across litters. Stacked bars show the total litter size at birth and colour differentiated by sex. Most litters had an approximately even sex ratio (0.50), except for litters F, G (male bias), L and N (female bias) which are indicated with *.

A3. Litter Size

A3.1 Average Weight and Growth Per Group

Table A3.1

Pup Weight Based on Litter Size

Litter Size	PND 7 Weight (g)	PND 14 Weight (g)	Weight Change (%)
5	21.20 ± 0.84	42.80 ± 2.77	102.00 ± 10.50
8	24.00 ± 1.69	41.10 ± 1.96	71.70 ± 5.87
9	21.60 ± 2.97	39.30 ± 2.52	70.90 ± 3.60
10	28.80 ± 1.93	38.60 ± 2.72	34.40 ± 11.20
11	19.59 ± 2.42	35.18 ± 1.77	81.83 ± 19.62
12	20.42 ± 1.55	34.00 ± 1.83	77.44 ± 3.36
14	16.67 ± 1.31	29.77 ± 2.64	79.84 ± 13.47
15	17.14 ± 2.47	28.36 ± 3.73	68.56 ± 33.04
16	15.27 ± 3.52	28.40 ± 4.51	92.58 ± 55.302
17	15.24 ± 2.35	25.10 ± 4.23	70.79 ± 22.22
18	14.00 ± 1.22	23.00 ± 2.40	64.60 ± 15.00

± indicates standard deviation.

Table A3.2*Pup Weight and Growth Across Groups*

Subjects	PND 7 Weight (g)	PND 14 Weight (g)	Weight Change (%)
DAD1 ^{+/+} males	19.10 ± 4.38	33.00 ± 5.54	77.40 ± 26.60
DAD1 ^{+/+} females	17.30 ± 5.10	29.10 ± 5.85	74.90 ± 44.30
DAD1 ^{-/-} males	16.20 ± 4.74	27.90 ± 7.23	77.40 ± 45.80
DAD1 ^{-/-} females	16.50 ± 4.32	27.20 ± 6.37	72.00 ± 32.00

± indicates standard deviation.

Appendix B – Chapter Four

A1. Pavlovian Instrumental Transfer Test

A1.1 PIT Test Descriptive Statistics

Table A1.1

Number of Lever Presses Per Block

Subjects	Block 1		Block 2		Block 3		Block 4	
	Pre-Cue	Cue	Pre-Cue	Cue	Pre-Cue	Cue	Pre-Cue	Cue
Active Lever								
DAD1 ^{+/+} males	33.30 ± 9.03	25.10 ± 7.82	28.90 ± 6.43	19.60 ± 6.34	18.00 ± 3.97	10.60 ± 5.13	10.60 ± 2.45	7.50 ± 6.85
DAD1 ^{+/+} females	18.90 ± 4.90	15.80 ± 1.21	15.20 ± 4.02	13.00 ± 6.06	11.00 ± 6.87	9.09 ± 5.09	7.00 ± 3.13	4.36 ± 3.43
DAD1 ^{-/-} males	11.30 ± 12.60	6.22 ± 16.70	6.11 ± 15.40	7.00 ± 8.65	4.33 ± 12.90	4.44 ± 8.55	3.00 ± 5.68	5.89 ± 5.81
DAD1 ^{-/-} females	2.00 ± 12.40	0.67 ± 8.49	2.17 ± 12.00	2.67 ± 9.54	3.00 ± 12.90	2.67 ± 6.07	1.83 ± 6.72	2.17 ± 3.93
Inactive Lever								
DAD1 ^{+/+} males	2.63 ± 1.99	0.38 ± 1.01	0.25 ± 0.44	0.63 ± 1.12	0.63 ± 3.33	0.25 ± 0.00	0.50 ± 1.41	1.63 ± 1.66
DAD1 ^{+/+} females	1.18 ± 0.41	3.18 ± 0.41	0.27 ± 0.00	0.09 ± 0.00	0.00 ± 0.00	0.27 ± 1.03	0.09 ± 0.00	0.00 ± 0.41
DAD1 ^{-/-} males	0.78 ± 5.21	0.44 ± 0.74	0.22 ± 0.71	0.67 ± 1.19	1.11 ± 1.77	0.00 ± 0.46	0.67 ± 1.07	0.67 ± 3.85
DAD1 ^{-/-} females	0.17 ± 2.14	0.17 ± 6.76	0.91 ± 1.90	0.30 ± 0.32	0.00 ± 0.00	0.67 ± 0.91	0.00 ± 0.30	0.17 ± 0.00

± indicates standard deviation.

Appendix C – Chapter Five

A1. Anticipation Phase

A1.1. 50kHz Vocalisations

Table A1.1

Additional 50kHz Vocalisation Statistics

	50kHz Vocalisation Measure	
	Frequency (kHz)	Duration
Genotype effect	$F_{1,27} = 0.13, p = 0.72, \eta_p^2 = 0.005$	$F_{1,27} = 0.10, p = 0.75, \eta_p^2 = 0.004$
Sex effect	$F_{1,27} = 1.22, p = 0.28, \eta_p^2 = 0.04$	$F_{1,27} = 0.04, p = 0.95, \eta_p^2 = 0.001$
Session effect	$F_{1,03,27.90} = 1.75, p = 0.11, \eta_p^2 = 0.06$	$F_{3,82,103.06} = 2.13, p = 0.12, \eta_p^2 = 0.07$

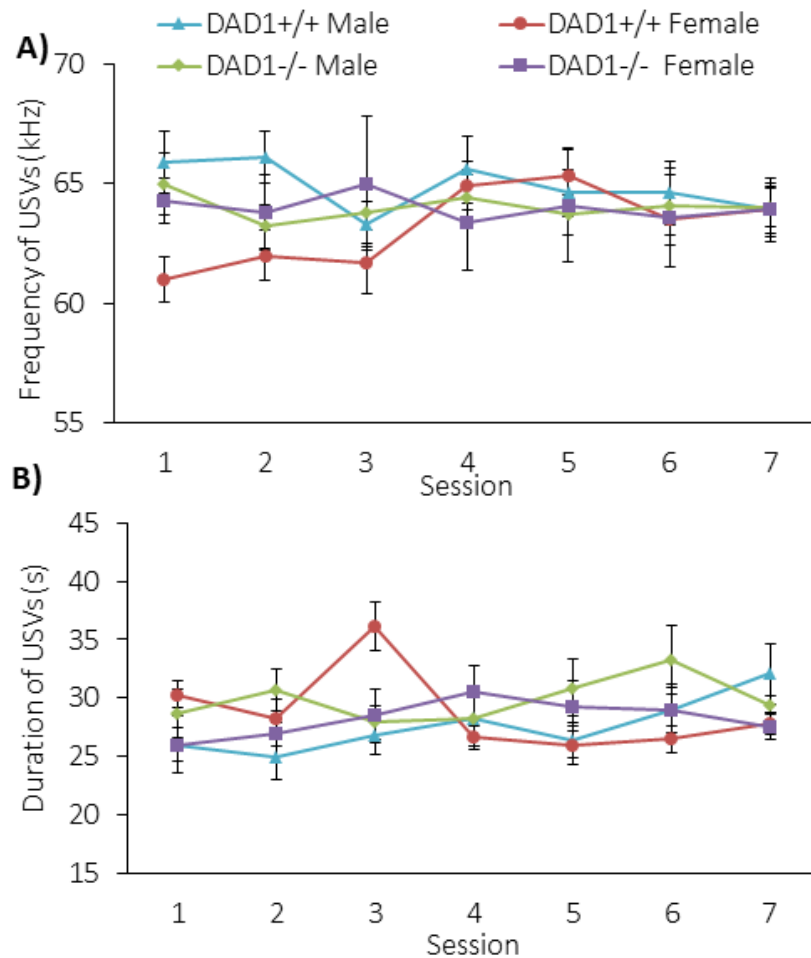


Figure A1.1. Average frequency and duration of 50 kHz vocalisations emitted per anticipatory session remained stable across sessions. Error bars represent $\pm$ SEM.

Appendix D – Chapter Six

A1. Social Play

A1.1 Chasing

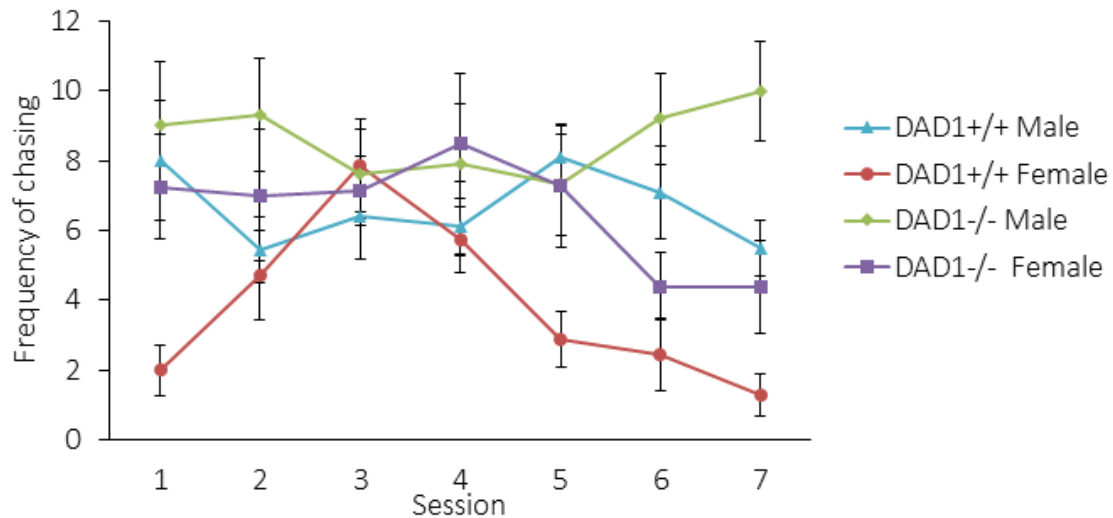


Figure A1.1. Average frequency of chasing events per session. Females' frequency of chasing varied across sessions. Error bars represent $\pm$ SEM.

A1.2 Wrestling

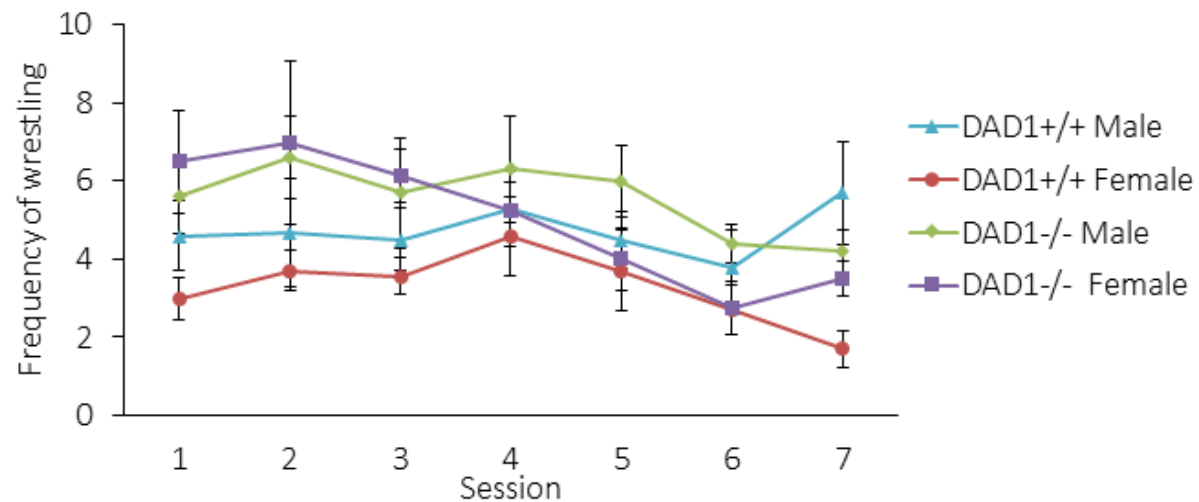


Figure A1.2. Average frequency of wrestling events per session. Subjects' frequency of wrestling decreased across sessions. Error bars represent $\pm$ SEM.

A1.3 Passive Social Behaviour

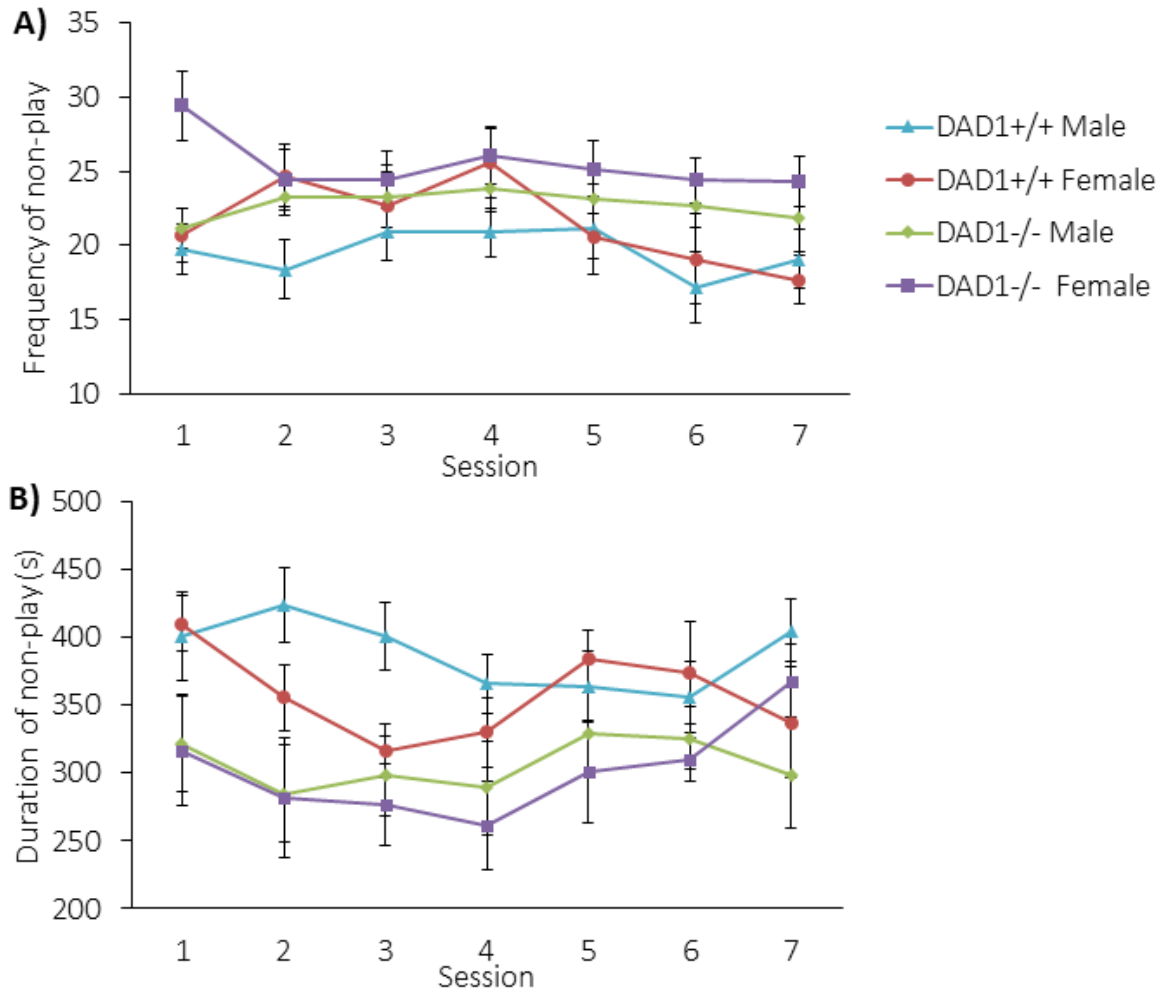
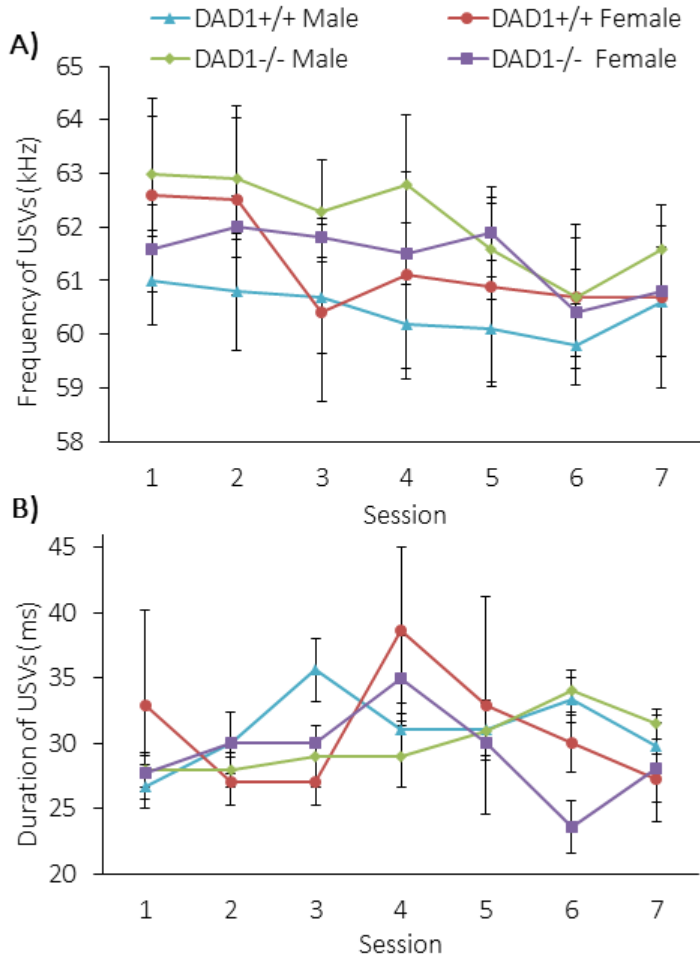


Figure A1.3. Average frequency (A) and duration (B) of passive social behaviour per session. Subjects' frequency of non-play behaviour varied across sessions. Error bars represent $\pm$ SD.

A1.4. 50kHz Vocalisations During Social Play

The average duration of 50kHz vocalisations remained stable across sessions (session effect: $F_{2.09, 62.59} = 1.67$, $p = 0.20$, $\eta_p^2 = 0.05$) while the average frequency (kHz) varied across sessions (session effect: $F_{3.06, 91.88} = 3.46$, $p < 0.05$, $\eta_p^2 = 0.10$), although it is important to note the difference is small (1-2 kHz) relative to the overall range of 50kHz vocalisations (50-80 kHz) ($p < 0.05$ for all post hoc tests; see figure A1.4).



A1.4. Average frequency and duration of 50 kHz vocalisations emitted per play session. The frequency (A) of vocalisations was greater during sessions 1 and 2 relative to 5 and 6. The duration (B) of vocalisations remained stable across sessions. Error bars represent $\pm$ SEM.

A2. Anticipation of Social Play

A2.1. Unsupported Rearing

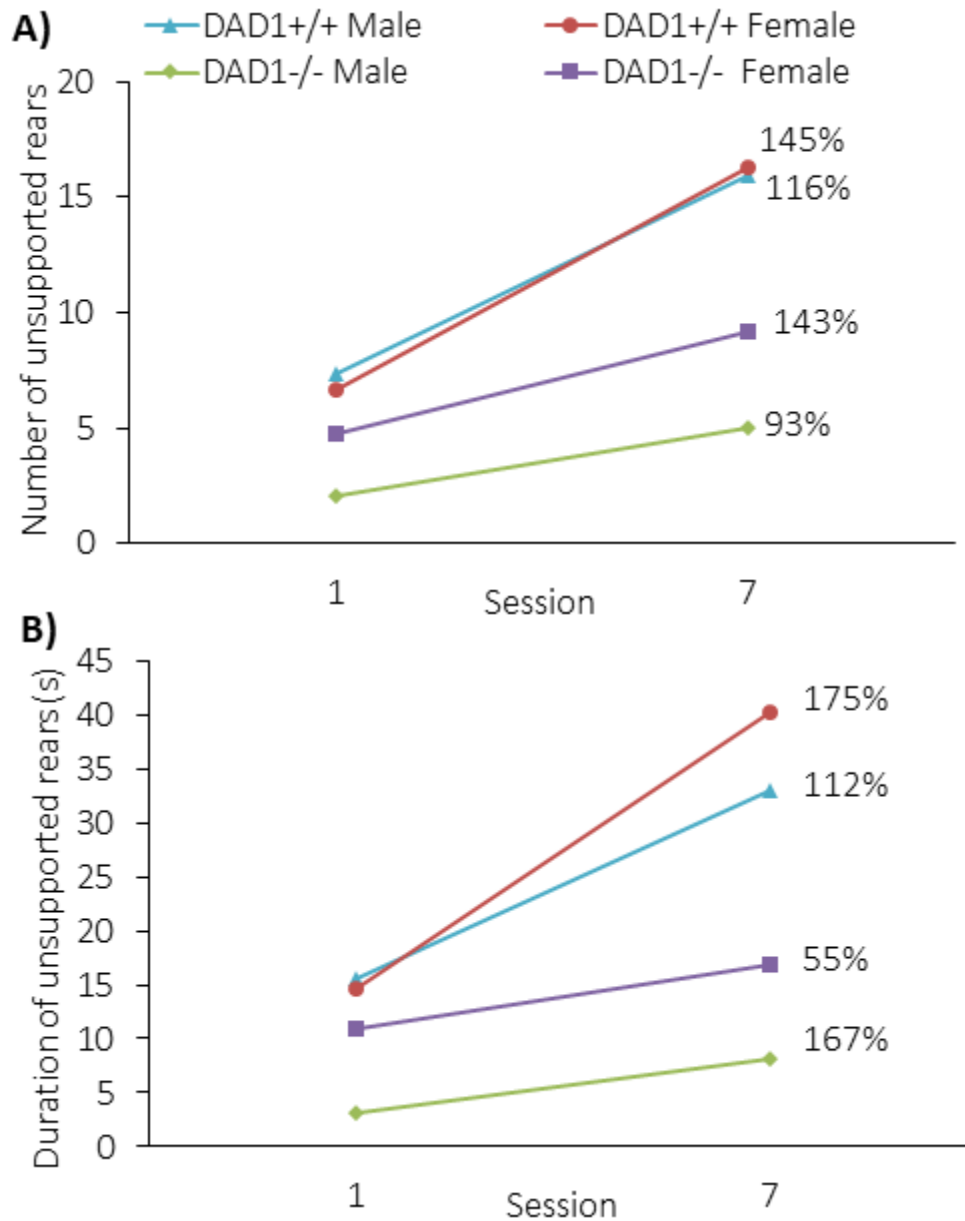


Figure A2.1. Percentage change in frequency (A) and duration (B) of unsupported rears from session 1 to 7. There were no significant sex or genotype differences in percentage change.

*denotes significance to $p < .05$.

A2.2. Supported Rearing

DAD1^{-/-} mutants had significantly lower frequency and duration of supported rears than wildtypes across sessions (frequency genotype effect: $F_{1,64} = 8.12$, $p < 0.01$, $\eta_p^2 = 0.11$; duration genotype effect: $F_{1,66} = 14.76$, $p < 0.001$, $\eta_p^2 = 0.18$; see figure A2.2). Females had a significantly longer duration of supported rears ($F_{1,66} = 4.68$, $p < 0.05$, $\eta_p^2 = 0.07$) than males but did not differ in frequency ($F_{1,64} = 0.96$, $p = 0.33$, $\eta_p^2 = 0.02$; see figure A2.3).

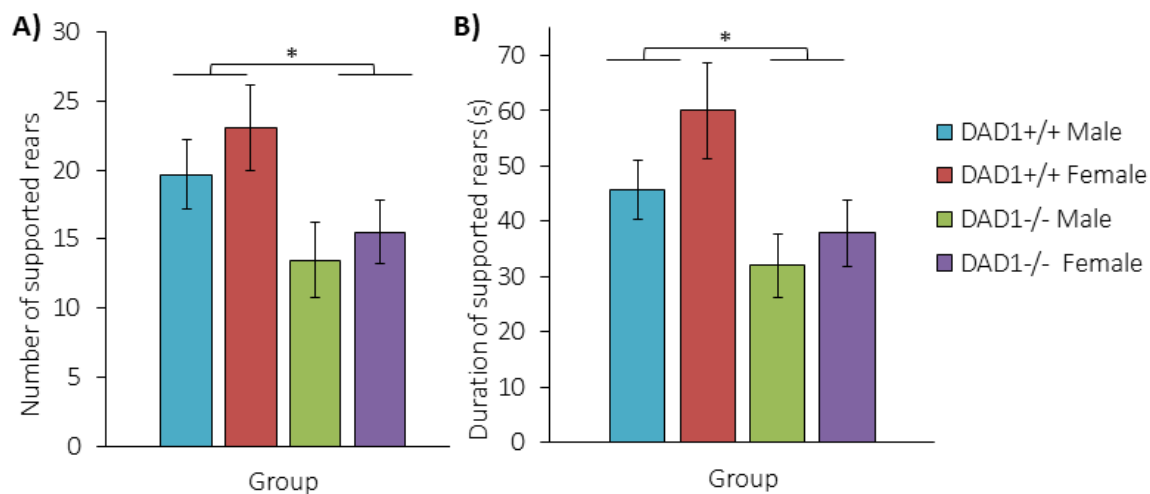


Figure A2.2. Average frequency and duration supported rears across anticipatory sessions. (A) DAD1^{-/-} mutants had fewer supported rears and a smaller duration (B) of unsupported rears than wildtypes. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

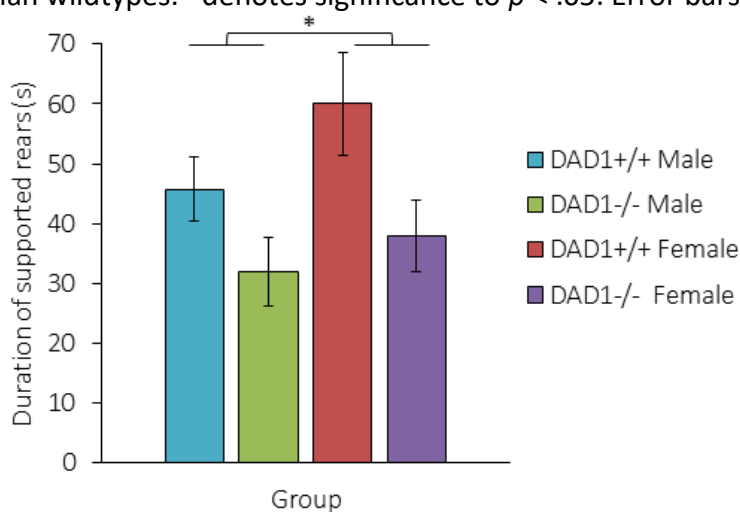


Figure A2.3. Average duration of supported rears across anticipatory sessions. Females had a longer duration of supported rears than males. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

Males significantly increased in frequency ($F_{5.36, 342.96} = 2.80, p < 0.05, \eta_p^2 = 0.04$) and duration ($F_{3.68, 243.02} = 3.74, p < 0.01, \eta_p^2 = 0.05$) of supported rearing across sessions. Females, on the other hand, only significantly increased their duration of supported rearing during session 7 compared to 4 ($p < 0.05$) and their frequency of supported rears did not significantly differ across sessions (all post hoc tests $p > 0.05$; see figure A2.4). There was no significant session or genotype interaction for supported rears (frequency session * genotype: $F_{1,35} = 1.95, p = 0.10, \eta_p^2 = 0.03$; duration session * genotype: $F_{3.68, 243.02} = 0.57, p = 0.70, \eta_p^2 = 0.01$).

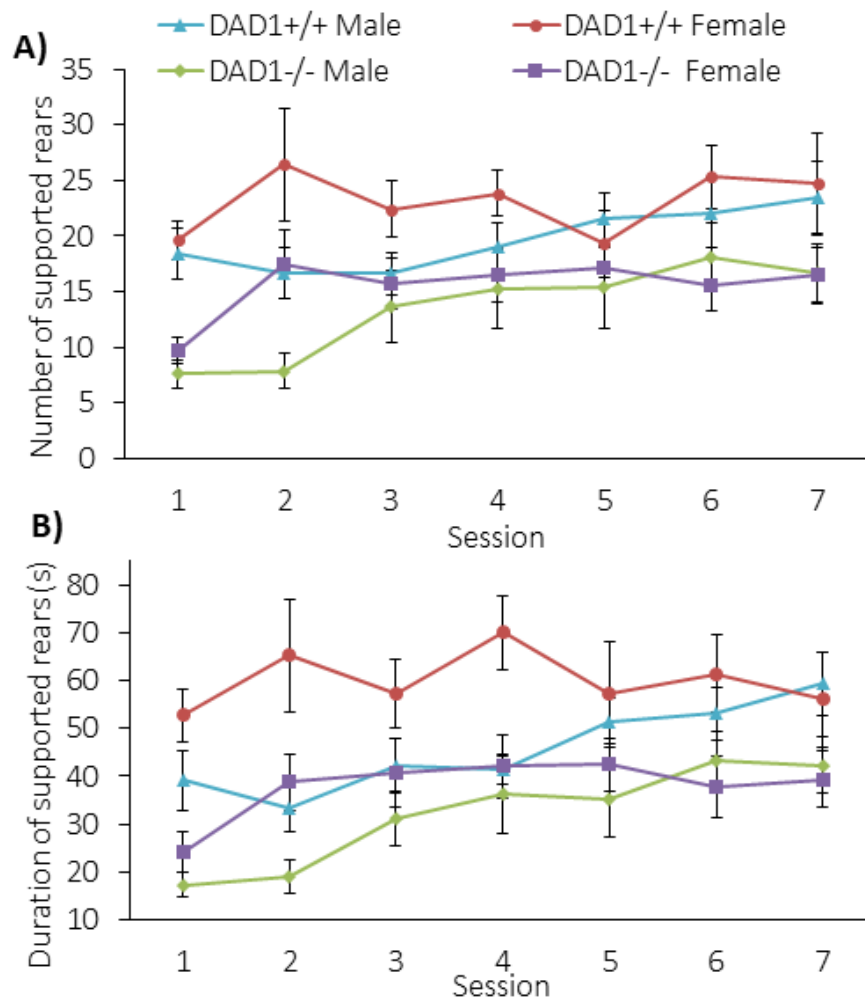


Figure A2.4. Frequency and duration of supported rears per anticipatory session. Males increased in frequency (A) and duration (B) of supported rears across sessions. Error bars represent $\pm$ SEM.

A2.3. Total Rearing

DAD1^{-/-} mutants had a significantly lower frequency ($F_{1,65} = 17.80, p < 0.001, \eta_p^2 = 0.21$) and duration ($F_{1,66} = 28.33, p < 0.001, \eta_p^2 = 0.30$) of total rears relative to wildtype rats across sessions (see figure A2.5). There was no sex effect for frequency or duration of total rears (frequency sex effect: $F_{1,66} = 1.35, p = 0.23, \eta_p^2 = 0.02$; duration sex effect: $F_{1,66} = 3.65, p = 0.07, \eta_p^2 = 0.05$).

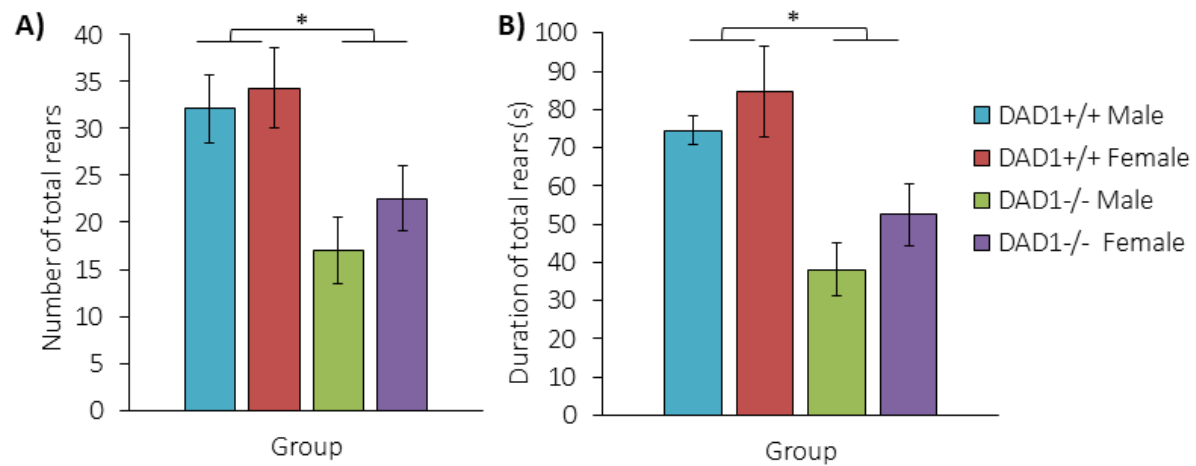


Figure A2.5. Average frequency and duration total rears across anticipatory sessions. (A) DAD1^{-/-} mutants had fewer rears than wildtypes. (B) DAD1^{-/-} mutants had a smaller duration of rears than wildtypes. *denotes significance to $p < .05$. Error bars represent $\pm$ SEM.

Subjects significantly increased in frequency ($F_{5, 357.17} = 11.47, p < 0.001, \eta_p^2 = 0.15$) and duration ($F_{4.13, 272.48} = 10.59, p < 0.001, \eta_p^2 = 0.14$) of total rears across sessions (see figure A2.6). There were no significant interactions between genotype and session for total rears (frequency session * genotype: $F_{5, 357.17} = 0.49, p = 0.74, \eta_p^2 = 0.007$; duration session * genotype: $F_{4.13, 272.48} = 0.29, p = 0.89, \eta_p^2 = 0.004$).

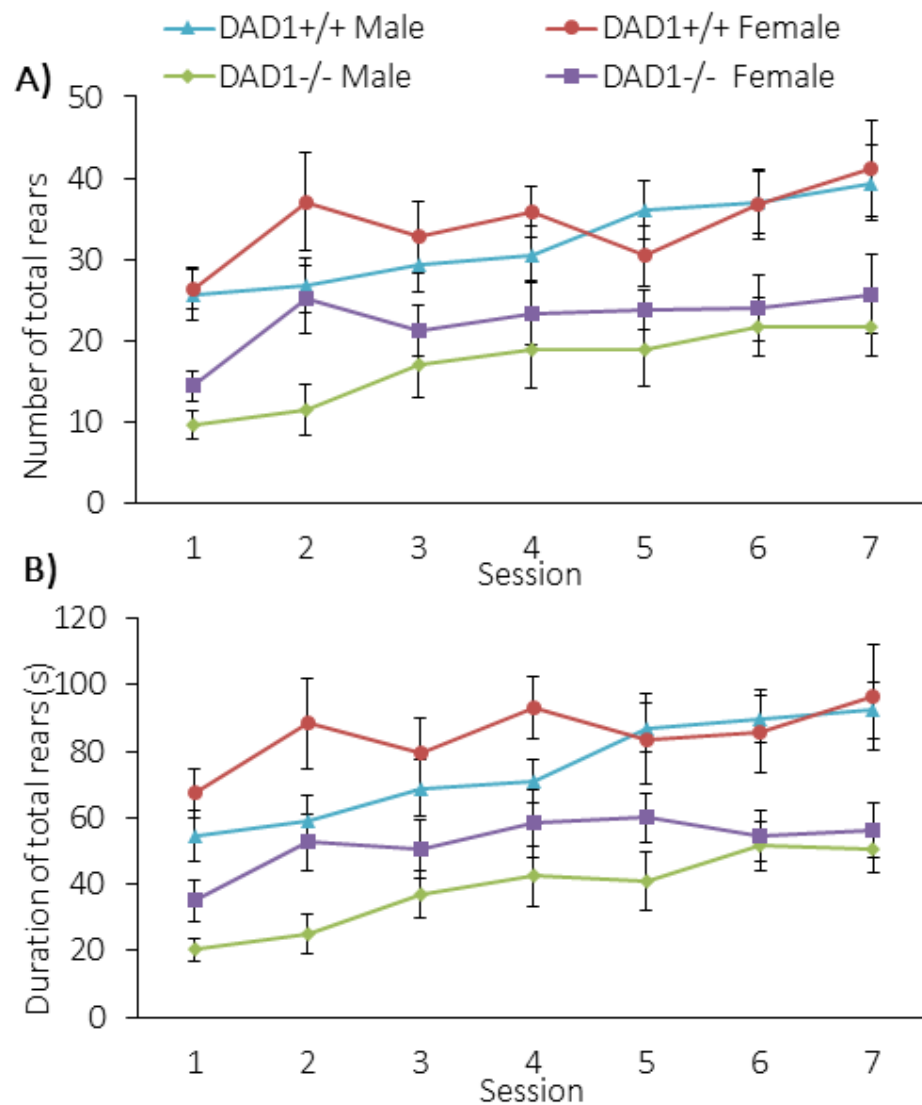


Figure A2.6. Frequency and duration of total rears per anticipatory session. Subjects increased in frequency (A) and duration (B) of total rears across sessions. Error bars represent $\pm$ SEM.

A2.4. 50kHz Vocalisations

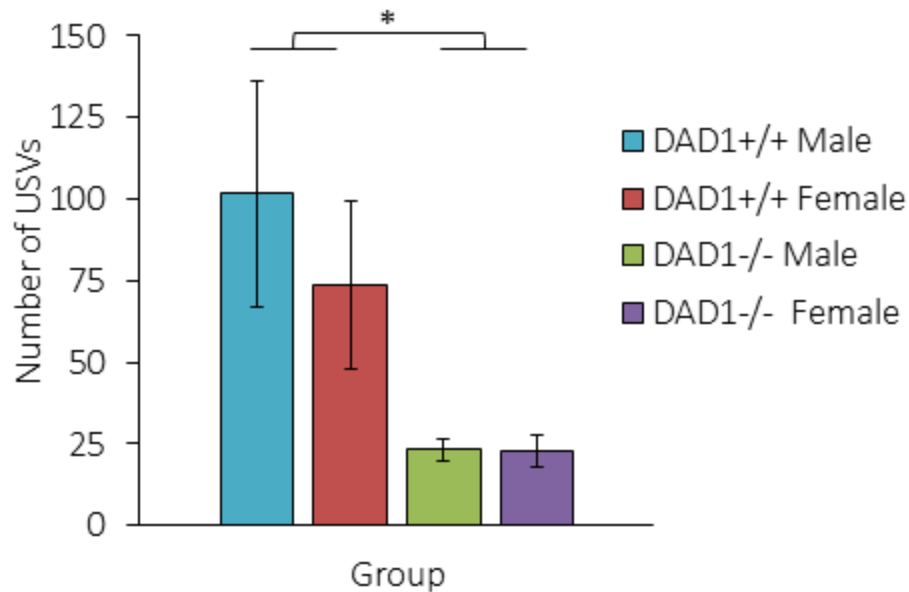


Figure A2.7. Average number of vocalisations emitted across anticipatory sessions. DAD1^{-/-} mutants emitted significantly fewer vocalisations than wildtypes. Error bars represent $\pm$ SEM. *denotes significance to $p < .05$.

The average *duration* of 50kHz vocalisations remained stable across anticipatory sessions (session effect: $F_{3.46, 58.57} = 1.20$, $p = 0.32$, $\eta_p^2 = 0.07$; see figure A2.8B). There was a significant effect of session for *frequency* ($F_{4.33, 73.58} = 2.88$, $p < 0.05$, $\eta_p^2 = 0.15$), however post hoc comparisons indicate that frequency did not significantly differ between sessions ($p > 0.05$ for all post hoc tests; see figure A2.8A).

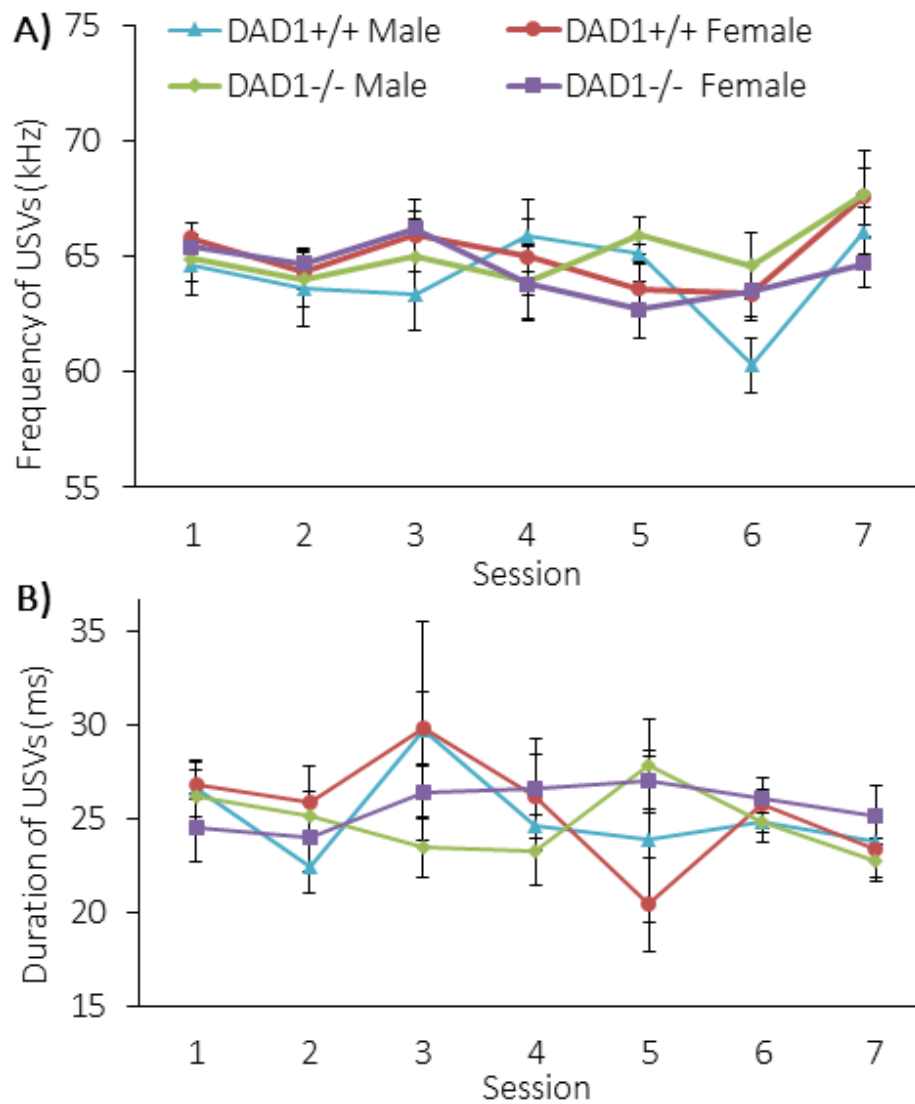


Figure A2.8. Average frequency and duration of 50 kHz vocalisations emitted per anticipatory session. The average frequency and duration of vocalisations remained stable across sessions. Error bars represent $\pm$ SEM.

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